TECHWISE

Smart E-Commerce & Product Intelligence Platform

System Analysis & Design

Comprehensive Working Draft

Team	Falcon Team
Program	Digital Egypt Pioneers Initiative (DEPI) — Round 5
Version	0.1
Date	5 November 2026
Target phase date	6 November 2026

Document Control

Field	Value
Project	Smart E-Commerce & Product Intelligence Platform
Team	Falcon Team
Program	Digital Egypt Pioneers Initiative (DEPI) — Round 5
Document title	System Analysis & Design
Version	0.1
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Document status	Comprehensive working draft for team/owner review; not approved or implemented
Physical schema	Deferred — DBMS not selected (OD-01)
Implementation	Not deployed; no tested code or approved UI prototype

This document is a formal project report. It presents the System Analysis & Design package together with detailed requirements, proposed API and UI/UX appendices, and the conceptual and logical database design. The component source documents retain their own dates and statuses; this volume integrates their substantive content into one cohesive technical report. Nothing in this document implies project-owner approval, implementation, deployment, testing, physical inspection, or institutional acceptance.

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1. Purpose, Problem, and Objectives

TechWise addresses fragmented shopping, technical product understanding, comparison, ownership lifecycle, and second-life resale workflows. The product direction is a smart e-commerce and product-intelligence platform that supports product discovery, evidence-aware product information, compatibility, purchase decision support, and lifecycle choices.

The authoritative problem statement, objectives, scope, and success criteria are maintained in Problem Statement & Objectives. This package does not replace that baseline. Its objectives are to:

- connect the requirements to system boundaries, actors, behaviors, and components;
- show where AI assistance ends and deterministic, authoritative validation begins;
- distinguish catalog products from physical product assets and ownership periods;
- document low-risk interface contracts and workflows without inventing transaction policy; and
- identify decisions that must be confirmed before implementation or physical deployment.

2. Scope and Design Status

2.1 In Scope

- Product discovery, filtering, comparison, product information, and compatibility assessment.
- Decision support for purchase and buy/upgrade/resell choices, with explicit limitations.
- Authorized product-asset lifecycle views and conceptually eligible second-life listing workflows.
- Traceable review, evidence, eligibility, and ownership concepts from the existing database-design baseline.
- Logical system design, proposed interfaces, and documentation-level validation scenarios.

2.2 Excluded or Unresolved

- Payment, escrow, refunds, returns, final sale confirmation, and complete dispute adjudication.
- Physical inspection or device-verification mechanisms not evidenced in the sources.
- Final category evidence matrices, condition rubric, eligibility authority, and appeal process.
- Final ownership-transfer trigger and the transaction semantics of Reserved and Sold.
- A selected DBMS, authentication provider, AI/search provider, cloud host, or deployment topology.
- Numeric service-level, response-time, availability, recovery, or retention commitments.

2.3 Status Vocabulary

Label	Meaning
Source baseline	Present in a cited requirements or design document; the source's own draft/working status still applies.
Proposed	A design choice for review; not approved or implemented.
Deferred	A decision is explicitly not made in the current baseline.
Out of scope	The current baseline excludes the capability unless the owner approves a scope change.

3. Actors and Use Cases

3.1 Actors

Actor	Goal / boundary
Shopper	Discover, inspect, compare, assess compatibility, and receive decision support for products.
Authorized product owner	View permitted information for an owned physical Product Asset and its lifecycle.
Current owner acting as seller	Submit a potential second-life listing and requested evidence, subject to eligibility rules.
Prospective second-life buyer	Review listing details, evidence summaries, condition, and stated status before choosing whether to proceed.
Authorized reviewer or administrator	Perform only the review or administrative actions granted by an approved policy; detailed authority is unresolved.
External AI capability (optional)	May help interpret a natural-language request or explain results; it is not authoritative and its provider is unselected.

3.2 Use-Case Diagram

The diagram uses Mermaid actor labels and oval use-case nodes; plain connectors mean participation, not `include` or `extend` relationships.

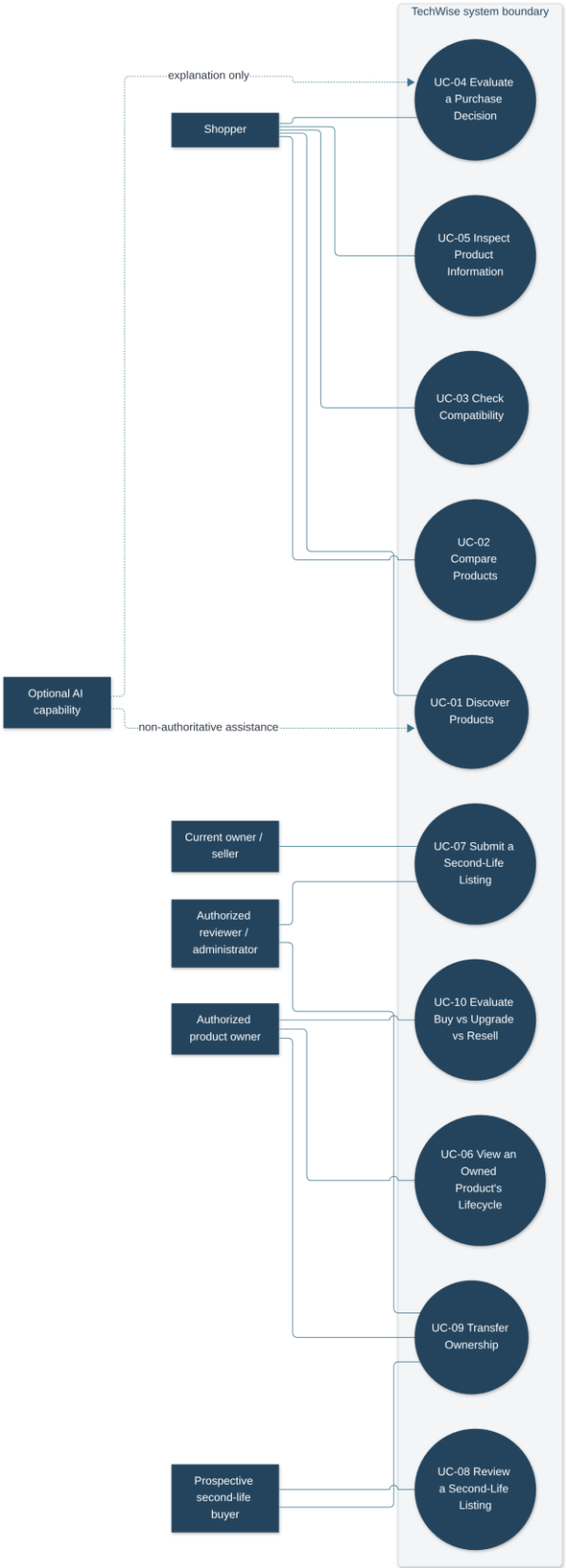


Figure 1. 3.2 Use-Case Diagram

3.3 Use-Case Descriptions

These summaries preserve the source actors, intent, key flow, and main exception. The Requirements Specification remains the detailed source for preconditions, alternatives, and postconditions.

ID / use case	Actor	Main flow and outcome	Important alternative / constraint	Traceability
UC-01 Discover Products	Shopper	Submit a query or supported filters; receive candidates and open a product.	Empty results or missing information must be explained; optional criteria may be relaxed.	FR-AI, FR-SF; AT-01, AT-02
UC-02 Compare Products	Shopper	Select at least two candidates; inspect comparable attributes and differences.	Unavailable values stay unavailable; never invent comparison data.	FR-PC; AT-02
UC-03 Check Compatibility	Shopper	Select items; evaluate available specifications and applicable compatibility rules; inspect rationale.	Missing specifications yield Unknown/insufficient data; failed mandatory constraints are identified.	FR-CG; AT-04
UC-04 Evaluate a Purchase Decision	Shopper	Provide requirements; validate criteria; evaluate candidates; review rankings, trade-offs, and limits.	Conflicts or missing inputs require clarification; a candidate violating a mandatory constraint is not labeled a full match.	FR-PD; AT-03, AT-09
UC-05 Inspect Product Information	Shopper	Open product details and review specifications, facts, sources, evidence, or confidence where available.	Conflicting, incomplete, seller-reported, or unknown information remains distinguishable.	FR-PT; AT-06
UC-06 View an Owned Product's Lifecycle	Authorized product owner	Pass authorization; view permitted Product Asset, ownership period, and lifecycle events.	Deny unauthorized requests without revealing another user's private information.	FR-PL, FR-AC; AT-08
UC-07 Submit a Second-Life Listing	Current owner / seller	Select an asset; evaluate initial eligibility; provide listing information and required evidence for review.	Ownership/origin failure or blocking conditions prevent or block progression; missing evidence is requested. Submission is not	FR-SL; AT-05, AT-06

			physical inspection.	
UC-08 Review a Second-Life Listing	Prospective second-life buyer	Inspect product identity, condition information, price, available evidence, and listing status.	Missing or unverified information must be disclosed; purchase workflow semantics are not defined here.	FR-SL, FR-PT; AT-06
UC-09 Transfer Ownership	Authorized current and eligible new owners	Validate transfer conditions; preserve the former period; record a new period for the same asset.	Failed validation blocks transfer. The effective trigger and required transaction conditions remain unresolved.	FR-PL, FR-AC; AT-07, AT-08
UC-10 Evaluate Buy vs Upgrade vs Resell	Product owner	Gather available condition, cost, and requirements; compare applicable options and assumptions.	Missing required inputs limit the result; do not present unsupported calculations as reliable.	FR-BUR; AT-10

4. Requirements Traceability

4.1 Functional Requirement Groups

Source requirement group	System capability in this package	Related use cases
FR-AI — AI Shopping Intelligence	Request interpretation and explanations remain subordinate to authoritative records and deterministic rules.	UC-01, UC-04
FR-SF — Smart Filtering	Search and filtering use available, typed criteria and disclose missing inputs.	UC-01
FR-PC — Product Comparison	Present comparable attributes, differences, and missing values.	UC-02
FR-CG — Product Compatibility Graph	Evaluate known rules and preserve Unknown/insufficient-data outcomes.	UC-03
FR-PD — Purchase Decision Engine	Validate constraints and explain rankings, trade-offs, and limits.	UC-04
FR-PT — Product Truth Layer	Keep product facts, source, seller claims, evidence, assessment, and confidence distinct.	UC-05, UC-08
FR-SL — Verified Second-Life Marketplace	Assess listing eligibility and present listing/evidence status transparently.	UC-07, UC-08
FR-PL — Product Lifecycle and Ownership	Preserve asset identity and historical ownership periods.	UC-06, UC-09
FR-BUR — Buy vs Upgrade vs Resell Engine	Compare available lifecycle choices from recorded inputs and disclosed assumptions.	UC-10

FR-AC — Access Control and Security	Protect owner-only and reviewer-only information; enforce authorization at the trusted boundary.	UC-06, UC-07, UC-09

4.2 Non-Functional Requirement Groups

Quality area in requirements baseline	Design response	Validation evidence
Performance	Numeric targets are not set in the source; measure discovery, filtering, comparison, and decision flows once scope and environment are confirmed.	AT-01, AT-02, AT-03; later measured performance tests
Security and privacy	Authorize private asset/evidence access; minimize exposed prior-owner and sensitive evidence data; validate untrusted external AI/provider output.	AT-08; later authorization/privacy tests
Reliability and data integrity	Keep ownership periods non-overlapping; prevent invalid state transitions and conflicting concurrent lifecycle changes.	AT-07, AT-08; later integration tests
Usability and accessibility	Responsive desktop/mobile layouts, clear missing/unknown/error states, keyboard and assistive-technology support.	UI review and accessibility tests; thresholds require confirmation
Maintainability	Separate UI, API, deterministic business logic, AI assistance, services, data access, and persistence.	Component boundaries, unit and integration test plan
Data quality and explainability	Show source and confidence context; preserve fact/claim/evidence/assessment distinctions.	AT-06, AT-09, AT-10
Scalability and availability	No provider or numeric commitment is selected; establish measurable targets after deployment decisions.	Deployment and operations review

5. Acceptance Scenario Coverage

The requirements source defines ten acceptance scenarios. This mapping is a test-planning aid, not evidence that tests have run.

ID	Scenario to validate	Main design area
AT-01	Empty search returns an informative empty state.	Discovery / DFD / UC-01
AT-02	A missing product specification is identified as unavailable rather than invented in discovery or comparison.	Product facts / comparison / UC-01–UC-02

AT-03	Purchase decision support applies mandatory constraints and explains the result.	Decision engine / UC-04
AT-04	Compatibility distinguishes a failed constraint from unknown due to missing specifications.	Compatibility / UC-03
AT-05	Second-life eligibility responds to required ownership, origin, evidence, and blocking inputs.	Eligibility / UC-07
AT-06	Seller claims, submitted evidence, assessment, and unverified information remain distinct.	Product truth / marketplace / UC-05, UC-07–UC-08
AT-07	Ownership history remains continuous and tied to the same physical asset across transfer.	Lifecycle / UC-09
AT-08	Unauthorized access does not disclose private ownership or evidence information.	Authorization / UC-06, UC-09
AT-09	AI output cannot override a conflicting authoritative record or deterministic decision.	AI boundary / UC-04
AT-10	Missing cost or other required decision input limits the buy/upgrade/resell conclusion.	Lifecycle decision / UC-10

6. Data Flow and System Boundaries

The following DFDs are logical data-flow views. Stores are conceptual data groups from the existing model, not claims about physical databases. The optional AI provider is an untrusted external integration; deterministic business rules remain inside the trusted application boundary.

External entities are rectangular, processes are rounded, conceptual data stores are cylinders, and labeled arrows identify flows. These views use Mermaid shapes to communicate DFD concepts; they do not prescribe implementation storage.

6.1 Context-Level DFD

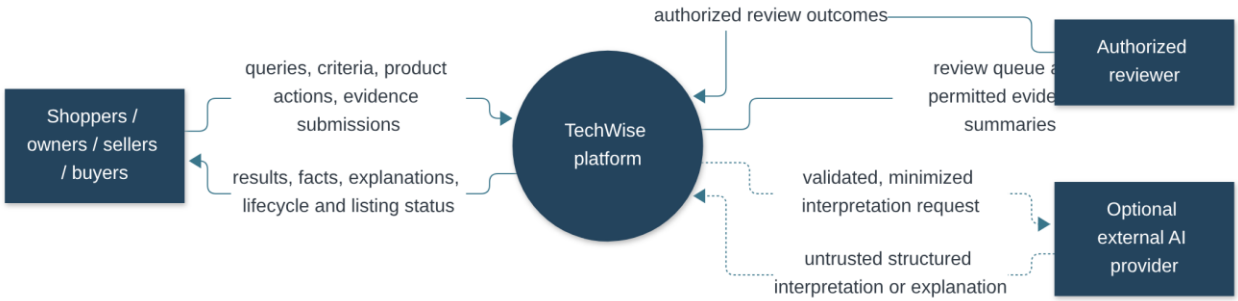


Figure 2. 6.1 Context-Level DFD

6.2 Level-1 DFD

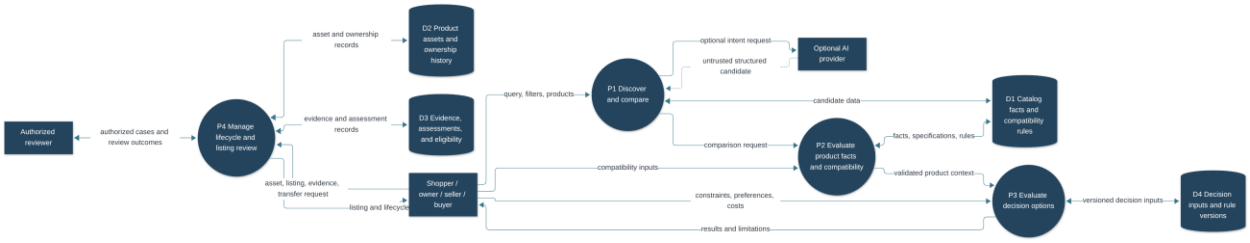


Figure 3. 6.2 Level-1 DFD

6.3 Data-Flow Controls

- Every client-supplied value is validated at the API boundary before it can affect business decisions.
- AI output is parsed and treated as untrusted input. It may suggest structured criteria or explanations but cannot directly write authoritative product, ownership, compatibility, eligibility, or listing state.
- Product facts retain provenance. Seller claims, submitted evidence, reviewer assessment, and confidence are separate information types.
- Ownership history is private; only authorized users receive permitted details. Public listing views expose approved summaries only.
- Evidence storage location, retention period, search infrastructure, and provider integrations remain undecided.

7. Software Architecture

7.1 Proposed Layered Architecture

Architecture style: layered application architecture, with a separate optional AI adapter and a deterministic domain layer.

Layer / component	Responsibility	Authority boundary
User interface	Responsive product discovery, comparison, lifecycle and listing screens; captures input and presents state.	Never authoritative for eligibility, access control, ownership, or compatibility.
API / application boundary	Accepts requests, validates external input, authorizes access, maps requests to application operations, and returns consistent responses.	Trusted boundary; does not accept AI output as authorization or policy.
Deterministic domain logic	Product/compatibility evaluation, decision constraints, ownership/lifecycle invariants, listing eligibility and state transitions.	Authoritative for business outcomes, subject to approved rules.
Application services	Orchestrate discovery, decision support, product truth, evidence review, and lifecycle workflows.	Coordinates domain logic; records which rule/data versions informed results where applicable.
AI assistance adapter (optional)	Requests structured interpretation or explanation from a provider.	Provider is unselected; output is untrusted and must be validated. No direct state writes.
Data-access boundary	Loads and persists approved logical records through a selected	DBMS, ORM, and deployment

	persistence technology.	behavior are deferred.
Logical persistence	Product/catalogue facts, assets/ownership, evidence/assessments, and decision context.	Existing logical model is the source; physical implementation is not established.

7.2 Component Diagram

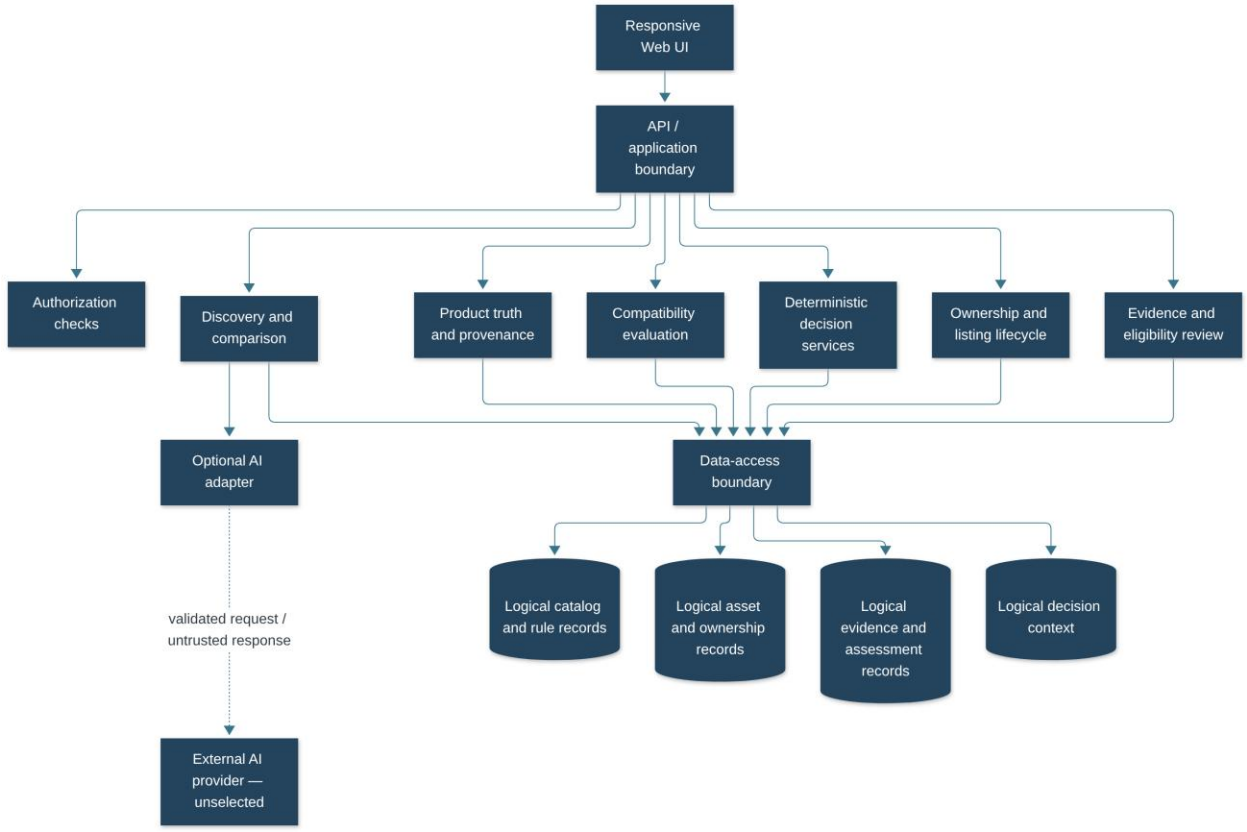


Figure 4. 7.2 Component Diagram

7.3 Proposed Deployment View

This is a logical deployment view for discussion, not a selected topology or infrastructure diagram. Runtime nodes, network controls, hosting, identity provider, database product, and evidence storage must be confirmed before deployment design.

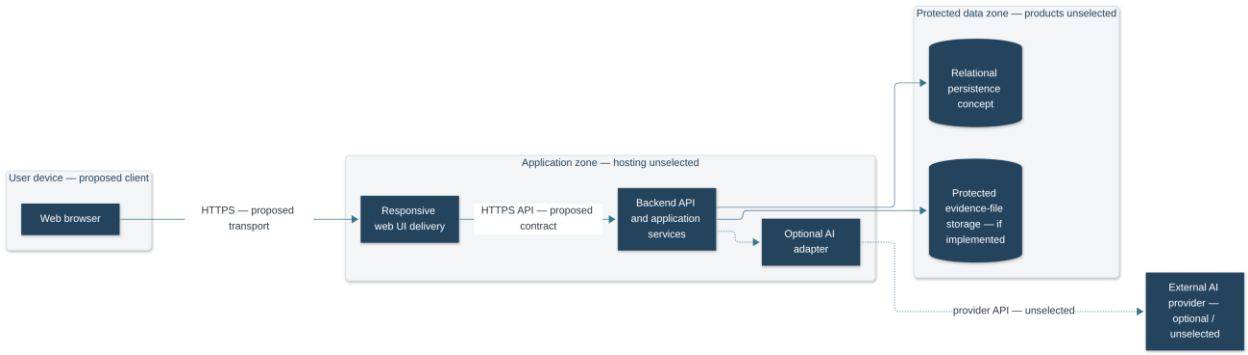


Figure 5. 7.3 Proposed Deployment View

No trust boundary here replaces authentication, authorization, privacy, or deployment threat analysis. The file store is conditional because file-storage product and evidence upload implementation are not confirmed.

7.4 Proposed Deployment Strategy

No hosting platform, CI/CD pipeline, execution environment, operational monitoring, backup policy, or scaling target was found in the inspected project sources. The following is a technology-neutral starting strategy for owner/team review, not a deployed or approved setup.

Stage	Proposed control	Decision still required
Build and verify	Build a versioned artifact; run static checks, unit tests, integration tests, and dependency checks when implementation exists.	Source repository workflow, tools, branch protection, and quality thresholds.
Development / test	Use isolated environments and synthetic or approved test data; keep credentials outside source control.	Hosting, identity, test data, external AI sandbox, and secrets-management choices.
Pre-release	Deploy the same versioned artifact to a representative staging environment; run migrations and smoke/acceptance checks only after DBMS and rollback behavior are approved.	Environment parity, schema migration policy, reviewer/approver, and release evidence.
Production release	Require an authorized release approval, health checks, monitoring, backup/recovery procedures, and a documented rollback path.	Production host, availability/recovery targets, data-retention controls, incident ownership.
Scaling	Measure workload first; consider scaling stateless UI/API instances and database capacity according to observed bottlenecks.	Workload assumptions, horizontal-scaling support, search/storage products, budgets, and numeric thresholds.

This design does not select a cloud provider or assert that CI/CD, backups, scaling, or monitoring are configured.

7.5 Technology Stack Status

Area	Proposal source	Current status used here
Frontend	React + TypeScript	Proposed in the draft proposal; confirmation required.
Backend	ASP.NET Core Web API + C#	Proposed in the draft proposal; confirmation required.
Database	Microsoft SQL Server appears in the proposal	**Not selected for this design.** OD-01 explicitly defers DBMS selection; use technology-neutral logical models.
AI	AI provider API with structured outputs	Capability is optional; provider, model, privacy terms, and integration are unselected.

Search	Search/filter capability appears in requirements	Search engine and indexing architecture are unselected.
Hosting/deployment	Cloud hosting is mentioned generically	Provider, topology, environments, network controls, and operations are not confirmed.
Testing	Testing tools are unspecified	Test levels and evidence are proposed below; toolchain and numeric thresholds require team confirmation.

8. Sequence Diagrams

These are representative logical interactions. They do not define final routes, payload schemas, provider behavior, or implementation classes.

8.1 Product Discovery and Decision Support

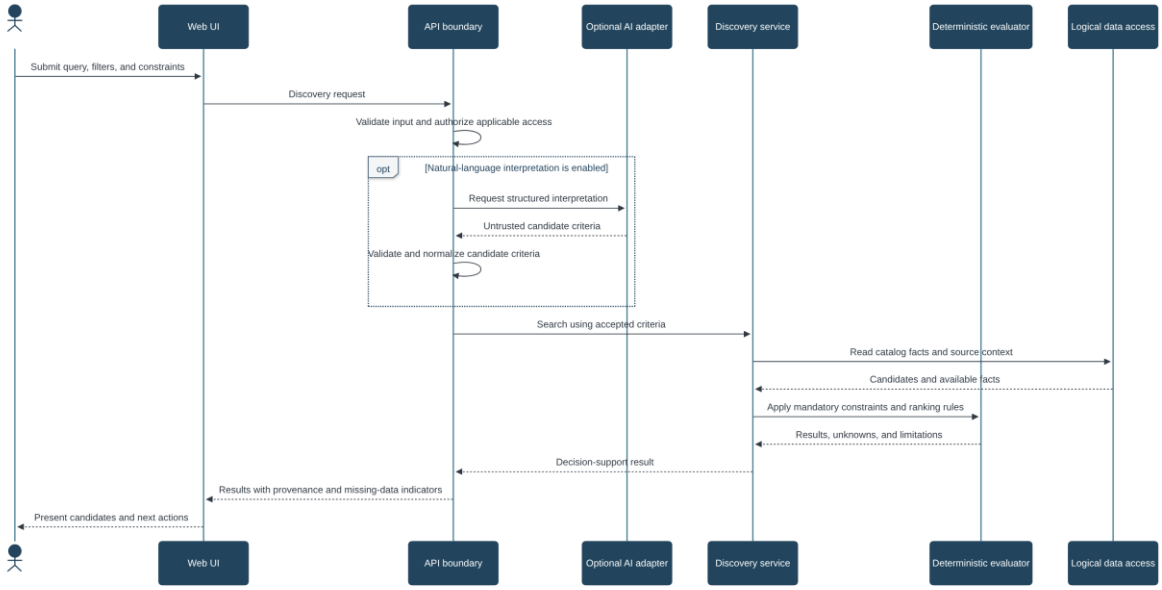


Figure 6. 8.1 Product Discovery and Decision Support

8.2 Compatibility Assessment

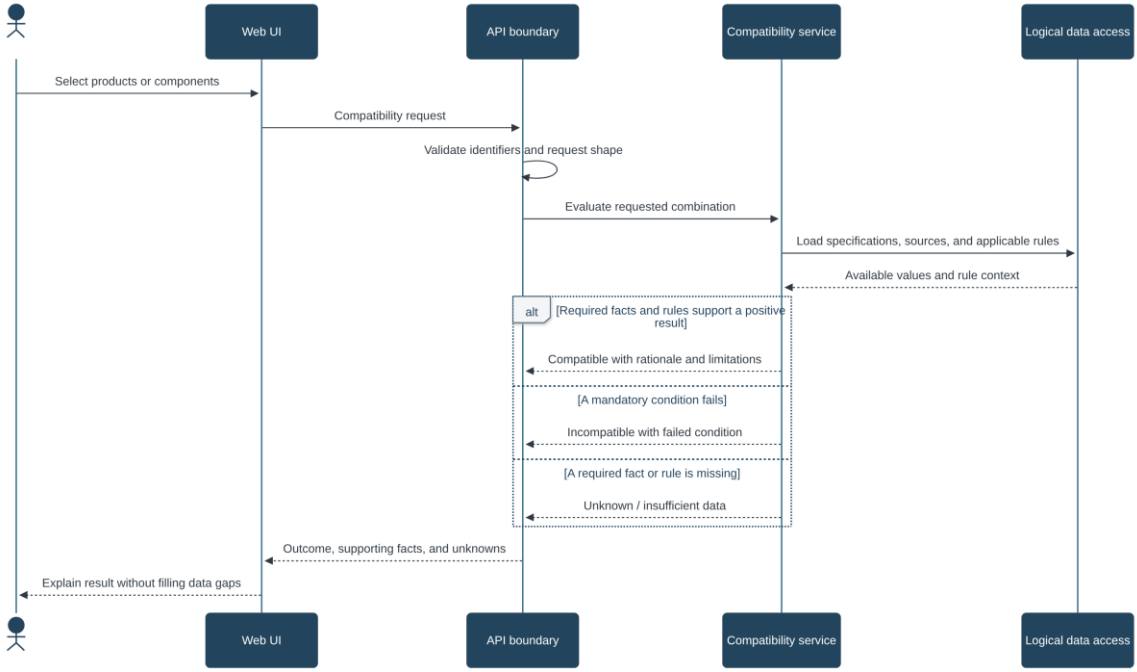


Figure 7. 8.2 Compatibility Assessment

8.3 Second-Life Submission and Review

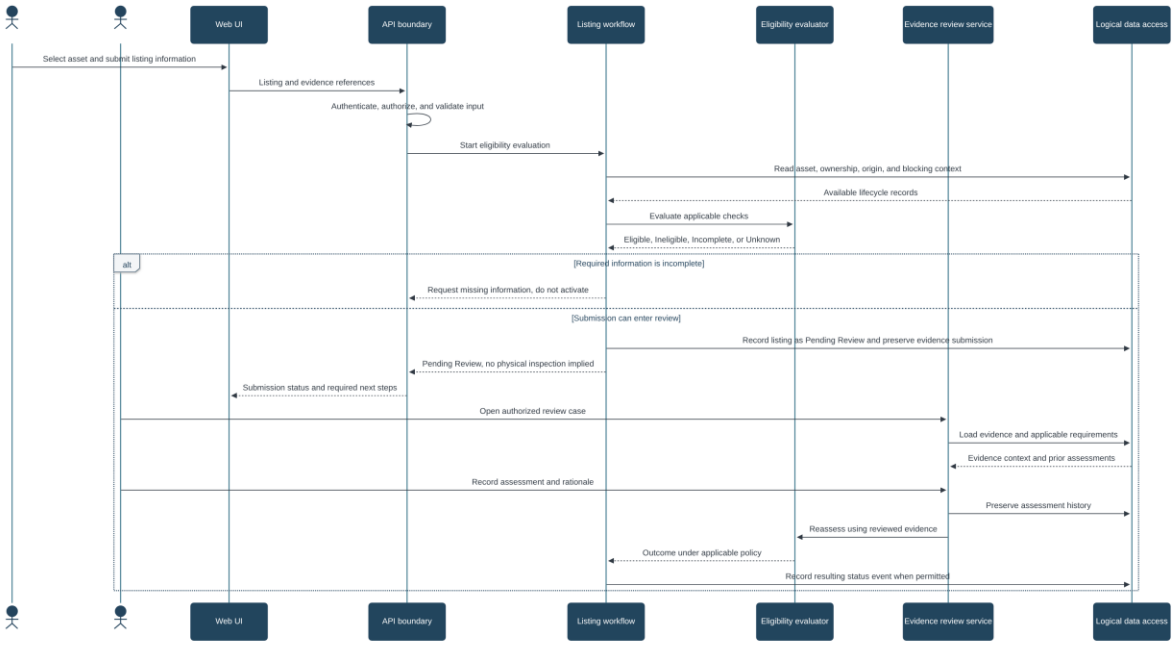


Figure 8. 8.3 Second-Life Submission and Review

Review authority, category evidence policy, and activation rules require owner confirmation. Evidence acceptance alone does not establish listing eligibility.

8.4 Ownership Transfer

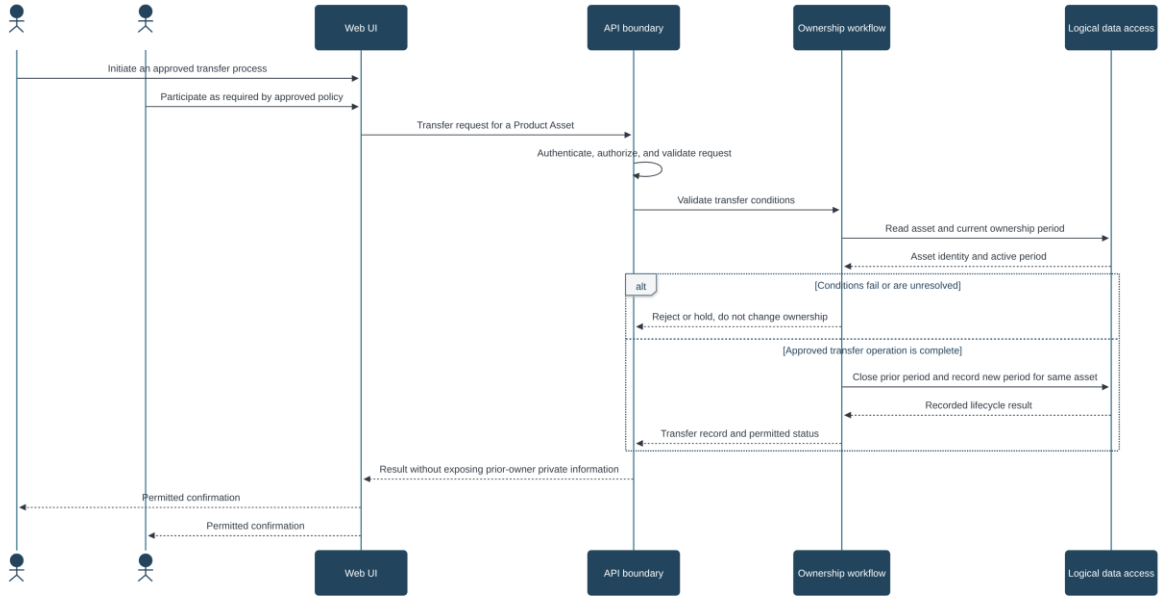


Figure 9. 8.4 Ownership Transfer

The exact event that makes the transfer effective is unresolved. The sequence does not imply payment, escrow, a sale-completion rule, or automatic transfer on a `Sold` listing.

9. Activity Diagram — Second-Life Listing Evaluation

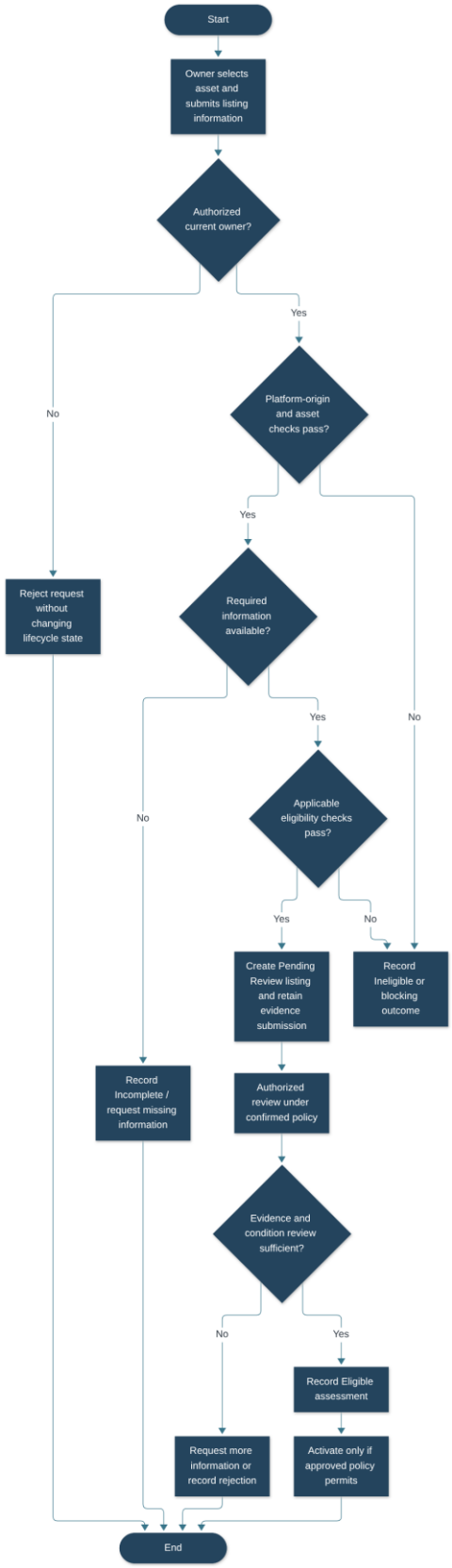


Figure 10. 9. Activity Diagram — Second-Life Listing Evaluation

Eligible does not mean physically inspected. The exact evidence matrix, reviewer authority, condition rubric, and policy for activation are unresolved. The diagram shows a review concept; it does not assert that every submission will be automatically published.

10. State Diagrams

State names follow the conceptual database baseline. State transitions are conceptual and do not define approval authority or transaction completion.

10.1 Resale Listing

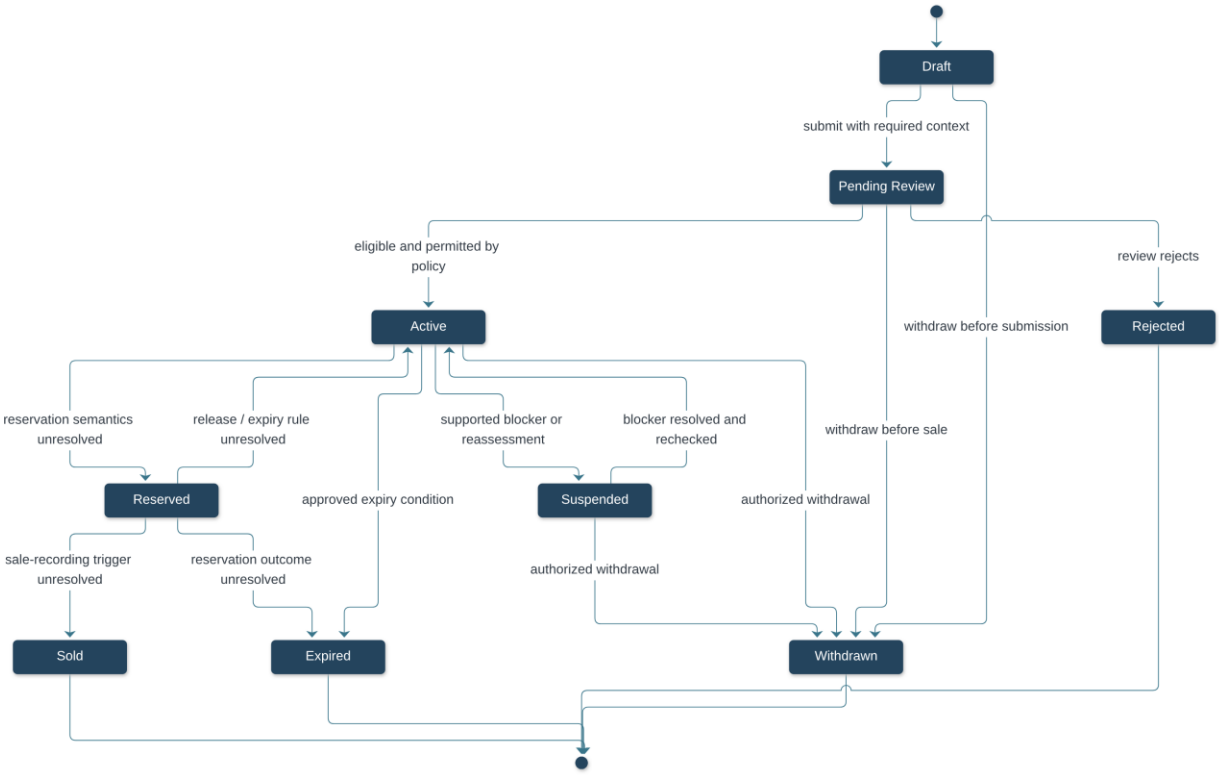


Figure 11. 10.1 Resale Listing

`Sold` does not prove payment or ownership transfer. No `Sold`-to-ownership transition is implied.

10.2 Evidence Submission and Assessment

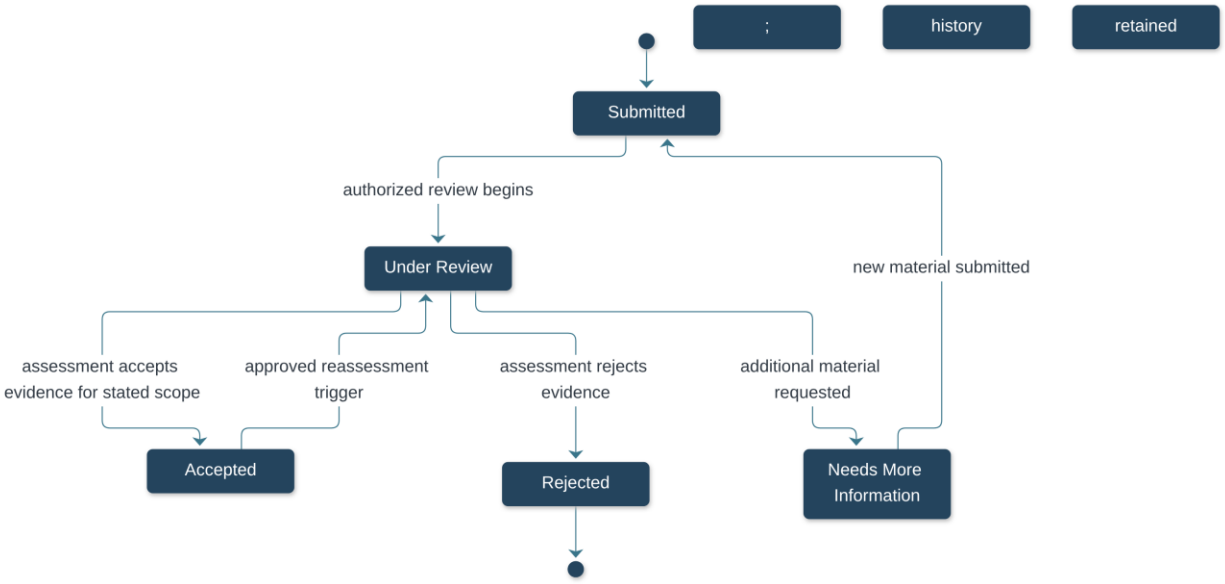


Figure 12. 10.2 Evidence Submission and Assessment

Evidence acceptance is scoped to an assessment; it does not automatically make a listing eligible. A resubmission preserves prior submission and assessment history.

10.3 Listing Eligibility Assessment

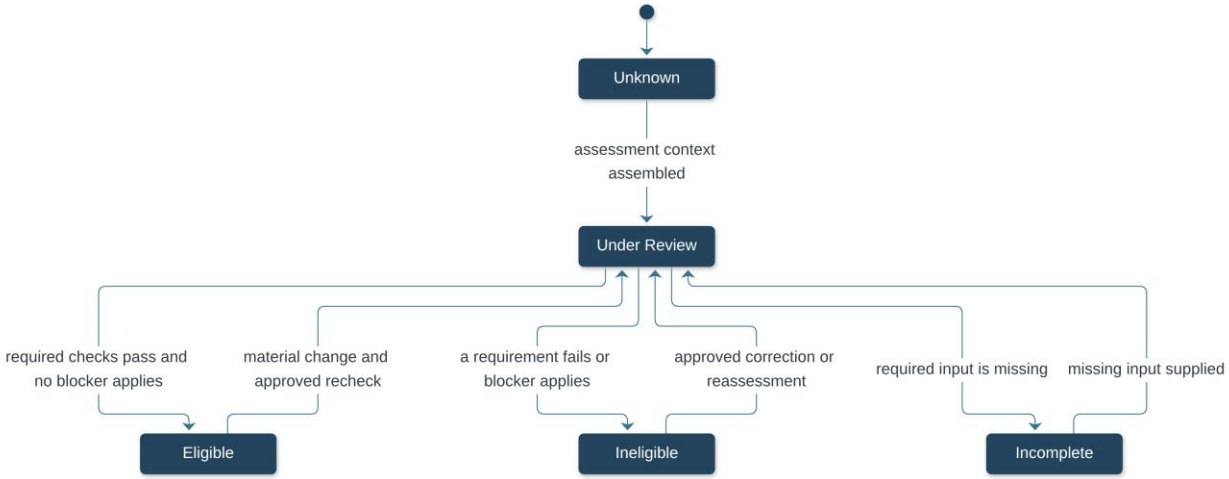


Figure 13. 10.3 Listing Eligibility Assessment

Eligibility is a separate determination from evidence acceptance, ownership state, and listing state. Policy, evidence matrix, condition rubric, and reviewer authority remain open.

10.4 Ownership Period

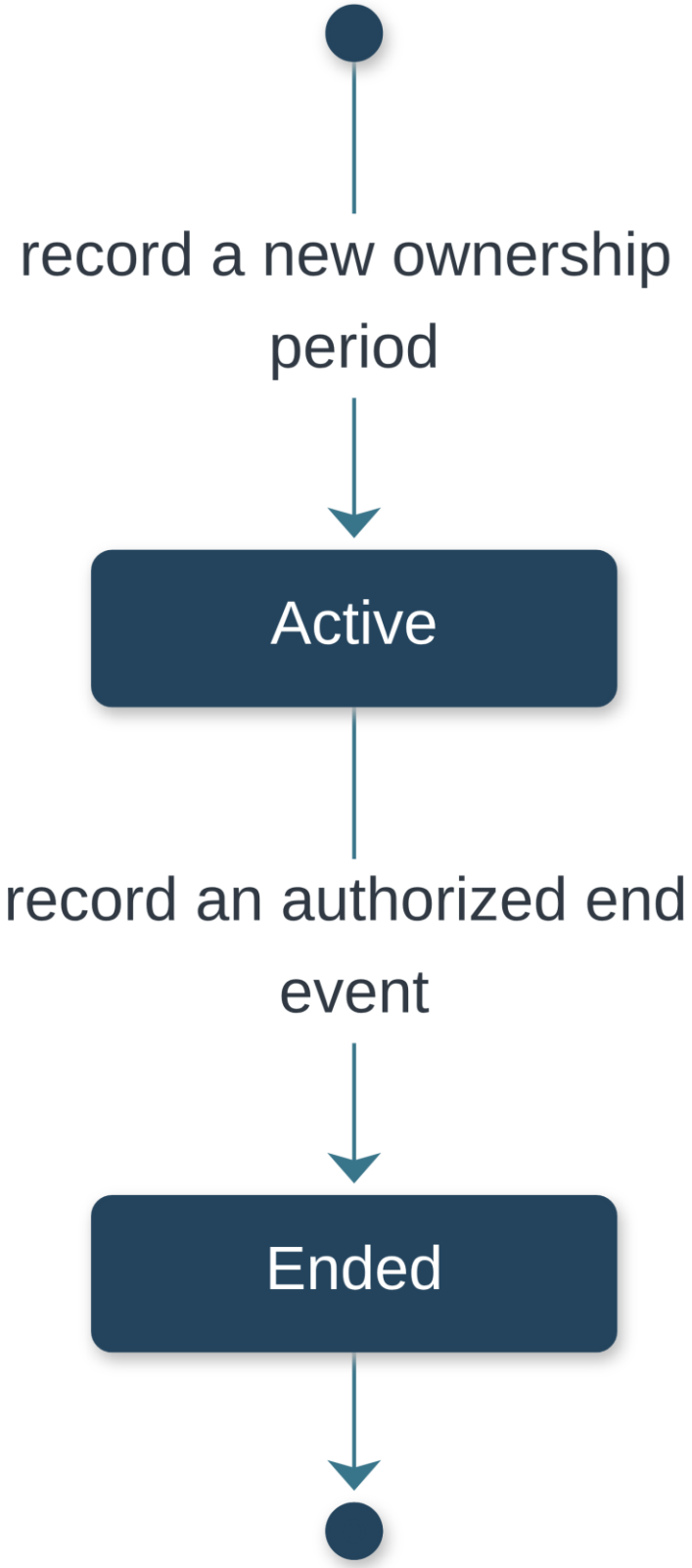


Figure 14. 10.4 Ownership Period

A transfer closes the former period and creates a **new** Active period for the **same Product Asset**. An Ended period is not reopened. The effective transfer trigger remains unresolved; a `Sold` listing alone does not change ownership.

11. Conceptual Class Diagram

This class-style diagram presents selected domain concepts and proposed service responsibilities. Entity attributes are conceptual groups, not physical column definitions. Service operations are proposed interface responsibilities, not implemented methods.

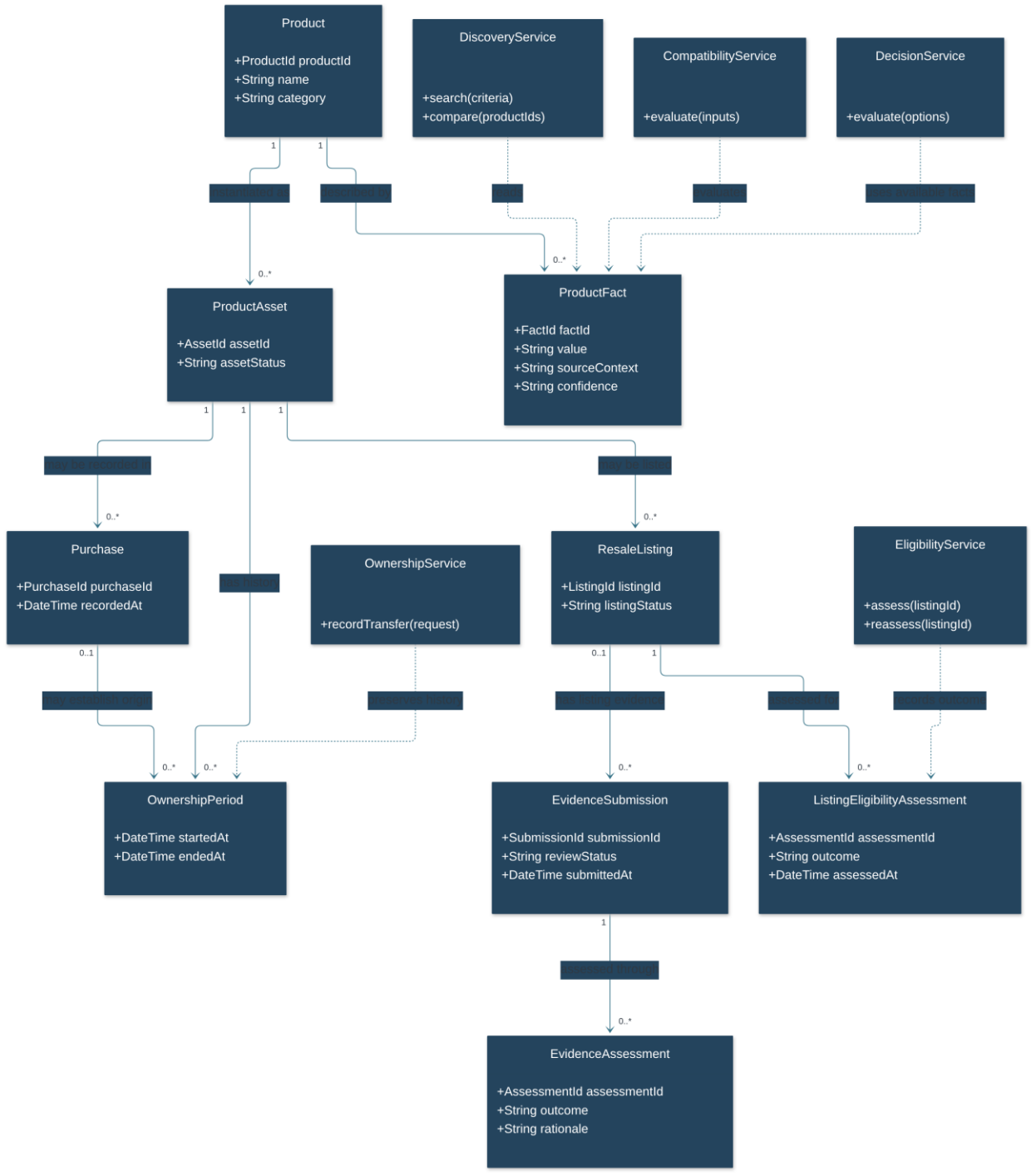


Figure 15. 11. Conceptual Class Diagram

Purchase is not ownership. A Sold listing does not prove a completed transfer. Evidence acceptance and listing eligibility are distinct assessments. The existing conceptual ERD remains authoritative for full entity relationships; this focused diagram is not a replacement.

12. Database Design and Schema Status

12.1 Existing Design Evidence

The database-design work is documented in the conceptual model, conceptual ERD source, logical design, logical relation catalogue, logical data dictionary, database schema documentation, and database decision log.

The validated logical baseline reports 54 relations, 502 attributes, 65 primary-key columns, 110 foreign keys, 37 candidate keys, 50 logical constraints, and 50 logical domains. These are logical-design measurements, not counts from a deployed or DBMS-specific schema.

12.2 Physical Schema

Status: deferred / not produced. OD-01 explicitly keeps DBMS selection open. The proposal's SQL Server reference is not sufficient to override that decision. Therefore this package does not invent SQL data types, vendor-specific DDL, indexes, migrations, ORM mappings, or physical deployment constraints. A physical schema can be prepared after the owner confirms the DBMS and the team validates implementation-specific requirements.

13. API and Interface Documentation

TechWise is designed to expose a backend API, but the project has no verified implemented endpoints. The proposed resource operations, request/response shapes, shared errors, validation, authorization, and retry considerations are documented separately in API Interface Contract Proposal. That companion document is a draft interface proposal, not an implementation or approved API. The proposed screen guidance and low-fidelity frames are in UI/UX Guidelines and Wireframes; no approved prototype or production UI is evidenced.

14. Testing and Validation Plan

No application source code or test suite is present in the inspected workspace. The following is a plan for later implementation and validation; none of these tests is claimed to have run.

Test level	Scope	Example evidence
Unit	Deterministic constraint evaluation, compatibility outcomes, eligibility conditions, state-transition guards, decision calculations, and input validation.	Results for valid, invalid, unknown, and incomplete inputs; rule/version references where applicable.
Integration	API-to-service-to-data-access interactions; persistence invariants; evidence and assessment history; authorization checks; AI adapter parsing.	Reproducible request/response records and database integrity checks in an approved test environment.
System / workflow	UC-01–UC-10 user journeys across responsive UI and backend boundary.	Scenario results mapped to AT-01–AT-10, including errors and missing-data behavior.
Security and privacy	Unauthorized asset/evidence access, prior-owner privacy, role checks, sensitive evidence visibility, untrusted provider response handling.	Access-control test records; no private data in unauthorized responses or logs.
Accessibility and responsive UI	Keyboard operation, focus order, semantics, contrast, screen-reader labels, mobile and desktop layout,	Manual/automated accessibility findings and viewport evidence; approved thresholds remain to be

	RTL/LTR preservation if Arabic is supported.	set.
Acceptance / UAT	Stakeholder review of intended scope, usability, limitation disclosure, and traceability.	Named reviewer, build/version, scenario outcomes, defects, and sign-off status; no sign-off is currently evidenced.

14.1 Minimum Scenario-to-Test Mapping

Test focus	Acceptance scenarios
Discovery, empty state, missing data, comparison	AT-01, AT-02
Mandatory constraints and compatibility unknowns	AT-03, AT-04
Listing eligibility and evidence distinction	AT-05, AT-06
Ownership continuity and private data access	AT-07, AT-08
AI/backend conflict and missing decision cost	AT-09, AT-10

Performance, availability, recovery, accessibility thresholds, retention periods, and production monitoring must be given measurable targets after the team confirms scope and environment.

15. Review Checklist and Open Decisions

Before this package is treated as an approved design:

- ☐ Confirm requirements baseline and actor/permission policy.
- ☐ Confirm final frontend/backend stack and DBMS; resolve OD-01.
- ☐ Confirm API style, operation names, request/response schemas, authorization, and retry semantics.
- ☐ Confirm evidence requirements, reviewer authority, condition rubric, eligibility and reassessment rules.
- ☐ Confirm reservation, sale, payment, cancellation, and ownership-transfer semantics; keep payment out of scope unless explicitly approved.
- ☐ Confirm hosting, identity, AI/search providers, data retention, and numeric quality targets.
- ☐ Review low-fidelity UI/UX proposal and verify whether an approved prototype already exists.
- ☐ Validate implementation and run the test strategy after application code and environments exist.

16. References

1. DEPI Project Documentation Guidelines, System Analysis & Design deliverables, pages 2–3.
2. Requirements Specification, draft requirements baseline.
3. Problem Statement & Objectives, current project scope and objectives.
4. Conceptual Database Design, conceptual relationships and state models.
5. Database Design Decision Log, including deferred technical and transaction decisions.
6. Project Proposal, initial architecture and proposed technology stack.
7. TECHWISE Document Design Standard, PDF visual identity and formatting requirements.
8. UI/UX Guidelines and Wireframes, proposed screen layouts and interaction guidance.
9. API Interface Contract Proposal, proposed operations and contract boundaries.

Appendix A: Detailed Requirements Baseline

6. Use Cases

The following use cases describe major user-system interactions. They are requirements-level descriptions, not final sequence diagrams or UI specifications.

UC-01 — Discover Products
Primary actor: Shopper

Preconditions:

- Product information is available to the search process.
- Any required access conditions have been satisfied.

Main flow:

10. The shopper opens product discovery.
11. The shopper enters a query or selects supported filters.
12. The system interprets the request and applies available search criteria.
13. The system returns relevant product candidates.
14. The shopper reviews the results and opens a product.

Alternative flows:

- If no results match, the system explains that no matching candidates were found and may suggest relaxing optional criteria.
- If required information is missing, the system requests clarification or indicates the limitation.

Postcondition: The shopper receives a result set or an informative empty state.

UC-02 — Compare Products
Primary actor: Shopper

Preconditions:

- At least two eligible product candidates are available for comparison.

Main flow:

15. The shopper selects products.
16. The system retrieves comparable attributes.
17. The system presents a comparison.
18. The shopper reviews similarities, differences, and missing information.
19. The shopper proceeds to a decision or returns to discovery.

Alternative flows:

- If a value is unavailable for one candidate, the system identifies it as unavailable rather than inventing a value.

Postcondition: The shopper can inspect a consistent comparison.

UC-03 — Check Compatibility
Primary actor: Shopper

Preconditions:

- Relevant product or component records and compatibility rules are available.

Main flow:

20. The shopper selects the items to evaluate.
21. The system retrieves their relevant specifications.
22. The compatibility service evaluates applicable constraints.
23. The system presents the result and its supporting reasoning.
24. The shopper reviews any limitations or alternatives.

Alternative flows:

- If required specifications are missing, the result is unknown or insufficient data.
- If a mandatory compatibility condition fails, the system identifies the failed condition.

Postcondition: A compatibility outcome is presented without confusing missing data with confirmed compatibility.

UC-04 — Evaluate a Purchase Decision
Primary actor: Shopper

Preconditions:

- Candidate products and the necessary decision inputs are available.

Main flow:

25. The shopper supplies requirements, constraints, and preferences.
26. The system interprets the request into structured criteria.
27. The system validates the criteria.
28. The decision engine evaluates candidates against mandatory constraints and preferences.
29. The system presents rankings, trade-offs, and relevant explanations.
30. The shopper reviews the outcome and chooses a next action.

Alternative flows:

- If criteria conflict, the system requests clarification or explains the conflict.
- If data is insufficient, the system discloses the limitation.
- If no candidate satisfies all mandatory constraints, the system must not label a violating candidate as a full match.

Postcondition: The shopper receives a decision-support result and understands its main limitations.

UC-05 — Inspect Product Information
Primary actor: Shopper

Preconditions:

- A product record exists.

Main flow:

31. The shopper opens a product's details.
32. The system presents available specifications and relevant product facts.
33. The system presents source, evidence, or confidence information where available.
34. The shopper reviews the information.

35. The shopper proceeds to comparison or decision support.

Alternative flows:

- Conflicting, incomplete, or seller-reported information is appropriately identified.

Postcondition: The shopper can distinguish available facts from claims and unknowns.

UC-06 — View an Owned Product's Lifecycle
Primary actor: Authorized product owner

Preconditions:

- The Product Asset and an authorized ownership record exist.

Main flow:

36. The owner opens the owned-products area.
37. The system checks access permissions.
38. The system retrieves the relevant Product Asset and ownership period.
39. The system presents authorized lifecycle events.
40. The owner reviews available history and relevant actions.

Alternative flows:

- Unauthorized requests are denied without disclosing another user's private information.

Postcondition: The authorized owner can view permitted information about the asset and their ownership period.

UC-07 — Submit a Second-Life Listing
Primary actor: Current product owner acting as seller

Preconditions:

- The seller can establish current ownership through the approved process.
- The product meets the platform-origin requirements.
- The Product Asset is eligible for evaluation.

Main flow:

41. The seller selects a Product Asset.
42. The system evaluates initial eligibility.
43. The system identifies required listing information and evidence.
44. The seller submits the listing and required evidence.
45. The system validates the submission.
46. The listing receives an appropriate status based on the result.

Alternative flows:

- If ownership or origin requirements fail, the listing is rejected or blocked according to the approved rules.
- If evidence is missing, the system requests the missing information.
- If a blocking dispute exists, eligibility is handled according to the approved dispute rules.

Postcondition: The listing is recorded with an eligibility or review status. Submission alone does not imply that the product has been physically inspected.

UC-08 — Review a Second-Life Listing
Primary actor: Prospective second-life buyer

Preconditions:

- A listing is available for viewing.

Main flow:

47. The buyer opens the listing.
48. The system presents product identity, condition information, price, and available evidence.
49. The buyer reviews the listing's status and supporting information.
50. The buyer decides whether to continue with the applicable purchase process.

Alternative flows:

- Missing or unverified information is clearly identified.

Postcondition: The buyer can assess the available information before proceeding.

UC-09 — Transfer Ownership
Primary actor: Authorized current owner and eligible new owner

Preconditions:

- The transfer is permitted by the approved business rules.
- Required transaction and eligibility conditions have been satisfied.

Main flow:

51. The system identifies the Product Asset involved.
52. The system validates the transfer conditions.
53. The authorized transfer process is completed.
54. The previous ownership period is preserved as historical data.
55. A new ownership period is recorded.
56. The product's original lifecycle and first-purchase date remain unchanged.

Alternative flows:

- If validation fails, the transfer is not completed.
- If a blocking dispute exists, the system follows the approved dispute process.

Postcondition: Ownership changes without resetting the physical asset's age or erasing historical ownership.

UC-10 — Evaluate Buy vs Upgrade vs Resell
Primary actor: Product owner

Preconditions:

- Relevant current-product information and enough data for the requested evaluation are available.

Main flow:

57. The owner selects a product or Product Asset.
58. The system collects available condition, cost, and requirement information.
59. The system evaluates applicable buy, upgrade, and resale options.
60. The system compares the options using defined calculations and assumptions.
61. The system presents the results and limitations.
62. The owner reviews the recommendation and chooses a next action.

Alternative flows:

- If a reliable comparison cannot be calculated because required inputs are missing, the system identifies those inputs and limits the conclusion.

Postcondition: The owner receives a transparent comparison of available lifecycle choices.

7. Functional Requirements

7.1 Requirement Classification

The following priority labels are used:

- Must: Necessary to satisfy the defined core capability or an essential business rule.
- Should: Important to the intended user experience but subject to implementation planning.
- Could: A useful enhancement if time and resources permit.

The priorities below are proposed for team validation. They are not a claim that DEPI has assigned these priorities.

7.2 AI Shopping Intelligence

FR-AI-01 — Natural-Language Input

The system shall accept shopping requests expressed in natural language.

Priority: Must

FR-AI-02 — Structured Requirement Extraction

The system shall convert interpretable shopping requests into structured requirements, preferences, and constraints before downstream evaluation.

Priority: Must

FR-AI-03 — Clarification of Ambiguous Requests

The system should request clarification when ambiguity materially affects the interpretation or evaluation of a shopping request.

Priority: Should

FR-AI-04 — Recommendation Explanation

The system shall provide understandable explanations for relevant recommendations using available product facts and decision results.

Priority: Must

FR-AI-05 — AI Boundary

AI-generated interpretation or explanation shall not independently override product records, authorization checks, compatibility rules, ownership state, or deterministic business decisions.

Priority: Must

FR-AI-06 — Uncertainty Disclosure

The system should communicate when a recommendation depends on incomplete data, ambiguous intent, or unsupported assumptions.

Priority: Must

7.3 Product Discovery and Smart Filtering
FR-SF-01 — Product Search

The system shall allow shoppers to search available product records using supported query criteria.

Priority: Must

FR-SF-02 — Structured Filtering

The system shall apply supported filters to available product attributes.

Priority: Must

FR-SF-03 — Mandatory Constraints

Where the user defines mandatory criteria, the system shall distinguish them from optional preferences.

Priority: Must

FR-SF-04 — Filter Combination

The system should support combining multiple supported filters.

Priority: Should

FR-SF-05 — Empty Results

The system shall communicate when no candidates satisfy the applied criteria.

Priority: Must

FR-SF-06 — Missing Attribute Handling

The system shall not treat an unknown attribute as proof that a product satisfies a filter.

Priority: Must

7.4 Product Comparison
FR-PC-01 — Attribute Comparison

The system shall present selected products using consistent comparison attributes when comparable data is available.

Priority: Must

FR-PC-02 — Missing Data Identification

The system shall identify unavailable comparison values.

Priority: Must

FR-PC-03 — Difference Explanation

The system should highlight relevant differences that affect the user's stated requirements.

Priority: Should

FR-PC-04 — Evidence-Aware Comparison

Where relevant evidence or source information is available, the system should expose that context for compared product facts.

Priority: Should

7.5 Product Compatibility Graph

FR-CG-01 — Compatibility Data Representation

The system shall represent supported compatibility relationships using structured product or component data.

Priority: Must

FR-CG-02 — Rule-Based Evaluation

The system shall evaluate compatibility using defined criteria and available structured facts.

Priority: Must

FR-CG-03 — Explicit Outcomes

The system shall distinguish supported compatible and incompatible outcomes from unknown or insufficient-data outcomes.

Priority: Must

FR-CG-04 — Reasoning Disclosure

The system should identify the relevant facts or constraints that support a compatibility result.

Priority: Should

FR-CG-05 — AI Boundary for Compatibility

AI may help interpret the user's question or explain a result, but shall not independently establish compatibility in place of the applicable rules and product data.

Priority: Must

7.6 Purchase Decision Engine

FR-PD-01 — Decision Input Collection

The system shall support the collection of user requirements, mandatory constraints, and preferences relevant to a purchase decision.

Priority: Must

FR-PD-02 — Candidate Evaluation

The system shall evaluate available candidates against the defined decision criteria.

Priority: Must

FR-PD-03 — Constraint Validation

The system shall evaluate mandatory constraints separately from optional preferences.

Priority: Must

FR-PD-04 — Ranking

The system shall rank eligible candidates using defined, reproducible evaluation rules.

Priority: Must

FR-PD-05 — Trade-Off Explanation

The system should explain material trade-offs between candidates.

Priority: Should

FR-PD-06 — No False Full Match

The system shall not label a candidate as fully satisfying the request when the candidate violates a mandatory constraint.

Priority: Must

FR-PD-07 — Insufficient Data

The system shall communicate when insufficient information prevents a reliable evaluation.

Priority: Must

FR-PD-08 — Decision Transparency

Where a result depends on assumptions, weights, or estimates, the system should disclose the relevant assumptions.

Priority: Should

7.7 Product Truth Layer

FR-PT-01 — Structured Product Facts

The system shall organize supported product attributes as structured data.

Priority: Must

FR-PT-02 — Source Context

Where source information is available, the system should associate relevant product facts with their sources or evidence.

Priority: Should

FR-PT-03 — Seller-Reported Claims

The system shall distinguish seller-reported information from information that has supporting evidence or an explicit platform assessment.

Priority: Must

FR-PT-04 — Evidence and Confidence

The system should present available evidence and platform confidence or assessment information without implying certainty beyond what the evidence supports.

Priority: Should

FR-PT-05 — Conflicting Information

The system should identify material conflicts or gaps in product information when they can be detected.

Priority: Should

FR-PT-06 — No Fabricated Facts

The system shall not present AI-generated or inferred values as authoritative product facts without the required data and validation.

Priority: Must

7.8 Verified Second-Life Marketplace

FR-SL-01 — Eligible Listing Discovery

The system shall allow users to discover second-life listings that satisfy the marketplace's applicable visibility and eligibility rules.

Priority: Must

FR-SL-02 — Platform-Origin Requirement

A product shall satisfy the applicable TechWise platform-purchase or origin requirement before becoming eligible for second-life listing.

Priority: Must

FR-SL-03 — Current Ownership Validation

The system shall validate the seller's current ownership through the approved ownership and purchase-history process.

Priority: Must

FR-SL-04 — Product Asset Association

A second-life listing shall be associated with the relevant Product Asset rather than treating the listing as an unrelated new product.

Priority: Must

FR-SL-05 — Evidence Submission

The system shall support submission of the evidence required by the approved listing rules. For electronic products, the current project direction includes screenshots and a product test video.

Priority: Must

FR-SL-06 — Eligibility Result

The system shall evaluate applicable listing conditions, including ownership, origin, listing state, evidence requirements, and blocking disputes where those rules are defined.

Priority: Must

FR-SL-07 — Listing Status

The system shall represent the listing's applicable status and prevent an ineligible listing from being presented as an eligible active listing.

Priority: Must

FR-SL-08 — Evidence-Aware Presentation

The system shall present relevant listing evidence and seller-reported condition information in a way that does not imply independent physical verification unless it has occurred.

Priority: Must

FR-SL-09 — Rejection Feedback

The system should explain missing eligibility requirements or evidence when a listing cannot proceed.

Priority: Should

FR-SL-10 — Authorized Listing Management

The system should allow an authorized seller to review and manage permitted listing information and status.

Priority: Should

7.9 Product Lifecycle and Ownership

FR-PL-01 — Separate Product and Product Asset

The system shall distinguish a product model or SKU from a specific physical Product Asset.

Priority: Must

FR-PL-02 — Ownership Periods

The system shall represent ownership separately from the Product Asset's identity.

Priority: Must

FR-PL-03 — Lifecycle Continuity

The Product Asset's identity and historical lifecycle shall remain associated with the same physical item across resale and ownership transfer.

Priority: Must

FR-PL-04 — Original First-Purchase Date

The system shall preserve the original first-purchase date after resale.

Priority: Must

FR-PL-05 — Historical Ownership

The system shall preserve previous ownership records as historical records when ownership changes.

Priority: Must

FR-PL-06 — Ownership Reference Rules

The system shall apply the approved ownership reference or code rules. Under the current project direction, the previous owner's code becomes invalid for the new ownership period, and the new owner receives a new code while the original first-purchase date remains unchanged.

Priority: Must

FR-PL-07 — Privacy of Previous Owners

The system shall not expose previous owners' personal information to unauthorized users.

Priority: Must

FR-PL-08 — Transfer Validation

The system shall validate authorization and applicable transfer conditions before recording an ownership change.

Priority: Must

7.10 Buy vs Upgrade vs Resell Engine

FR-BUR-01 — Lifecycle Option Comparison

The system shall support evaluation of applicable buy, upgrade, and resell options using available data.

Priority: Must

FR-BUR-02 — Requirement-Based Evaluation

The system should evaluate options against the user's current requirements and constraints.

Priority: Should

FR-BUR-03 — Cost and Condition Inputs

The system shall use available relevant cost, product-condition, and lifecycle information in applicable evaluations.

Priority: Must

FR-BUR-04 — Deterministic Calculations

Calculations used for the decision shall be performed using defined rules and validated inputs rather than relying solely on AI-generated numerical conclusions.

Priority: Must

FR-BUR-05 — Assumption Disclosure

The system should disclose important assumptions, missing inputs, or estimates that affect the comparison.

Priority: Should

FR-BUR-06 — Explainable Outcome

The system should explain the main factors influencing the comparison.

Priority: Should

7.11 Access Control and Administrative Requirements

FR-AC-01 — Authorization

The system shall restrict protected operations to authorized users and roles.

Priority: Must

FR-AC-02 — Ownership Access

The system shall restrict access to ownership and lifecycle information according to approved permissions.

Priority: Must

FR-AC-03 — Eligibility Enforcement

The system shall enforce listing and ownership rules on the backend rather than relying only on frontend controls or AI output.

Priority: Must

FR-AC-04 — Status Change Traceability

Where required by the approved business process, important ownership and listing status changes should be recorded for traceability.

Priority: Should

Detailed role definitions and permission matrices must be finalized during requirements validation and system design.

8. Non-Functional Requirements

Non-functional requirements define quality expectations and constraints. The requirements below establish intended quality attributes. Numeric performance targets, availability targets, and testing thresholds remain proposed until validated against the project scope and available infrastructure.

8.1 Performance

NFR-PER-01 —

Common user interactions should return feedback within an acceptable time under the agreed test environment.

Priority: Proposed

NFR-PER-02 —

Product search and filtering should remain responsive for the dataset and load defined by the project test plan.

Priority: Proposed

NFR-PER-03 —

Purchase decision and compatibility evaluations should complete within agreed test thresholds for representative inputs.

Priority: Proposed

NFR-PER-04 —

Performance claims shall be supported by measurements under documented conditions rather than assumed from the design.

Priority: Proposed

8.2 Security and Privacy

NFR-SEC-01 —

Protected information and operations shall require appropriate authentication and authorization according to the approved access model.

Priority: Must

NFR-SEC-02 —

Personal, purchase, and ownership information shall be protected against unauthorized access.

Priority: Must

NFR-SEC-03 —

Critical business rules and permissions shall be enforced by trusted backend logic, not solely by client-side controls.

Priority: Must

NFR-SEC-04 —

User inputs and submitted evidence metadata shall be validated before being processed or persisted.

Priority: Must

NFR-SEC-05 —

The system shall prevent unauthorized disclosure of previous owners' personal information.

Priority: Must

NFR-SEC-06 —

Error messages shall provide useful feedback without exposing credentials, sensitive records, or internal implementation details.

Priority: Must

8.3 Reliability and Data Integrity

NFR-REL-01 —

Ownership changes shall not reset the Product Asset's original first-purchase date or erase historical ownership records.

Priority: Must

NFR-REL-02 —

The system shall maintain consistent relationships among products, Product Assets, purchases, ownership periods, and second-life listings.

Priority: Must

NFR-REL-03 —

Failed validation or processing shall produce an appropriate error or status without falsely reporting success.

Priority: Must

NFR-REL-04 —

The system should handle repeated or conflicting operations in a way that prevents inconsistent ownership or listing states.

Priority: Must

8.4 Usability and Accessibility

NFR-USA-01 —

The system should explain technical product information and decision outcomes in language appropriate for the intended user.

Priority: Should

NFR-USA-02 —

Common actions, feedback messages, and navigation patterns should behave consistently across supported screens.

Priority: Should

NFR-USA-03 —

The user interface should remain usable across the desktop and mobile screen sizes included in the approved scope.

Priority: Should

NFR-USA-04 —

The system should explain validation errors and provide a clear way to correct user input where possible.

Priority: Should

NFR-USA-05 —

Recommendations, compatibility results, and eligibility outcomes should communicate their main reasons and limitations.

Priority: Should

8.5 Maintainability
NFR-MNT-01 —

AI interpretation, deterministic business logic, data access, and user-interface responsibilities should be separated sufficiently to support testing and maintenance.

Priority: Should

NFR-MNT-02 —

Core business rules should be testable independently of AI-generated explanations where technically appropriate.

Priority: Should

NFR-MNT-03 —

Implementation and documentation shall use the defined distinctions among Product, Product Asset, Purchase, and Ownership.

Priority: Should

NFR-MNT-04 —

Requirements and related implementation changes should be traceable through the team's agreed development workflow.

Priority: Should

8.6 Data Quality and Explainability
NFR-DQ-01 —

Where source information is available, relevant product facts should retain sufficient source context for appropriate review.

Priority: Must

NFR-DQ-02 —

Unknown or unavailable data shall not be silently converted into a positive claim.

Priority: Must

NFR-DQ-03 —

Deterministic calculations and rule-based evaluations should produce consistent results for the same validated inputs and rule version.

Priority: Must

NFR-DQ-04 —

AI-generated explanations shall remain consistent with the underlying facts and evaluation results.

Priority: Must

NFR-DQ-05 —

The system shall not describe submitted evidence as proof of a fact beyond what the evidence and review process actually establish.

Priority: Must

8.7 Scalability and Availability

NFR-SCA-01 —

The system design should allow the product catalogue and lifecycle records to grow without requiring unnecessary redesign of core business concepts.

Priority: Proposed

NFR-AVL-01 —

Availability requirements and recovery targets shall be defined according to the approved deployment environment and project scope.

Priority: Proposed

No uptime percentage or recovery-time guarantee is claimed until a deployment environment and measurement method are confirmed.

9. Business Rules and Constraints

The following rules reflect the current project direction and must remain consistent across requirements, design, implementation, and testing.

9.1 AI and Source-of-Truth Rules

BR-01: AI is not the authoritative source of product facts or database state.

BR-02: AI may interpret intent, extract requirements, explain results, summarize information, and support review intelligence.

BR-03: Validation, authorization, eligibility, compatibility evaluation, ranking calculations, ownership state, and other critical business rules must be enforced by deterministic backend logic or other explicitly defined authoritative systems.

BR-04: AI explanations must not override a validated business result.

9.2 Product and Ownership Rules

BR-05: A Product is distinct from a Product Asset.

BR-06: A Product Asset represents a specific physical item and retains its identity across ownership changes.

BR-07: Ownership represents a particular user's ownership period and is not the same as the Product Asset.

BR-08: Product age and original first-purchase date do not reset after resale.

BR-09: Previous ownership records remain part of the lifecycle history, while previous owners' personal information remains protected.

9.3 Second-Life Rules

BR-10: Second-life listing eligibility depends on the applicable TechWise platform-origin or purchase-history requirements.

BR-11: The current seller must satisfy the approved current-ownership validation process.

BR-12: A listing must be associated with the relevant Product Asset.

BR-13: Required evidence must be supplied and evaluated according to the approved listing rules. For electronics, the current direction includes screenshots and a product test video.

BR-14: Eligibility may depend on listing state, evidence completeness, current ownership, platform origin, and blocking disputes as defined by the approved business process.

BR-15: Seller-reported claims, submitted evidence, and platform confidence or assessment must remain distinguishable.

BR-16: No physical platform inspection or verification shall be claimed unless that activity is genuinely performed by an implemented process.

BR-17: After an approved resale, the previous ownership code becomes invalid for the new ownership period and the new owner receives a new code, while the original first-purchase date and asset history remain intact.

Appendix B: Proposed API Interface Contract

> **Contract boundary.** The requirements and proposal describe a backend API, but the inspected workspace contains no implemented endpoints. This document proposes a REST/JSON interface shape for discussion. It does not select a framework, API host, authentication provider, persistence technology, or external provider.

1. Principles

- Treat this document as a draft contract. Approve the API style, operations, schemas, and authorization policy before implementation.
- Use a versioned resource-oriented route pattern and consistent camelCase JSON fields in examples.
- Validate all client input and external provider responses at trust boundaries.
- Keep authorization, ownership, compatibility, eligibility, and decision rules in deterministic backend services.
- Return missing/unknown information explicitly; never invent facts or coerce unknown values to `false`, zero, or a confident result.
- Do not expose private evidence files or prior-owner data through public listing responses.
- Use consistent structured errors and document state-changing retry behavior before clients rely on it.
- Do not expose payment, escrow, refund, or purchase-completion operations; they are outside the approved baseline.

2. Proposed Base and Shared Conventions

Item	Proposed convention	Status
Base path	`/api/v1`	Proposed; approval required
Transport	HTTPS in deployed environments	Required for deployment; environment unselected
Media type	`application/json; charset=utf-8`	Proposed
Identifiers	Opaque resource IDs; format not selected	Proposed
Timestamps	ISO 8601 UTC strings	Proposed
Pagination	`page`, `pageSize`; bounded page size to be selected	Proposed
Sorting	Explicit allow-listed `sortBy` and `sortOrder`	Proposed
Authentication	Authenticated identity required for private asset, listing-management, evidence, and transfer operations; provider unselected	Deferred
Authorization	Server-side role/resource checks on every protected operation	Required principle; role policy needs approval
Unknown values	Explicit outcome/status and missing-data reasons	Required behavior

2.1 Common Error Shape

Every error response should use the same envelope and avoid internal stack traces, secrets, raw evidence, or unnecessary personal data:

Field / Path	Type	Example	Meaning
error	object	{...}	
error.code	string	VALIDATION_ERROR	Stable machine-readable error code.
error.message	string	The request is invalid.	Short human-readable summary.
error.details	array of objects	[1 item]	Per-field validation details, when applicable.
error.details[].field	string	productIds	Request field or parameter that failed.
error.details[].reason	string	At least two product identifiers are required.	Plain-language reason for the failure.
error.correlationId	string	server-generated-reference	Server-generated reference for tracing.

HTTP status	Proposed use
400	Malformed request syntax or unsupported parameter shape.
401	Authentication is required or invalid.
403	Authenticated caller lacks permission for the operation.
404	Resource not found; whether private-resource existence should be concealed must be confirmed.
409	Conflicting state or operation cannot be applied in the current state.
422	Syntactically valid request fails semantic validation or mandatory business constraints.
500	Unexpected server failure; return no implementation details.
502 / 503	Optional external integration unavailable or returned an unusable response; only if such integration is implemented.

These are proposed mappings, not claims about existing behavior.

3. Operation Catalogue

All routes below are proposals. {id} denotes an opaque resource identifier. Authorization and data visibility must be applied on the server for every request.

Proposed operation	Use case	Input / query	Response intent	Status / constraint
GET /api/v1/products	UC-01	q, supported filters, page, pageSize, sortBy, sortOrder	Paged candidates, visible facts, and match limitations.	Search engine and final filters unselected.
GET /api/v1/products/{productId}	UC-05	Product identifier	Product details, facts, provenance, and explicit unknowns.	Do not expose private evidence.
POST /api/v1/comparisons	UC-02	Two or more product IDs	Comparable values, differences, and unavailable fields.	Never substitute missing facts.
POST /api/v1/compatibility-assessments	UC-03	Product/component identifiers and applicable constraints	COMPATIBLE, INCOMPATIBLE, or UNKNOWN, with rationale and limitations.	Outcome rules must remain deterministic; rule catalogue/version is not selected.
POST /api/v1/purchase-decisions	UC-04	Requirements, mandatory constraints, and preferences	Candidate ranking, constraint outcomes, trade-offs, sources, and limitations.	Does not place an order or authorize purchase.
GET /api/v1/assets	UC-06	Authenticated caller; paging parameters	Only Product Assets the caller may access.	Private, owner-scoped; pagination required for lists.
GET /api/v1/assets/{assetId}/lifecycle	UC-06	Asset identifier	Permitted ownership period and lifecycle information.	Deny unauthorized access without disclosing private history.
POST /api/v1/assets/{assetId}/resale-listings	UC-07	Listing details and evidence references	Recorded submission and current review/eligibility status.	Upload mechanism, evidence policy, and activation rules unresolved.
GET /api/v1/resale-listings/{listingId}	UC-08	Listing identifier	Public-safe listing details, evidence summary, condition and status.	Raw private evidence excluded.
POST /api/v1/resale-listings/{listingId}/evidence-assessments	UC-07	Authorized reviewer outcome, reason, scope, and evidence context	Recorded assessment and resulting review state.	Reviewer authority and allowed outcomes require approval.
POST /api/v1/assets/{assetId}/ownership-transfers	UC-09	Proposed transfer participants and required approved context	Proposed transfer result or pending state.	**Blocked from finalization** until effective trigger and validation policy are confirmed.

POST /api/v1/lifecycle-decisions	UC-10	Asset/product, costs, condition, requirements, and available options	Buy/upgrade/resell comparison with assumptions and missing inputs.	No unsupported prices or calculations; no transaction is executed.

The ownership-transfer operation is intentionally marked blocked: publishing its final request/response contract would falsely imply that transaction completion semantics are settled.

4. Representative Schemas

These examples are illustrative contract shapes, not generated types or deployed payloads.

4.1 Product Fact and Provenance

Field / Path	Type	Example	Meaning
productId	string (opaque id)	opaque-product-id	Opaque product identifier.
name	string	Product name	Market-facing product name.
facts	array of objects	[1 item]	List of structured facts for the product.
facts[].attribute	string	exampleAttribute	Governed attribute code from the logical data dictionary.
facts[].value	null	null	Asserted value; null means not supplied.
facts[].availability	enum (UNKNOWN)	UNKNOWN	Availability state: UNKNOWN, AVAILABLE, or NOT_APPLICABLE.
facts[].source	null	null	Source reference; null when no source is recorded.
facts[].assessment	string	NOT_ASSESSED	Platform assessment state; NOT_ASSESSED when no assessment exists.
facts[].confidence	string	INSUFFICIENT_EVIDENCE	Confidence classification; INSUFFICIENT_EVIDENCE when evidence is missing.

The actual product attributes and controlled values must follow the approved logical data dictionary. null, UNKNOWN, NOT_ASSESSED, and NOT_APPLICABLE are not interchangeable.

4.2 Compatibility Result

Field / Path	Type	Example	Meaning
outcome	enum (UNKNOWN)	UNKNOWN	Top-level outcome: COMPATIBLE, INCOMPATIBLE, or

			UNKNOWN.
evaluatedAt	string (ISO 8601 UTC)	2026-11-04T12:00:00Z	ISO 8601 UTC timestamp of evaluation.
ruleVersion	string	approved-version-reference	Reference to the rule version used to evaluate.
checks	array of objects	[1 item]	List of constraint checks performed.
checks[].constraint	string	exampleConstraint	Constraint identifier evaluated.
checks[].outcome	enum (UNKNOWN)	UNKNOWN	Per-check outcome: COMPATIBLE, INCOMPATIBLE, or UNKNOWN.
checks[].reason	string	A required product specification is unavailable.	Plain-language reason for the check outcome.
limitations	array	[1 item]	List of limitations affecting the result.

The timestamp and version strings are examples only. Do not invent a numeric confidence score or a compatibility conclusion when required evidence is absent.

4.3 Decision Result

Field / Path	Type	Example	Meaning
evaluatedAt	string (ISO 8601 UTC)	2026-11-04T12:00:00Z	ISO 8601 UTC timestamp of evaluation.
results	array of objects	[1 item]	Per-candidate decision results.
results[].productId	string (opaque id)	opaque-product-id	Opaque product identifier of the candidate.
results[].mandatoryConstraints	array of objects	[1 item]	Per-constraint outcome list.
results[].mandatoryConstraints[].constraint	string	exampleConstraint	Constraint identifier.
results[].mandatoryConstraints[].outcome	string	NOT_EVALUABLE	Constraint outcome, including NOT_EVALUABLE when required input is missing.
results[].mandatoryConstraints[].reason	string	Required input is missing.	Reason for the outcome.

results[].rank	null	null	Rank position; null when not evaluable.
results[].explanation	string	A full-match ranking is unavailable until required input ...	Plain-language explanation of the result.
assumptions	array	[0 items]	List of assumptions applied during evaluation.
missingInputs	array	[1 item]	List of inputs required but not supplied.

4.4 Listing Submission Result

Field / Path	Type	Example	Meaning
listingId	string (opaque id)	opaque-listing-id	Opaque listing identifier.
listingStatus	enum (Pending Review)	Pending Review	Current listing state. Pending Review means the submission is awaiting review.
eligibility	object	{...}	Eligibility determination for the submission.
eligibility.outcome	enum (Incomplete)	Incomplete	Eligibility outcome: Eligible, Ineligible, Incomplete, or Unknown.
eligibility.missingInputs	array	[1 item]	List of missing inputs that prevented a complete determination.
evidenceAssessment	enum (Not Assessed)	Not Assessed	Current evidence assessment state; Not Assessed when no assessment has been performed.
physicalInspection	enum (Not Claimed)	Not Claimed	Physical inspection state; Not Claimed when no inspection is asserted.

The representation of physicalInspection is illustrative; it must not be used to imply that an inspection occurred.

5. Authorization and Privacy Requirements

- Authenticate and authorize each private operation on the backend, regardless of UI visibility.
- Scope asset and lifecycle reads to an authorized relationship and approved role.
- Limit reviewer operations to an approved reviewer role and assigned review scope.

- Return only approved public-safe evidence summaries for second-life listings; raw uploaded files require separate authorization and storage controls.
- Do not reveal previous-owner identity, account details, private evidence, access tokens, or unnecessary device identifiers in public responses or error details.
- Treat AI/provider input and output as untrusted. Validate schemas, lengths, allowed fields, and use in downstream logic.
- Do not log raw evidence, credentials, full private payloads, or sensitive provider responses.

The identity provider, role/permission catalogue, file-upload mechanism, retention policy, and audit retention are not selected here.

6. Validation, Pagination, and Concurrency

6.1 Boundary Validation

Validate request shape, identifiers, lengths, supported filters, enum values, ownership context, and required fields at the API boundary. Then invoke deterministic business validation. External provider responses must be parsed against an explicit schema before use.

6.2 List Responses

Paged list responses should use one consistent envelope:

Field / Path	Type	Example	Meaning
`data`	array	[0 items]	Array of result items for the current page.
`pagination`	object	{...}	Pagination metadata.
`pagination.page`	integer	1	Current page number (1-indexed).
`pagination.pageSize`	integer	20	Number of items per page.
`pagination.totalItems`	integer	0	Total number of matching items.
`pagination.totalPages`	integer	0	Total number of pages.

Default/max page size, stable sorting, filtering allow-list, and large-result behavior require implementation-time decisions.

6.3 State Changes and Retry Safety

Listing submission, evidence assessment, and ownership operations can create or change lifecycle records. Before clients are told that retries are safe, the implementation must define duplicate-request behavior and demonstrate an atomic deduplication mechanism, including same-key/different-payload handling. This proposal does **not** advertise an `Idempotency-Key` header or promise idempotent effects. Do not infer sale or ownership transfer from a listing status.

7. API Review and Approval Checklist

- ☐ Confirm REST/JSON, base version, and naming with the project owner/team.
- ☐ Confirm final request/response schemas and controlled enum values against the logical data dictionary.

- ☐ Confirm authentication, authorization, privacy, reviewer permissions, and resource-existence disclosure.
- ☐ Confirm evidence upload and secure file-access contract before accepting evidence references.
- ☐ Resolve eligibility, review, listing activation, and ownership-transfer semantics before approving state-changing routes.
- ☐ Define consistent validation, error, pagination, sorting, and external-provider failure behavior.
- ☐ Define and test concurrency and retry semantics before promising idempotency.
- ☐ Add an OpenAPI specification and contract tests when routes are approved and implementation begins.

8. Source References

- Requirements Specification, UC-01–UC-10, FR groups, business rules, and NFRs.
- System Analysis & Design Package, system boundaries, diagrams, and status.
- Logical Database Design, logical attributes and relationships.
- Database Design Decision Log, especially deferred DBMS and transaction decisions.
- TECHWISE Document Design Standard, PDF conversion visual identity.

Appendix C: Proposed UI/UX Guidelines and Wireframes

> **Design boundary.** These wireframes describe an initial responsive interface for the requirements baseline. They are not production screens, a verified prototype, or a claim that specific components exist. Confirm the latest Figma/prototype and product-owner direction before implementation.

1. Design Brief

Dimension	Direction
Purpose	Support product discovery, technical comparison, compatibility understanding, ownership lifecycle, and a transparent second-life workflow.
Audience	Shoppers, current product owners, sellers, prospective second-life buyers, and authorized reviewers on mobile and desktop.
Tone	**Commerce Editorial** — a clear product catalogue with evidence-aware technical detail.
Reference	A technical product catalogue, prioritizing readable specifications and provenance over promotional treatment.
Palette	Light ivory and white surfaces, deep navy ink, one restrained teal accent; semantic status colors only where needed.
Type	Geometric, legible headings and an Arabic/Latin-compatible body stack, with RTL/LTR preservation.
Memorable	Product comparison shows **Compatible / Incompatible / Unknown** alongside the evidence and criteria behind the result.
Restraint	No invented logo, provider, payment flow, physical verification, fake telemetry, decorative gradients, or unnecessary ornaments.

2. Design Principles

63. **Evidence before persuasion.** Explain why a result appears and distinguish platform facts, seller claims, submitted evidence, and assessment.
64. **Unknown is a real state.** Missing specifications, unassessed evidence, and not-applicable values have different labels and are never silently shown as a negative or positive fact.
65. **Deterministic outcomes stay legible.** AI may help interpret or explain; it does not replace compatibility, authorization, eligibility, or ownership decisions.
66. **Lifecycle continuity.** Show the physical Product Asset and its ownership periods without implying that a listing or sale status changes ownership.
67. **Responsive, not reduced.** Mobile layouts stack and prioritize content; they do not hide mandatory constraints, source labels, or access limitations.
68. **Plain language.** Explain technical terms in context and avoid implying physical inspection or certification that did not occur.
69. **Progressive detail.** Present decision-critical facts first and make provenance, caveats, and further specifications available without cluttering every result.

3. Proposed Visual Tokens

The following values are a UI proposal, not an approved brand extension. Keep the official PDF-document palette in TECHWISE Document Design Standard; do not substitute the document palette for application-state semantics.

Token	Value	Purpose
Typography		
--font-display	"IBM Plex Sans", "Noto Sans Arabic", "Segoe UI", sans-serif	Display and heading typeface stack; Arabic/Latin-compatible.
--font-body	"IBM Plex Sans", "Noto Sans Arabic", "Segoe UI", sans-serif	Body typeface stack; Arabic/Latin-compatible.
--type--1	0.833rem	Smallest type size (0.833 rem). Used for fine print and labels.
--type-0	1rem	Base body size (1 rem).
--type-1	1.2rem	Heading-4 / subheading size (1.2 rem).
--type-2	1.44rem	Heading-3 size (1.44 rem).
--type-3	1.728rem	Heading-2 size (1.728 rem).
--type-4	2.074rem	Heading-1 size (2.074 rem).
--type-5	2.488rem	Display size (2.488 rem).
Spacing		
--space-1	0.25rem	0.25 rem (4 px). Tightest spacing unit.
--space-2	0.5rem	0.5 rem (8 px).
--space-3	0.75rem	0.75 rem (12 px).
--space-4	1rem	1 rem (16 px). Base spacing unit.
--space-6	1.5rem	1.5 rem (24 px).
--space-8	2rem	2 rem (32 px).
--space-12	3rem	3 rem (48 px).
--space-16	4rem	4 rem (64 px). Largest spacing unit.
Surfaces and ink		
--color-page	#F7F5EF	Page background; light ivory.
--color-surface	#FFFFFF	Card or panel surface; white.
--color-ink	#24445D	Primary heading and accent ink; deep navy.

--color-body	#303943	Body text color; dark gray.
--color-muted	#66727D	Supporting text color; secondary gray.
--color-border	#D9E0E5	Border and rule color; light border gray.
--color-accent	#39758A	Functional accent color; muted teal.
Control styling		
--radius-control	4px	Control corner radius (4 px).
--border-default	1px solid var(--color-border)	Default border specification.
--shadow-default	none	Default elevation shadow; none in the proposal.
Motion		
--motion-fast	120ms	Fast transition duration (120 ms).
--motion-standard	220ms	Standard transition duration (220 ms).
--motion-slow	420ms	Slow transition duration (420 ms).
--ease-standard	cubic-bezier(0.2, 0, 0, 1)	Standard easing curve.

3.1 Status Communication

Use semantic status colors only as a secondary cue, paired with text, shape, and accessible names. The palette is illustrative and must be checked for contrast in implementation.

Meaning	Required label	UI behavior
Positive / compatible	“Compatible” or the specific positive outcome	Include applicable rule/fact summary; do not imply universal compatibility.
Failed constraint	“Incompatible” or “Constraint not met”	Name the failed mandatory condition and source context.
Insufficient information	“Unknown” / “Not enough information”	Say what is missing and what action, if any, can resolve it.
Not assessed	“Not assessed”	Do not use “verified” or “approved.”
Seller statement	“Seller reported”	Visually separate the claim from platform evidence and assessment.
Evidence accepted	“Evidence accepted for [scope]”	Do not infer overall listing eligibility or physical inspection.

4. Navigation and Content Structure

4.1 Proposed Primary Navigation

- **Discover** — search, filters, product candidates, and comparison.

- **My Products** — private asset and ownership lifecycle views when authenticated.
- **Second-Life** — public-safe listings for buyers; listing management for authorized sellers.
- **Saved / Profile** — only if the corresponding persistence and account features are approved.

Do not expose administrative review navigation to ordinary shoppers. Reviewer workflows need a confirmed role and access policy.

4.2 Content Hierarchy

70. User's task and current object (query, product, Product Asset, or listing).
71. Decision-critical outcome and its scope.
72. Supporting facts, provenance, constraints, and missing inputs.
73. Next safe action.
74. Additional technical detail, history, or limitations.

5. Low-Fidelity Wireframes

All frames are conceptual. Text blocks show hierarchy and interaction intent, not pixel-perfect sizing or finalized copy.

5.1 Product Discovery — Desktop

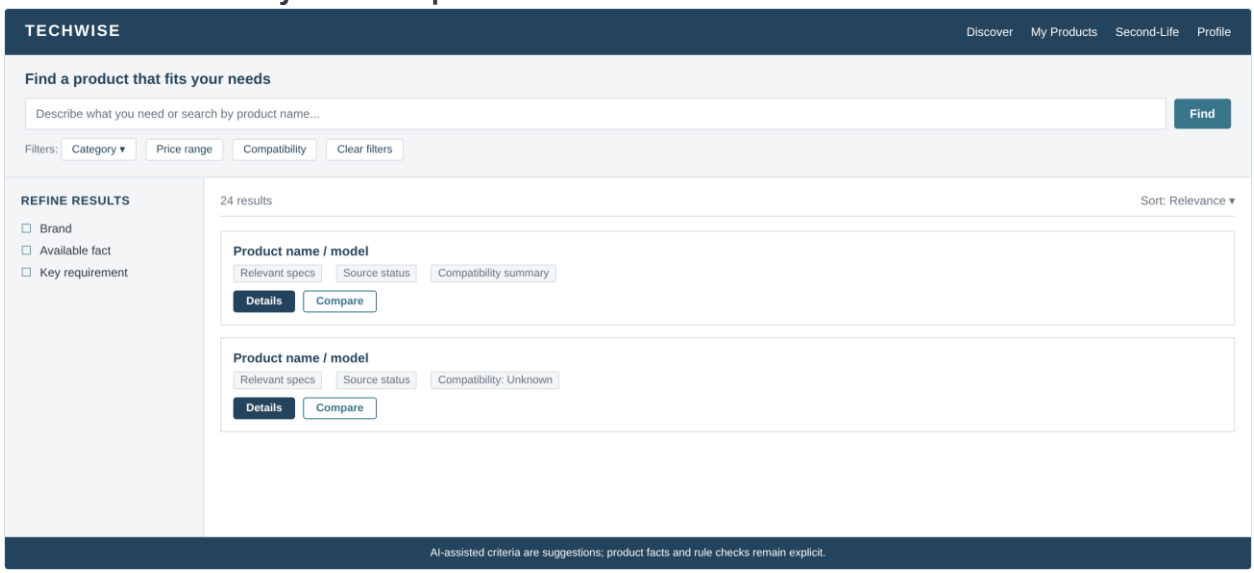


Figure 16. 5.1 Product Discovery — Desktop

5.2 Product Discovery — Mobile

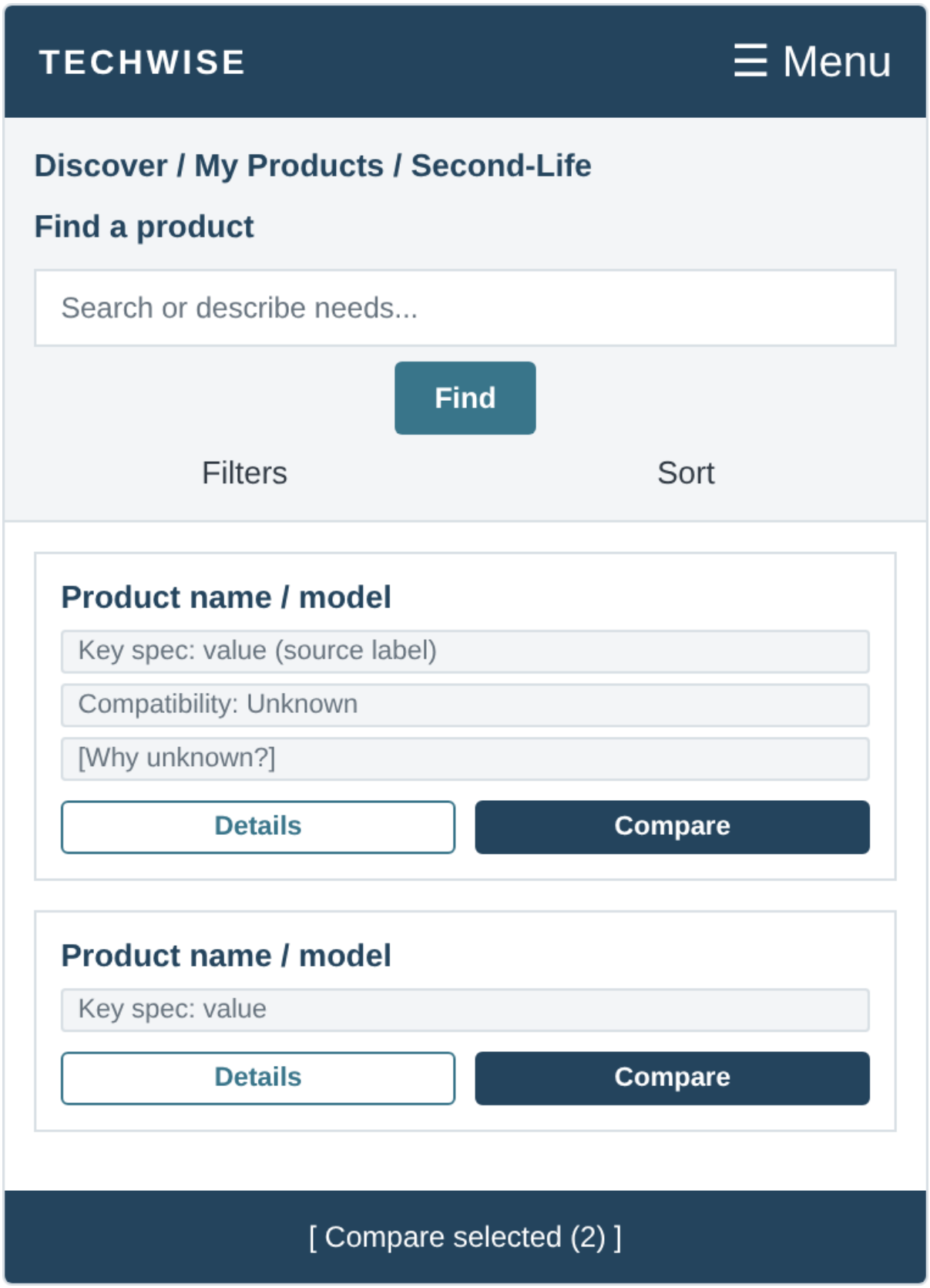


Figure 17. 5.2 Product Discovery — Mobile

On narrow screens, filters open in a labelled dialog or drawer with keyboard focus management; no essential constraint or status is hidden behind a color-only icon.

5.3 Product Comparison and Compatibility

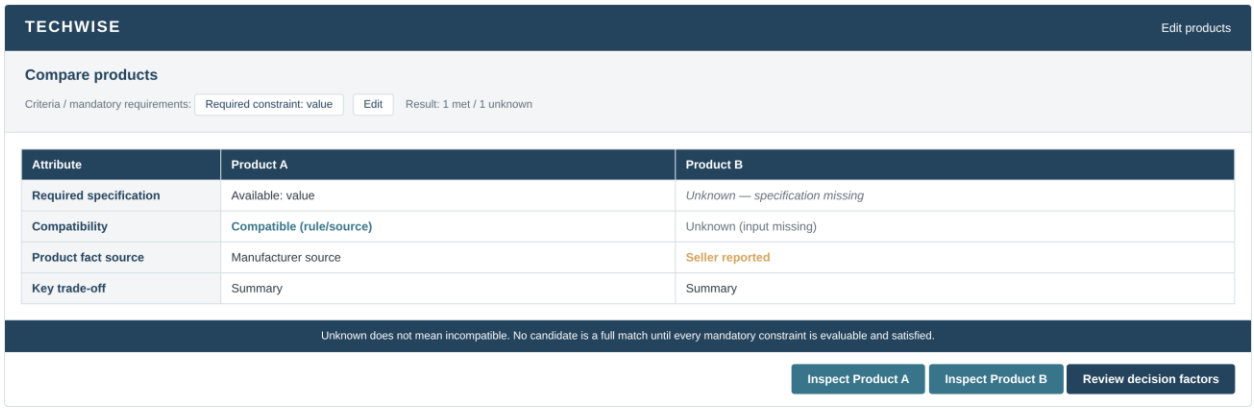


Figure 18. 5.3 Product Comparison and Compatibility

At mobile widths, preserve the selected products and comparison attribute labels. Use one product per panel or a labelled horizontal comparison with keyboard-accessible controls; do not rely on unlabeled horizontal scrolling.

5.4 Product Information and Second-Life Listing

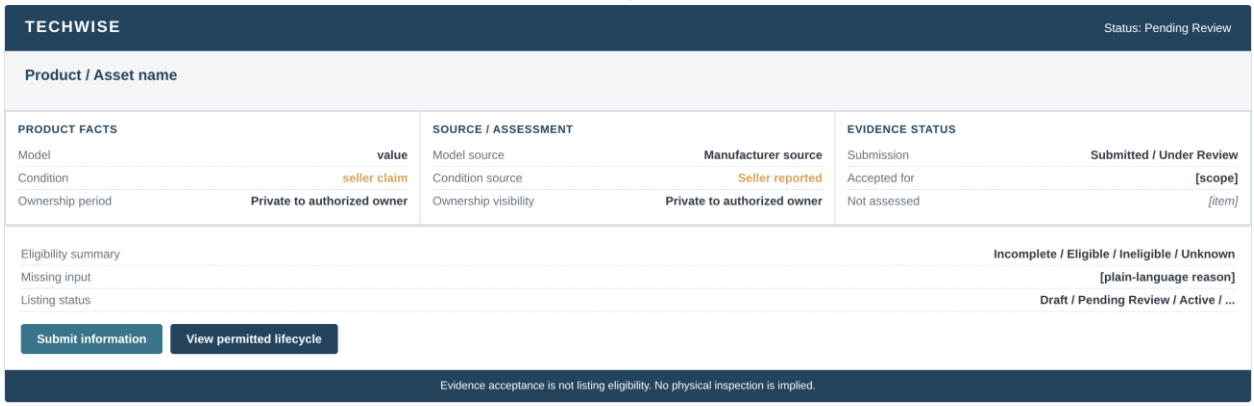


Figure 19. 5.4 Product Information and Second-Life Listing

Public buyer view omits private ownership details and raw evidence. Seller and reviewer views are role-gated. Show a direct explanation for every blocked or incomplete step.

5.5 Buy, Upgrade, or Resell Decision

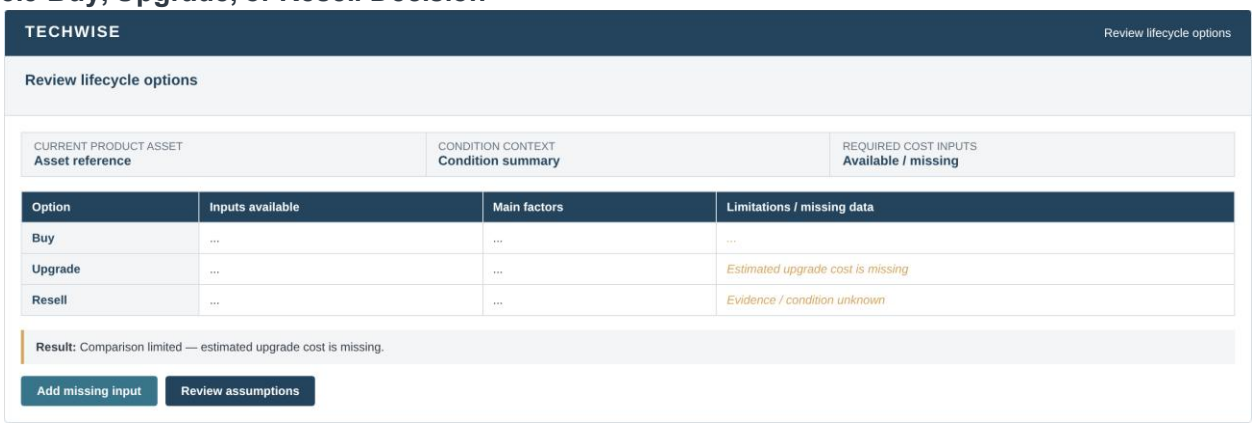


Figure 20. 5.5 Buy, Upgrade, or Resell Decision

Do not invent market prices, savings, condition scores, or resale eligibility. Label estimates, sources, currency, and timestamps where approved data exists.

6. Interaction and State Guidelines

Every interactive control must provide clear resting, hover, active, keyboard-focus, disabled, and busy/loading states as applicable. Hover must not be the only indication of interaction. Keep focus visible and do not collapse layout during loading.

State	Guidance
Loading	Preserve page structure; name the task in progress; announce meaningful updates to assistive technology.
Empty	Explain what was searched and offer a safe way to change optional criteria.
Error	State what failed and a safe next action; do not expose stack traces or private provider details.
Unknown / incomplete	Identify the missing fact or evidence and how it affects the result.
Unauthorized	Do not show protected details; provide a generic access message where disclosure is sensitive.
Conflict	Show the conflicting sources and that the issue is unresolved; never silently choose one as fact.
Success	Confirm only what actually happened, such as “submitted for review,” not “verified” or “sold” unless that state is established.

7. Responsive, Accessibility, and Localization Guidance

- Design for the approved mobile and desktop scope; proposed responsive breakpoints should follow content needs and be confirmed during implementation.
- Use semantic headings, landmarks, labels, accessible names, descriptive validation, and logical keyboard order.
- Target WCAG 2.2 AA for implementation review; this target is proposed and no conformance audit has been performed.
- Preserve visible focus, sufficient text contrast, scalable text, and non-color status distinctions.
- Use a minimum 44 × 44 CSS-pixel target as a proposed touch affordance; validate in the implemented interface.
- Respect reduced-motion preferences. Avoid motion that is not needed to communicate a meaningful state.
- Support English LTR and Arabic RTL if required by the approved audience. Set document language and direction correctly; preserve technical IDs, serials, numbers, and code order when mixed with Arabic.
- Ensure tables remain understandable on narrow screens and assistive technologies; provide a labelled alternative to visually dense comparisons.
- Treat evidence uploads, device identifiers, addresses, and owner history as sensitive; collect and display only what the approved workflow needs.

8. Content and Trust Rules

Prefer precise labels:

- “Product fact — source: …”
- “Seller reported”
- “Evidence submitted”

- “Evidence accepted for [scope]”
- “Eligibility: Incomplete / Eligible / Ineligible / Unknown”
- “Not assessed”
- “Unknown — [required specification] is unavailable”

Do not use “verified product,” “physically inspected,” “guaranteed compatible,” “certified,” or equivalent claims without approved evidence and a matching process. AI-generated explanations must be labelled as assistance and remain consistent with deterministic results.

9. Review Checklist

- ☐ Confirm an approved prototype does not already supersede these low-fidelity frames.
- ☐ Validate screens against UC-01–UC-10 and AT-01–AT-10.
- ☐ Confirm Arabic/RTL scope, platform support, and preferred typefaces.
- ☐ Review color contrast and status semantics in a rendered prototype.
- ☐ Test responsive layouts, keyboard/focus, screen reader labels, loading/error/empty states, and accessibility target.
- ☐ Confirm no interface claims physical inspection, sale completion, or ownership change based only on listing state.
- ☐ Obtain product-owner/team review; no approval is implied by this draft.

10. References

- System Analysis & Design Package, use cases and system boundaries.
- API Interface Contract Proposal, proposed backend operations and data boundaries.
- Requirements Specification, draft requirements and workflows.
- TECHWISE Document Design Standard, project-wide PDF identity (distinct from proposed app UI tokens).

Appendix D: Conceptual Database ERD Source

Smart E-Commerce & Product Intelligence Platform

System Analysis & Design — Task 02, Phase 02.2

1. Purpose and Status

This file is the maintainable diagram source for the Phase 02.2 conceptual model. It is a companion to Conceptual_Database_Design.md; it does not replace the entity catalogue, relationship catalogue, cardinality matrix, state models, or traceability.

- Status: Working conceptual baseline; not institutionally approved.
- Diagram language: Mermaid erDiagram.
- Scope: Conceptual entities and relationships only.
- Excluded: SQL, DBML, physical tables, physical column types, indexes, DBMS features, provider selections.
- Completeness: The overview diagram contains all 42 accepted entities and all 78 relationships. The seven focused diagrams are subsets of the same model.

2. How to Read the Diagrams

Symbol	Conceptual meaning
<code>||--o{</code>	One instance on the left relates to zero or more on the right.
<code>||--||</code>	One-to-one, both mandatory.
<code>|o--||</code>	Zero or one on the left relates to exactly one on the right.
<code>||--o|</code>	One on the left relates to zero or one on the right.
<code>}o--o{</code>	Many-to-many, both optional at the instance level.

The diagrams show conceptual participation. They do not declare physical foreign keys, indexes, or constraints. Relationship IDs are defined in Conceptual_Database_Design.md, Section 8.

3. Complete Overview ERD

This overview is the authoritative completeness view. It includes all 42 accepted entities and all 78 relationships. Domain diagrams in Sections 4–10 are extracted views; they do not add concepts or relationships.

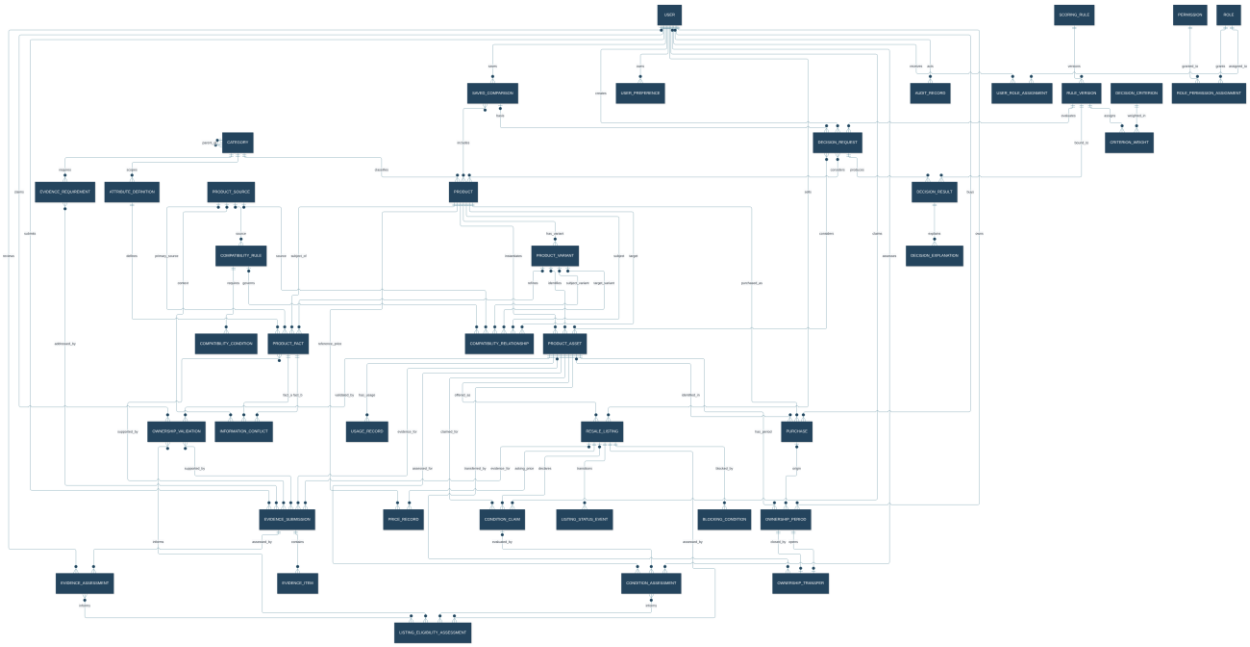


Figure 21. 3. Complete Overview ERD

Notes:

- All 42 accepted entities appear. `RetentionCategory` is a controlled classification, not an entity, and therefore does not appear.
- All 78 relationships appear. The nine original listing states are values, not separate entities.
- Product and ProductAsset are separate; a listing and a transfer are separate.
- AI is not an entity or an authority; explanation provenance is represented on `DECISION_EXPLANATION`.

4. Domain Diagram — Product Catalogue and Attributes

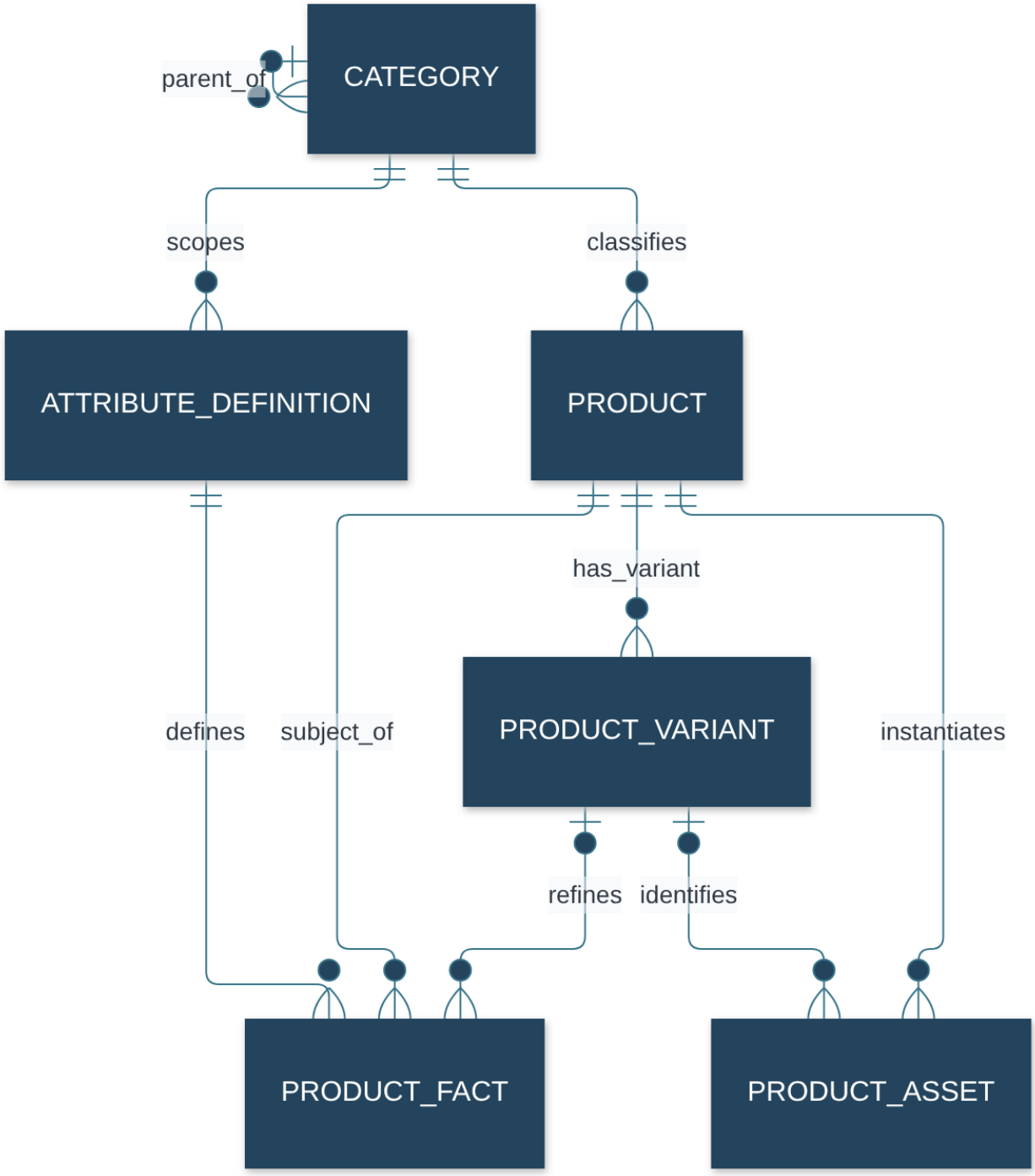


Figure 22. 4. Domain Diagram — Product Catalogue and Attributes

Notes: Category scopes controlled definitions; Product is not a physical asset. The self-relationship must remain acyclic. Single-category applicability is a working assumption (A-01, A-02).

5. Domain Diagram — Product Truth and Evidence

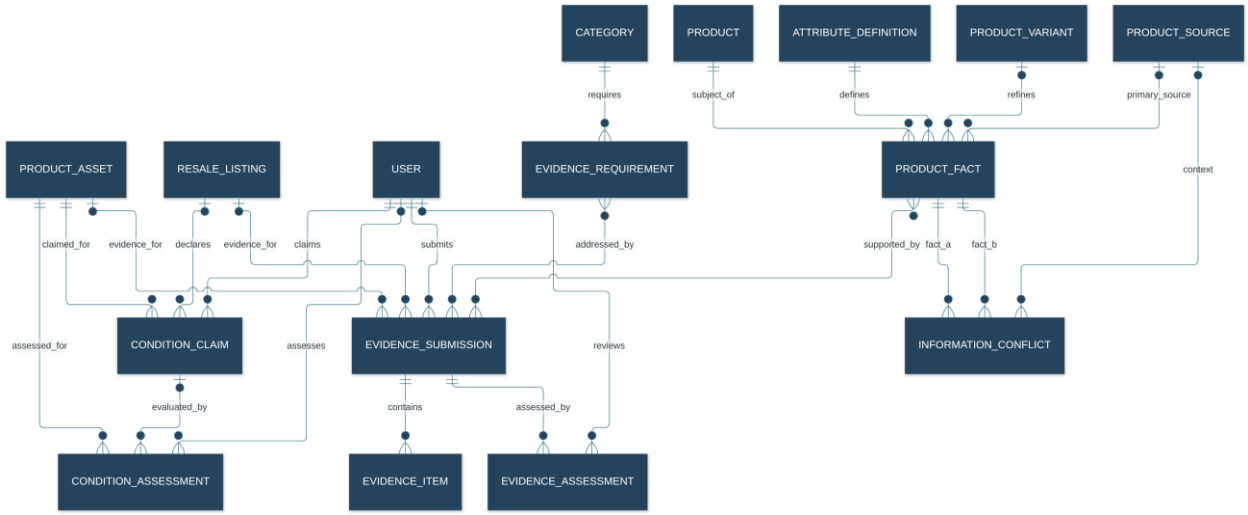


Figure 23. 5. Domain Diagram — Product Truth and Evidence

Notes: A fact is an assertion with provenance, not truth by existence. Evidence submission is not assessment, and assessment is not eligibility. ConditionClaim and ConditionAssessment are separate. Unknown and Insufficient Evidence are not negative facts.

6. Domain Diagram — Product Asset and Ownership Lifecycle

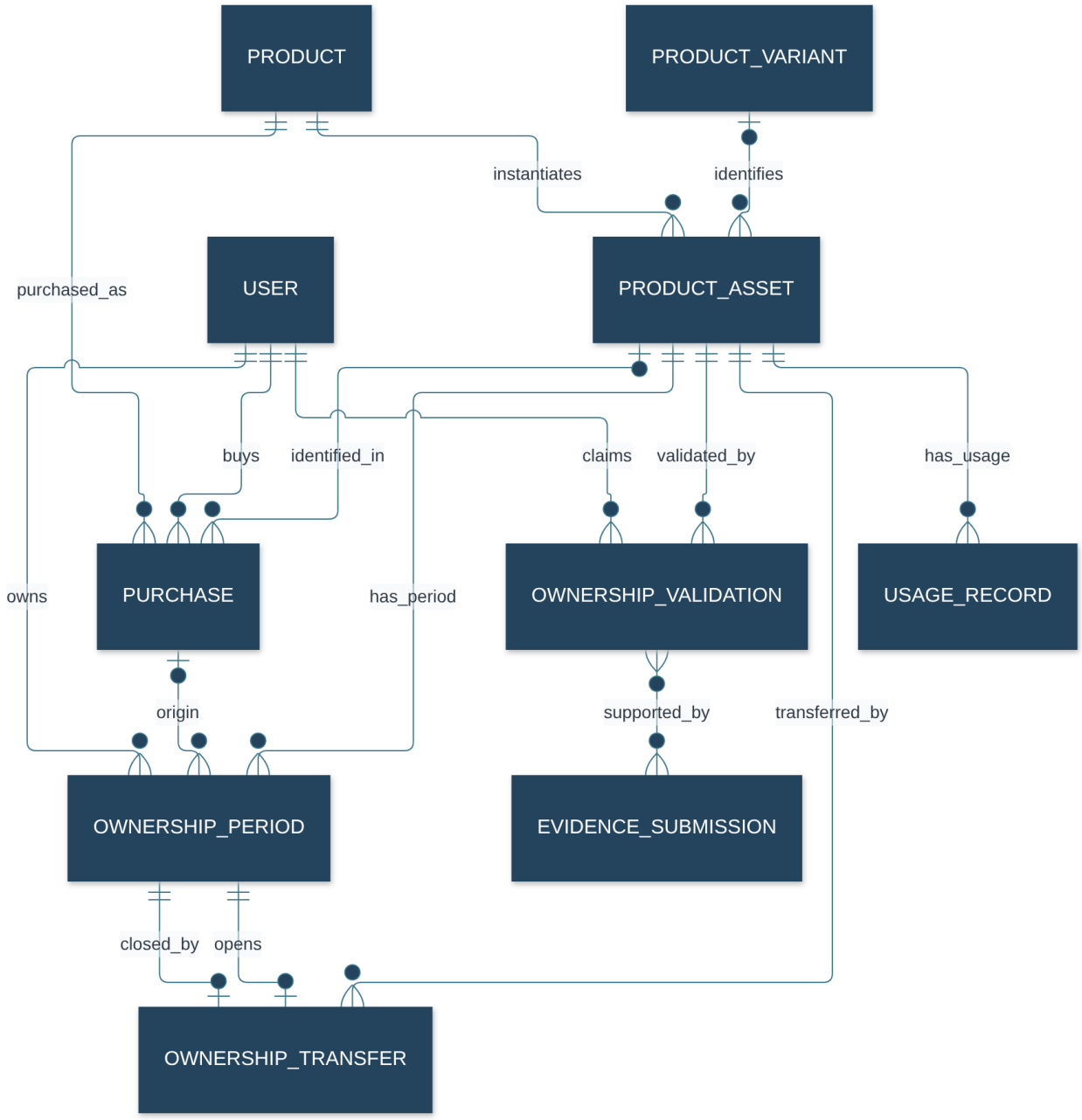


Figure 24. 6. Domain Diagram — Product Asset and Ownership Lifecycle

Notes: The asset identity is stable; purchase is not ownership; ownership is temporal. A transfer closes one period and opens another for the same asset. Sold is not modeled here as a transfer. Prior-owner privacy is a constraint (IC-26), not a diagram edge.

7. Domain Diagram — Second-Life Marketplace

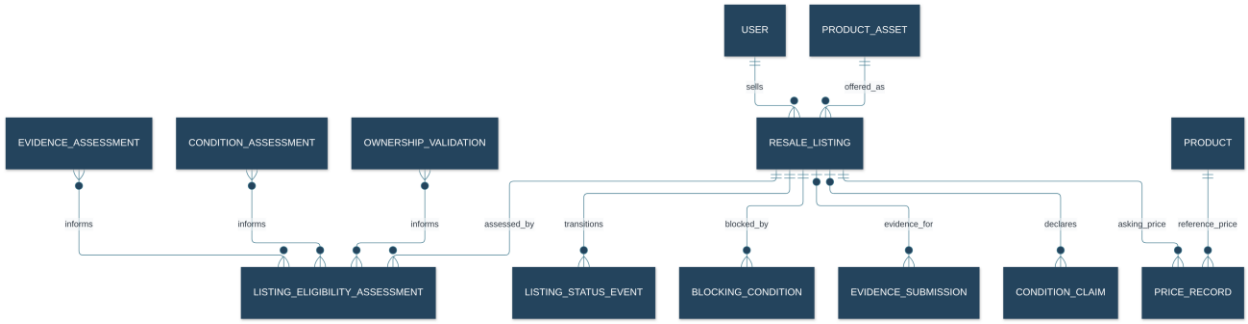


Figure 25. 7. Domain Diagram — Second-Life Marketplace

Notes: ACTIVE requires an applicable eligibility outcome under SM-01, not merely accepted evidence. RESERVED and SOLD transitions remain transaction-policy dependent. BLOCKING_CONDITION is deliberately minimal; payment, escrow, refund, return, and full dispute processing are excluded.

8. Domain Diagram — Compatibility Graph

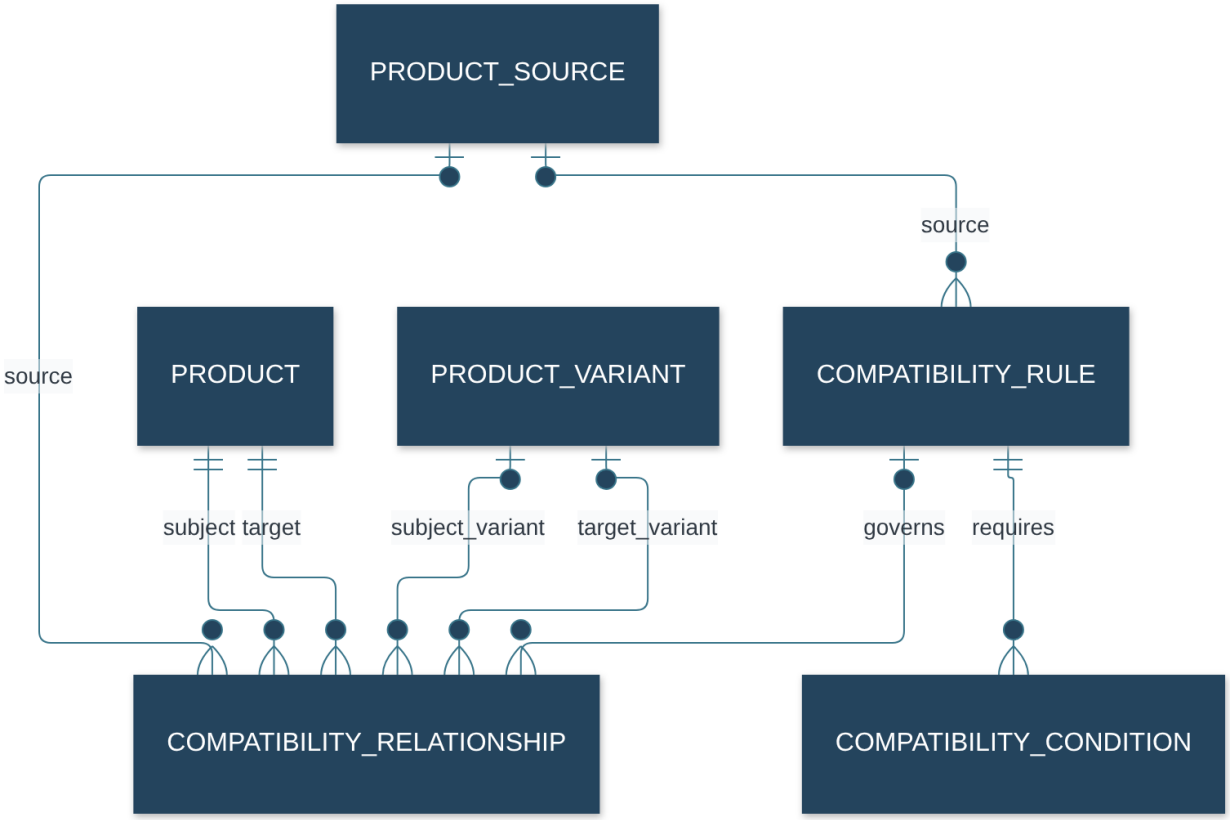


Figure 26. 8. Domain Diagram — Compatibility Graph

Notes: The relationship is directed. Compatible, Incompatible, Conditional, and Unknown are outcomes; absence means Unknown, never Incompatible. Conditional requires attributable conditions.

9. Domain Diagram — Purchase Decision Engine

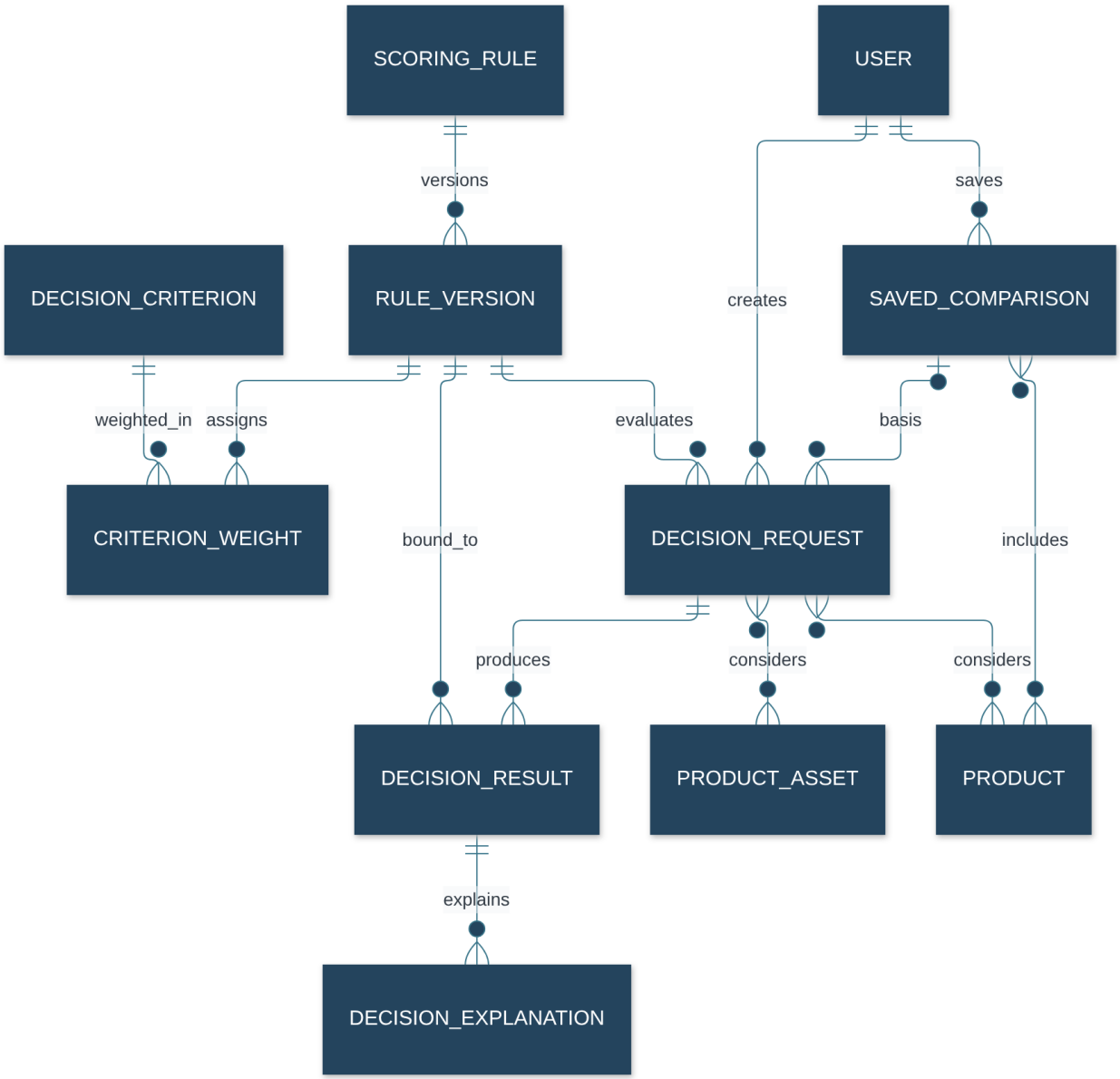


Figure 27. 9. Domain Diagram — Purchase Decision Engine

Notes: Results are bound to a rule version. Weights are version-specific. No formula, threshold, or tie-break rule is represented or approved. An AI-assisted DECISION_EXPLANATION is labelled and subordinate to the deterministic result.

10. Domain Diagram — Users, Roles, Permissions, and Audit

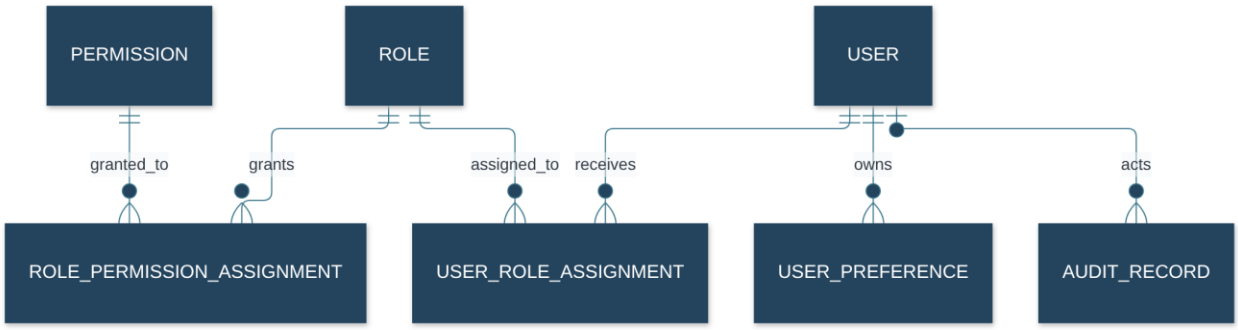


Figure 28. 10. Domain Diagram — Users, Roles, Permissions, and Audit

Notes: Role and permission assignment are associative entities with their own history. A role does not prove ownership. Audit targets are recorded references rather than a single polymorphic foreign key; audit content must not contain secrets.

11. Diagram Consistency and Limitations

Check	Result
Overview entity count	42 accepted conceptual entities
Overview relationship count	78 relationships
Domain diagrams	Seven focused subsets plus overview
Catalogue relationship coverage	Every R-01–R-78 appears in the overview
State values	Listing states are not modeled as entities; they remain controlled values
Physical design	No SQL, DBML, physical type, index, or DBMS feature
Provider selection	None
Referential assumption	Cardinalities are conceptual and require Phase 02.3 logical resolution

The diagram source must remain synchronized with the relationship catalogue and cardinality matrix. If a relationship is added, removed, or renamed, all three views and the traceability section must be updated together.

End of Conceptual Database ERD Source.

Appendix E: Logical Database Design

Smart E-Commerce & Product Intelligence Platform

System Analysis & Design - Task 02, Phase 02.3 Logical Database Design

1. Document Control

Field	Value
Project Name	TechWise
Official Description	Smart E-Commerce & Product Intelligence Platform
Team	Falcon Team
Program	Digital Egypt Pioneers Initiative (DEPI) - Round 5
Document Title	TechWise Logical Database Design
Deliverable	System Analysis & Design - Task 02, Phase 02.3
Version	1.0
Preparation Date	4 November 2026
Status	Logical baseline complete; technology-neutral; not institutionally approved
Baseline Input	Conceptual_Database_Design.md version 1.0 and Conceptual_Database_ERD.md version 1.0, Phase 02.2
Related Outputs	Logical_Relation_Catalogue.md, Logical_Relationship_and_Key_Mapping.md, Logical_Data_Dictionary.md, Logical_Normalization_and_Integrity_Report.md, Phase_02.3_Execution_Report.md
Phase 02.4 Readiness	READY WITH DOCUMENTED DEPENDENCIES

1.1 Status Statement

This document is a logical relational blueprint. It defines relations, attributes, keys, foreign keys, domains, constraints, normalization, controlled values, temporal structures, and logical indexes. It does not select a DBMS, hosting, storage engine, search engine, AI provider or model, authentication provider, or payment provider. It contains no SQL, DBML, migration, ORM, API, or physical schema.

The model is a working engineering baseline. It is not institutionally approved, and it does not resolve the business-policy dependencies carried forward from Phase 02.1 and Phase 02.2.

1.2 Source-of-Truth Priority Applied

Priority	Source class	Application in this phase
1	Latest explicit project-owner decisions	No new owner decision was supplied; none was invented.
2	Latest approved project	No document is institutionally approved; the working baseline is

	documentation	treated as the current reference.
3	Latest confirmed project decisions	Phase 02.1 decision statuses preserved exactly.
4	Phase 02.1 requirements and decision baseline	Database_Requirements_Analysis.md version 1.1 and Database_Design_Decision_Log.md.
5	Phase 02.2 conceptual model	42 entities and 78 relationships are the mapping baseline.
6	Earlier source documents	R1, R2, R4, and Instruction.md used for wording, scope, and format.
7	Clearly labelled recommendations	Referential actions, indexes, and structural refinements are labelled as recommendations.

2. Purpose and Scope

2.1 Purpose

Transform the reviewed conceptual model into an internally consistent, technology-neutral logical relational design detailed enough to drive physical design and implementation later. The design preserves every conceptual distinction that the baseline requires, including product versus asset, purchase versus ownership, listing versus transfer, evidence versus assessment, confidence versus condition, and AI output versus authority.

2.2 In Scope

Area	Coverage
Relations and attributes	54 relations; 502 attributes with logical types, nullability, and validation
Keys	54 primary key strategies; 65 primary key columns; 37 candidate or alternate keys
Relationships	78 conceptual relationships mapped; 110 foreign key constraints
Associative structures	9 conceptual M:N resolutions plus 3 attribute-group decompositions
Normalization	1NF, 2NF, 3NF analysis and a controlled-redundancy register
Integrity	50 logical constraints classified by enforcement mechanism
Controlled values	15 baseline vocabularies plus a deferred-domain register
Temporal modeling	Ownership periods, event history, assessment supersession, effective dating, and decision snapshots
Indexes	40 logical index recommendations, conditional where workload evidence is absent

Traceability	56 DRs, 17 BRs, 6 DBRs, 42 entities, 78 relationships, 10 ATs, 12 USs, 10 UCs

2.3 Out of Scope

DBMS selection, hosting, physical data types, storage parameters, partitioning, index syntax, executable SQL, DBML, migrations, ORM models, APIs, backend code, search engines, AI providers or models, authentication providers, payment or escrow design, retention periods, reviewer policy, condition rubric, scoring formulas, transaction completion semantics, and media storage.

3. Sources and Design Baseline

3.1 Sources Inspected

ID	Source	Inspection outcome	Limit
R1	Requirements Specification/TechWise_Requirements_Specification.docx	Requirements families, 17 business rules, user stories, use cases, and acceptance scenarios reviewed	Version 1.0 draft with incomplete approval record (CF-01)
R2	project-plan/TechWise_ProjectProposal.docx	Product vision, lifecycle, and conceptual domain framing reviewed	Initial consolidated draft
R3	LITERATURE REVIEW/TechWise_LiteratureReview.pdf	Partial text inspection only	Image-based PDF; no complete inspection claimed
R4	System-Analysis&Design/01-problem-statement-objectives/Problem_Statement_and_Objectives.md	Finalized scope and invariants reviewed	Working baseline
DRA	Database_Requirements_Analysis.md	56 DRs, 17 BRs, 6 DBRs, 40 candidates, 18 decisions verified	Not modified
DL	Database_Design_Decision_Log.md	Decision statuses and qualifications verified	Not modified
CD	Conceptual_Database_Design.md	42 entities, 78 relationships, 40 constraints, 15 vocabularies, 5 state models verified	Not modified
ERD	Conceptual_Database_ERD.md	8 diagrams, 42 overview nodes, 78 overview edges verified	Structural check only
ER	Phase_02.2_Execution_Report.md	Counts and validation results	Report is not treated as a

		cross-checked against the conceptual files	substitute for the files
INS	Instruction.md	Document structure, formal English, stable identifiers, readable tables	No PDF generated

3.2 Baseline Counts Verified Directly

Baseline element	Verified count	Verification source
Accepted conceptual entities	42	42 CE-* sections in Conceptual_Database_Design.md Section 7
Conceptual relationships	78	R-01 to R-78 in Section 8; 78 matrix rows in Section 9
Conceptual diagrams	8	Overview plus seven domain diagrams
State-model assessments	5	SM-01 to SM-05
Conceptual integrity constraints	40	IC-01 to IC-40
Controlled vocabularies	15	CV-01 to CV-15
Candidate entities accounted for	40	25 accepted, 6 merged, 2 value objects, 1 external reference, 4 deferred, 2 rejected
Data requirements	56	DR-001 to DR-056
Original business rules	17	BR-01 to BR-17
Derived business rules	6	DBR-01 to DBR-06
Acceptance scenarios	10	AT-01 to AT-10
User stories	12	US-01 to US-12
Use cases	10	UC-01 to UC-10

3.3 Decision Statuses Preserved

Status	Count	Decisions
Resolved	14	OD-02 to OD-09, OD-13 to OD-18
Resolved - Deferred Technical Selection	3	OD-01 DBMS, OD-10 search, OD-11 AI provider
Conditionally Resolved	1	OD-12 retention direction
Not Resolved	0	-

No deferred technical selection was converted into a confirmed choice. No new decision status was created.

4. Logical Modeling Principles

ID	Principle	Logical consequence
LP-01	Technology neutrality	Only eleven logical type families are used. Physical types, lengths where precision matters, and index syntax are deferred.
LP-02	Product is not ProductAsset	RL-07 describes a model or SKU; RL-14 identifies a physical device. Product facts never attach to an asset as if the asset were a model.
LP-03	Asset identity is stable	ProductAssetId is generated once and never replaced on resale. External identifiers live in RL-43.
LP-04	Purchase is not ownership	RL-15 records acquisition; current ownership is derived from RL-16.
LP-05	Ownership is temporal	Periods have start, optional end, and state. No mutable owner column exists on the asset.
LP-06	Product Truth is provenance-aware	RL-11 separates the claim from RL-12 source, RL-13 conflict, and RL-45 evidence support.
LP-07	Evidence is not assessment and assessment is not eligibility	RL-20, RL-22, RL-30, and the basis relations remain distinct.
LP-08	Listing state is not transfer	RL-28 and RL-29 record the offer and its lifecycle; RL-17 records transfer. Sold does not complete a transfer, a payment, or a sale automatically.
LP-09	Compatibility is directed and may be unknown	RL-25 keeps subject, target, type, outcome, and direction. Absence of a row means Unknown.
LP-10	AI is not authority	RL-39 labels producer type; LC-29 prevents an explanation from overriding deterministic output or facts.
LP-11	Unknown is not negative	Nullable attributes and explicit Unknown or Not Assessed values are never coerced to zero, false, or Incompatible.
LP-12	History is append-oriented	Assessments and status events are superseded, not overwritten. Audit is selective and privacy-aware.

5. Conceptual-to-Logical Mapping Summary

5.1 Counts

Measure	Count
Conceptual entities	42
Entities with a documented disposition	42
Entities mapped to one primary relation	39
Entities mapped to multiple relations	3
Conceptual relationships	78
Relationships mapped to foreign keys	69
Relationships resolved by associative relations	9
Logical relations	54
Net additional relations	12

The logical model has 54 relations rather than 42 because three conceptual entities are decomposed to satisfy First Normal Form and nine conceptual M:N relationships require associative structures. The increase is fully accounted for: 42 entity relations plus 3 decompositions plus 9 M:N resolutions equals 54. No relation is an accidental duplicate.

5.2 Entity dispositions

Conceptual entity	Name	Disposition	Logical relation(s)
CE-01	User	One logical relation	RL-01
CE-02	Role	One logical relation	RL-02
CE-03	Permission	One logical relation	RL-03
CE-04	UserRoleAssignment	One logical relation	RL-04
CE-05	RolePermissionAssignment	One logical relation	RL-05
CE-06	UserPreference	One logical relation	RL-06
CE-07	Product	One logical relation	RL-07
CE-08	ProductVariant	One logical relation	RL-08
CE-09	Category	One logical relation	RL-09
CE-10	AttributeDefinition	Multiple logical relations	RL-10, RL-44
CE-11	ProductFact	One logical relation	RL-11
CE-12	ProductSource	One logical relation	RL-12
CE-13	InformationConflict	One logical relation	RL-13
CE-14	ProductAsset	Multiple logical relations	RL-14, RL-43
CE-15	Purchase	One logical relation	RL-15

CE-16	OwnershipPeriod	One logical relation	RL-16
CE-17	OwnershipTransfer	One logical relation	RL-17
CE-18	OwnershipValidation	One logical relation	RL-18
CE-19	EvidenceRequirement	One logical relation	RL-19
CE-20	EvidenceSubmission	One logical relation	RL-20
CE-21	EvidenceItem	One logical relation	RL-21
CE-22	EvidenceAssessment	One logical relation	RL-22
CE-23	ConditionClaim	One logical relation	RL-23
CE-24	ConditionAssessment	One logical relation	RL-24
CE-25	CompatibilityRelationship	One logical relation	RL-25
CE-26	CompatibilityRule	One logical relation	RL-26
CE-27	CompatibilityCondition	One logical relation	RL-27
CE-28	ResaleListing	One logical relation	RL-28
CE-29	ListingStatusEvent	One logical relation	RL-29
CE-30	ListingEligibilityAssessment	One logical relation	RL-30
CE-31	BlockingCondition	One logical relation	RL-31
CE-32	PriceRecord	One logical relation	RL-32
CE-33	DecisionCriterion	One logical relation	RL-33
CE-34	CriterionWeight	One logical relation	RL-34
CE-35	ScoringRule	One logical relation	RL-35
CE-36	RuleVersion	One logical relation	RL-36
CE-37	DecisionRequest	One logical relation	RL-37
CE-38	DecisionResult	Multiple logical relations	RL-38, RL-54
CE-39	DecisionExplanation	One logical relation	RL-39
CE-40	UsageRecord	One logical relation	RL-40
CE-41	SavedComparison	One logical relation	RL-41
CE-42	AuditRecord	One logical relation	RL-42

6. Relation Catalogue Summary

The detailed relation catalogue is Logical_Relation_Catalogue.md. This summary lists every relation, its domain, its conceptual source, its primary key, and its size. Attribute-level detail is not duplicated here, to avoid contradictory duplication.

ID	Relation	Domain	Conceptual	Primary key	Attribut	FK	Candida

			source		es	s	te keys
RL-01	User	Access and identity	CE-01	UserId	10	0	1
RL-02	Role	Access and identity	CE-02	RoleId	7	0	1
RL-03	Permission	Access and identity	CE-03	PermissionId	8	0	1
RL-04	UserRoleAssignment	Access and identity	CE-04	UserRoleAssignmentId	8	3	1
RL-05	RolePermissionAssignment	Access and identity	CE-05	RolePermissionAssignmentId	8	3	1
RL-06	UserPreference	Access and identity	CE-06	UserPreferenceId	8	1	1
RL-07	Product	Product catalogue and attributes	CE-07	ProductId	11	1	1
RL-08	ProductVariant	Product catalogue and attributes	CE-08	ProductVariantId	7	1	1
RL-09	Category	Product catalogue and attributes	CE-09	CategoryId	10	1	1
RL-10	AttributeDefinition	Product catalogue and attributes	CE-10	AttributeDefinitionId	17	1	2
RL-11	ProductFact	Product Truth and provenance	CE-11	ProductFactId	23	4	1
RL-12	ProductSource	Product Truth and provenance	CE-12	ProductSourceId	12	0	1
RL-13	InformationConflict	Product Truth and provenance	CE-13	InformationConflictId	11	3	0

RL-14	ProductAsset	Product asset and ownership	CE-14	ProductAssetId	8	2	0
RL-15	Purchase	Product asset and ownership	CE-15	PurchaseId	10	3	1
RL-16	OwnershipPeriod	Product asset and ownership	CE-16	OwnershipPeriodId	13	5	2
RL-17	OwnershipTransfer	Product asset and ownership	CE-17	OwnershipTransferId	11	5	0
RL-18	OwnershipValidation	Product asset and ownership	CE-18	OwnershipValidationId	13	4	0
RL-19	EvidenceRequirement	Evidence and condition	CE-19	EvidenceRequirementId	11	1	1
RL-20	EvidenceSubmission	Evidence and condition	CE-20	EvidenceSubmissionId	10	4	0
RL-21	EvidenceItem	Evidence and condition	CE-21	EvidenceItemId	10	1	0
RL-22	EvidenceAssessment	Evidence and condition	CE-22	EvidenceAssessmentId	12	3	0
RL-23	ConditionClaim	Evidence and condition	CE-23	ConditionClaimId	10	3	0
RL-24	ConditionAssessment	Evidence and condition	CE-24	ConditionAssessmentId	13	4	0
RL-25	CompatibilityRelationship	Compatibility	CE-25	CompatibilityRelationshipId	15	6	1
RL-26	CompatibilityRule	Compatibility	CE-26	CompatibilityRuleId	11	1	1
RL-27	CompatibilityCondition	Compatibility	CE-27	CompatibilityConditionId	11	2	0
RL-28	ResaleListing	Second-Life marketplace	CE-28	ResaleListingId	10	2	1

RL-29	ListingStatusEvent	Second-Life marketplace	CE-29	ListingStatusEventId	11	3	0
RL-30	ListingEligibilityAssessment	Second-Life marketplace	CE-30	ListingEligibilityAssessmentId	11	3	0
RL-31	BlockingCondition	Second-Life marketplace	CE-31	BlockingConditionId	10	1	0
RL-32	PriceRecord	Second-Life marketplace	CE-32	PriceRecordId	12	2	0
RL-33	DecisionCriterion	Decision support	CE-33	DecisionCriterionId	10	0	1
RL-34	CriterionWeight	Decision support	CE-34	CriterionWeightId	7	2	1
RL-35	ScoringRule	Decision support	CE-35	ScoringRuleId	9	0	1
RL-36	RuleVersion	Decision support	CE-36	RuleVersionId	10	1	1
RL-37	DecisionRequest	Decision support	CE-37	DecisionRequestId	10	3	0
RL-38	DecisionResult	Decision support	CE-38	DecisionResultId	10	3	0
RL-39	DecisionExplanation	Decision support	CE-39	DecisionExplanationId	9	1	0
RL-40	UsageRecord	Product asset and ownership	CE-40	UsageRecordId	13	2	0
RL-41	SavedComparison	Decision support	CE-41	SavedComparisonId	7	1	0
RL-42	AuditRecord	Audit and governance	CE-42	AuditRecordId	12	1	0

RL-43	ProductAssetIdentifier	Product asset and ownership	CE-14 (identifier attribute group)	ProductAssetIdentifierId	9	1	2
RL-44	AttributeAllowedValue	Associative and reference support	CE-10 (value-domain attribute group)	AttributeDefinitionId, AllowedValueCode	7	1	1
RL-45	FactEvidenceLink	Associative and reference support	R-19 conceptual M:N	ProductFactId, EvidenceSubmissionId	3	2	1
RL-46	ValidationEvidenceLink	Associative and reference support	R-31 conceptual M:N	OwnershipValidationId, EvidenceSubmissionId	3	2	1
RL-47	SubmissionRequirementLink	Associative and reference support	R-33 conceptual M:N	EvidenceSubmissionId, EvidenceRequirementId	3	2	1
RL-48	EligibilityEvidenceBasis	Associative and reference support	R-47 conceptual M:N	ListingEligibilityAssessmentId, EvidenceAssessmentId	3	2	1
RL-49	EligibilityOwnershipBasis	Associative and reference support	R-48 conceptual M:N	ListingEligibilityAssessmentId, OwnershipValidationId	3	2	1
RL-50	EligibilityConditionBasis	Associative and reference support	R-49 conceptual M:N	ListingEligibilityAssessmentId, ConditionAssessmentId	3	2	1
RL-51	SavedComparisonItem	Associative and reference support	R-74 conceptual M:N	SavedComparisonId, ProductId	4	2	1
RL-52	DecisionRequestCandidateProduct	Associative and reference support	R-76 conceptual M:N	DecisionRequestId, ProductId	4	2	1
RL-53	DecisionRequestCandidateAsset	Associative and reference support	R-77 conceptual M:N	DecisionRequestId, ProductAssetId	4	2	1
RL-54	DecisionResultItem	Decision support	CE-38 (ranking attribute group)	DecisionResultId, CandidateSequence	7	3	1

6.1 Domain distribution

Domain	Relations
Access and identity	6
Associative and reference support	10
Audit and governance	1
Compatibility	3
Decision support	9
Evidence and condition	6
Product Truth and provenance	3
Product asset and ownership	7
Product catalogue and attributes	4
Second-Life marketplace	5

7. Primary, Candidate, and Alternate Key Strategy

7.1 Strategy

Topic	Decision	Reason
Entity identity	Opaque surrogate `Identifier` primary key on every entity-bearing relation.	External identifiers are mutable, sensitive, or not universally available. A surrogate key keeps identity stable across resale and account changes.
Associative identity	Composite primary key formed from the parent keys.	The pair is the natural identity; it prevents duplicate links and avoids an artificial key with no business meaning.
Business codes	Stable codes such as `RoleCode`, `PermissionCode`, `CategoryCode`, `AttributeCode`, `RuleCode`, and `CriterionCode` are candidate keys.	They are stable, non-sensitive, and useful for integration.
Scoped natural keys	`(PrimaryCategoryId, AttributeCode)`, `(ProductId, VariantCode)`, `(ScoringRuleId, VersionLabel)`, and similar scoped pairs.	The business identifier is only unique within its scope.
Partial uniqueness	Open role assignments, open permission grants, one active listing per asset, and open block conditions.	A simple unique constraint would block legitimate history.
External values	Email, phone, serial number, IMEI, and order reference are never primary keys.	They are mutable, sensitive, or non-unique across systems.

7.2 Primary key summary

Key strategy	Relations	Primary key columns
Single surrogate key	43	43
Composite associative key	11	22
Total	54	65

7.3 Candidate and alternate keys

Key ID	Relation	Attributes	Status	Justification
AK-USER-01	RL-01	ContactEmail	Conditional	Alternate key only if email becomes the authentication contact; personal data, not a universal key.
AK-ROLE-01	RL-02	RoleCode	Confirmed	Stable business code; safe alternate key for integration and reference.
AK-PERM-01	RL-03	PermissionCode	Confirmed	Stable capability code.
AK-URA-01	RL-04	UserId, RoleId	Conditional	Unique only for assignments that are still open (EndedAt is absent); partial uniqueness enforced in the application.
AK-RPA-01	RL-05	RoleId, PermissionId	Conditional	Unique only for open grants (EndedAt absent); partial uniqueness enforced in the application.
AK-PREF-01	RL-06	UserId, PreferenceType, PreferenceScope	Conditional	Scope may be absent; effective uniqueness needs a documented scope default before physical enforcement.
AK-PROD-01	RL-07	BrandReference, ModelNumber	Conditional	Candidate business key only after brand governance and model-number normalization are approved (D-01).
AK-VAR-01	RL-08	ProductId, VariantCode	Confirmed	Scoped natural key consistent with the

				conceptual unique variant code within a product.
AK-CAT-01	RL-09	CategoryCode	Confirmed	Stable classification code.
AK-ATTR-01	RL-10	PrimaryCategoryId, AttributeCode	Confirmed	Scoped business key; supports category-specific definitions without a global taxonomy.
AK-ATTR-02	RL-10	PrimaryCategoryId, AttributeCode, DefinitionVersion	Conditional	Version-aware key if definition versioning becomes effective-dated.
AK-FACT-01	RL-11	SubjectProductId, SubjectVariantId, AttributeDefinitionId, PrimarySourceId, ObservedAt	Deferred	Context-dependent uniqueness only; duplicate assertions and conflicts are legitimate. No universal fact key is asserted.
AK-SRC-01	RL-12	SourceTypeCode, SourceReference, RetrievedAt	Conditional	Candidate deduplication key; a source may legitimately be re-retrieved at a different time.
AK-PUR-01	RL-15	AcquisitionChannel, ExternalOrderReference	Conditional	Only where an external order reference is present and integration semantics are approved.
AK-OP-01	RL-16	ProductAssetId, PeriodStartAt	Confirmed	Prevents two periods for the same asset from starting at the same instant.
AK-OP-02	RL-16	ProductAssetId, PeriodReference	Conditional	Only when the BR-17 reference-code policy is approved; reference format not invented.
AK-EVR-01	RL-19	PrimaryCategoryId, PurposeCode, EvidenceTypeCode, PolicyReference	Conditional	Candidate uniqueness pending the category evidence matrix (D-02).
AK-COMPAT-01	RL-25	SubjectProductId, SubjectVariantId,	Deferred	Conditional deduplication only.

		TargetProductId, TargetVariantId, CompatibilityTypeCode, PrimarySourceId		Multiple attributable assertions must remain possible, so no unconditional unique constraint is imposed.
AK-CRULE-01	RL-26	RuleCode, VersionLabel	Confirmed	Versioned rule identity.
AK-LIST-01	RL-28	ProductAssetId, CreatedAt	Conditional	Candidate business key for listing identity; not required for physical implementation.
AK-CRIT-01	RL-33	CriterionCode	Confirmed	Stable criterion code.
AK-WEIGHT-01	RL-34	RuleVersionId, DecisionCriterionId	Confirmed	One weight per criterion within a rule version.
AK-SRULE-01	RL-35	RuleCode	Confirmed	Stable rule code.
AK-RVER-01	RL-36	ScoringRuleId, VersionLabel	Confirmed	Version identity within a scoring rule.
AK-ASSETID-01	RL-43	IdentifierTypeCode, IdentifierValueNormalized	Conditional	Global uniqueness only if the identifier policy confirms identifier scope; otherwise scoped uniqueness applies.
AK-ASSETID-02	RL-43	ProductAssetId, IdentifierTypeCode, IdentifierValueNormalized	Confirmed	Prevents the same value being attached twice to one asset.
AK-ALLOWED-01	RL-44	AttributeDefinitionId, AllowedValueCode	Confirmed	Composite primary key is the natural relationship key.
AK-FACTEV-01	RL-45	ProductFactId, EvidenceSubmissionId	Confirmed	Prevents duplicate support links.
AK-VALEV-01	RL-46	OwnershipValidationId, EvidenceSubmissionId	Confirmed	Prevents duplicate support links.
AK-SUBREQ-01	RL-47	EvidenceSubmissionId, EvidenceRequirementId	Confirmed	Prevents duplicate requirement links.
AK-ELIGEV-01	RL-48	ListingEligibilityAssessmentId, EvidenceAssessmentId	Confirmed	Prevents duplicate basis links.
AK-ELIGOWN-01	RL-49	ListingEligibilityAssessmentId, OwnershipValidationId	Confirmed	Prevents duplicate basis links.

AK-ELIGCOND-01	RL-50	ListingEligibilityAssessmentId, ConditionAssessmentId	Confirmed	Prevents duplicate basis links.
AK-SCITEM-01	RL-51	SavedComparisonId, DisplaySequence	Confirmed	Stable display ordering within a comparison.
AK-DRCP-01	RL-52	DecisionRequestId, CandidateSequence	Confirmed	Stable supply order within a request.
AK-DRCA-01	RL-53	DecisionRequestId, CandidateSequence	Confirmed	Stable supply order within a request.
AK-DRI-01	RL-54	DecisionResultId, RankPosition	Conditional	Unique only when a rank is present; ties and tie-breaking policy remain deferred.

No unconditional candidate key is asserted for RL-11 ProductFact, RL-14 ProductAsset, RL-25 CompatibilityRelationship, or the pure evidence and decision structures. RL-16 OwnershipPeriod has the period-start key and a conditional period reference. Duplicate fact and compatibility assertions are legitimate, so no universal natural key is invented.

8. Foreign Key and Relationship Mapping

Measure	Count
Conceptual relationships	78
Mapped by foreign key	69
Mapped by associative relation	9
Total foreign key constraints	110
Mandatory foreign keys	68
Optional foreign keys	42

8.1 Referential actions

Recommended action	Count	Scope
Cascade (link rows only)	15	Pure link rows only
Restrict	95	Master and historical data

Restrict is the default because the model retains ownership, assessment, listing, decision, and audit history. Cascade (link rows only) is recommended only for pure associative rows whose parents are intentionally deleted under an approved retention policy. No cascade reaches a historical business record.

8.2 Relationship rules

Rule	Application
Role-specific relationship	Conflict fact A and fact B, and compatibility subject and target, use two role-specific foreign keys. No redundant junction is created.

Inverse one-to-one	OwnershipPeriod.OpeningTransferId is unique and optional, implementing the transfer-to-new-period link without a circular foreign key.
Polymorphic target	Audit target is a type and identifier pair (A-12). Referential integrity to every possible target is deliberately not claimed.
Optional provenance	A fact may have no source only when IsUnsourced is explicitly true (LC-06).
Cross-product checks	Variant-to-product consistency for assets, facts, and compatibility endpoints is a transactional check, not a simple foreign key.

The complete 78-row mapping and the full foreign key catalogue are in Logical_Relationship_and_Key_Mapping.md.

9. Associative Relations

9.1 Conceptual M:N resolutions

R ID	Associative relation	Key	Key strategy	Business purpose
R-19	RL-45 FactEvidenceLink	Composite PK (ProductFactId, EvidenceSubmissionId)	Composite primary key	Support does not make a fact authoritative.
R-31	RL-46 ValidationEvidenceLink	Composite PK (OwnershipValidationId, EvidenceSubmissionId)	Composite primary key	Validation remains a distinct decision.
R-33	RL-47 SubmissionRequirementLink	Composite PK (EvidenceSubmissionId, EvidenceRequirementId)	Composite primary key	Requirement satisfaction is assessed separately.
R-47	RL-48 EligibilityEvidenceBasis	Composite PK (ListingEligibilityAssessmentId, EvidenceAssessmentId)	Composite primary key	Evidence acceptance alone is insufficient.
R-48	RL-49 EligibilityOwnershipBasis	Composite PK (ListingEligibilityAssessmentId, OwnershipValidationId)	Composite primary key	Ownership validation is not the listing state.
R-49	RL-50 EligibilityConditionBasis	Composite PK (ListingEligibilityAssessmentId, ConditionAssessmentId)	Composite primary key	Condition grade is not evidence confidence.
R-74	RL-51 SavedComparisonItem	Composite PK (SavedComparisonId, ProductId)	Composite primary key	Comparison is not a product fact.
R-76	RL-52 DecisionRequestCandidateProduct	Composite PK (DecisionRequestId, ProductId)	Composite primary key	Candidate absence is not a negative preference.

R-77	RL-53 DecisionRequestCandidateAsset	Composite PK (DecisionRequestId, ProductAssetId)	Composite primary key	Asset context does not replace product facts.

No associative relation duplicates a first-class business entity. Ownership transfer, evidence assessment, eligibility assessment, and decision result already exist as entities; their relationships are therefore foreign keys, not junctions.

9.2 Supporting decomposed relations

Relation	Parent relation	Why a separate relation is required
RL-43 ProductAssetIdentifier	RL-14 ProductAsset	External identifier values are multivalued and policy-controlled.
RL-44 AttributeAllowedValue	RL-10 AttributeDefinition	An enumerated value set is multivalued; storing it in one row would violate 1NF.
RL-54 DecisionResultItem	RL-38 DecisionResult	Ranking produces one row per candidate; a single row would contain a repeating group.

10. Attribute Catalogue Summary

The full attribute catalogue is in Logical_Relation_Catalogue.md Sections 6.1 to 6.54 and Logical_Data_Dictionary.md Section 4. It contains **502 attributes** with stable identifiers LAT-001 to LAT-502.

10.1 Logical type distribution

Logical type	Attributes
Identifier	154
Timestamp	109
Short text	77
Enumerated/reference value	66
Long text	42
Date	17
Boolean	12
Structured value	9
Integer	8
Decimal	6
Binary/file reference	2

10.2 Attribute counts by relation

Relation	Name	Attributes	Required	Optional
RL-01	User	10	3	7

RL-02	Role	7	5	2
RL-03	Permission	8	4	4
RL-04	UserRoleAssignment	8	5	3
RL-05	RolePermissionAssignment	8	5	3
RL-06	UserPreference	8	6	2
RL-07	Product	11	4	7
RL-08	ProductVariant	7	4	3
RL-09	Category	10	5	5
RL-10	AttributeDefinition	17	10	7
RL-11	ProductFact	23	8	15
RL-12	ProductSource	12	6	6
RL-13	InformationConflict	11	7	4
RL-14	ProductAsset	8	4	4
RL-15	Purchase	10	5	5
RL-16	OwnershipPeriod	13	6	7
RL-17	OwnershipTransfer	11	5	6
RL-18	OwnershipValidation	13	9	4
RL-19	EvidenceRequirement	11	8	3
RL-20	EvidenceSubmission	10	6	4
RL-21	EvidenceItem	10	4	6
RL-22	EvidenceAssessment	12	8	4
RL-23	ConditionClaim	10	6	4
RL-24	ConditionAssessment	13	6	7
RL-25	CompatibilityRelationship	15	7	8
RL-26	CompatibilityRule	11	6	5
RL-27	CompatibilityCondition	11	5	6
RL-28	ResaleListing	10	6	4
RL-29	ListingStatusEvent	11	6	5
RL-30	ListingEligibilityAssessment	11	8	3
RL-31	BlockingCondition	10	7	3
RL-32	PriceRecord	12	6	6

RL-33	DecisionCriterion	10	5	5
RL-34	CriterionWeight	7	5	2
RL-35	ScoringRule	9	4	5
RL-36	RuleVersion	10	4	6
RL-37	DecisionRequest	10	7	3
RL-38	DecisionResult	10	5	5
RL-39	DecisionExplanation	9	6	3
RL-40	UsageRecord	13	5	8
RL-41	SavedComparison	7	4	3
RL-42	AuditRecord	12	6	6
RL-43	ProductAssetIdentifier	9	7	2
RL-44	AttributeAllowedValue	7	4	3
RL-45	FactEvidenceLink	3	3	0
RL-46	ValidationEvidenceLink	3	3	0
RL-47	SubmissionRequirementLink	3	3	0
RL-48	EligibilityEvidenceBasis	3	3	0
RL-49	EligibilityOwnershipBasis	3	3	0
RL-50	EligibilityConditionBasis	3	3	0
RL-51	SavedComparisonItem	4	4	0
RL-52	DecisionRequestCandidateProduct	4	4	0
RL-53	DecisionRequestCandidateAsset	4	4	0
RL-54	DecisionResultItem	7	4	3

Total: 502 attributes, 286 required, 216 optional.

11. Normalization Analysis

Full analysis is in `Logical_Normalization_and_Integrity_Report.md`. The findings are summarized here.

Normal form	Finding	Evidence
1NF	Satisfied	All attributes are atomic. Multivalued identifier values, allowed attribute values, evidence links, eligibility bases, comparison items, decision candidates, and ranking items are separate relations. The typed value group in `ProductFact` is an alternative representation of one governed value, not a repeating group.

2NF	Satisfied	Every composite-key relation has non-key attributes that depend on the whole key. Linkage timestamps and sequence numbers describe the relationship, not one parent.
3NF	Satisfied for identifiable dependencies	No non-key attribute determines another non-key attribute. Category, source, definition, and rule metadata are referenced, not copied. Four controlled redundancies are registered with enforcement rules.
4NF	No separate issue identified	Multivalued dependencies are decomposed into associative relations; no remaining independent multivalued fact is stored in one relation.

11.1 Controlled redundancies

ID	Redundancy	Justification	Enforcement
CR-01	Listing current state versus status events	Active discovery and eligibility gating need a current value	Transactional write pair plus reconciliation
CR-02	Ownership period state versus end time	Lifecycle state is explicit in the conceptual model	Check plus transactional write pair
CR-03	Result rule version versus request rule version	Reproducibility requires the exact version used	Match at generation; request immutable after result
CR-04	Asset first acquisition date versus purchase dates	The original date must survive resale and imported history	Set once from origin evidence; audited correction only

No performance denormalization is proposed, because no query workload or measurement exists.

12. Integrity Constraints

The model defines **50 logical constraints**. LC-01 to LC-40 correspond one-to-one with Phase 02.2 IC-01 to IC-40. LC-41 to LC-50 are structural constraints for associative relations and ranking.

ID	Constraint	Enforcement classification	Source
LC-01	A ProductAsset retains one stable platform identity across purchase, ownership, listing, resale, and transfer; resale never creates a new ProductAsset.	Transactional/application-level	IC-01
LC-02	Every ProductAsset references exactly one Product, and any ProductVariant reference must belong to that Product.	Standard relational + transactional check	IC-02
LC-03	Every ProductVariant references exactly one Product and cannot	Standard relational	IC-03

	exist without it.		
LC-04	Every AttributeDefinition declares a value type, an allowed domain, a scope, and a unit where applicable; category-specific definitions are not unrestricted free text.	Standard relational + policy-dependent	IC-04
LC-05	Every ProductFact has exactly one Product subject and exactly one AttributeDefinition; a variant-scoped fact uses a variant of that Product and a value conforming to the definition.	Standard relational + transactional check	IC-05
LC-06	A ProductFact retains provenance class and recorded time; a source is optional only when the fact is explicitly marked unsourced; AI-generated content is not authoritative.	Standard relational + transactional check	IC-06
LC-07	An InformationConflict references two distinct ProductFacts, and resolution never deletes or overwrites either fact.	Standard relational (check) + transactional	IC-07
LC-08	A ProductAsset may exist with zero ownership periods until an acquisition or origin is recorded; an initial ownership start references a purchase or accepted origin evidence.	Transactional/application-level	IC-08
LC-09	A ProductAsset has at most one Active OwnershipPeriod at a time, and active periods do not overlap.	Transactional/application-level	IC-09
LC-10	An OwnershipPeriod end time does not precede its start time; an ended period is retained and corrected only through a controlled, auditable correction.	Standard relational (check) + transactional	IC-10
LC-11	An OwnershipTransfer references the same ProductAsset through its prior and new periods, closes the prior period, opens the new period, and never creates overlapping active ownership.	Transactional/application-level	IC-11
LC-12	Current ownership is determined from the applicable Active OwnershipPeriod; a role, listing, purchase, or claim is not by itself proof of ownership.	Transactional/application-level	IC-12
LC-13	A Purchase references a recorded buyer and a Product, may reference a ProductAsset, and	Standard relational + transactional	IC-13

	does not by itself prove current ownership.		
LC-14	A ResaleListing references exactly one ProductAsset and one recorded seller; seller authorization is validated and not inferred from the listing.	Standard relational + policy-dependent	IC-14
LC-15	A ProductAsset has at most one Active or Reserved ResaleListing at the same time.	Transactional/application-level	IC-15
LC-16	An Active listing has an applicable Eligible ListingEligibilityAssessment and no unresolved blocking condition.	Transactional/application-level	IC-16
LC-17	A Sold listing is not proof that a sale, payment, or ownership transfer completed; listing state and ownership state remain separate.	Transactional/application-level	IC-17
LC-18	An EvidenceSubmission has a submitter and an explicit context; a complete submission contains at least one EvidenceItem, and each item belongs to exactly one submission.	Standard relational + transactional	IC-18
LC-19	Every EvidenceAssessment references one EvidenceSubmission and records an outcome, rationale, time, and reviewer or process; prior assessments are retained when superseded.	Standard relational + transactional	IC-19
LC-20	Evidence acceptance is not listing eligibility; ownership validation and eligibility determination remain separate records.	Structural (separate relations) + transactional	IC-20
LC-21	Evidence confidence uses Low, Medium, High, or Insufficient Evidence; condition uses 5 to 1, Not Assessed, or Unknown; the two classifications are never conflated.	Standard relational (domain check)	IC-21
LC-22	A ConditionClaim records claimed condition; a ConditionAssessment records assessed condition; neither overwrites the other.	Structural + transactional	IC-22
LC-23	A CompatibilityRelationship has a subject and target, uses Compatible, Incompatible, Conditional, or Unknown, and preserves direction; absence	Standard relational + transactional	IC-23

	means Unknown, not Incompatible.		
LC-24	A Conditional compatibility outcome has attributable conditions, and rule and source attribution is retained where used.	Transactional/application-level	IC-24
LC-25	A role or permission does not establish ownership, product-truth authority, or transaction completion.	Policy-dependent	IC-25
LC-26	Prior-owner personal information is not exposed to subsequent owners through ordinary ownership or listing views.	Transactional/application-level + policy-dependent	IC-26
LC-27	AuditRecord is append-oriented for sensitive or lifecycle-critical events, contains no secrets or unnecessary personal data, and is not proof of transaction completion.	Transactional/application-level + policy-dependent	IC-27
LC-28	A DecisionResult references the exact RuleVersion used, criterion weights are version-scoped, and no weight, formula, threshold, normalization, or tie-break rule is invented.	Standard relational + policy-dependent	IC-28
LC-29	An AI-assisted DecisionExplanation is labelled, cannot override deterministic output or authoritative facts, and cannot be the sole basis for a product-truth claim.	Standard relational (check) + transactional	IC-29
LC-30	Unknown, Not Assessed, missing, and Insufficient Evidence are not converted into a negative fact, a zero value, or Incompatible.	Transactional/application-level	IC-30
LC-31	A PriceRecord has exactly one approved scope, a non-negative amount, a currency, and a source context; a missing price is not zero and market observation is not implied.	Standard relational (check)	IC-31
LC-32	An EvidenceRequirement identifies its scope, evidence type, mandatory or optional intent, and policy or version context; no category matrix is invented.	Standard relational + policy-dependent	IC-32
LC-33	An Eligible determination requires the applicable ownership, evidence, condition, and blocking checks to pass; Incomplete and	Transactional/application-level + policy-dependent	IC-33

	Ineligible are distinct and never treated as eligible.		
LC-34	A BlockingCondition is limited to approved blocking types and creates no payment, escrow, refund, remedy, or full dispute subsystem.	Standard relational + policy-dependent	IC-34
LC-35	RetentionCategory is a controlled classification, not a relation, and no retention period is invented.	Policy-dependent	IC-35
LC-36	A UserRoleAssignment is unique for an open (UserId, RoleId) pair; a RolePermissionAssignment is unique for an open (RoleId, PermissionId) pair; historical assignments remain distinct.	Transactional/application-level	IC-36
LC-37	A UserPreference is user-scoped and is not a verified fact, a decision result, or a permanent requirement.	Standard relational + transactional	IC-37
LC-38	A ProductSource retains attributable source context; freshness is optional only when explicitly unknown; a source record does not make its claim true.	Standard relational + transactional	IC-38
LC-39	A UsageRecord is optional and evidence-limited; unknown usage is not zero and does not override assessed condition.	Standard relational + transactional	IC-39
LC-40	A SavedComparison belongs to one user and references selected products; it is not a DecisionResult and does not freeze authoritative product facts.	Standard relational + transactional	IC-40
LC-41	FactEvidenceLink is unique on (ProductFactId, EvidenceSubmissionId).	Standard relational (composite PK)	R-19
LC-42	ValidationEvidenceLink is unique on (OwnershipValidationId, EvidenceSubmissionId).	Standard relational (composite PK)	R-31
LC-43	SubmissionRequirementLink is unique on (EvidenceSubmissionId, EvidenceRequirementId).	Standard relational (composite PK)	R-33
LC-44	EligibilityEvidenceBasis is unique on (ListingEligibilityAssessmentId, EvidenceAssessmentId).	Standard relational (composite PK)	R-47
LC-45	EligibilityOwnershipBasis is unique on (ListingEligibilityAssessmentId,	Standard relational (composite PK)	R-48

	OwnershipValidationId).		
LC-46	EligibilityConditionBasis is unique on (ListingEligibilityAssessmentId, ConditionAssessmentId).	Standard relational (composite PK)	R-49
LC-47	SavedComparisonItem is unique on (SavedComparisonId, ProductId) and display order is unique within the comparison.	Standard relational + transactional	R-74
LC-48	DecisionRequestCandidateProduct is unique on (DecisionRequestId, ProductId) and supply order is unique within the request.	Standard relational + transactional	R-76
LC-49	DecisionRequestCandidateAsset is unique on (DecisionRequestId, ProductAssetId) and supply order is unique within the request.	Standard relational + transactional	R-77
LC-50	DecisionResultItem is unique on (DecisionResultId, CandidateSequence); a rank position, where present, is unique within the result and no tie-break rule is invented.	Standard relational + transactional	CE-38 ranking

12.1 Enforcement summary

Enforcement classification	Count
Policy-dependent	2
Standard relational	1
Standard relational (check)	1
Standard relational (check) + transactional	3
Standard relational (composite PK)	6
Standard relational (domain check)	1
Standard relational + policy-dependent	5
Standard relational + transactional	12
Standard relational + transactional check	3
Structural (separate relations) + transactional	1
Structural + transactional	1
Transactional/application-level	11
Transactional/application-level + policy-dependent	3

Standard relational constraints cover identity, referential integrity, mandatory participation, declared uniqueness, and simple domain checks. Temporal overlap, listing exclusivity, eligibility gating, cross-period transfer consistency, and current-state reconciliation require transactional logic. Retention,

reviewer authority, ownership proof, evidence content, condition rubric, scoring, market price, identifier policy, audit catalogue, and transaction completion remain policy-dependent.

13. Controlled Values and Reference Data

13.1 Baseline vocabularies

ID	Vocabulary	Values	Status	Source
CV-01	Listing state	Draft, Pending Review, Active, Rejected, Reserved, Sold, Suspended, Withdrawn, Expired	Documented	OD-14; SM-01
CV-02	Ownership period state	Active, Ended	Documented conceptual	OD-04; SM-03
CV-03	Evidence confidence	Low, Medium, High, Insufficient Evidence	Documented	OD-06
CV-04	Condition grade	5 - Excellent, 4 - Very Good, 3 - Good, 2 - Fair, 1 - Poor, Not Assessed, Unknown	Documented	OD-13
CV-05	Compatibility outcome	Compatible, Incompatible, Conditional, Unknown	Documented	OD-08
CV-06	Evidence review status	Submitted, Under Review, Accepted, Rejected, Needs More Information	Working conceptual	OD-06
CV-07	Eligibility outcome	Unknown, Under Review, Eligible, Ineligible, Incomplete	Mixed documented and working	OD-06; OD-14
CV-08	Price type	Asking, Reference	Working conceptual	OD-15
CV-09	Producer type	BackendRule, Human, AI-assisted, SystemProcess	Working conceptual	OD-11; OD-16
CV-10	Audit outcome	Success, Failure, Denied	Working conceptual	OD-18
CV-11	Verification status	Unverified, Supported, Verified, Conflicted	Working conceptual	OD-09; OD-11
CV-12	Conflict status	Open, Under Review, Resolved	Working conceptual	OD-09
CV-13	Evidence item type	Content deferred	Deferred	OD-05; D-02
CV-14	Blocking condition	Content deferred	Deferred	OD-07; OD-14

	type			
CV-15	Compatibility type	Content deferred	Deferred	OD-08

13.2 Deferred domains

DV-16 to DV-50 define 35 logical domains for statuses and categories that the inspected sources mention but do not enumerate. Thirty-four of them have no approved value list. No value is invented. DV-22 is fully defined because it is the technology-neutral logical type vocabulary used by this document, not a business taxonomy.

ID	Domain	Values	Dependency
DV-16	Account status	Content deferred	D-17
DV-17	Role assignment status	Content deferred	OD-02; D-11
DV-18	Permission grant status	Content deferred	OD-02; D-11
DV-19	Preference type	Content deferred	OD-03
DV-20	Product catalogue status	Content deferred	OD-09
DV-21	Variant status	Content deferred	OD-09
DV-22	Logical value type family	Identifier, Short text, Long text, Integer, Decimal, Boolean, Date, Timestamp, Structured value, Binary/file reference, Enumerated/reference value	Technology-neutral vocabulary of this document
DV-23	Product source type	Content deferred	OD-09; source policy
DV-24	Source freshness status	Content deferred	OD-09; freshness policy
DV-25	Conflict type	Content deferred	OD-09
DV-26	Acquisition channel	Content deferred	OD-07
DV-27	Purchase record status	Content deferred	OD-07
DV-28	Ownership transfer status	Content deferred	OD-04; OD-07
DV-29	Ownership validation outcome	Content deferred	OD-04; D-04
DV-30	Evidence requirement purpose	Content deferred	OD-05; D-02
DV-31	Evidence submission purpose	Content deferred	OD-05
DV-32	Evidence privacy classification	Content deferred	OD-12; D-12
DV-33	Evidence assessment outcome	Content deferred	OD-06; D-11

DV-34	Compatibility rule status	Content deferred	OD-08
DV-35	Compatibility condition type	Content deferred	OD-08
DV-36	Blocking condition status	Content deferred	OD-07; OD-14
DV-37	Price record status	Content deferred	OD-15
DV-38	Criterion preference direction	Content deferred	OD-16; D-06
DV-39	Decision criterion status	Content deferred	OD-16; D-06
DV-40	Scoring rule status	Content deferred	OD-16; D-06
DV-41	Rule version status	Content deferred	OD-16; D-06
DV-42	Decision request type	Content deferred	OD-16
DV-43	Decision request status	Content deferred	OD-16
DV-44	Decision result status	Content deferred	OD-16
DV-45	Validation status	Content deferred	OD-11; OD-16
DV-46	Usage metric	Content deferred	OD-17
DV-47	Audit action type	Content deferred	OD-18; D-10
DV-48	Audit target type	Content deferred	OD-18; D-10
DV-49	Asset identifier type	Content deferred	D-09
DV-50	Decision result item outcome	Content deferred	OD-16; D-06

13.3 Reference-data rules

Rule	Application
Documented values only	Listing state, ownership state, evidence confidence, condition grade, compatibility outcome, evidence review status, eligibility outcome, price type, producer type, audit outcome, verification status, and conflict status use exactly the Phase 02.2 values.
Deferred content	CV-13, CV-14, and CV-15 remain empty. No evidence type, blocking type, or compatibility type is invented.
Extensibility	A documented vocabulary may change only through its named decision record. Working vocabularies are marked as such.
Reference relations	RL-44 AttributeAllowedValue holds governed permitted values for enumerated attributes. The baseline vocabularies may be implemented as domains or reference relations at physical design; that choice is deferred.

14. Temporal and Historical Modeling

Concern	Logical representation	Rule
Ownership history	RL-16 OwnershipPeriod with PeriodStartAt, optional PeriodEndAt, PeriodStatusCode	No overlapping active periods; current owner derived from the active period.
Ownership transfer	RL-17 OwnershipTransfer plus OwnershipPeriod.OpeningTransferId	Transfer links one prior period to one new period; effective point deferred.
Listing history	RL-29 ListingStatusEvent plus current state on RL-28	Every transition is recorded; current state is a documented derived value.
Evidence assessment history	RL-22 EvidenceAssessment.SupersedesAssessmentId	A newer assessment supersedes rather than deletes.
Condition assessment history	RL-24 ConditionAssessment.SupersedesAssessmentId	Claimed and assessed grades remain separate across versions.
Eligibility history	RL-30 ListingEligibilityAssessment.SupersedesAssessmentId	Latest applicable determination is identified by assessment time and supersession.
Fact provenance and observation time	RL-11 RecordedAt, ObservedAt, PrimarySourceId, ProducerTypeCode	A later claim does not overwrite an earlier one; conflicts are recorded in RL-13.
Decision rule versions	RL-36 RuleVersion with effective dates and status	Results bind to the exact version used.
Saved decision context	RL-37 RequirementSnapshot and RL-38 InputSnapshotReference	Inputs are preserved without becoming product truth.
Audit history	RL-42 AuditRecord	Append-oriented, selective, privacy-safe; not proof of transaction completion.
Category and requirement effective dating	RL-09, RL-19, RL-26, RL-36 effective dates	Effective dating is recorded where the conceptual model defines it; no automatic expiry job is invented.

14.1 State-model preservation

The five Phase 02.2 state-model assessments are preserved. No new state, transition, or guard is invented. The logical model supplies the attributes and history relations that the assessments require.

Assessment	Subject	Logical representation	Preservation note
SM-01	Resale listing lifecycle	RL-28 CurrentStateCode (CV-01), RL-29 ListingStatusEvent	The nine documented states remain; current state plus event history supports recorded transitions. Transition authority and expiry

			semantics remain deferred (D-05).
SM-02	Evidence submission and review lifecycle	RL-20 ReviewStatusCode (CV-06), RL-22 EvidenceAssessment	Review status and assessment history are separate. Reviewer authority and appeal policy remain deferred (D-11).
SM-03	Ownership lifecycle	RL-16 OwnershipPeriod (CV-02), RL-17 OwnershipTransfer	Active and Ended are preserved; transfer closes one period and opens the next. Effective-point semantics remain deferred (D-05).
SM-04	Listing eligibility determination lifecycle	RL-30 OutcomeCode (CV-07), basis relations RL-48 to RL-50	Unknown, Under Review, Eligible, Ineligible, and Incomplete are preserved. Eligibility policy remains deferred (D-04).
SM-05	Purchase, sale, and transfer lifecycle assessment	RL-15 Purchase, RL-16 OwnershipPeriod, RL-17 OwnershipTransfer, RL-28 ResaleListing	The assessment is represented as separate purchase, listing, and ownership records. Sold is not treated as a completed sale, payment, or transfer. No lifecycle stage is added.

No Mermaid diagram was added or altered in Phase 02.3. The eight conceptual diagrams remain the visual baseline; the logical design expresses relationships in tables.

14.2 Temporal attributes deliberately not added

No DeletedAt or soft-delete flag is defined, because retention and deletion policy is unresolved (OD-12). No blanket UpdatedAt is added to append-only relations such as RL-29, RL-42, RL-45 to RL-50, and RL-54; their event or linkage time is sufficient. CreatedAt is present where record creation time supports audit or lifecycle reconstruction, not by default on every relation.

15. Logical Index Recommendations

Indexes are logical recommendations only. No DBMS syntax is specified, no index is claimed to be measured, and every conditional recommendation depends on future workload evidence or an unresolved policy.

ID	Relation	Attribute(s)	Expected access pattern	Why it may help	Conditional
IX-01	RL-01 User	UserId	Primary key lookup; actor resolution	Every FK to User resolves by primary key.	No
IX-02	RL-01 User	ContactEmail	Account contact lookup	Conditional on email-based authentication or contact	Conditional (D-17)

				workflow.	
IX-03	RL-04 UserRoleAssignment	UserId, RoleId	Open assignments for a user	Supports authorization checks and assignment history views.	No
IX-04	RL-04 UserRoleAssignment	RoleId	Assignments for a role	Supports role administration and audit.	No
IX-05	RL-05 RolePermissionAssignment	RoleId, PermissionId	Permission grants for a role	Supports authorization resolution.	No
IX-06	RL-06 UserPreference	UserId, PreferenceType	Preference retrieval for a user	Supports session and decision input assembly.	No
IX-07	RL-07 Product	PrimaryCategoryId	Products in a category	Supports catalogue browsing and category filters.	No
IX-08	RL-07 Product	BrandReference, ModelNumber	Product identification	Supports catalogue matching and duplicate review.	Conditional (D-01)
IX-09	RL-08 ProductVariant	ProductId, VariantCode	Variant lookup within a product	Supports variant selection and candidate assembly.	No
IX-10	RL-09 Category	ParentCategoryId	Hierarchy traversal	Supports category tree retrieval.	No
IX-11	RL-10 AttributeDefinition	PrimaryCategoryId, AttributeCode	Attribute definitions for a category	Supports comparison, filtering, and fact validation.	No
IX-12	RL-11 ProductFact	SubjectProductId, AttributeDefinitionId	Facts for a product attribute	Supports comparison and truth inspection.	No
IX-13	RL-11 ProductFact	SubjectVariantId	Facts for a variant	Supports variant-level comparison.	No
IX-14	RL-11 ProductFact	PrimarySourceId	Facts attributed to a source	Supports provenance review.	No

IX-15	RL-13 InformationConflict	ConflictFactAId	Conflicts involving a fact	Supports conflict inspection from a fact.	No
IX-16	RL-13 InformationConflict	ConflictFactBId	Conflicts involving a fact	Second role path for the same access pattern.	No
IX-17	RL-14 ProductAsset	ProductId	Assets of a product model	Supports installed-base and asset lookup.	No
IX-18	RL-16 OwnershipPeriod	ProductAssetId, PeriodStatusCode, PeriodEndAt	Current ownership for an asset	Core ownership lookup; supports non-overlap verification.	No
IX-19	RL-16 OwnershipPeriod	OwnerUserId, PeriodEndAt	Assets owned by a user	Supports owner dashboards and privacy-scoped views.	No
IX-20	RL-17 OwnershipTransfer	ProductAssetId, RecordedAt	Transfer history for an asset	Supports lifecycle reconstruction.	No
IX-21	RL-18 OwnershipValidation	ProductAssetId, ValidatedAt	Validation history for an asset	Supports eligibility basis review.	No
IX-22	RL-20 EvidenceSubmission	ProductAssetId, SubmittedAt	Submissions for an asset	Supports evidence review queues.	No
IX-23	RL-20 EvidenceSubmission	ResaleListingId, SubmittedAt	Submissions for a listing	Supports listing review.	No
IX-24	RL-22 EvidenceAssessment	EvidenceSubmissionId, AssessedAt	Assessment history for a submission	Supports latest-assessment retrieval.	No
IX-25	RL-23 ConditionClaim	ProductAssetId, ClaimedAt	Claims for an asset	Supports claim and disclosure review.	No
IX-26	RL-24 ConditionAssessment	ProductAssetId, AssessedAt	Assessments for an asset	Supports latest assessed condition retrieval.	No
IX-27	RL-25 CompatibilityRelationship	SubjectProductId, TargetProductId	Compatibility lookup for a subject	Core compatibility graph traversal.	No

IX-28	RL-25 CompatibilityRelationship	TargetProductId, SubjectProductId	Reverse compatibility lookup	Supports lookups without assuming symmetry.	No
IX-29	RL-28 ResaleListing	ProductAssetId, CurrentStateCode	Active or reserved listing for an asset	Supports the one-active-listing rule and discovery.	No
IX-30	RL-28 ResaleListing	CurrentStateCode, CreatedAt	Active listing discovery	Supports marketplace browsing by state.	No
IX-31	RL-29 ListingStatusEvent	ResaleListingId, EventAt	Listing state history	Supports history reconstruction and current-state verification.	No
IX-32	RL-30 ListingEligibilityAssessment	ResaleListingId, AssessedAt	Latest eligibility determination	Supports eligibility gating.	No
IX-33	RL-31 BlockingCondition	ResaleListingId, ResolvedAt	Open blocking conditions	Supports eligibility checks.	No
IX-34	RL-32 PriceRecord	ResaleListingId, ObservedAt	Asking price history	Supports listing price display.	No
IX-35	RL-32 PriceRecord	ProductId, PriceTypeCode, ObservedAt	Reference price lookup	Supports decision inputs.	No
IX-36	RL-34 CriterionWeight	RuleVersionId, DecisionCriterionId	Weights for a rule version	Supports reproducible evaluation.	No
IX-37	RL-37 DecisionRequest	RequestingUserId, RequestedAt	Requests by user	Supports decision history.	No
IX-38	RL-38 DecisionResult	DecisionRequestId, GeneratedAt	Results for a request	Supports result retrieval.	No
IX-39	RL-42 AuditRecord	TargetTypeCode, TargetIdentifier, EventAt	Audit trail for a target	Supports investigation without a polymorphic FK.	No
IX-40	RL-42 AuditRecord	ActorUserId, EventAt	Audit trail for an actor	Supports accountability review.	No

Primary key access is not separately indexed in this list because every relation has a primary key by definition. Foreign key columns are covered where the expected join or lifecycle lookup justifies a recommendation. No index is recommended for a column whose access pattern is unknown.

16. Logical Data Dictionary

The full attribute-level dictionary is Logical_Data_Dictionary.md. It contains all 502 attributes with meaning, type, required status, key role, uniqueness, and validation, plus the domain register. The dictionary is generated from the same specification as the relation catalogue, so identifiers, types, and key roles agree.

17. Requirements Traceability

17.1 Coverage status definitions

Status	Meaning
Fully Mapped	A logical structure represents the requirement, subject to later physical design and unresolved policy.
Partially Mapped	A logical structure exists, but a named policy, source, or mechanism is unresolved.
Deferred by Design	The requirement is intentionally not modeled now because scope or persistence is unapproved.
Not Mapped	No logical representation exists.
Not Applicable	The requirement creates no logical data need.

17.2 Data requirements

DR	Requirement	Logical relations	Coverage	Note
DR-001	Acting user reference	RL-01	Fully Mapped	Authentication remains a separate deferred concern.
DR-002	Role and permission assignment	RL-02, RL-03, RL-04, RL-05	Fully Mapped	Permission catalogue content deferred (D-11).
DR-003	Saved user preferences	RL-06	Fully Mapped	Preference catalogue and consent policy deferred.
DR-004	Structured user requirements	RL-37, RL-52, RL-53	Fully Mapped	EN-04 is represented by the request snapshot, not a separate entity.
DR-005	AI interpretation with validation status	RL-11, RL-39	Partially Mapped	EN-05 deferred; AI persistence policy D-07 unresolved.
DR-006	Search or recommendation history	None	Deferred by Design	No approved session-persistence requirement; no entity created.
DR-007	Product model record	RL-07	Fully Mapped	Product is not an asset.

DR-008	Brand reference	RL-07	Fully Mapped	Brand is a controlled external reference attribute (EN-07).
DR-009	Category hierarchy	RL-09	Fully Mapped	Taxonomy content deferred (D-01).
DR-010	Product variant or SKU identity	RL-08	Fully Mapped	Variant-defining attributes deferred.
DR-011	Category-specific attribute definitions	RL-10, RL-44	Fully Mapped	Attribute catalogue content deferred.
DR-012	Product attribute and specification values	RL-11	Fully Mapped	EN-11 is merged into ProductFact.
DR-013	Product media references	None	Deferred by Design	EN-14 deferred; media scope D-15 unresolved.
DR-014	Product descriptions	RL-07	Fully Mapped	Description is a product attribute, not a media entity.
DR-015	Product price information	RL-32	Partially Mapped	Price structures exist; market source and refresh policy deferred (D-08).
DR-016	Product information source	RL-12	Fully Mapped	Source-quality policy deferred.
DR-017	Fact-to-source association	RL-11, RL-12, RL-45	Fully Mapped	Single primary source plus evidence linkage.
DR-018	Evidence record and file reference	RL-19, RL-20, RL-21, RL-47	Fully Mapped	Evidence types and media handling deferred.
DR-019	Verification or confidence assessment	RL-22	Fully Mapped	Assessment rubric and reviewer policy deferred.
DR-020	Conflicting information record	RL-13	Fully Mapped	Resolution authority deferred; no silent winner.
DR-021	Source freshness or retrieval date	RL-12	Fully Mapped	Freshness thresholds deferred.
DR-022	Compatibility relationship	RL-25	Fully Mapped	Direction and Unknown preserved.
DR-023	Compatibility rule, condition, and version	RL-26, RL-27	Fully Mapped	Rule expression format deferred.
DR-024	Compatibility evaluation result	RL-25, RL-38	Partially Mapped	Independent persistence of an evaluation is not

				approved; decision output may retain it.
DR-025	Component or accessory identity	RL-07, RL-08	Fully Mapped	Component classification content deferred.
DR-026	Purchase record	RL-15	Fully Mapped	Completion semantics deferred; purchase is not ownership.
DR-027	Purchase-to-product association	RL-15	Fully Mapped	No specific asset required.
DR-028	Purchase-to-asset association	RL-15	Fully Mapped	Nullable by design at creation.
DR-029	Product asset stable identity	RL-14	Fully Mapped	Device identifier policy D-09 remains open.
DR-030	Asset-specific identifier	RL-43	Partially Mapped	Identifier values are modeled separately; visibility and identifier policy deferred.
DR-031	Asset acquisition or origin link	RL-15, RL-16, RL-18	Fully Mapped	Acceptable origin evidence standard deferred.
DR-032	Ownership period with start, end, and status	RL-16	Fully Mapped	Joint ownership not supported under A-05.
DR-033	Ownership transfer record	RL-17	Fully Mapped	Effective-point semantics unresolved (D-05).
DR-034	Ownership reference code per period	RL-16	Partially Mapped	PeriodReference exists but issuance, format, and invalidation policy remain pending (BR-17).
DR-035	Ownership validation result or evidence	RL-18, RL-46	Fully Mapped	Validation rubric and reviewer authority deferred.
DR-036	Current ownership lookup	RL-16	Fully Mapped	Derived from the active period; no mutable owner column.
DR-037	Asset condition record	RL-23, RL-24	Fully Mapped	Rubric and assessor authority deferred.
DR-038	Usage record	RL-40	Fully Mapped	Usage metrics and evidence policy

				deferred.
DR-039	Resale listing	RL-28	Fully Mapped	Listing rules and transaction semantics deferred.
DR-040	Listing-to-asset association	RL-28	Fully Mapped	No product-level orphan listing.
DR-041	Listing state and status history	RL-29	Fully Mapped	Transition authority and expiry policy deferred.
DR-042	Listing eligibility assessment	RL-30, RL-48, RL-49, RL-50	Fully Mapped	Eligibility policy inputs deferred.
DR-043	Listing-evidence association	RL-20, RL-21, RL-47	Fully Mapped	Evidence taxonomy and matrix deferred.
DR-044	Seller-reported condition claim	RL-23	Fully Mapped	Claim disclosure policy deferred.
DR-045	Asking price	RL-32	Fully Mapped	Market observation separately deferred.
DR-046	Dispute or blocking status	RL-31	Partially Mapped	Minimal blocking condition only; full dispute subsystem excluded.
DR-047	Product review record	None	Deferred by Design	EN-31 deferred; review scope D-16 unresolved.
DR-048	Review source and metadata	None	Deferred by Design	Dependent on DR-047; no review entity.
DR-049	Comparison set	RL-41, RL-51	Fully Mapped	Snapshot and sharing policy deferred.
DR-050	Decision request inputs	RL-37, RL-52, RL-53	Fully Mapped	Input snapshot policy and AI extraction persistence deferred.
DR-051	Decision result and ranking	RL-38, RL-54, RL-36	Fully Mapped	Weights and tie-break rules not invented.
DR-052	Decision explanation	RL-39	Fully Mapped	AI provider and validation policy deferred.
DR-053	Buy, upgrade, resell input set	RL-37, RL-07, RL-14	Fully Mapped	Cost and condition input completeness policy deferred.
DR-054	Asset lifecycle event	RL-16, RL-17, RL-28,	Partially Mapped	History is derived from authoritative

		RL-29, RL-42		records; no generic event entity (EN-36 rejected).
DR-055	Ownership and listing audit trail	RL-42	Partially Mapped	Audit event catalogue and mandatory scope pending (D-10).
DR-056	Retention and deletion metadata	None	Partially Mapped	RetentionCategory is a classification, not a relation; no period invented (OD-12).

Coverage	Count
Deferred by Design	4
Fully Mapped	43
Partially Mapped	9

17.3 Original business rules

BR	Rule	Logical relations	Coverage	Note
BR-01	AI is not authoritative for facts or database state	RL-11, RL-12, RL-39	Fully Mapped	AI provenance is labelled; no direct fact authority.
BR-02	AI may interpret, extract, explain, summarize, and support review	RL-39, RL-11	Fully Mapped	Producer type distinguishes AI.
BR-03	Validation, authorization, eligibility, compatibility, ranking, and ownership are enforced by the backend	RL-04, RL-05, RL-25, RL-28, RL-30, RL-38	Fully Mapped	Logical constraints are classified by enforcement mechanism.
BR-04	AI explanations must not override validated results	RL-38, RL-39	Fully Mapped	Explanation is subordinate to result.
BR-05	Product is distinct from Product Asset	RL-07, RL-08, RL-14	Fully Mapped	Separate identities and relationships.
BR-06	Asset identity is retained across ownership changes	RL-14	Fully Mapped	Resale never inserts a replacement asset.
BR-07	Ownership is a user ownership period, not the asset	RL-16, RL-01	Fully Mapped	Temporal model with no owner column.

BR-08	Product age and original first-purchase date do not reset	RL-14, RL-15, RL-16	Fully Mapped	FirstAcquisitionDate and origin links are retained.
BR-09	Previous ownership records and privacy are retained	RL-16, RL-26	Fully Mapped	Retention plus privacy constraints; access control deferred.
BR-10	Eligibility depends on platform origin or purchase history	RL-15, RL-18, RL-30	Fully Mapped	Exact origin-proof standard deferred.
BR-11	A seller must satisfy current-ownership validation	RL-18, RL-16, RL-30	Fully Mapped	Validation authority deferred.
BR-12	A listing is associated with the relevant Product Asset	RL-28, RL-14	Fully Mapped	No orphan listing.
BR-13	Required evidence is supplied and evaluated	RL-19, RL-20, RL-22	Fully Mapped	Evidence matrix content deferred.
BR-14	Eligibility may depend on state, evidence, ownership, origin, and blocking conditions	RL-30, RL-31, RL-48, RL-49, RL-50	Fully Mapped	Full dispute subsystem excluded.
BR-15	Seller claims, evidence, and platform assessment remain distinguishable	RL-23, RL-24, RL-22	Fully Mapped	No silent promotion between them.
BR-16	No physical inspection claim unless performed	RL-22, RL-24	Fully Mapped	Assessment scope and limitations are retained.
BR-17	Previous ownership reference is invalid; the new owner receives a new reference; history is retained	RL-16, RL-17, RL-18	Partially Mapped	PeriodReference exists; format and invalidation policy pending.

17.4 Derived business rules

DBR	Rule	Logical relations	Coverage	Note
DBR-01	No overlapping ownership periods for the same asset	RL-16	Fully Mapped	Enforced transactionally (LC-09).
DBR-02	Listing requires validated ownership	RL-18, RL-30, RL-49	Fully Mapped	Exact policy inputs

	and origin before eligibility			deferred.
DBR-03	Product facts retain source context or are explicitly unsourced	RL-11, RL-12	Fully Mapped	IsUnsourced is explicit.
DBR-04	AI-derived values are non-authoritative until validated and origin is visible	RL-11, RL-39	Fully Mapped	Producer type and verification status are retained.
DBR-05	Asset identity and first-purchase date are immutable through transfer	RL-14, RL-17	Fully Mapped	Enforced by transactional rules.
DBR-06	Evidence must not be relabeled as verification or inspection	RL-22, RL-24	Fully Mapped	Assessment type and scope are retained.

Original BR-* and derived DBR-* identifiers remain separate; no derived rule is presented as an original requirement.

17.5 Acceptance scenarios

AT	Scenario	Logical relations	Coverage	Note
AT-01	Search request has no matches; the system does not invent products	RL-07, RL-11	Fully Mapped	Empty results are an application behaviour; no invented rows.
AT-02	A required specification is missing and is not treated as a confirmed match	RL-10, RL-11	Fully Mapped	Missing facts remain absent.
AT-03	A candidate violates a mandatory constraint and is not labelled a full match	RL-37, RL-38, RL-54	Fully Mapped	Constraint catalogue remains deferred.
AT-04	Compatibility cannot be determined and is identified as unknown	RL-25	Fully Mapped	Absence of a row means Unknown.
AT-05	A seller lacks origin or ownership; the listing is blocked or handled by eligibility	RL-18, RL-30, RL-31	Fully Mapped	Exact eligibility policy deferred.
AT-06	Seller condition claims are distinguishable from evidence and assessment	RL-23, RL-24, RL-22	Fully Mapped	Separate relations and vocabularies.

AT-07	Ownership changes after resale while asset and first-purchase date stay unchanged	RL-14, RL-17, RL-15	Fully Mapped	Identity and first acquisition date are immutable.
AT-08	An unauthorized user requests private ownership information and is denied	RL-01, RL-16	Partially Mapped	Privacy constraint exists; access-control implementation is out of scope.
AT-09	An AI explanation conflicts with the validated backend result and the result remains authoritative	RL-38, RL-39	Fully Mapped	Producer type and precedence constraint.
AT-10	A buy-versus-upgrade comparison lacks cost input and the limitation is identified	RL-37, RL-38, RL-32	Fully Mapped	No unsupported definitive conclusion.

17.6 User stories

US	Story	Logical relations	Coverage	Note
US-01	Natural-language product discovery	RL-11, RL-10, RL-37	Partially Mapped	AI interpretation persistence and search history deferred.
US-02	Structured product filtering	RL-10, RL-11, RL-09	Fully Mapped	Search infrastructure deferred.
US-03	Product comparison	RL-11, RL-41	Fully Mapped	Comparison rendering is later.
US-04	Compatibility assessment	RL-25, RL-26, RL-27	Fully Mapped	Unknown is supported.
US-05	Purchase decision support	RL-37, RL-38, RL-36	Partially Mapped	Weights and formula deferred.
US-06	Product information assessment	RL-11, RL-12, RL-22	Fully Mapped	Source-quality policy deferred.
US-07	Ownership lifecycle tracking	RL-14, RL-16, RL-17	Fully Mapped	Privacy and access control later.
US-08	Resale eligibility	RL-18, RL-30	Fully Mapped	Policy inputs deferred.
US-09	Second-Life listing evaluation	RL-28, RL-20, RL-22	Fully Mapped	Transaction semantics deferred.
US-10	Ownership transfer	RL-17, RL-16	Partially Mapped	Transfer effective

				point unresolved.
US-11	Buy, upgrade, or resell evaluation	RL-37, RL-14, RL-32, RL-40	Partially Mapped	Cost source and weights deferred.
US-12	Authorized administration	RL-02, RL-03, RL-04, RL-42	Partially Mapped	Audit catalogue and permission catalogue deferred.

17.7 Use cases

UC	Use case	Logical relations	Coverage	Note
UC-01	Discover products	RL-07, RL-09, RL-11	Partially Mapped	Search history and AI interpretation deferred.
UC-02	Compare products	RL-11, RL-41	Fully Mapped	No open logical dependency.
UC-03	Check compatibility	RL-25, RL-26, RL-27	Fully Mapped	No open logical dependency.
UC-04	Evaluate a purchase decision	RL-37, RL-36, RL-38, RL-39	Partially Mapped	Formula and weights deferred.
UC-05	Inspect product information	RL-11, RL-12, RL-22	Fully Mapped	No open logical dependency.
UC-06	View owned product lifecycle	RL-14, RL-16, RL-40	Fully Mapped	No open logical dependency.
UC-07	Submit a second-life listing	RL-28, RL-20, RL-23	Fully Mapped	Evidence matrix and transaction semantics deferred.
UC-08	Review a second-life listing	RL-30, RL-22, RL-31	Fully Mapped	Reviewer policy deferred.
UC-09	Transfer ownership	RL-17, RL-16, RL-18	Partially Mapped	Completion and effective-point semantics unresolved.
UC-10	Evaluate buy versus upgrade versus resell	RL-37, RL-14, RL-32, RL-24	Partially Mapped	Cost, condition rubric, and weights deferred.

17.8 Traceability summary

Requirement set	Total	Fully Mapped	Partially Mapped	Deferred by Design	Not Mapped
Data requirements	56	43	9	4	0
Original business rules	17	16	1	0	0
Derived business rules	6	6	0	0	0

Acceptance scenarios	10	9	1	0	0
User stories	12	7	5	0	0
Use cases	10	6	4	0	0
Conceptual entities	42	42	0	0	0
Conceptual relationships	78	78	0	0	0

All 56 data requirements are individually traced. The nine partial mappings are DR-005, DR-015, DR-024, DR-030, DR-034, DR-046, DR-054, DR-055, and DR-056. The four deferred requirements are DR-006, DR-013, DR-047, and DR-048. No requirement is silently dropped.

17.9 Entity and relationship traceability

The 42-row entity disposition register is in Section 5.2. The 78-row relationship mapping is in Logical_Relationship_and_Key_Mapping.md Section 6.2. Both registers cover every conceptual element exactly once.

18. Assumptions and Dependencies

18.1 Working assumptions preserved

ID	Assumption	Logical effect
A-01	Each product has one primary category.	Product.PrimaryCategoryId is mandatory.
A-02	Each attribute definition has one primary category.	AttributeDefinition.PrimaryCategoryId is mandatory.
A-03	A fact has at most one primary source.	ProductFact.PrimarySourceId is optional and single-valued; additional support uses evidence links.
A-04	An asset may exist with zero ownership periods.	Ownership participation is not forced at asset creation.
A-05	One asset has at most one active owner; joint ownership is unsupported.	OwnershipPeriod.OwnerUserId is single-valued.
A-06	An asset has at most one Active or Reserved listing.	Partial uniqueness is recorded as a transactional rule.
A-07	A purchase may link to an asset; origin evidence may substitute.	OriginPurchaseId is optional.
A-08	A transfer links a prior and a new period; the effective point is unresolved.	TransferAt is optional and semantics are deferred.
A-09	Variant is optional for asset, fact, and compatibility endpoints.	All variant foreign keys are nullable.
A-10	A submission may contain multiple items and should contain one when	Items are a separate relation; completeness is a lifecycle rule.

	complete.	
A-11	Eligibility may aggregate multiple assessments.	Basis relations are 1:N from the determination.
A-12	Audit target is a recorded type and identifier.	No polymorphic foreign key is created.
A-13	A price record has exactly one approved scope.	Check constraint enforces one of listing or product.
A-14	A decision request may reference multiple products and assets.	Candidate relations are separate.
A-15	Assignments track active uniqueness and history.	Partial uniqueness plus ended time.
A-16	Brand is a controlled external reference; media is deferred.	Brand is an attribute; no media relation exists.
A-17	Payment, escrow, refund, return, and complete dispute processing are out of scope.	No such relation exists.
A-18	RetentionCategory is a classification, not an entity.	No retention relation or period is created.

18.2 Dependency register

ID	Dependency	Logical impact	Next action
D-01	Final category taxonomy	Category and attribute content; candidate keys	Project owner supplies or approves taxonomy.
D-02	Category evidence matrix	RL-19 content and eligibility basis	Define per category and purpose.
D-03	Condition rubric	RL-23 and RL-24 value application	Approve rubric and assessor authority.
D-04	Ownership verification and origin rules	RL-18, seller authorization, eligibility	Approve proof standards.
D-05	Transaction completion semantics	Reserved, Sold, transfer effective point, expiry	Project-owner decision required.
D-06	Decision weights, formula, thresholds, tie-break	RL-33 to RL-38, RL-54	Approve criteria and formulas.
D-07	AI provider, persistence, and validation policy	RL-11, RL-39	Defer provider; define persistence policy.
D-08	Market-price source and refresh policy	RL-32	Approve source or keep observation deferred.
D-09	Device identifier policy	RL-43 visibility and uniqueness	Define identifier types, masking, retention.
D-10	Audit event catalogue and mandatory scope	RL-42 content	Approve catalogue and privacy rules.

D-11	Reviewer authority and separation of duties	RL-22, RL-24, RL-30, RL-18 authorization	Approve reviewer policy.
D-12	Retention and legal policy	Deletion, cascade, retention metadata	Formal review; keep conditional.
D-13	Search infrastructure	Index and retrieval implementation	Defer engine selection.
D-14	DBMS and hosting selection	Physical types, indexing, concurrency	Defer to physical design.
D-15	Media scope, storage, moderation, serving	RL-21 file references	Confirm scope.
D-16	Review-intelligence scope	No review relation exists	Confirm whether review is in scope.
D-17	Authentication and account-state policy	RL-01 status and contact uniqueness	Defer provider; define minimal policy.
D-18	Joint or organizational ownership policy	A-05 and ownership cardinality	Confirm before physical design.

19. Deferred Technical Decisions

Decision	Status	Logical treatment
OD-01 DBMS and persistence technology	Resolved - Deferred Technical Selection	No DBMS, type, storage, or hosting choice appears.
OD-10 Search and retrieval infrastructure	Resolved - Deferred Technical Selection	Search-supporting data exists; no engine or index product is chosen.
OD-11 AI provider, model, and interaction persistence	Resolved - Deferred Technical Selection	Producer type and non-authority are modeled; no provider or model is chosen.
OD-12 Retention and deletion policy	Conditionally Resolved	No period or deletion rule is invented; no soft-delete flag exists.
OD-05 Evidence per category	Resolved direction; content deferred	RL-19 structure exists; matrix content is empty.
OD-13 Condition scale and authority	Resolved direction; rubric deferred	Documented grades are used; rubric reference remains free of invented criteria.
OD-14 Listing state machine	Resolved direction; semantics deferred	Nine states are enforced; transitions not defined by the baseline are not invented.
OD-15 Price and market data	Resolved direction; source deferred	Asking and reference scopes only.
OD-16 Decision scoring	Resolved direction; formula deferred	Rules, versions, weights, results, and ranking positions exist; no formula or weight value is supplied.
OD-17 Usage data	Resolved direction; metrics deferred	RL-40 is optional and evidence-limited.

OD-18 Audit scope	Resolved direction; catalogue deferred	RL-42 exists; action and target types are deferred domains.

20. Validation Results

#	Validation check	Result	Evidence
1	All 42 conceptual entities have a documented disposition	PASS	42 unique CE rows in the entity disposition register
2	All 78 conceptual relationships have a documented mapping	PASS	78 unique R rows across the mapping and M:N tables
3	Every relation has exactly one defined primary key strategy	PASS	54 relations, each with one declared PK; all PK attributes exist
4	All foreign key targets exist and reference a declared key	PASS	All 110 foreign keys resolve to an existing primary key attribute
5	Foreign key types are logically compatible with referenced key types	PASS	All FK and referenced key attributes use the Identifier logical type
6	Candidate and alternate keys are justified	PASS	37 candidate keys, each with a written justification and status
7	No accidental duplicate relations exist	PASS	54 unique relation IDs, 54 unique names, 54 unique attribute sets
8	All M:N relationships are correctly resolved	PASS	9 M:N relationships each mapped to a composite-key associative relation
9	No unjustified repeating groups exist	PASS	Atomic attributes only; multivalued concepts decomposed into 12 supporting relations
10	Normalization analysis is evidence-based	PASS	1NF, 2NF, and 3NF sections with per-relation assessments and a controlled-redundancy register
11	Functional dependencies are documented where identifiable	PASS	36 functional dependency rows recorded; no dependency is invented for fact and compatibility assertions
12	All controlled values match the approved working baseline	PASS	15 vocabularies match Phase 02.2; CV-13 to CV-15 remain explicitly deferred

13	Ownership history remains temporal and no mutable owner column exists	PASS	RL-16 has start, optional end, and state; RL-14 has no owner attribute
14	Product Asset identity remains stable	PASS	RL-14 uses a single surrogate stable key; RL-43 holds external values under its own key and never replaces the asset key
15	Sold remains distinct from ownership transfer completion	PASS	LC-17 and LP-08 state that Sold neither completes a sale nor a transfer; no listing-to-transfer key exists
16	Product Truth provenance is preserved	PASS	Fact stores source reference, producer type, explicit unsourced marker, recorded time, confidence, and conflict participation
17	Evidence confidence remains distinct from condition	PASS	Confidence is an evidence and fact domain (CV-03); condition grade is a separate domain (CV-04)
18	Compatibility Unknown is preserved	PASS	OutcomeCode references CV-05 including Unknown; LC-23 states absence is not Incompatible
19	No DBMS-specific types or syntax appear	PASS	No vendor type, storage clause, or engine name found in the new deliverables
20	No executable SQL, DBML, migration, or ORM implementation appears	PASS	No SQL or DBML code block, DDL statement, migration, or ORM fragment found
21	All 56 data requirements are traceable	PASS	56 unique DR rows in the main traceability table
22	Original and derived business rules remain distinguishable	PASS	17 BR rows and 6 DBR rows with separate identifiers and separate tables
23	All attribute dictionaries match the relation catalogue	PASS	502 catalogue attributes and 502 dictionary attributes agree on ID, name, type, and key role
24	Relationship mappings match the conceptual ERD	PASS	78 unique R identifiers; endpoints retain the conceptual source and target entities

25	No unsupported approval or verification claims appear	PASS	Documents state that the model is a working baseline and not institutionally approved
26	All internal references are valid	PASS	All RL, LAT, LC, CE, DR, BR, DBR, AT, US, UC, OD, A, D, CV, DV, IC, and SM references resolve
27	Markdown tables and diagrams are structurally valid where tooling permits	PASS	All tables have consistent column counts and code fences are balanced; no new Mermaid diagram was required
28	No unfinished placeholders remain	PASS	No placeholder or unfinished note found; the validation insertion marker is replaced after this run
29	Total logical relation count reconciles	PASS	42 entity relations + 3 decompositions + 9 M:N resolutions = 54
30	All 40 conceptual constraints are carried forward	PASS	LC-01 to LC-40 map one-to-one to IC-01 to IC-40; LC-41 to LC-50 are structural
31	No conceptual entity was renamed or removed	PASS	42 CE identifiers unchanged; no merged, deferred, or rejected entity re-entered the logical model
32	Every attribute has a validation statement and a source rationale	PASS	502 attributes each have validation text and a source or design rationale

Summary: PASS: 32. Checks marked NOT PERFORMED are explained in the following subsection.

20.1 Not performed

Item	Reason	Effect
Visual diagram rendering	No Mermaid compiler or CLI is installed in this environment.	No rendered diagram is claimed. No new Mermaid diagram is required by Phase 02.3; relationship mapping is expressed in tables.
Physical schema validation	Out of logical scope.	No physical types, indexes, or DBMS behaviour exist to validate.
Workload-based index validation	No query workload or measurement exists.	Index recommendations are logical and conditional, not measured.
Complete literature inspection	R3 is image-based with limited machine-readable text.	No complete literature claim; R3 is not used to justify a key or constraint.

Policy validation	Business policies remain unresolved.	No policy is claimed to be approved.

21. Phase 02.4 Readiness Assessment

Status: READY WITH DOCUMENTED DEPENDENCIES.

Dimension	Assessment	Basis
Relation completeness	Ready	All 42 entities and 78 relationships mapped into 54 relations.
Key strategy	Ready	Every relation has one primary key strategy; candidate keys are justified or explicitly conditional.
Foreign key mapping	Ready	110 FKs; all targets exist; types are compatible (Identifier to Identifier).
Attribute definitions	Ready	502 attributes with type, nullability, validation, and rationale.
Normalization	Ready	1NF, 2NF, and 3NF assessed; four controlled redundancies registered.
Integrity constraints	Ready with dependencies	50 constraints classified; policy-dependent constraints cannot be finalized.
Data dictionary consistency	Ready	Dictionary and catalogue share one specification and one attribute identifier set.
Traceability	Ready	56 DRs, 17 BRs, 6 DBRs, 42 entities, 78 relationships, 10 ATs, 12 USs, 10 UCs individually traced.

21.1 Decisions required before physical design

The following must be resolved or explicitly accepted as deferred before physical schema, indexing, and implementation: OD-01, OD-10, OD-11, OD-12, D-01 to D-12, D-15, D-17, and D-18. In particular, category taxonomy, evidence matrix, condition rubric, ownership proof, transaction completion, scoring formula, retention, reviewer authority, device identifier policy, audit catalogue, media scope, authentication policy, and joint ownership affect physical keys, constraints, partitioning, and access control.

22. Conclusion

Phase 02.3 produced a complete technology-neutral logical relational design: 54 relations, 502 attributes, 54 primary key strategies, 65 primary key columns, 37 candidate or alternate keys, 110 foreign key constraints, 9 M:N resolutions, 50 logical constraints, 15 preserved vocabularies, and full traceability to 56 data requirements, 17 business rules, 6 derived rules, 42 entities, and 78 relationships.

The design preserves product versus asset, purchase versus ownership, temporal ownership, stable asset identity, Product Truth provenance, evidence and assessment separation, listing versus transfer, Unknown compatibility, and AI non-authority. It does not invent taxonomies, formulas, weights, retention periods, reviewer policy, verification operations, or transaction semantics.

Phase 02.4 readiness is **READY WITH DOCUMENTED DEPENDENCIES**.

23. References

ID	Reference
R1	Requirements Specification/TechWise_Requirements_Specification.docx
R2	project-plan/TechWise_ProjectProposal.docx
R3	LITERATURE REVIEW/TechWise_LiteratureReview.pdf
R4	System-Analysis&Design/01-problem-statement-objectives/Problem_Statement_and_Objectives.md
DRA	System-Analysis&Design/02-database-design/Database_Requirements_Analysis.md
DL	System-Analysis&Design/02-database-design/Database_Design_Decision_Log.md
CD	System-Analysis&Design/02-database-design/Conceptual_Database_Design.md
ERD	System-Analysis&Design/02-database-design/Conceptual_Database_ERD.md
ER2	System-Analysis&Design/02-database-design/Phase_02.2_Execution_Report.md
PREP	System-Analysis&Design/02-database-design/Conceptual_Database_Design_Preparation.md
INS	Instruction.md
LRC	Logical_Relation_Catalogue.md
LRKM	Logical_Relationship_and_Key_Mapping.md
LDD	Logical_Data_Dictionary.md
LNIR	Logical_Normalization_and_Integrity_Report.md
ER3	Phase_02.3_Execution_Report.md

24. Revision History

Version	Date	Author	Change
1.0	4 November 2026	Falcon Team	Created the logical relational design, mapping, normalization, constraints, controlled values, temporal rules, indexes, and traceability

End of Logical Database Design.

Appendix F: Complete Logical ERD

Smart E-Commerce & Product Intelligence Platform

System Analysis & Design — Task 02, Phase 02.4 Complete Database ERD

1. ERD Overview

1.1 Purpose and Status

Field	Value
Document Title	TechWise Complete Database ERD
Version	1.0
Date	4 November 2026
Status	Phase 02.4 working ERD baseline; not institutionally approved
Scope	54 logical relations and 110 foreign-key relationships; 78 conceptual relationships are mapped to their logical implementation
Diagram language	Mermaid erDiagram
Companion documents	Database_Schema_Documentation.md, Database_Schema_Reference.md, Phase_02.4_Execution_Report.md

This file is generated from the same authoritative data layer as the schema documentation and the concise reference. It is not manually reconstructed from memory. It contains no SQL, DBML, physical table definition, or DBMS-specific feature.

1.2 Modeling Scope

- Every one of the 54 logical relations appears in the overview.
- Every one of the 110 foreign-key constraints appears in the overview and in the domain diagram of the relation that holds it.
- Every one of the 78 conceptual relationships is mapped to an FK path or an associative relation and appears in the relationship coverage matrix.
- Attributes are shown only as PK/FK key attributes in domain diagrams. The complete 502-attribute detail is in Database_Schema_Documentation.md, Section 6, and Database_Schema_Reference.md.

1.3 Notation

Notation	Meaning		
`|	|	` on the parent side	Each child record must reference exactly one parent record.
`|	o` on the parent side	Each child record may reference zero or one parent record (nullable foreign key).	

o{ on the child side	A parent record may be referenced by zero or more child records.		
`o\	` on the child side	A parent record may be referenced by zero or one child record (unique foreign key).	
PK	Attribute participates in the primary key.		
FK	Attribute is a foreign key.		
PK, FK	Attribute is both a primary-key column and a foreign key.		
RL-nn_Name	Mermaid-safe entity identifier. The hyphen in RL-nn is replaced by an underscore; the exact catalogue identifier is RL-nn and the name follows.		
Edge label	FK-nnn_Attribute_R-nn where applicable. This is traceability metadata, not a physical constraint name.		

The diagrams show logical foreign-key relationships. They do not declare physical indexes, storage, triggers, or DBMS referential actions. The recorded Restrict and Cascade (link rows only) labels remain recommendations.

1.4 Known Limitations

- All 11 diagrams rendered with Mermaid CLI 12.2.0 using Microsoft Edge. No renderer error markers were present. The overview and ERD-D10 are dense and should be viewed at full size; layout aesthetics were not optimized.
- The overview is intentionally dense because it is the completeness view. Domain diagrams are the readable views.
- Domain diagrams show outgoing foreign keys held by the domain and external parents as context. Incoming references are visible in the overview and in the referencing relation's own domain diagram.
- The diagrams do not show all 502 attributes; they show key attributes only.
- No physical cardinality, optionality, or referential action is inferred beyond the logical documentation.

2. Complete Logical ERD

2.1 Overview Diagram `ERD-OV`

ERD-OV is the completeness view. It contains all 54 relations and all 110 foreign-key edges. The edge label carries the FK identifier, the source attribute, and the conceptual relationship identifier where one exists.

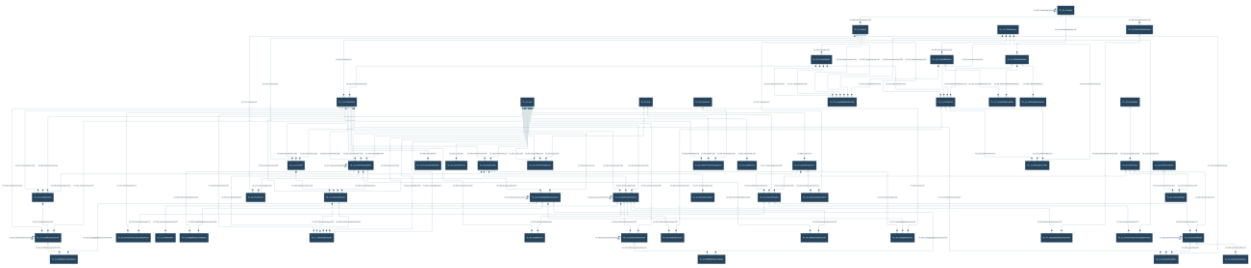


Figure 29. 2.1 Overview Diagram `ERD-OV`

2.2 Reading the Overview

The overview is grouped by domain with Mermaid comments. Because Mermaid does not render comments as visible group boxes, the comments are documentation aids only. Use the domain diagrams in Section 3 for readable subsets.

3. Domain-Level ERDs

Each domain diagram includes all relations allocated to that domain and every foreign key held by those relations. External parent relations are included only as context and are marked as such. No diagram adds a relation or a foreign key.

3.1 Access and identity — `ERD-D1`

Relations in domain: 6. *Outgoing foreign keys shown:* 7. *Attributes shown:* PK and FK only.

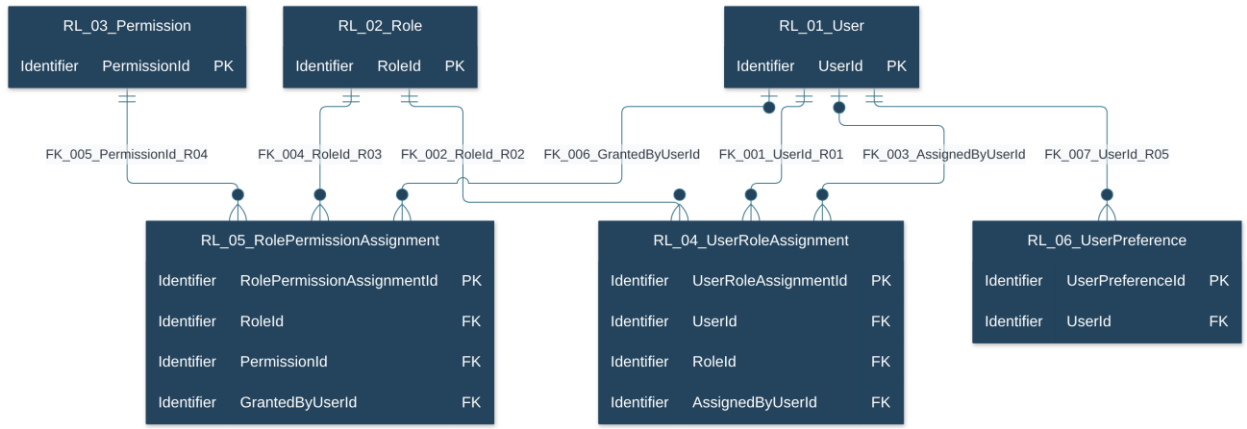


Figure 30. 3.1 Access and identity — `ERD-D1`

Coverage note. Incoming references from other domains are not redrawn here. They appear in `ERD-OV` and in the domain diagram of the relation that holds the foreign key.

3.2 Product catalogue and attributes — `ERD-D2`

Relations in domain: 4. *Outgoing foreign keys shown:* 4. *Attributes shown:* PK and FK only.

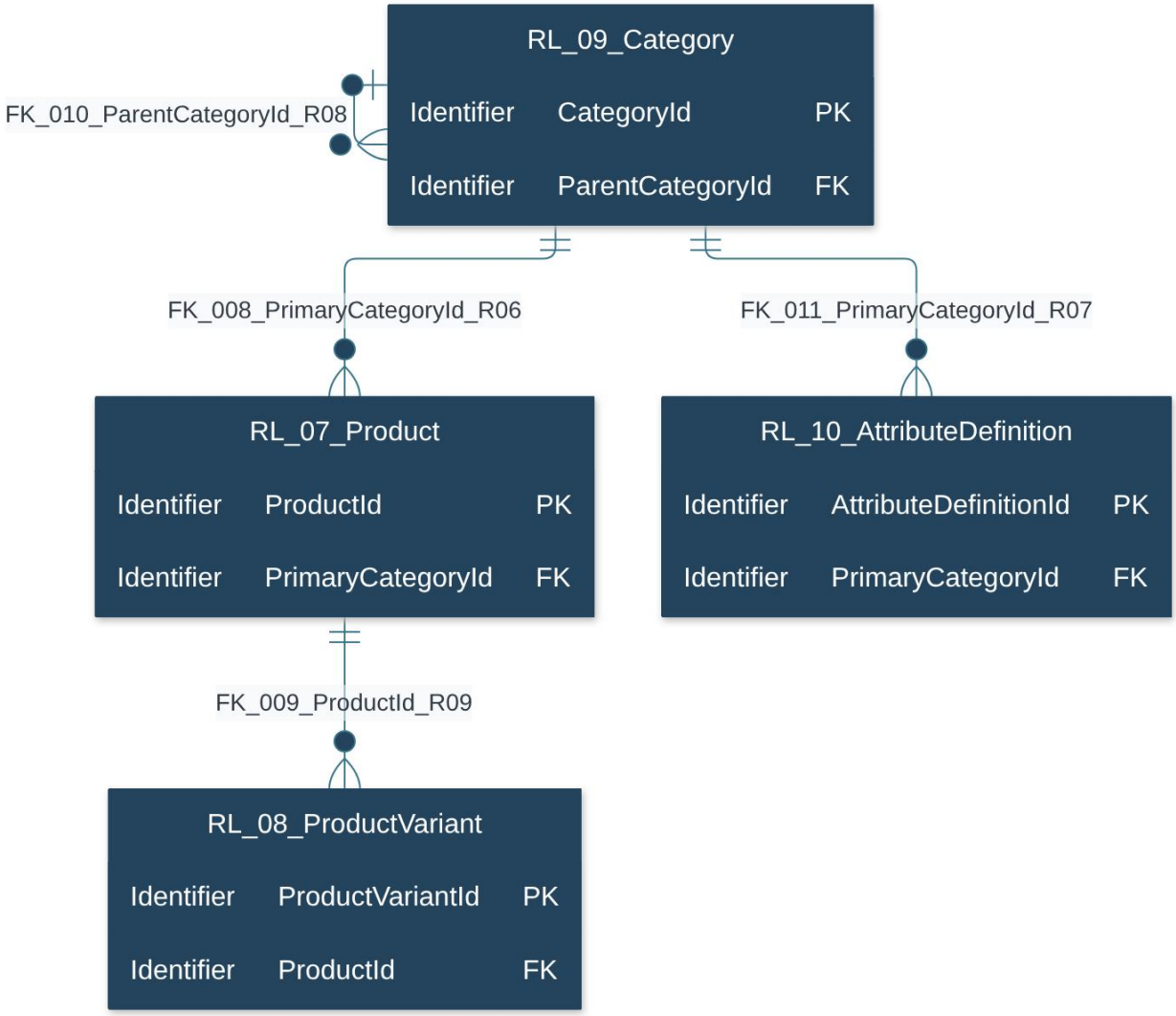


Figure 31. 3.2 Product catalogue and attributes — `ERD-D2`

Coverage note. Incoming references from other domains are not redrawn here. They appear in ERD-OV and in the domain diagram of the relation that holds the foreign key.

3.3 Product Truth and provenance — `ERD-D3`

Relations in domain: 3. *Outgoing foreign keys shown:* 7. *Attributes shown:* PK and FK only.

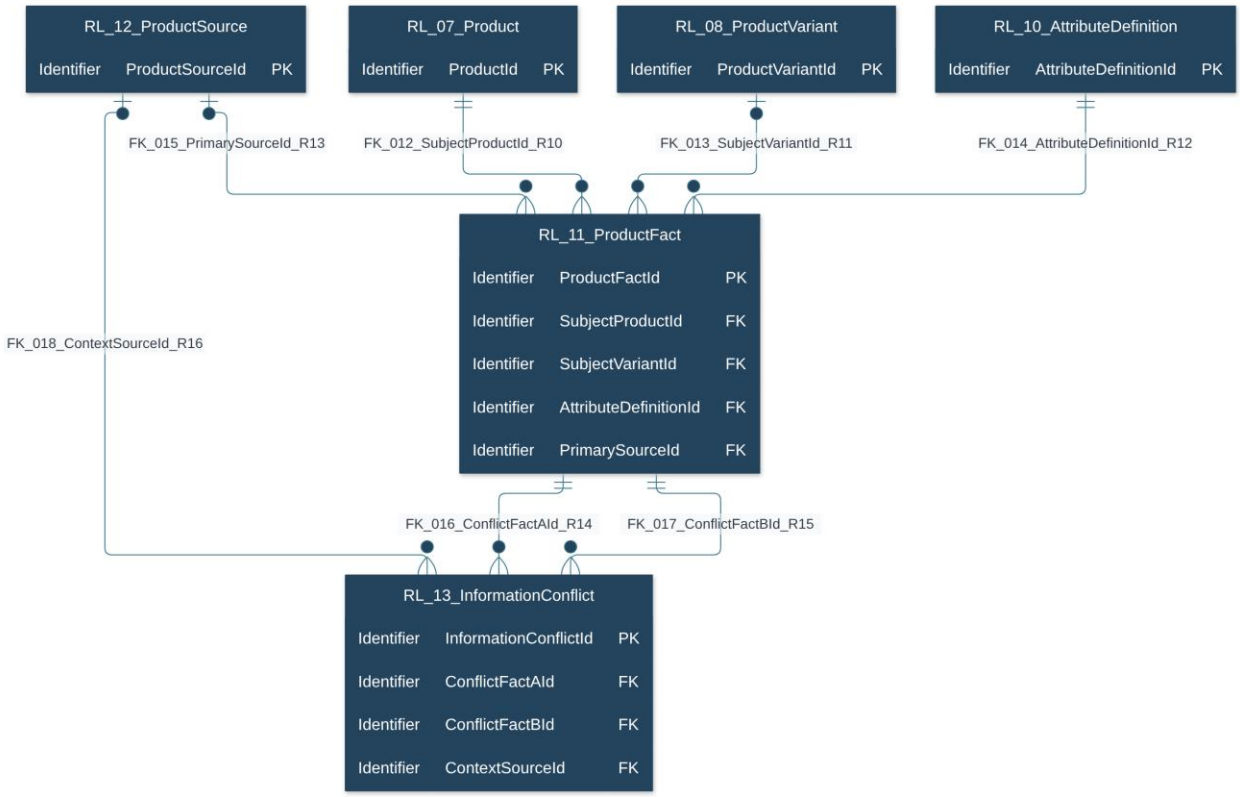


Figure 32. 3.3 Product Truth and provenance — `ERD-D3`

Coverage note. Incoming references from other domains are not redrawn here. They appear in ERD-OV and in the domain diagram of the relation that holds the foreign key.

3.4 Product asset and ownership — `ERD-D4`

Relations in domain: 7. *Outgoing foreign keys shown:* 22. *Attributes shown:* PK and FK only.

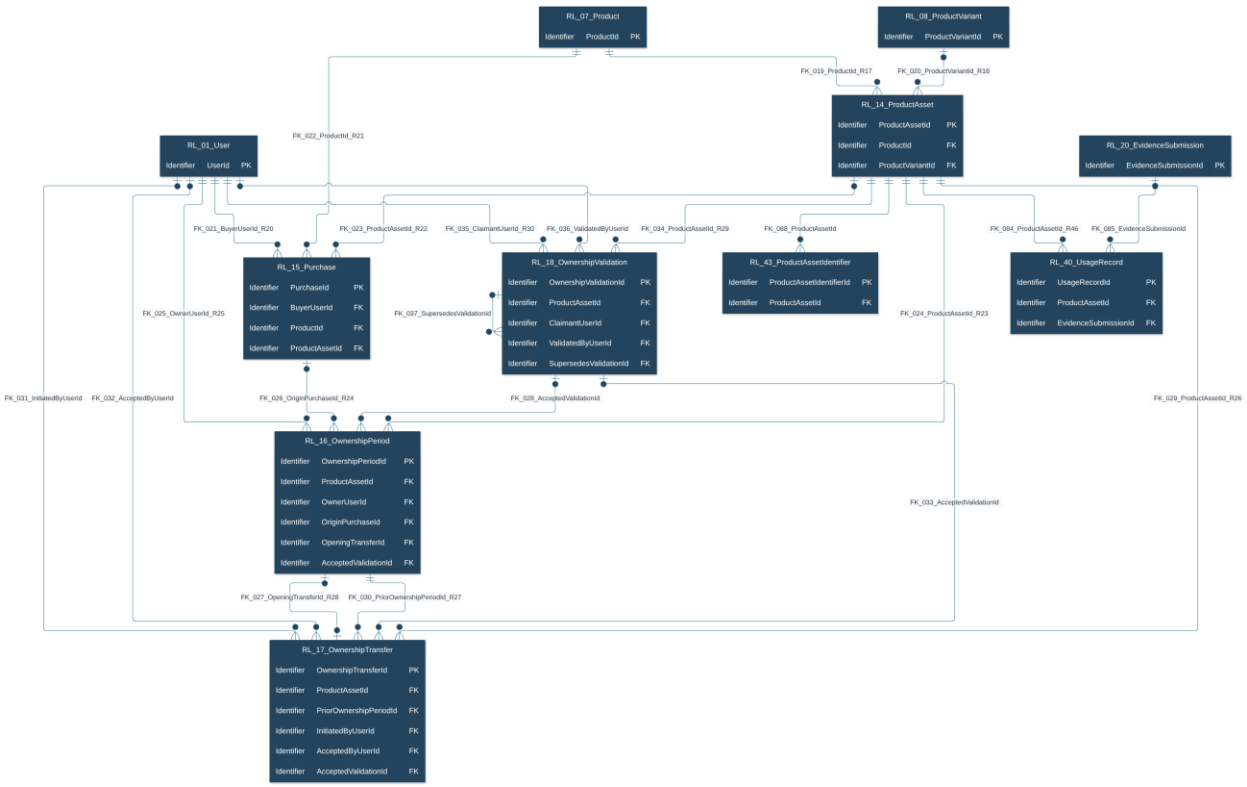


Figure 33. 3.4 Product asset and ownership — `ERD-D4`

Coverage note. Incoming references from other domains are not redrawn here. They appear in ERD-OV and in the domain diagram of the relation that holds the foreign key.

3.5 Evidence and condition — `ERD-D5`

Relations in domain: 6. *Outgoing foreign keys shown:* 16. *Attributes shown:* PK and FK only.

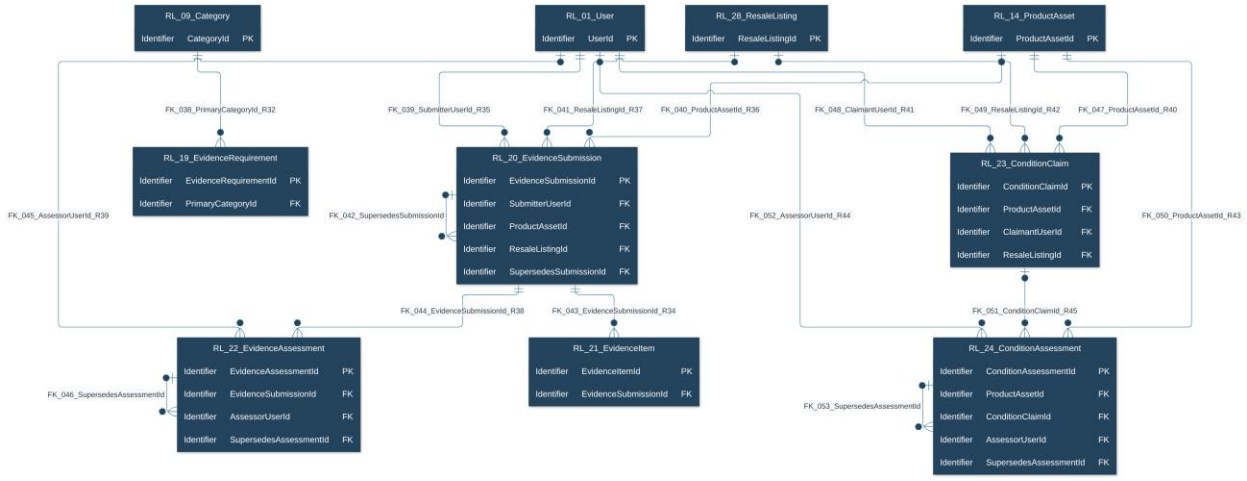


Figure 34. 3.5 Evidence and condition — `ERD-D5`

Coverage note. Incoming references from other domains are not redrawn here. They appear in ERD-OV and in the domain diagram of the relation that holds the foreign key.

3.6 Second-Life marketplace — `ERD-D6`

Relations in domain: 5. *Outgoing foreign keys shown:* 11. *Attributes shown:* PK and FK only.

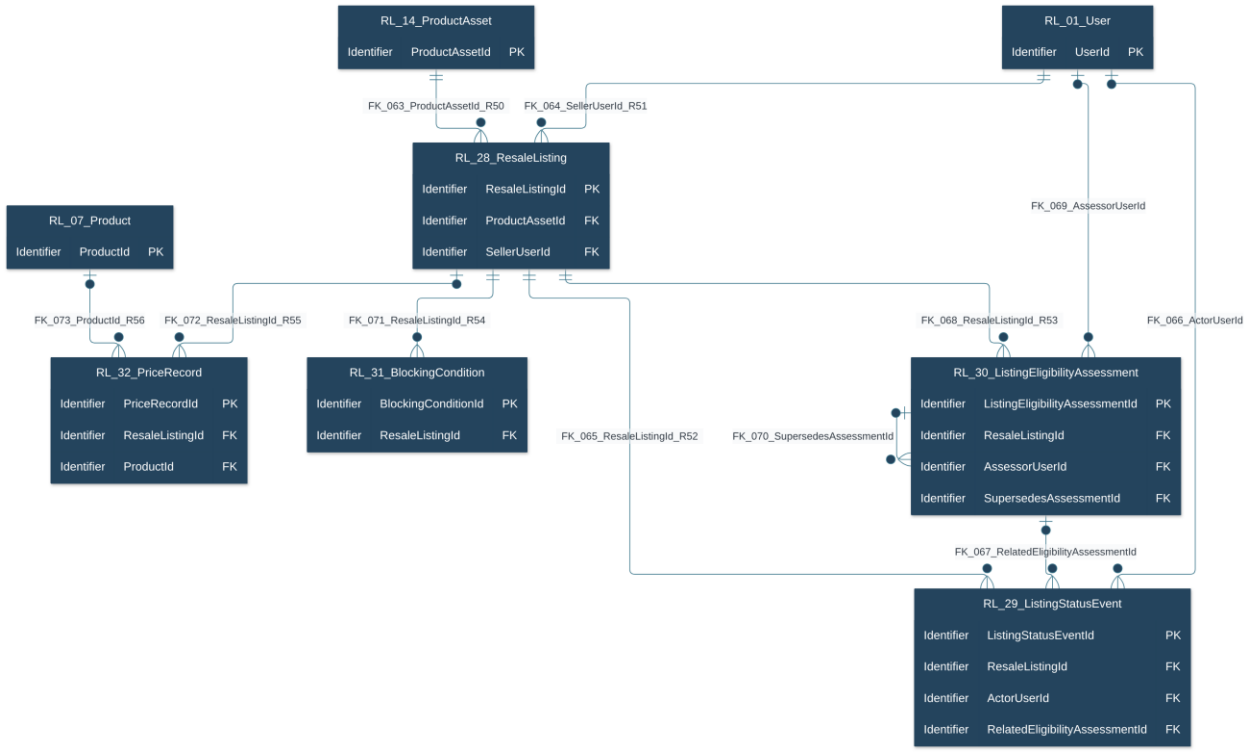


Figure 35. 3.6 Second-Life marketplace — `ERD-D6`

Coverage note. Incoming references from other domains are not redrawn here. They appear in ERD-OV and in the domain diagram of the relation that holds the foreign key.

3.7 Compatibility — `ERD-D7`

Relations in domain: 3. *Outgoing foreign keys shown:* 9. *Attributes shown:* PK and FK only.

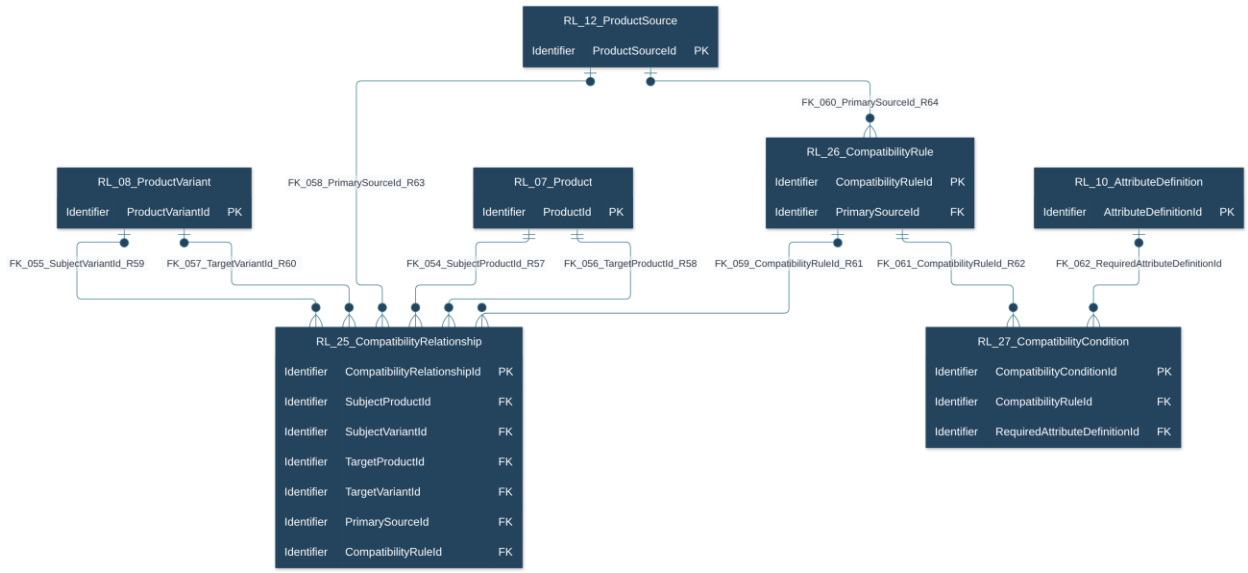


Figure 36. 3.7 Compatibility — `ERD-D7`

Coverage note. Incoming references from other domains are not redrawn here. They appear in ERD-OV and in the domain diagram of the relation that holds the foreign key.

3.8 Decision support — `ERD-D8`

Relations in domain: 9. *Outgoing foreign keys shown:* 14. *Attributes shown:* PK and FK only.

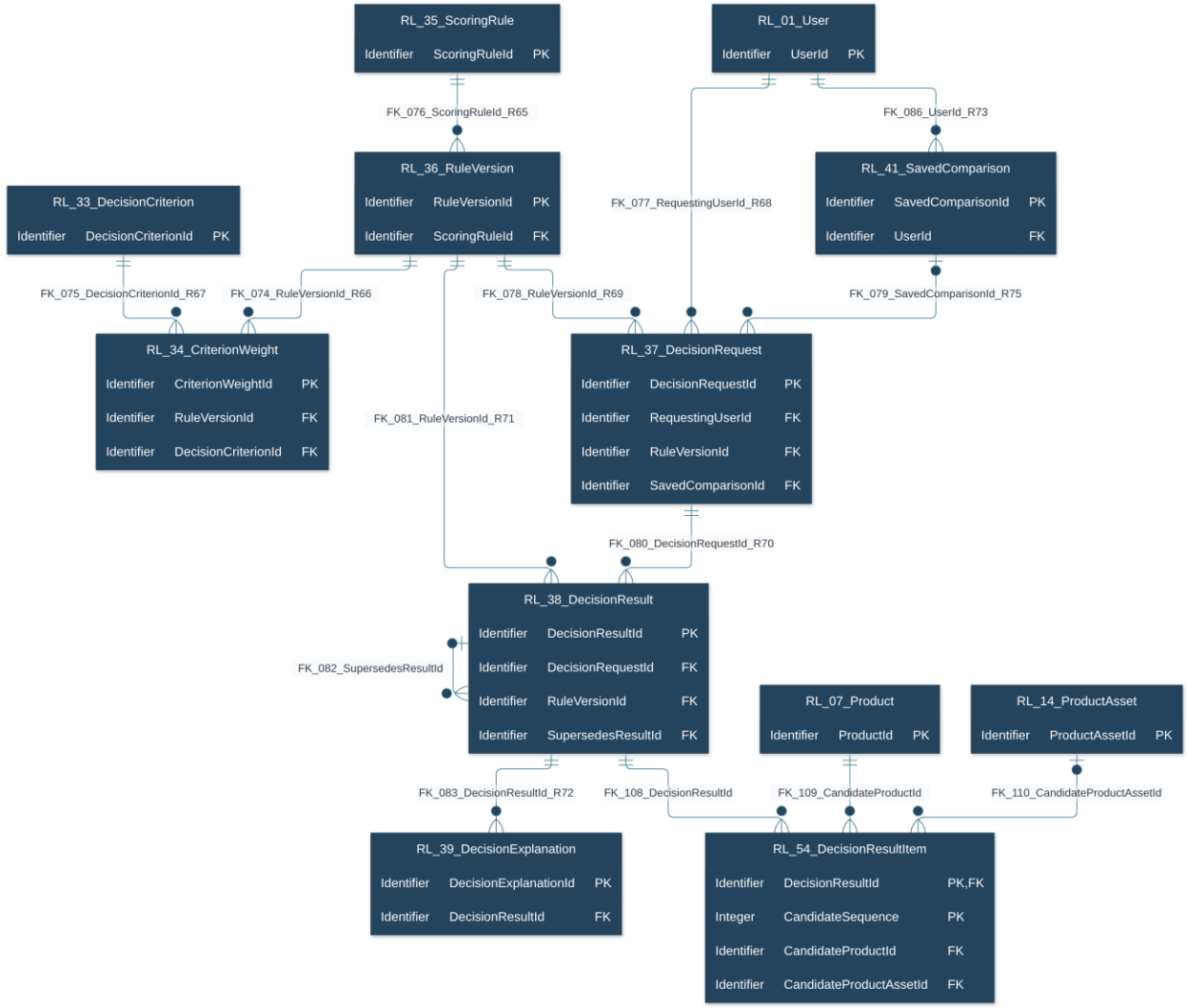


Figure 37. 3.8 Decision support — `ERD-D8`

Coverage note. Incoming references from other domains are not redrawn here. They appear in ERD-OV and in the domain diagram of the relation that holds the foreign key.

3.9 Audit and governance — `ERD-D9`

Relations in domain: 1. *Outgoing foreign keys shown:* 1. *Attributes shown:* PK and FK only.

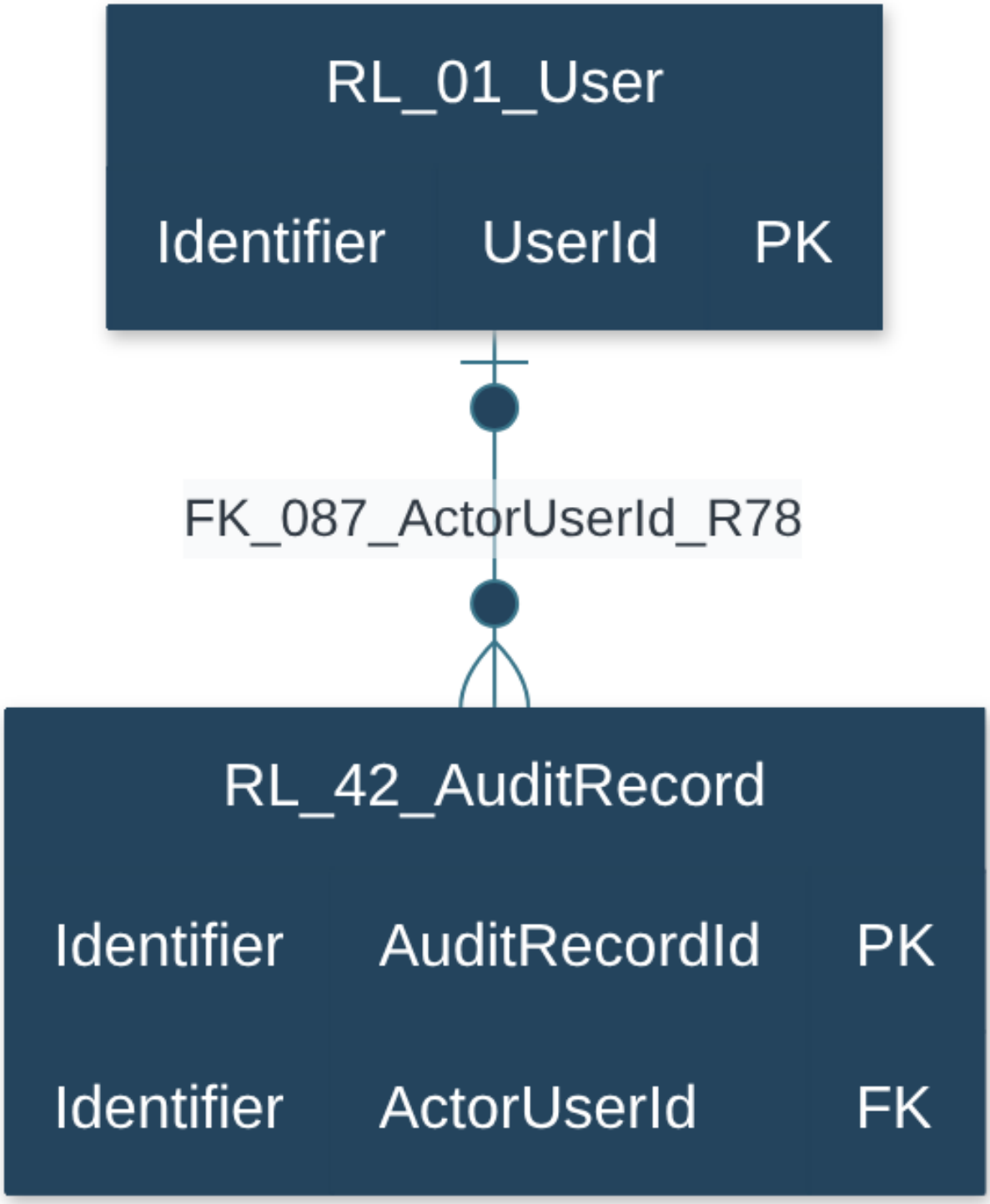


Figure 38. 3.9 Audit and governance — `ERD-D9`

Coverage note. Incoming references from other domains are not redrawn here. They appear in ERD-OV and in the domain diagram of the relation that holds the foreign key.

3.10 Associative and reference support — `ERD-D10`
Relations in domain: 10. *Outgoing foreign keys shown:* 19. *Attributes shown:* PK and FK only.

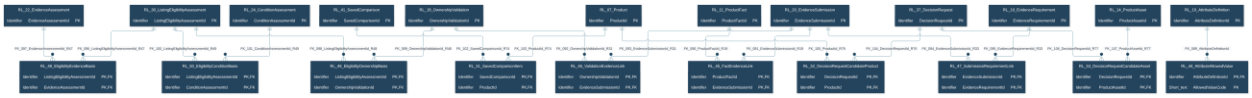


Figure 39. 3.10 Associative and reference support — `ERD-D10`

Coverage note. Incoming references from other domains are not redrawn here. They appear in ERD-OV and in the domain diagram of the relation that holds the foreign key.

4. ERD Coverage Matrix

4.1 Relation Coverage

Relation	Name	Domain	Overview	Domain diagram	FKs held	Status	Exclusion reason
RL-01	User	Access and identity	ERD-OV	ERD-D1	0	Included	None
RL-02	Role	Access and identity	ERD-OV	ERD-D1	0	Included	None
RL-03	Permission	Access and identity	ERD-OV	ERD-D1	0	Included	None
RL-04	UserRoleAssignment	Access and identity	ERD-OV	ERD-D1	3	Included	None
RL-05	RolePermissionAssignment	Access and identity	ERD-OV	ERD-D1	3	Included	None
RL-06	UserPreference	Access and identity	ERD-OV	ERD-D1	1	Included	None
RL-07	Product	Product catalogue and attributes	ERD-OV	ERD-D2	1	Included	None
RL-08	ProductVariant	Product catalogue and attributes	ERD-OV	ERD-D2	1	Included	None
RL-09	Category	Product catalogue and attributes	ERD-OV	ERD-D2	1	Included	None
RL-10	AttributeDefinition	Product catalogue and attributes	ERD-OV	ERD-D2	1	Included	None
RL-11	ProductFact	Product Truth and provenance	ERD-OV	ERD-D3	4	Included	None
RL-12	ProductSource	Product Truth and provenance	ERD-OV	ERD-D3	0	Included	None

RL-13	InformationConflict	Product Truth and provenance	ERD-OV	ERD-D3	3	Included	None
RL-14	ProductAsset	Product asset and ownership	ERD-OV	ERD-D4	2	Included	None
RL-15	Purchase	Product asset and ownership	ERD-OV	ERD-D4	3	Included	None
RL-16	OwnershipPeriod	Product asset and ownership	ERD-OV	ERD-D4	5	Included	None
RL-17	OwnershipTransfer	Product asset and ownership	ERD-OV	ERD-D4	5	Included	None
RL-18	OwnershipValidation	Product asset and ownership	ERD-OV	ERD-D4	4	Included	None
RL-19	EvidenceRequirement	Evidence and condition	ERD-OV	ERD-D5	1	Included	None
RL-20	EvidenceSubmission	Evidence and condition	ERD-OV	ERD-D5	4	Included	None
RL-21	EvidenceItem	Evidence and condition	ERD-OV	ERD-D5	1	Included	None
RL-22	EvidenceAssessment	Evidence and condition	ERD-OV	ERD-D5	3	Included	None
RL-23	ConditionClaim	Evidence and condition	ERD-OV	ERD-D5	3	Included	None
RL-24	ConditionAssessment	Evidence and condition	ERD-OV	ERD-D5	4	Included	None
RL-25	CompatibilityRelationship	Compatibility	ERD-OV	ERD-D7	6	Included	None
RL-26	CompatibilityRule	Compatibility	ERD-OV	ERD-D7	1	Included	None
RL-27	CompatibilityCondition	Compatibility	ERD-OV	ERD-D7	2	Included	None
RL-28	ResaleListing	Second-Life marketplace	ERD-OV	ERD-D6	2	Included	None
RL-29	ListingStatusEvent	Second-Life marketplace	ERD-OV	ERD-D6	3	Included	None
RL-30	ListingEligibilityAssessment	Second-Life	ERD-OV	ERD-D6	3	Included	None

		marketplace					
RL-31	BlockingCondition	Second-Life marketplace	ERD-OV	ERD-D6	1	Included	None
RL-32	PriceRecord	Second-Life marketplace	ERD-OV	ERD-D6	2	Included	None
RL-33	DecisionCriterion	Decision support	ERD-OV	ERD-D8	0	Included	None
RL-34	CriterionWeight	Decision support	ERD-OV	ERD-D8	2	Included	None
RL-35	ScoringRule	Decision support	ERD-OV	ERD-D8	0	Included	None
RL-36	RuleVersion	Decision support	ERD-OV	ERD-D8	1	Included	None
RL-37	DecisionRequest	Decision support	ERD-OV	ERD-D8	3	Included	None
RL-38	DecisionResult	Decision support	ERD-OV	ERD-D8	3	Included	None
RL-39	DecisionExplanation	Decision support	ERD-OV	ERD-D8	1	Included	None
RL-40	UsageRecord	Product asset and ownership	ERD-OV	ERD-D4	2	Included	None
RL-41	SavedComparison	Decision support	ERD-OV	ERD-D8	1	Included	None
RL-42	AuditRecord	Audit and governance	ERD-OV	ERD-D9	1	Included	None
RL-43	ProductAssetIdentifier	Product asset and ownership	ERD-OV	ERD-D4	1	Included	None
RL-44	AttributeAllowedValue	Associative and reference support	ERD-OV	ERD-D10	1	Included	None
RL-45	FactEvidenceLink	Associative and reference support	ERD-OV	ERD-D10	2	Included	None
RL-46	ValidationEvidenceLink	Associative and reference support	ERD-OV	ERD-D10	2	Included	None
RL-47	SubmissionRequirementLink	Associative and reference	ERD-OV	ERD-D10	2	Included	None

		support					
RL-48	EligibilityEvidenceBasis	Associative and reference support	ERD-OV	ERD-D10	2	Included	None
RL-49	EligibilityOwnershipBasis	Associative and reference support	ERD-OV	ERD-D10	2	Included	None
RL-50	EligibilityConditionBasis	Associative and reference support	ERD-OV	ERD-D10	2	Included	None
RL-51	SavedComparisonItem	Associative and reference support	ERD-OV	ERD-D10	2	Included	None
RL-52	DecisionRequestCandidateProduct	Associative and reference support	ERD-OV	ERD-D10	2	Included	None
RL-53	DecisionRequestCandidateAsset	Associative and reference support	ERD-OV	ERD-D10	2	Included	None
RL-54	DecisionResultItem	Decision support	ERD-OV	ERD-D8	3	Included	None

All 54 relations are included. There are zero exclusions.

4.2 Foreign-Key Coverage

FK ID	Source	Target	Relationship	Diagram(s)	Verified
FK-001	RL-04.UserId	RL-01.UserId	R-01	ERD-OV; ERD-D1	Verified
FK-002	RL-04.RoleId	RL-02.RoleId	R-02	ERD-OV; ERD-D1	Verified
FK-003	RL-04.AssignedByUserId	RL-01.UserId	Structural	ERD-OV; ERD-D1	Verified
FK-004	RL-05.RoleId	RL-02.RoleId	R-03	ERD-OV; ERD-D1	Verified
FK-005	RL-05.PermissionId	RL-03.PermissionId	R-04	ERD-OV; ERD-D1	Verified
FK-006	RL-05.GrantedByUserId	RL-01.UserId	Structural	ERD-OV; ERD-D1	Verified
FK-	RL-06.UserId	RL-01.UserId	R-05	ERD-OV;	Verified

007				ERD-D1	
FK-008	RL-07.PrimaryCategoryId	RL-09.CategoryId	R-06	ERD-OV; ERD-D2	Verified
FK-009	RL-08.ProductId	RL-07.ProductId	R-09	ERD-OV; ERD-D2	Verified
FK-010	RL-09.ParentCategoryId	RL-09.CategoryId	R-08	ERD-OV; ERD-D2	Verified
FK-011	RL-10.PrimaryCategoryId	RL-09.CategoryId	R-07	ERD-OV; ERD-D2	Verified
FK-012	RL-11.SubjectProductId	RL-07.ProductId	R-10	ERD-OV; ERD-D3	Verified
FK-013	RL-11.SubjectVariantId	RL-08.ProductVariantId	R-11	ERD-OV; ERD-D3	Verified
FK-014	RL-11.AttributeDefinitionId	RL-10.AttributeDefinitionId	R-12	ERD-OV; ERD-D3	Verified
FK-015	RL-11.PrimarySourceId	RL-12.ProductSourceId	R-13	ERD-OV; ERD-D3	Verified
FK-016	RL-13.ConflictFactAId	RL-11.ProductFactId	R-14	ERD-OV; ERD-D3	Verified
FK-017	RL-13.ConflictFactBId	RL-11.ProductFactId	R-15	ERD-OV; ERD-D3	Verified
FK-018	RL-13.ContextSourceId	RL-12.ProductSourceId	R-16	ERD-OV; ERD-D3	Verified
FK-019	RL-14.ProductId	RL-07.ProductId	R-17	ERD-OV; ERD-D4	Verified
FK-020	RL-14.ProductVariantId	RL-08.ProductVariantId	R-18	ERD-OV; ERD-D4	Verified
FK-021	RL-15.BuyerUserId	RL-01.UserId	R-20	ERD-OV; ERD-D4	Verified
FK-022	RL-15.ProductId	RL-07.ProductId	R-21	ERD-OV; ERD-D4	Verified
FK-023	RL-15.ProductAssetId	RL-14.ProductAssetId	R-22	ERD-OV; ERD-D4	Verified
FK-024	RL-16.ProductAssetId	RL-14.ProductAssetId	R-23	ERD-OV; ERD-D4	Verified
FK-025	RL-16.OwnerUserId	RL-01.UserId	R-25	ERD-OV; ERD-D4	Verified
FK-026	RL-16.OriginPurchaseId	RL-15.PurchaseId	R-24	ERD-OV; ERD-D4	Verified
FK-027	RL-16.OpeningTransferId	RL-17.OwnershipTransferId	R-28	ERD-OV; ERD-D4	Verified

FK-028	RL-16.AcceptedValidationId	RL-18.OwnershipValidationId	Structural	ERD-OV; ERD-D4	Verified
FK-029	RL-17.ProductAssetId	RL-14.ProductAssetId	R-26	ERD-OV; ERD-D4	Verified
FK-030	RL-17.PriorOwnershipPeriodId	RL-16.OwnershipPeriodId	R-27	ERD-OV; ERD-D4	Verified
FK-031	RL-17.InitiatedByUserId	RL-01.UserId	Structural	ERD-OV; ERD-D4	Verified
FK-032	RL-17.AcceptedByUserId	RL-01.UserId	Structural	ERD-OV; ERD-D4	Verified
FK-033	RL-17.AcceptedValidationId	RL-18.OwnershipValidationId	Structural	ERD-OV; ERD-D4	Verified
FK-034	RL-18.ProductAssetId	RL-14.ProductAssetId	R-29	ERD-OV; ERD-D4	Verified
FK-035	RL-18.ClaimantUserId	RL-01.UserId	R-30	ERD-OV; ERD-D4	Verified
FK-036	RL-18.ValidatedByUserId	RL-01.UserId	Structural	ERD-OV; ERD-D4	Verified
FK-037	RL-18.SupersedesValidationId	RL-18.OwnershipValidationId	Structural	ERD-OV; ERD-D4	Verified
FK-038	RL-19.PrimaryCategoryId	RL-09.CategoryId	R-32	ERD-OV; ERD-D5	Verified
FK-039	RL-20.SubmitterUserId	RL-01.UserId	R-35	ERD-OV; ERD-D5	Verified
FK-040	RL-20.ProductAssetId	RL-14.ProductAssetId	R-36	ERD-OV; ERD-D5	Verified
FK-041	RL-20.ResaleListingId	RL-28.ResaleListingId	R-37	ERD-OV; ERD-D5	Verified
FK-042	RL-20.SupersedesSubmissionId	RL-20.EvidenceSubmissionId	Structural	ERD-OV; ERD-D5	Verified
FK-043	RL-21.EvidenceSubmissionId	RL-20.EvidenceSubmissionId	R-34	ERD-OV; ERD-D5	Verified
FK-044	RL-22.EvidenceSubmissionId	RL-20.EvidenceSubmissionId	R-38	ERD-OV; ERD-D5	Verified
FK-045	RL-22.AssessorUserId	RL-01.UserId	R-39	ERD-OV; ERD-D5	Verified
FK-046	RL-22.SupersedesAssessmentId	RL-22.EvidenceAssessmentId	Structural	ERD-OV; ERD-D5	Verified
FK-047	RL-23.ProductAssetId	RL-14.ProductAssetId	R-40	ERD-OV; ERD-D5	Verified
FK-	RL-23.ClaimantUserId	RL-01.UserId	R-41	ERD-OV;	Verified

048				ERD-D5	
FK-049	RL-23.ResaleListingId	RL-28.ResaleListingId	R-42	ERD-OV; ERD-D5	Verified
FK-050	RL-24.ProductAssetId	RL-14.ProductAssetId	R-43	ERD-OV; ERD-D5	Verified
FK-051	RL-24.ConditionClaimId	RL-23.ConditionClaimId	R-45	ERD-OV; ERD-D5	Verified
FK-052	RL-24.AssessorUserId	RL-01.UserId	R-44	ERD-OV; ERD-D5	Verified
FK-053	RL-24.SupersedesAssessmentId	RL-24.ConditionAssessmentId	Structural	ERD-OV; ERD-D5	Verified
FK-054	RL-25.SubjectProductId	RL-07.ProductId	R-57	ERD-OV; ERD-D7	Verified
FK-055	RL-25.SubjectVariantId	RL-08.ProductVariantId	R-59	ERD-OV; ERD-D7	Verified
FK-056	RL-25.TargetProductId	RL-07.ProductId	R-58	ERD-OV; ERD-D7	Verified
FK-057	RL-25.TargetVariantId	RL-08.ProductVariantId	R-60	ERD-OV; ERD-D7	Verified
FK-058	RL-25.PrimarySourceId	RL-12.ProductSourceId	R-63	ERD-OV; ERD-D7	Verified
FK-059	RL-25.CompatibilityRuleId	RL-26.CompatibilityRuleId	R-61	ERD-OV; ERD-D7	Verified
FK-060	RL-26.PrimarySourceId	RL-12.ProductSourceId	R-64	ERD-OV; ERD-D7	Verified
FK-061	RL-27.CompatibilityRuleId	RL-26.CompatibilityRuleId	R-62	ERD-OV; ERD-D7	Verified
FK-062	RL-27.RequiredAttributeDefinitionId	RL-10.AttributeDefinitionId	Structural	ERD-OV; ERD-D7	Verified
FK-063	RL-28.ProductAssetId	RL-14.ProductAssetId	R-50	ERD-OV; ERD-D6	Verified
FK-064	RL-28.SellerUserId	RL-01.UserId	R-51	ERD-OV; ERD-D6	Verified
FK-065	RL-29.ResaleListingId	RL-28.ResaleListingId	R-52	ERD-OV; ERD-D6	Verified
FK-066	RL-29.ActorUserId	RL-01.UserId	Structural	ERD-OV; ERD-D6	Verified
FK-067	RL-29.RelatedEligibilityAssessmentId	RL-30.ListingEligibilityAssessmentId	Structural	ERD-OV; ERD-D6	Verified
FK-068	RL-30.ResaleListingId	RL-28.ResaleListingId	R-53	ERD-OV; ERD-D6	Verified

FK-069	RL-30.AssessorUserId	RL-01.UserId	Structural	ERD-OV; ERD-D6	Verified
FK-070	RL-30.SupersedesAssessmentId	RL-30.ListingEligibilityAssessmentId	Structural	ERD-OV; ERD-D6	Verified
FK-071	RL-31.ResaleListingId	RL-28.ResaleListingId	R-54	ERD-OV; ERD-D6	Verified
FK-072	RL-32.ResaleListingId	RL-28.ResaleListingId	R-55	ERD-OV; ERD-D6	Verified
FK-073	RL-32.ProductId	RL-07.ProductId	R-56	ERD-OV; ERD-D6	Verified
FK-074	RL-34.RuleVersionId	RL-36.RuleVersionId	R-66	ERD-OV; ERD-D8	Verified
FK-075	RL-34.DecisionCriterionId	RL-33.DecisionCriterionId	R-67	ERD-OV; ERD-D8	Verified
FK-076	RL-36.ScoringRuleId	RL-35.ScoringRuleId	R-65	ERD-OV; ERD-D8	Verified
FK-077	RL-37.RequestingUserId	RL-01.UserId	R-68	ERD-OV; ERD-D8	Verified
FK-078	RL-37.RuleVersionId	RL-36.RuleVersionId	R-69	ERD-OV; ERD-D8	Verified
FK-079	RL-37.SavedComparisonId	RL-41.SavedComparisonId	R-75	ERD-OV; ERD-D8	Verified
FK-080	RL-38.DecisionRequestId	RL-37.DecisionRequestId	R-70	ERD-OV; ERD-D8	Verified
FK-081	RL-38.RuleVersionId	RL-36.RuleVersionId	R-71	ERD-OV; ERD-D8	Verified
FK-082	RL-38.SupersedesResultId	RL-38.DecisionResultId	Structural	ERD-OV; ERD-D8	Verified
FK-083	RL-39.DecisionResultId	RL-38.DecisionResultId	R-72	ERD-OV; ERD-D8	Verified
FK-084	RL-40.ProductAssetId	RL-14.ProductAssetId	R-46	ERD-OV; ERD-D4	Verified
FK-085	RL-40.EvidenceSubmissionId	RL-20.EvidenceSubmissionId	Structural	ERD-OV; ERD-D4	Verified
FK-086	RL-41.UserId	RL-01.UserId	R-73	ERD-OV; ERD-D8	Verified
FK-087	RL-42.ActorUserId	RL-01.UserId	R-78	ERD-OV; ERD-D9	Verified
FK-088	RL-43.ProductAssetId	RL-14.ProductAssetId	Structural	ERD-OV; ERD-D4	Verified
FK-	RL-44.AttributeDefinitionId	RL-10.AttributeDefinitionId	Structural	ERD-OV;	Verified

089				ERD-D10	
FK-090	RL-45.ProductFactId	RL-11.ProductFactId	R-19	ERD-OV; ERD-D10	Verified
FK-091	RL-45.EvidenceSubmissionId	RL-20.EvidenceSubmissionId	R-19	ERD-OV; ERD-D10	Verified
FK-092	RL-46.OwnershipValidationId	RL-18.OwnershipValidationId	R-31	ERD-OV; ERD-D10	Verified
FK-093	RL-46.EvidenceSubmissionId	RL-20.EvidenceSubmissionId	R-31	ERD-OV; ERD-D10	Verified
FK-094	RL-47.EvidenceSubmissionId	RL-20.EvidenceSubmissionId	R-33	ERD-OV; ERD-D10	Verified
FK-095	RL-47.EvidenceRequirementId	RL-19.EvidenceRequirementId	R-33	ERD-OV; ERD-D10	Verified
FK-096	RL-48.ListingEligibilityAssessmentId	RL-30.ListingEligibilityAssessmentId	R-47	ERD-OV; ERD-D10	Verified
FK-097	RL-48.EvidenceAssessmentId	RL-22.EvidenceAssessmentId	R-47	ERD-OV; ERD-D10	Verified
FK-098	RL-49.ListingEligibilityAssessmentId	RL-30.ListingEligibilityAssessmentId	R-48	ERD-OV; ERD-D10	Verified
FK-099	RL-49.OwnershipValidationId	RL-18.OwnershipValidationId	R-48	ERD-OV; ERD-D10	Verified
FK-100	RL-50.ListingEligibilityAssessmentId	RL-30.ListingEligibilityAssessmentId	R-49	ERD-OV; ERD-D10	Verified
FK-101	RL-50.ConditionAssessmentId	RL-24.ConditionAssessmentId	R-49	ERD-OV; ERD-D10	Verified
FK-102	RL-51.SavedComparisonId	RL-41.SavedComparisonId	R-74	ERD-OV; ERD-D10	Verified
FK-103	RL-51.ProductId	RL-07.ProductId	R-74	ERD-OV; ERD-D10	Verified
FK-104	RL-52.DecisionRequestId	RL-37.DecisionRequestId	R-76	ERD-OV; ERD-D10	Verified
FK-105	RL-52.ProductId	RL-07.ProductId	R-76	ERD-OV; ERD-D10	Verified
FK-106	RL-53.DecisionRequestId	RL-37.DecisionRequestId	R-77	ERD-OV; ERD-D10	Verified
FK-107	RL-53.ProductAssetId	RL-14.ProductAssetId	R-77	ERD-OV; ERD-D10	Verified
FK-108	RL-54.DecisionResultId	RL-38.DecisionResultId	Structural	ERD-OV; ERD-D8	Verified
FK-109	RL-54.CandidateProductId	RL-07.ProductId	Structural	ERD-OV; ERD-D8	Verified

FK-110	RL-54.CandidateProductAssetId	RL-14.ProductAssetId	Structural	ERD-OV; ERD-D8	Verified

All 110 foreign keys are represented. There are zero uncovered foreign keys.

4.3 Relationship Coverage

Relationship	Conceptual source	Conceptual target	Diagram	Implementing FK(s)	Verified status
R-01	CE-01 User	CE-04 UserRoleAssignment	ERD-D1	FK-001	Verified: endpoint, cardinality, and diagram resolved
R-02	CE-02 Role	CE-04 UserRoleAssignment	ERD-D1	FK-002	Verified: endpoint, cardinality, and diagram resolved
R-03	CE-02 Role	CE-05 RolePermissionAssignment	ERD-D1	FK-004	Verified: endpoint, cardinality, and diagram resolved
R-04	CE-03 Permission	CE-05 RolePermissionAssignment	ERD-D1	FK-005	Verified: endpoint, cardinality, and diagram resolved
R-05	CE-01 User	CE-06 UserPreference	ERD-D1	FK-007	Verified: endpoint, cardinality, and diagram resolved
R-06	CE-09 Category	CE-07 Product	ERD-D2	FK-008	Verified: endpoint, cardinality, and diagram resolved
R-07	CE-09 Category	CE-10 AttributeDefinition	ERD-D2	FK-011	Verified: endpoint, cardinality, and diagram resolved
R-08	CE-09 Category	CE-09 Category	ERD-D2	FK-010	Verified: endpoint, cardinality, and diagram resolved
R-09	CE-07 Product	CE-08 ProductVariant	ERD-D2	FK-009	Verified: endpoint, cardinality, and diagram

					resolved
R-10	CE-07 Product	CE-11 ProductFact	ERD-D3	FK-012	Verified: endpoint, cardinality, and diagram resolved
R-11	CE-08 ProductVariant	CE-11 ProductFact	ERD-D3	FK-013	Verified: endpoint, cardinality, and diagram resolved
R-12	CE-10 AttributeDefinition	CE-11 ProductFact	ERD-D3	FK-014	Verified: endpoint, cardinality, and diagram resolved
R-13	CE-12 ProductSource	CE-11 ProductFact	ERD-D3	FK-015	Verified: endpoint, cardinality, and diagram resolved
R-14	CE-11 ProductFact	CE-13 InformationConflict	ERD-D3	FK-016	Verified: endpoint, cardinality, and diagram resolved
R-15	CE-11 ProductFact	CE-13 InformationConflict	ERD-D3	FK-017	Verified: endpoint, cardinality, and diagram resolved
R-16	CE-12 ProductSource	CE-13 InformationConflict	ERD-D3	FK-018	Verified: endpoint, cardinality, and diagram resolved
R-17	CE-07 Product	CE-14 ProductAsset	ERD-D4	FK-019	Verified: endpoint, cardinality, and diagram resolved
R-18	CE-08 ProductVariant	CE-14 ProductAsset	ERD-D4	FK-020	Verified: endpoint, cardinality, and diagram resolved
R-19	CE-11 ProductFact	CE-20 EvidenceSubmission	ERD-D10	FK-090, FK-091	Verified: endpoint, cardinality, and diagram

					resolved
R-20	CE-01 User	CE-15 Purchase	ERD-D4	FK-021	Verified: endpoint, cardinality, and diagram resolved
R-21	CE-07 Product	CE-15 Purchase	ERD-D4	FK-022	Verified: endpoint, cardinality, and diagram resolved
R-22	CE-14 ProductAsset	CE-15 Purchase	ERD-D4	FK-023	Verified: endpoint, cardinality, and diagram resolved
R-23	CE-14 ProductAsset	CE-16 OwnershipPeriod	ERD-D4	FK-024	Verified: endpoint, cardinality, and diagram resolved
R-24	CE-15 Purchase	CE-16 OwnershipPeriod	ERD-D4	FK-026	Verified: endpoint, cardinality, and diagram resolved
R-25	CE-01 User	CE-16 OwnershipPeriod	ERD-D4	FK-025	Verified: endpoint, cardinality, and diagram resolved
R-26	CE-14 ProductAsset	CE-17 OwnershipTransfer	ERD-D4	FK-029	Verified: endpoint, cardinality, and diagram resolved
R-27	CE-16 OwnershipPeriod	CE-17 OwnershipTransfer	ERD-D4	FK-030	Verified: endpoint, cardinality, and diagram resolved
R-28	CE-17 OwnershipTransfer	CE-16 OwnershipPeriod	ERD-D4	FK-027	Verified: endpoint, cardinality, and diagram resolved
R-29	CE-14 ProductAsset	CE-18 OwnershipValidation	ERD-D4	FK-034	Verified: endpoint, cardinality, and diagram

					resolved
R-30	CE-01 User	CE-18 OwnershipValidation	ERD-D4	FK-035	Verified: endpoint, cardinality, and diagram resolved
R-31	CE-18 OwnershipValidation	CE-20 EvidenceSubmission	ERD-D10	FK-092, FK-093	Verified: endpoint, cardinality, and diagram resolved
R-32	CE-09 Category	CE-19 EvidenceRequirement	ERD-D5	FK-038	Verified: endpoint, cardinality, and diagram resolved
R-33	CE-19 EvidenceRequirement	CE-20 EvidenceSubmission	ERD-D10	FK-094, FK-095	Verified: endpoint, cardinality, and diagram resolved
R-34	CE-20 EvidenceSubmission	CE-21 EvidenceItem	ERD-D5	FK-043	Verified: endpoint, cardinality, and diagram resolved
R-35	CE-01 User	CE-20 EvidenceSubmission	ERD-D5	FK-039	Verified: endpoint, cardinality, and diagram resolved
R-36	CE-14 ProductAsset	CE-20 EvidenceSubmission	ERD-D5	FK-040	Verified: endpoint, cardinality, and diagram resolved
R-37	CE-28 ResaleListing	CE-20 EvidenceSubmission	ERD-D5	FK-041	Verified: endpoint, cardinality, and diagram resolved
R-38	CE-20 EvidenceSubmission	CE-22 EvidenceAssessment	ERD-D5	FK-044	Verified: endpoint, cardinality, and diagram resolved
R-39	CE-01 User	CE-22 EvidenceAssessment	ERD-D5	FK-045	Verified: endpoint, cardinality, and diagram

					resolved
R-40	CE-14 ProductAsset	CE-23 ConditionClaim	ERD-D5	FK-047	Verified: endpoint, cardinality, and diagram resolved
R-41	CE-01 User	CE-23 ConditionClaim	ERD-D5	FK-048	Verified: endpoint, cardinality, and diagram resolved
R-42	CE-28 ResaleListing	CE-23 ConditionClaim	ERD-D5	FK-049	Verified: endpoint, cardinality, and diagram resolved
R-43	CE-14 ProductAsset	CE-24 ConditionAssessment	ERD-D5	FK-050	Verified: endpoint, cardinality, and diagram resolved
R-44	CE-01 User	CE-24 ConditionAssessment	ERD-D5	FK-052	Verified: endpoint, cardinality, and diagram resolved
R-45	CE-23 ConditionClaim	CE-24 ConditionAssessment	ERD-D5	FK-051	Verified: endpoint, cardinality, and diagram resolved
R-46	CE-14 ProductAsset	CE-40 UsageRecord	ERD-D4	FK-084	Verified: endpoint, cardinality, and diagram resolved
R-47	CE-22 EvidenceAssessment	CE-30 ListingEligibilityAssessment	ERD-D10	FK-096, FK-097	Verified: endpoint, cardinality, and diagram resolved
R-48	CE-18 OwnershipValidation	CE-30 ListingEligibilityAssessment	ERD-D10	FK-098, FK-099	Verified: endpoint, cardinality, and diagram resolved
R-49	CE-24 ConditionAssessment	CE-30 ListingEligibilityAssessment	ERD-D10	FK-100, FK-101	Verified: endpoint, cardinality, and diagram

					resolved
R-50	CE-14 ProductAsset	CE-28 ResaleListing	ERD-D6	FK-063	Verified: endpoint, cardinality, and diagram resolved
R-51	CE-01 User	CE-28 ResaleListing	ERD-D6	FK-064	Verified: endpoint, cardinality, and diagram resolved
R-52	CE-28 ResaleListing	CE-29 ListingStatusEvent	ERD-D6	FK-065	Verified: endpoint, cardinality, and diagram resolved
R-53	CE-28 ResaleListing	CE-30 ListingEligibilityAssessment	ERD-D6	FK-068	Verified: endpoint, cardinality, and diagram resolved
R-54	CE-28 ResaleListing	CE-31 BlockingCondition	ERD-D6	FK-071	Verified: endpoint, cardinality, and diagram resolved
R-55	CE-28 ResaleListing	CE-32 PriceRecord	ERD-D6	FK-072	Verified: endpoint, cardinality, and diagram resolved
R-56	CE-07 Product	CE-32 PriceRecord	ERD-D6	FK-073	Verified: endpoint, cardinality, and diagram resolved
R-57	CE-07 Product	CE-25 CompatibilityRelationship	ERD-D7	FK-054	Verified: endpoint, cardinality, and diagram resolved
R-58	CE-07 Product	CE-25 CompatibilityRelationship	ERD-D7	FK-056	Verified: endpoint, cardinality, and diagram resolved
R-59	CE-08 ProductVariant	CE-25 CompatibilityRelationship	ERD-D7	FK-055	Verified: endpoint, cardinality, and diagram

					resolved
R-60	CE-08 ProductVariant	CE-25 CompatibilityRelationship	ERD-D7	FK-057	Verified: endpoint, cardinality, and diagram resolved
R-61	CE-26 CompatibilityRule	CE-25 CompatibilityRelationship	ERD-D7	FK-059	Verified: endpoint, cardinality, and diagram resolved
R-62	CE-26 CompatibilityRule	CE-27 CompatibilityCondition	ERD-D7	FK-061	Verified: endpoint, cardinality, and diagram resolved
R-63	CE-12 ProductSource	CE-25 CompatibilityRelationship	ERD-D7	FK-058	Verified: endpoint, cardinality, and diagram resolved
R-64	CE-12 ProductSource	CE-26 CompatibilityRule	ERD-D7	FK-060	Verified: endpoint, cardinality, and diagram resolved
R-65	CE-35 ScoringRule	CE-36 RuleVersion	ERD-D8	FK-076	Verified: endpoint, cardinality, and diagram resolved
R-66	CE-36 RuleVersion	CE-34 CriterionWeight	ERD-D8	FK-074	Verified: endpoint, cardinality, and diagram resolved
R-67	CE-33 DecisionCriterion	CE-34 CriterionWeight	ERD-D8	FK-075	Verified: endpoint, cardinality, and diagram resolved
R-68	CE-01 User	CE-37 DecisionRequest	ERD-D8	FK-077	Verified: endpoint, cardinality, and diagram resolved
R-69	CE-36 RuleVersion	CE-37 DecisionRequest	ERD-D8	FK-078	Verified: endpoint, cardinality, and diagram

					resolved
R-70	CE-37 DecisionRequest	CE-38 DecisionResult	ERD-D8	FK-080	Verified: endpoint, cardinality, and diagram resolved
R-71	CE-36 RuleVersion	CE-38 DecisionResult	ERD-D8	FK-081	Verified: endpoint, cardinality, and diagram resolved
R-72	CE-38 DecisionResult	CE-39 DecisionExplanation	ERD-D8	FK-083	Verified: endpoint, cardinality, and diagram resolved
R-73	CE-01 User	CE-41 SavedComparison	ERD-D8	FK-086	Verified: endpoint, cardinality, and diagram resolved
R-74	CE-41 SavedComparison	CE-07 Product	ERD-D10	FK-102, FK-103	Verified: endpoint, cardinality, and diagram resolved
R-75	CE-41 SavedComparison	CE-37 DecisionRequest	ERD-D8	FK-079	Verified: endpoint, cardinality, and diagram resolved
R-76	CE-37 DecisionRequest	CE-07 Product	ERD-D10	FK-104, FK-105	Verified: endpoint, cardinality, and diagram resolved
R-77	CE-37 DecisionRequest	CE-14 ProductAsset	ERD-D10	FK-106, FK-107	Verified: endpoint, cardinality, and diagram resolved
R-78	CE-01 User	CE-42 AuditRecord	ERD-D9	FK-087	Verified: endpoint, cardinality, and diagram resolved

All 78 relationships are represented. Nine are represented by associative relations; 69 are represented by foreign keys.

4.4 Diagram Inventory

Diagram	Title	In-domain relations	FK edges	Conceptual relationships mapped
ERD-OV	Complete overview	54	110	78
ERD-D1	Access and identity	6	7	5
ERD-D2	Product catalogue and attributes	4	4	4
ERD-D3	Product Truth and provenance	3	7	7
ERD-D4	Product asset and ownership	7	22	14
ERD-D5	Evidence and condition	6	16	13
ERD-D6	Second-Life marketplace	5	11	7
ERD-D7	Compatibility	3	9	8
ERD-D8	Decision support	9	14	10
ERD-D9	Audit and governance	1	1	1
ERD-D10	Associative and reference support	10	19	9

The overview plus the ten domain diagrams collectively cover all relations, all foreign keys, and all conceptual relationships.

5. Diagram Validation

5.1 Checks Performed

#	Check	Result	Evidence				
1	All relation names in diagrams exist in the catalogue	PASS	Entity identifiers are generated from RL-nn and the catalogue name; no free-text entity name is used.				
2	All relationship endpoints exist	PASS	Every FK source and target resolves through REL_BY_ID;				

			generation asserts 110 valid rows.				
3	All foreign-key endpoints are valid	PASS	Every target attribute is the declared primary key of the target relation.				
4	No relationship is invented	PASS	Every edge is generated from one of the 110 recorded foreign keys. No extra edge is added.				
5	No cardinality contradicts the logical mapping	PASS	Nullable FKs use `\	o; unique non-PK FKs use o\	; all others use \	\	and o{` from the recorded nullability and uniqueness.
6	No relation is duplicated unintentionally	PASS	Mermaid identifiers are unique; each relation appears once in the overview and once in its own domain diagram.				
7	Every relation is covered by at least one diagram	PASS	54/54 relations are in ERD-OV and in their domain diagram.				
8	Every foreign key is covered by at least one diagram	PASS	110/110 FKs are in ERD-OV and in the holding relation's domain diagram.				
9	Every conceptual relationship is	PASS	78/78 relationships map to an				

	covered		FK or an associative relation and to a diagram.				
10	Mermaid code fences and structural edge syntax	PASS	11 Mermaid blocks; balanced fences; all edge lines match the expected `entity cardinality--cardinality entity : label` structure.				
11	Visual Mermaid rendering	PASS	Mermaid CLI 12.2.0 with Microsoft Edge rendered 11/11 SVGs without error markers. `ERD-D1`, `ERD-D10`, and `ERD-OV` were visually inspected.				

5.2 Validation Boundary

The checks above combine structural, data-driven checks with an actual Mermaid render. They verify that the diagram source is generated from the authoritative catalogue, that every relation, foreign key, and conceptual relationship is represented, and that the renderer produced 11 diagrams without error markers. Visual inspection was limited to `ERD-D1`, `ERD-D10`, and `ERD-OV`; the dense overview and `ERD-D10` are not claimed to be aesthetically optimized.

6. Revision History

Version	Date	Author	Change
1.0	4 November 2026	Falcon Team	Created the complete logical ERD with overview, ten domain diagrams, full relation/FK/relationship coverage matrices, and structural validation.

End of Complete Database ERD.

Appendix G: Logical Data Dictionary

Smart E-Commerce & Product Intelligence Platform

System Analysis & Design - Task 02, Phase 02.3 Logical Database Design

1. Document Control

Field	Value
Document Title	TechWise Logical Data Dictionary
Version	1.0
Preparation Date	4 November 2026
Status	Logical baseline; technology-neutral
Scope	All 54 logical relations and 502 attributes
Companion Document	Logical_Relation_Catalogue.md (authoritative relation metadata)

2. Purpose

This dictionary defines every logical attribute: its meaning, logical type, required status, key role, uniqueness, domain or validation rule, and the relation that owns it. It is generated from the same logical specification as the relation catalogue, so attribute identifiers, types, and key roles agree by construction.

3. Global Conventions

Topic	Rule
Identifiers	Every relation has a surrogate Identifier primary key unless it is a pure associative relation, which uses a composite natural key.
Timestamps	All timestamps are logical UTC instants; precision is a physical-design decision.
Text	Logical maximum lengths are stated; character set and collation are deferred.
Nulls	0 means the attribute may be absent. Absence must not be interpreted as a negative value, zero, or Incompatible (IC-30).
Defaults	Defaults are stated only where the conceptual model justifies them; no business value is invented.
Enumerated values	Enumerated attributes reference a controlled domain. Domain content is listed in the controlled-value catalogue; deferred domains contain no invented values.
Deletion	No soft-delete flag is defined. Retention and deletion policy remains deferred (OD-12).

Personal data	Prior-owner and account personal data are subject to privacy constraints and access control; no secret is stored in these relations.

4. Attribute Dictionary

RL-01 - User

Source: CE-01. *Primary key:* UserId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-001	UserId	Stable platform identity of the account.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-002	DisplayName	Name presented in ordinary product and account contexts.	Short text	Required	—	No	Not blank.
LAT-003	AccountStatus	Lifecycle state of the account.	Enumerated/reference value	Optional	—	No	Content pending D-17; no value invented.
LAT-004	LocaleCode	Preferred presentation locale.	Short text	Optional	—	No	Format pending; language-tag-style reference only.
LAT-005	ContactEmail	Contact address where collection is approved.	Short text	Optional	—	Conditional	Personal data; uniqueness conditional on authentication policy D-17.
LAT-006	ContactPhone	Contact number where collection is approved.	Short text	Optional	—	No	Personal data; format pending.
LAT-007	ProfileReference	Minimal profile attributes permitted by privacy policy.	Structured value	Optional	—	No	Must exclude authentication secrets and unnecessary personal data.
LAT-008	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform

							epoch.
LAT-009	UpdatedAt	Last permitted profile update time.	Timestamp	Optional	—	No	Not earlier than CreatedAt.
LAT-010	DeactivatedAt	Time the account became inactive, if applicable.	Timestamp	Optional	—	No	Not earlier than CreatedAt; no physical deletion implied.

RL-02 - Role

Source: CE-02. *Primary key:* RoleId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-011	RoleId	Stable role identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-012	RoleCode	Stable business code for the role.	Short text	Required	—	Yes	Unique; not reused after retirement.
LAT-013	RoleName	Human-readable role name.	Short text	Required	—	No	Not blank.
LAT-014	Description	Purpose and intended scope of the role.	Long text	Optional	—	No	No authorization logic embedded in text.
LAT-015	IsActive	Whether the role may receive new assignments.	Boolean	Required	—	No	Default true at creation; historical assignments are unaffected.
LAT-016	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.
LAT-017	UpdatedAt	Last permitted role update.	Timestamp	Optional	—	No	Not earlier than CreatedAt.

RL-03 - Permission

Source: CE-03. *Primary key:* PermissionId.

AT ID	Attribute	Meaning	Logical	Required	Key role	Unique	Domain /

			type				validation
LAT-018	PermissionId	Stable permission identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-019	PermissionCode	Stable capability code.	Short text	Required	—	Yes	Unique; resource and action semantics documented separately.
LAT-020	Description	Meaning and intended use of the capability.	Long text	Optional	—	No	Must not imply ownership or product-truth authority.
LAT-021	ResourceScope	Resource family addressed by the capability.	Short text	Optional	—	No	Content pending permission catalogue D-11.
LAT-022	ActionScope	Action family addressed by the capability.	Short text	Optional	—	No	Content pending permission catalogue D-11.
LAT-023	IsActive	Whether the permission may be granted.	Boolean	Required	—	No	Default true at creation.
LAT-024	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.
LAT-025	UpdatedAt	Last permitted permission update.	Timestamp	Optional	—	No	Not earlier than CreatedAt.

RL-04 - UserRoleAssignment

Source: CE-04. *Primary key:* UserRoleAssignmentId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-026	UserRoleAssignmentId	Stable assignment identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-	UserId	Assigned	Identifier	Required	FK	No	Must reference

027		user.					an existing user.
LAT-028	RoleId	Assigned role.	Identifier	Required	FK	No	Must reference an existing role.
LAT-029	AssignmentStatus	Lifecycle state of the assignment.	Enumerated/reference value	Optional	—	No	Content deferred; EndedAt is authoritative for closure.
LAT-030	AssignedAt	Time the assignment became effective.	Timestamp	Required	—	No	Not earlier than the user creation time.
LAT-031	EndedAt	Time the assignment ceased to be effective.	Timestamp	Optional	—	No	Not earlier than AssignedAt.
LAT-032	AssignedByUserId	Actor who performed the assignment, where recorded.	Identifier	Optional	FK	No	Must be an existing user.
LAT-033	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-05 - RolePermissionAssignment

Source: CE-05. *Primary key:* RolePermissionAssignmentId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-034	RolePermissionAssignmentId	Stable grant identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-035	RoleId	Granted role.	Identifier	Required	FK	No	Must reference an existing role.
LAT-036	PermissionId	Granted permission.	Identifier	Required	FK	No	Must reference an existing permission.

LAT-037	GrantStatus	Lifecycle state of the grant.	Enumerated/reference value	Optional	—	No	Content deferred; EndedAt is authoritative for closure.
LAT-038	GrantedAt	Time the grant became effective.	Timestamp	Required	—	No	Not earlier than role creation time.
LAT-039	EndedAt	Time the grant ceased to be effective.	Timestamp	Optional	—	No	Not earlier than GrantedAt.
LAT-040	GrantedByUserId	Actor who performed the grant, where recorded.	Identifier	Optional	FK	No	Must be an existing user.
LAT-041	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-06 - UserPreference

Source: CE-06. *Primary key:* UserPreferenceId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-042	UserPreferenceId	Stable preference identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-043	UserId	Preference owner.	Identifier	Required	FK	No	Must reference an existing user.
LAT-044	PreferenceType	Controlled preference category.	Enumerated/reference value	Required	—	No	Content pending preference catalogue and consent policy.
LAT-045	PreferenceScope	Optional scope qualifier for the preference.	Short text	Optional	—	No	Uniqueness applies per user, type, and scope where present.

LAT-046	PreferenceValue	Stored preference value.	Structured value	Required	—	No	Must conform to the declared preference type; no free-form blob.
LAT-047	SourceNote	Origin of the preference.	Short text	Optional	—	No	Must not imply verification.
LAT-048	UpdatedAt	Last update time.	Timestamp	Required	—	No	Not earlier than CreatedAt.
LAT-049	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-07 - Product

Source: CE-07. *Primary key:* ProductId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-050	ProductId	Stable product identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-051	ProductName	Market-facing product name.	Short text	Required	—	No	Not blank.
LAT-052	BrandReference	Brand attribution without inventing a brand-governance entity.	Short text	Optional	—	No	Must resolve through the approved brand reference list when one exists (A-16).
LAT-053	ModelNumber	Manufacturer model designation.	Short text	Optional	—	No	Not a unique key on its own.
LAT-054	ModelVersion	Model revision or market version context.	Short text	Optional	—	No	No variant semantics implied.
LAT-055	PrimaryCategoryId	Primary category under working assumption	Identifier	Required	FK	No	Must reference an existing category.

		A-01.					
LAT-056	Description	Descriptive product information.	Long text	Optional	—	No	Must not be treated as a verified fact.
LAT-057	ReleaseDate	Known market release date.	Date	Optional	—	No	Must not predate manufacturer existence; no validation inferred.
LAT-058	ProductStatus	Catalogue lifecycle state.	Enumerated/reference value	Optional	—	No	Content deferred; no value invented.
LAT-059	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.
LAT-060	UpdatedAt	Last permitted catalogue update.	Timestamp	Optional	—	No	Not earlier than CreatedAt.

RL-08 - ProductVariant

Source: CE-08. *Primary key:* ProductVariantId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-061	ProductVariantId	Stable variant identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-062	ProductId	Parent product.	Identifier	Required	FK	No	Must reference an existing product.
LAT-063	VariantCode	Variant code unique within its product.	Short text	Required	—	No	Not blank; unique within ProductId.
LAT-064	VariantLabel	Human-readable variant label.	Short text	Optional	—	No	Not blank when present.
LAT-065	VariantStatus	Variant lifecycle state.	Enumerated/reference value	Optional	—	No	Content deferred; no value invented.

LAT-066	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the parent product creation.
LAT-067	UpdatedAt	Last permitted variant update.	Timestamp	Optional	—	No	Not earlier than CreatedAt.

RL-09 - Category

Source: CE-09. *Primary key:* CategoryId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-068	CategoryId	Stable category identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-069	CategoryCode	Stable category code.	Short text	Required	—	Yes	Unique; not reused after retirement.
LAT-070	CategoryName	Human-readable category name.	Short text	Required	—	No	Not blank.
LAT-071	ParentCategoryId	Parent category for hierarchy.	Identifier	Optional	FK	No	Must not create a cycle; root categories leave this empty.
LAT-072	TaxonomyVersionReference	Reference to the approved taxonomy version.	Short text	Optional	—	No	Content deferred (D-01); no version invented.
LAT-073	IsActive	Whether the category may be used for new products.	Boolean	Required	—	No	Default true; historical use retained.
LAT-074	EffectiveFrom	Start of category applicability.	Date	Optional	—	No	Not later than EffectiveTo when both present.
LAT-075	EffectiveTo	End of category applicability.	Date	Optional	—	No	Not earlier than EffectiveFrom.

LAT-076	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.
LAT-077	UpdatedAt	Last permitted category update.	Timestamp	Optional	—	No	Not earlier than CreatedAt.

RL-10 - AttributeDefinition

Source: CE-10. *Primary key:* AttributeDefinitionId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-078	AttributeDefinitionId	Stable attribute definition identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-079	AttributeCode	Stable attribute code within its primary category.	Short text	Required	—	No	Unique within PrimaryCategoryId.
LAT-080	AttributeName	Human-readable attribute name.	Short text	Required	—	No	Not blank.
LAT-081	Description	Definition and intended interpretation.	Long text	Optional	—	No	Must not encode category policy that belongs to EvidenceRequirement.
LAT-082	ValueTypeCode	Logical value family of the attribute.	Enumerated/reference value	Required	—	No	One of the documented logical type families.
LAT-083	UnitCode	Unit of measure where applicable.	Short text	Optional	—	No	Required when ValueTypeCode is numeric and the attribute is measurable.
LAT-084	PrimaryCategoryId	Primary category scope under working assumption A-02.	Identifier	Required	FK	No	Must reference an existing category.
LAT-	ApplicabilityScope	Whether the attribute	Enumerated/referen	Require	—	No	Must be Product or Variant; no other

085		applies at product or variant level.	ce value	d			value invented.
LAT-086	IsExpectationRequired	Whether the attribute is expected for the category.	Boolean	Required	—	No	Default false; does not by itself make a fact mandatory for eligibility.
LAT-087	MinNumericValue	Optional inclusive lower bound for numeric values.	Decimal	Optional	—	No	Only meaningful for numeric value types.
LAT-088	MaxNumericValue	Optional inclusive upper bound for numeric values.	Decimal	Optional	—	No	Not less than MinNumericValue when both present.
LAT-089	MaxLength	Optional maximum character length for text values.	Integer	Optional	—	No	Must be positive when present.
LAT-090	ValidationRuleReference	Reference to an approved validation rule where one exists.	Short text	Optional	—	No	Must not embed executable logic in the data model.
LAT-091	DefinitionVersion	Version label of the definition.	Short text	Required	—	No	Not blank.
LAT-092	IsActive	Whether new facts may use the definition.	Boolean	Required	—	No	Default true; existing facts remain valid.
LAT-093	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.
LAT-094	UpdatedAt	Last permitted definition update.	Timestamp	Optional	—	No	Not earlier than CreatedAt.

RL-11 - ProductFact

Source: CE-11. *Primary key:* ProductFactId.

AT	Attribute	Meaning	Logical type	Require	Key	Uniqu	Domain /

ID				d	rol e	e	validation
LAT -095	ProductFactId	Stable fact assertion identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT -096	SubjectProductId	Product that is the subject of the assertion.	Identifier	Required	FK	No	Must reference an existing product.
LAT -097	SubjectVariantId	Optional more specific variant subject.	Identifier	Optional	FK	No	Must belong to SubjectProductId.
LAT -098	AttributeDefinitionId	Governed attribute used to express the fact.	Identifier	Required	FK	No	Must reference an existing, applicable definition.
LAT -099	ValueShortText	Short-text value representation.	Short text	Optional	—	No	Populated only when the definition value type is short text.
LAT -100	ValueLongText	Long-text value representation.	Long text	Optional	—	No	Populated only when the definition value type is long text.
LAT -101	ValueInteger	Integer value representation.	Integer	Optional	—	No	Populated only for integer value types.
LAT -102	ValueDecimal	Decimal value representation.	Decimal	Optional	—	No	Populated only for decimal value types.
LAT -103	ValueBoolean	Boolean value representation.	Boolean	Optional	—	No	Populated only for boolean value types.
LAT -104	ValueDate	Date value representation.	Date	Optional	—	No	Populated only for date value types.
LAT -105	ValueTimestamp	Timestamp value representation.	Timestamp	Optional	—	No	Populated only for timestamp value types.
LAT -106	ValueReferenceCode	Reference or enumerated value code.	Short text	Optional	—	No	Must be a permitted value when the definition declares an enumerated

							domain.
LAT-107	ValueStructured	Structured value representation.	Structured value	Optional	—	No	Must conform to a governed definition; not a free-form dump.
LAT-108	ValueFileReference	File-backed value reference.	Binary/file reference	Optional	—	No	Only when the definition permits a file reference.
LAT-109	PrimarySourceId	Primary provenance record for the assertion.	Identifier	Optional	FK	No	Required unless IsUnsourced is true (IC-06).
LAT-110	ConfidenceCode	Confidence classification of the assertion.	Enumerated/reference value	Optional	—	No	Low, Medium, High, or Insufficient Evidence only.
LAT-111	VerificationStatusCode	Verification state of the assertion.	Enumerated/reference value	Required	—	No	Unverified, Supported, Verified, or Conflicted only.
LAT-112	ProducerTypeCode	Class of producer that created the assertion.	Enumerated/reference value	Required	—	No	BackendRule, Human, AI-assisted, or SystemProcess only.
LAT-113	IsUnsourced	Explicit marker for an assertion with no source record.	Boolean	Required	—	No	Default false; true requires PrimarySourceId to be empty.
LAT-114	RecordedAt	Time the assertion was recorded.	Timestamp	Required	—	No	Not earlier than the subject product creation.
LAT-115	ObservedAt	Time the asserted state was observed or published.	Timestamp	Optional	—	No	Must not be later than RecordedAt without an explicit justification.
LAT-116	Notes	Free-text clarification of the assertion.	Long text	Optional	—	No	Must not be used to smuggle an ungoverned value.
LAT-117	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-12 - ProductSource

Source: CE-12. *Primary key:* ProductSourceId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-118	ProductSourceId	Stable source record identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-119	SourceTypeCode	Controlled source category.	Enumerated/reference value	Required	—	No	Content pending source-quality policy; no type invented.
LAT-120	SourceName	Name of the source.	Short text	Required	—	No	Not blank.
LAT-121	SourceReference	Citation or locator for the source.	Short text	Required	—	No	Not blank; must not embed credentials.
LAT-122	RetrievedAt	Time the source was retrieved.	Timestamp	Required	—	No	Not earlier than the platform epoch.
LAT-123	PublicationAt	Publication or issue time where known.	Timestamp	Optional	—	No	Must not be later than RetrievedAt when the source is already published.
LAT-124	EffectiveFrom	Start of source validity context.	Date	Optional	—	No	Not later than EffectiveTo when both present.
LAT-125	EffectiveTo	End of source validity context.	Date	Optional	—	No	Not earlier than EffectiveFrom.
LAT-126	FreshnessStatus	Freshness classification where a policy exists.	Enumerated/reference value	Optional	—	No	Content and thresholds deferred; unknown is permitted.
LAT-127	ReliabilityNote	Attributable reliability observation.	Long text	Optional	—	No	Must not be treated as a verified fact.
LAT-128	ProvenanceMetadata	Additional provenance metadata permitted by policy.	Structured value	Optional	—	No	Must not contain secrets or unnecessary personal data.

LAT-129	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-13 - InformationConflict

Source: CE-13. *Primary key:* InformationConflictId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-130	InformationConflictId	Stable conflict identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-131	ConflictFactAId	First fact participating in the conflict.	Identifier	Required	FK	No	Must differ from ConflictFactBId.
LAT-132	ConflictFactBId	Second fact participating in the conflict.	Identifier	Required	FK	No	Must differ from ConflictFactAId.
LAT-133	ConflictTypeCode	Controlled conflict category.	Enumerated/reference value	Required	—	No	Content deferred; no type invented.
LAT-134	ConflictStatusCode	Lifecycle state of the conflict.	Enumerated/reference value	Required	—	No	Open, Under Review, or Resolved only.
LAT-135	ContextSourceId	Optional source providing conflict context.	Identifier	Optional	FK	No	Must reference an existing source; does not decide the conflict.
LAT-136	OpenedAt	Time the conflict was recorded.	Timestamp	Required	—	No	Not earlier than either fact recording time.
LAT-137	ClosedAt	Time the conflict was closed.	Timestamp	Optional	—	No	Not earlier than OpenedAt.
LAT-138	ResolutionSummary	Recorded resolution rationale.	Long text	Optional	—	No	Must not delete or overwrite either original fact.
LAT-139	ResolvedByUserId	Actor who recorded the resolution.	Identifier	Optional	—	No	Must be an existing user.
LAT-140	CreatedAt	Record creation	Timestamp	Required	—	No	Not earlier than the platform

		time.					epoch.

RL-14 - ProductAsset

Source: CE-14. *Primary key:* ProductAssetId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-141	ProductAssetId	Stable platform identity of the physical asset.	Identifier	Required	PK	Yes	Unique and immutable across all lifecycle events.
LAT-142	ProductId	Product model instantiated by the asset.	Identifier	Required	FK	No	Must reference an existing product; immutable after ownership exists.
LAT-143	ProductVariantId	Optional specific variant of the product.	Identifier	Optional	FK	No	Must belong to ProductId.
LAT-144	FirstAcquisitionDate	Original first acquisition date used for age context.	Date	Optional	—	No	Must not reset on transfer; no date derived from resale.
LAT-145	EnrollmentAt	Time the asset was enrolled in the platform.	Timestamp	Required	—	No	Not earlier than the product creation.
LAT-146	Notes	Operational notes about the asset.	Long text	Optional	—	No	Must not contain prior-owner personal data beyond policy.
LAT-147	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.
LAT-148	UpdatedAt	Last permitted asset update.	Timestamp	Optional	—	No	Not earlier than CreatedAt; identity and first

							acquisition date are not mutable.

RL-15 - Purchase

Source: CE-15. *Primary key:* PurchaseId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-149	PurchaseId	Stable purchase record identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-150	BuyerUserId	User recorded as buyer.	Identifier	Required	FK	No	Must reference an existing user; buyer status is not current ownership.
LAT-151	ProductId	Product acquired.	Identifier	Required	FK	No	Must reference an existing product.
LAT-152	ProductAssetId	Specific asset acquired, when identified.	Identifier	Optional	FK	No	Must be an asset of the referenced product.
LAT-153	AcquisitionAt	Time of acquisition.	Timestamp	Required	—	No	Not earlier than the buyer account creation.
LAT-154	AcquisitionChannel	Channel or origin type of the acquisition.	Enumerated/reference value	Optional	—	No	Content deferred; no channel invented.
LAT-155	ExternalOrderReference	External order or receipt reference where available.	Short text	Optional	—	No	Must not embed payment credentials.
LAT-156	PurchaseStatus	Recorded status of the purchase record.	Enumerated/reference value	Optional	—	No	Content deferred; no transaction completion implied.

LAT-157	ProvenanceNote	Note on how the purchase was recorded.	Long text	Optional	—	No	Must not assert verification that did not occur.
LAT-158	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-16 - OwnershipPeriod

Source: CE-16. *Primary key:* OwnershipPeriodId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-159	OwnershipPeriodId	Stable ownership-period identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-160	ProductAssetId	Asset whose ownership is recorded.	Identifier	Required	FK	No	Must reference an existing asset.
LAT-161	OwnerUserId	User recorded as owner during the period.	Identifier	Required	FK	No	Must reference an existing user; joint ownership is not supported under A-05.
LAT-162	PeriodStartAt	Start of the ownership period.	Timestamp	Required	—	No	Must precede or equal PeriodEndAt.
LAT-163	PeriodEndAt	End of the ownership period; empty means the current period.	Timestamp	Optional	—	No	Not earlier than PeriodStartAt.
LAT-164	PeriodStatusCode	Lifecycle state of the period.	Enumerated/reference value	Required	—	No	Active or Ended only; must agree with PeriodEndAt.
LAT-165	OriginPurchaseId	Purchase providing the origin basis for the period.	Identifier	Optional	FK	No	Must concern the same asset when present.

LAT-166	OpeningTransferId	Transfer that opened this period, where one did.	Identifier	Optional	FK	Yes	Unique; empty only for an initial or imported acquisition.
LAT-167	AcceptedValidationId	Ownership validation accepted as basis for the period.	Identifier	Optional	FK	No	Must concern the same asset and owner claimant.
LAT-168	PeriodReference	Opaque period reference for the owner-facing ownership code.	Short text	Optional	—	Conditional	Unique within the asset; format and issuance policy deferred (BR-17).
LAT-169	ClosureReason	Reason the period ended.	Short text	Optional	—	No	No transaction outcome inferred.
LAT-170	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the asset enrollment.
LAT-171	ClosedAt	Time the period record was closed.	Timestamp	Optional	—	No	Not earlier than PeriodStartAt.

RL-17 - OwnershipTransfer

Source: CE-17. *Primary key:* OwnershipTransferId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-172	OwnershipTransferId	Stable transfer identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-173	ProductAssetId	Asset being transferred.	Identifier	Required	FK	No	Must reference an existing asset.
LAT-174	PriorOwnershipPeriodId	Period closed by the transfer.	Identifier	Required	FK	No	Must belong to ProductAssetId and be the applicable

							prior period.
LAT-175	TransferAt	Recorded effective time of transfer.	Timestamp	Optional	—	No	Effective-point semantics remain unresolved (D-05).
LAT-176	InitiatedByUserId	Actor who initiated the transfer.	Identifier	Optional	FK	No	Must be an existing user.
LAT-177	AcceptedByUserId	Actor who accepted the transfer, where recorded.	Identifier	Optional	FK	No	Must be an existing user.
LAT-178	TransferStatus	Recorded status of the transfer.	Enumerated/reference value	Optional	—	No	Content deferred; Sold state does not set this status.
LAT-179	AcceptedValidationId	Validation supporting the transfer.	Identifier	Optional	FK	No	Must concern the same asset.
LAT-180	SourceReference	Reference for the transfer basis.	Short text	Optional	—	No	Must not embed payment credentials.
LAT-181	RecordedAt	Time the transfer was recorded.	Timestamp	Required	—	No	Not earlier than the prior period start.
LAT-182	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-18 - OwnershipValidation

Source: CE-18. *Primary key:* OwnershipValidationId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-183	OwnershipValidationId	Stable validation identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-184	ProductAssetId	Asset whose claim is validated.	Identifier	Required	FK	No	Must reference an existing asset.

LAT-185	ClaimantUserId	User making the claim.	Identifier	Required	FK	No	Must reference an existing user.
LAT-186	ValidationScope	Scope of the validation decision.	Short text	Required	—	No	Content semantics pending D-04.
LAT-187	OutcomeCode	Validation outcome.	Enumerated/reference value	Required	—	No	Content and permitted values deferred; no outcome invented.
LAT-188	Rationale	Reason for the outcome.	Long text	Required	—	No	Must not claim physical verification unless performed.
LAT-189	ValidatedByUserId	Actor who performed the validation.	Identifier	Optional	FK	No	Required unless the validation was system-generated.
LAT-190	ProducerTypeCode	Class of producer of the validation.	Enumerated/reference value	Required	—	No	BackendRule, Human, AI-assisted, or SystemProcess only.
LAT-191	PolicyReference	Approved policy applied.	Short text	Optional	—	No	Content pending D-04.
LAT-192	ValidatedAt	Time of the validation decision.	Timestamp	Required	—	No	Not earlier than the asset enrollment.
LAT-193	Limitations	Scope limits and unknowns.	Long text	Optional	—	No	Must not be omitted when evidence is partial.
LAT-194	SupersedesValidationId	Prior validation replaced by this record.	Identifier	Optional	FK	No	Must concern the same asset.
LAT-195	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-19 - EvidenceRequirement

Source: CE-19. *Primary key:* EvidenceRequirementId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-196	EvidenceRequirementId	Stable requirement identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-197	PrimaryCategoryId	Category scoping the requirement.	Identifier	Required	FK	No	Must reference an existing category.
LAT-198	PurposeCode	Purpose the requirement serves.	Enumerated/reference value	Required	—	No	Content deferred; no purpose invented.
LAT-199	EvidenceTypeCode	Controlled evidence type required.	Enumerated/reference value	Required	—	No	Values deferred; no evidence operation invented.
LAT-200	IsMandatory	Whether the requirement is mandatory for the scope.	Boolean	Required	—	No	Default true; mandatory intent does not itself grant eligibility.
LAT-201	ConditionContext	Condition or claim context in which the requirement applies.	Short text	Optional	—	No	Content pending D-03.
LAT-202	PolicyReference	Policy or version supporting the requirement.	Short text	Required	—	No	Not blank; traceability required.
LAT-203	EffectiveFrom	Start of requirement applicability.	Date	Optional	—	No	Not later than EffectiveTo when both present.
LAT-204	EffectiveTo	End of requirement applicability.	Date	Optional	—	No	Not earlier than EffectiveFrom.
LAT-205	IsActive	Whether the requirement applies to new submissions.	Boolean	Required	—	No	Default true; historical submissions unaffected.

LAT-206	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-20 - EvidenceSubmission

Source: CE-20. *Primary key:* EvidenceSubmissionId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-207	EvidenceSubmissionId	Stable submission identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-208	SubmitterUserId	User who submitted the package.	Identifier	Required	FK	No	Must reference an existing user.
LAT-209	SubmissionPurpose	Declared purpose of the submission.	Enumerated/reference value	Required	—	No	Content deferred; context must be explicit (IC-18).
LAT-210	ProductAssetId	Asset context of the submission.	Identifier	Optional	FK	No	Optional but at least one explicit context is required.
LAT-211	ResaleListingId	Listing context of the submission.	Identifier	Optional	FK	No	Optional but at least one explicit context is required.
LAT-212	ReviewStatusCode	Current review state of the submission.	Enumerated/reference value	Required	—	No	Submitted, Under Review, Accepted, Rejected, or Needs More Information only.
LAT-213	SubmittedAt	Time the package was submitted.	Timestamp	Required	—	No	Not earlier than the submitter account creation.
LAT-214	ScopeNote	Scope and content summary of the	Long text	Optional	—	No	Must not assert verification.

		submission.					
LAT-215	SupersedesSubmissionId	Prior submission replaced by this one.	Identifier	Optional	FK	No	Must have the same submitter and context.
LAT-216	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-21 - EvidenceItem

Source: CE-21. *Primary key:* EvidenceItemId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-217	EvidenceItemId	Stable evidence item identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-218	EvidenceSubmissionId	Submission containing the item.	Identifier	Required	FK	No	Must reference an existing submission.
LAT-219	EvidenceItemTypeCode	Controlled item type.	Enumerated/reference value	Required	—	No	Values deferred; no capture operation invented.
LAT-220	FileReference	Stored file reference for file-backed evidence.	Binary/file reference	Optional	—	No	Either FileReference or ContentDescription must be present.
LAT-221	ContentDescription	Description or structured content of the item.	Long text	Optional	—	No	Required when no file reference exists.
LAT-222	CapturedAt	Time the item was captured or issued.	Timestamp	Optional	—	No	Must not be later than submission time without justification.
LAT-223	SourceNote	Origin of the item.	Short text	Optional	—	No	Must not imply verification.
LAT-224	PrivacyClassification	Access classification of the item.	Enumerated/reference value	Optional	—	No	Content pending D-12 and privacy policy.

LAT-225	IntegrityReference	Integrity reference where an approved mechanism exists.	Short text	Optional	—	No	No hashing algorithm or mechanism is selected here.
LAT-226	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-22 - EvidenceAssessment

Source: CE-22. *Primary key:* EvidenceAssessmentId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-227	EvidenceAssessmentId	Stable assessment identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-228	EvidenceSubmissionId	Submission being assessed.	Identifier	Required	FK	No	Must reference an existing submission.
LAT-229	AssessorUserId	Reviewer who performed the assessment.	Identifier	Optional	FK	No	Required unless ProducerTypeCode is a system process.
LAT-230	ProducerTypeCode	Class of producer of the assessment.	Enumerated/reference value	Required	—	No	BackendRule, Human, AI-assisted, or SystemProcess only.
LAT-231	OutcomeCode	Assessment outcome.	Enumerated/reference value	Required	—	No	Content deferred; acceptance does not imply eligibility.
LAT-232	ConfidenceCode	Confidence classification of the assessment.	Enumerated/reference value	Required	—	No	Low, Medium, High, or Insufficient Evidence only.
LAT-233	Rationale	Reasoning for the outcome.	Long text	Required	—	No	Must not claim inspection that did not occur.
LAT-234	AssessedAt	Time of assessment.	Timestamp	Required	—	No	Not earlier than submission time.

LAT-235	ScopeLimitations	Limits on what the assessment covers.	Long text	Optional	—	No	Required when confidence is Insufficient Evidence.
LAT-236	PolicyReference	Rubric or policy applied.	Short text	Optional	—	No	Content pending D-11.
LAT-237	SupersedesAssessmentId	Prior assessment replaced by this one.	Identifier	Optional	FK	No	Must concern the same submission.
LAT-238	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-23 - ConditionClaim

Source: CE-23. *Primary key:* ConditionClaimId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-239	ConditionClaimId	Stable claim identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-240	ProductAssetId	Asset the claim concerns.	Identifier	Required	FK	No	Must reference an existing asset.
LAT-241	ClaimantUserId	User making the claim.	Identifier	Required	FK	No	Must reference an existing user.
LAT-242	ResaleListingId	Listing context of the claim.	Identifier	Optional	FK	No	Must concern the same asset when present.
LAT-243	ClaimedGradeCode	Claimed condition grade.	Enumerated/reference value	Required	—	No	5, 4, 3, 2, 1, Not Assessed, or Unknown only.
LAT-244	ClaimNotes	Claimant explanation.	Long text	Optional	—	No	Must remain labelled as a claim.
LAT-245	ClaimedAt	Time the claim was made.	Timestamp	Required	—	No	Not earlier than the asset

							enrollment.
LAT-246	ClaimSource	Origin of the claim.	Short text	Optional	—	No	Must not imply verification.
LAT-247	DisclosureLimitations	Known gaps in the claim.	Long text	Optional	—	No	Must not be omitted when the claimant declares partial knowledge.
LAT-248	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-24 - ConditionAssessment

Source: CE-24. *Primary key:* ConditionAssessmentId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-249	ConditionAssessmentId	Stable condition assessment identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-250	ProductAssetId	Asset being assessed.	Identifier	Required	FK	No	Must reference an existing asset.
LAT-251	ConditionClaimId	Specific claim evaluated, where applicable.	Identifier	Optional	FK	No	Must concern the same asset.
LAT-252	AssessorUserId	Assessor who performed the assessment.	Identifier	Optional	FK	No	Required unless ProducerTypeCode is a system process.
LAT-253	ProducerTypeCode	Class of producer of the assessment.	Enumerated/reference value	Required	—	No	BackendRule, Human, AI-assisted, or SystemProcess only.
LAT-254	AssessedGradeCode	Assessed condition grade.	Enumerated/reference value	Required	—	No	5, 4, 3, 2, 1, Not Assessed, or Unknown only.

LAT-255	RubricReference	Rubric or policy applied.	Short text	Optional	—	No	Content pending D-03; no rubric invented.
LAT-256	EvidenceContextNote	Evidence context considered.	Long text	Optional	—	No	Must not claim physical inspection unless performed.
LAT-257	AssessedAt	Time of assessment.	Timestamp	Required	—	No	Not earlier than the asset enrollment.
LAT-258	Rationale	Reasoning for the assessed grade.	Long text	Optional	—	No	Must not derive condition solely from evidence confidence.
LAT-259	Limitations	Known limits of the assessment.	Long text	Optional	—	No	Required when grade is Not Assessed or Unknown.
LAT-260	SupersedesAssessmentId	Prior assessment replaced by this one.	Identifier	Optional	FK	No	Must concern the same asset.
LAT-261	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-25 - CompatibilityRelationship

Source: CE-25. *Primary key:* CompatibilityRelationshipId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-262	CompatibilityRelationshipId	Stable assertion identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-263	SubjectProductId	Product that is the subject of the assertion.	Identifier	Required	FK	No	Must reference an existing product.
LAT-264	SubjectVariantId	Optional more specific subject variant.	Identifier	Optional	FK	No	Must belong to SubjectProductId.
LAT-265	TargetProductId	Product that is the target of the	Identifier	Required	FK	No	Must reference an existing product.

		assertion.					
LAT-266	TargetVariantId	Optional more specific target variant.	Identifier	Optional	FK	No	Must belong to TargetProductId.
LAT-267	CompatibilityTypeCode	Controlled compatibility type.	Enumerated/reference value	Required	—	No	Values deferred; no type invented.
LAT-268	OutcomeCode	Compatibility outcome.	Enumerated/reference value	Required	—	No	Compatible, Incompatible, Conditional, or Unknown only.
LAT-269	IsDeclaredSymmetric	Whether the assertion is explicitly declared to hold in both directions.	Boolean	Optional	—	No	Interpretation and symmetry policy remain deferred (D-08).
LAT-270	PrimarySourceId	Primary source attributed to the assertion.	Identifier	Optional	FK	No	A source does not establish truth.
LAT-271	CompatibilityRuleId	Rule governing the assertion, where one applies.	Identifier	Optional	FK	No	Must reference an existing rule.
LAT-272	ValidFrom	Start of assertion validity.	Date	Optional	—	No	Not later than ValidTo when both present.
LAT-273	ValidTo	End of assertion validity.	Date	Optional	—	No	Not earlier than ValidFrom.
LAT-274	RecordedAt	Time the assertion was recorded.	Timestamp	Required	—	No	Not earlier than either product creation.
LAT-275	Notes	Additional attribution or scope notes.	Long text	Optional	—	No	Must not hide a conditional requirement in free text.
LAT-276	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-26 - CompatibilityRule

Source: CE-26. *Primary key:* CompatibilityRuleId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-277	CompatibilityRuleId	Stable rule identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-278	RuleCode	Stable rule code.	Short text	Required	—	No	Unique with VersionLabel.
LAT-279	RuleName	Human-readable rule name.	Short text	Required	—	No	Not blank.
LAT-280	CompatibilityTypeCode	Compatibility type the rule addresses.	Enumerated/reference value	Required	—	No	Values deferred.
LAT-281	Description	Rule intent and evaluation description.	Long text	Optional	—	No	Must not contain executable code.
LAT-282	PrimarySourceId	Source attributed to the rule.	Identifier	Optional	FK	No	A source does not establish truth.
LAT-283	VersionLabel	Rule version label.	Short text	Required	—	No	Unique within RuleCode.
LAT-284	RuleStatus	Lifecycle state of the rule version.	Enumerated/reference value	Optional	—	No	Content deferred; published rules are not silently mutated.
LAT-285	EffectiveFrom	Start of rule applicability.	Date	Optional	—	No	Not later than EffectiveTo when both present.
LAT-286	EffectiveTo	End of rule applicability.	Date	Optional	—	No	Not earlier than EffectiveFrom.
LAT-287	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-27 - CompatibilityCondition

Source: CE-27. *Primary key:* CompatibilityConditionId.

AT ID	Attribute	Meaning	Logical type	Required	Key rol	Unique	Domain / validation

					e		
LAT-288	CompatibilityConditionId	Stable condition identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-289	CompatibilityRuleId	Rule to which the condition belongs.	Identifier	Required	FK	No	Must reference an existing rule.
LAT-290	ConditionTypeCode	Controlled condition category.	Enumerated/reference value	Required	—	No	Content deferred; no type invented.
LAT-291	SubjectContext	Which side or component the condition constrains.	Short text	Optional	—	No	Must be attributable to the rule.
LAT-292	RequiredAttributeDefinitionId	Governed attribute the condition depends on.	Identifier	Optional	FK	No	Must reference an existing definition.
LAT-293	OperatorCode	Comparison operator code.	Short text	Optional	—	No	Content and allowed operators deferred.
LAT-294	RequiredValue	Value required by the condition.	Short text	Optional	—	No	Must conform to the referenced attribute type when present.
LAT-295	UnitCode	Unit for a numeric required value.	Short text	Optional	—	No	Required when the referenced attribute is measurable.
LAT-296	IsMandatory	Whether the condition is mandatory for the rule.	Boolean	Required	—	No	Default true.
LAT-297	Description	Human-readable condition	Long text	Optional	—	No	Must not replace the structured fields for

		description.					machine evaluation.
LAT-298	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-28 - ResaleListing

Source: CE-28. *Primary key:* ResaleListingId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-299	ResaleListingId	Stable listing identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-300	ProductAssetId	Specific asset offered for resale.	Identifier	Required	FK	No	Must reference an existing asset.
LAT-301	SellerUserId	User recorded as seller.	Identifier	Required	FK	No	Must reference an existing user; seller status does not prove ownership.
LAT-302	ListingTitle	Marketplace listing title.	Short text	Required	—	No	Not blank.
LAT-303	ListingDescription	Seller-provided listing description.	Long text	Optional	—	No	Must not be presented as a verified fact.
LAT-304	CurrentStateCode	Current lifecycle state of the listing.	Enumerated/reference value	Required	—	No	One of the nine documented states only.
LAT-305	ExpiryAt	Planned expiry time.	Timestamp	Optional	—	No	Content semantics pending D-05.
LAT-306	WithdrawnAt	Time the seller withdrew the listing.	Timestamp	Optional	—	No	Not earlier than listing creation.
LAT-307	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the asset enrollment.

LAT-308	UpdatedAt	Last permitted listing update.	Timestamp	Optional	—	No	Not earlier than CreatedAt.

RL-29 - ListingStatusEvent

Source: CE-29. *Primary key:* ListingStatusEventId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-309	ListingStatusEventId	Stable event identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-310	ResaleListingId	Listing whose state changed.	Identifier	Required	FK	No	Must reference an existing listing.
LAT-311	PreviousStateCode	State before the transition; empty for the initial event.	Enumerated/reference value	Optional	—	No	Must be a documented listing state.
LAT-312	NewStateCode	State after the transition.	Enumerated/reference value	Required	—	No	Must be a documented listing state and must differ from PreviousStateCode.
LAT-313	EventAt	Time of the transition.	Timestamp	Required	—	No	Not earlier than listing creation.
LAT-314	ActorUserId	User who caused the transition, where applicable.	Identifier	Optional	FK	No	Required unless ProducerTypeCode is a system process.
LAT-315	ProducerTypeCode	Class of producer of the transition.	Enumerated/reference value	Required	—	No	BackendRule, Human, AI-assisted, or SystemProcess only.
LAT-316	ReasonCode	Controlled reason code where one exists.	Short text	Optional	—	No	Content deferred; no workflow reason invented.

LAT-317	ReasonNote	Explanation of the transition.	Long text	Optional	—	No	Must not claim an approval that was not recorded.
LAT-318	RelatedEligibilityAssessmentId	Eligibility assessment relevant to the transition.	Identifier	Optional	FK	No	Must concern the same listing.
LAT-319	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than EventAt.

RL-30 - ListingEligibilityAssessment

Source: CE-30. *Primary key:* ListingEligibilityAssessmentId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-320	ListingEligibilityAssessmentId	Stable eligibility assessment identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-321	ResaleListingId	Listing being assessed.	Identifier	Required	FK	No	Must reference an existing listing.
LAT-322	OutcomeCode	Eligibility outcome.	Enumerated/reference value	Required	—	No	Unknown, Under Review, Eligible, Ineligible, or Incomplete only.
LAT-323	PolicyReference	Policy or rule set applied.	Short text	Required	—	No	Not blank; content pending D-04.
LAT-324	AssessorUserId	User who performed the determination.	Identifier	Optional	FK	No	Required unless ProducerTypeCode is a system process.
LAT-325	ProducerTypeCode	Class of producer of the determination.	Enumerated/reference value	Required	—	No	BackendRule, Human, AI-assisted, or SystemProcess only.
LAT-326	Rationale	Reasoning for the outcome.	Long text	Required	—	No	Must not claim checks that were not performed.
LAT-327	AssessedAt	Time of the determination.	Timestamp	Required	—	No	Not earlier than listing creation.

LAT-328	Limitations	Scope limits and missing inputs.	Long text	Optional	—	No	Required when outcome is Incomplete or Unknown.
LAT-329	SupersedesAssessmentId	Prior determination replaced by this one.	Identifier	Optional	FK	No	Must concern the same listing.
LAT-330	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-31 - BlockingCondition

Source: CE-31. *Primary key:* BlockingConditionId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-331	BlockingConditionId	Stable blocking condition identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-332	ResaleListingId	Listing affected by the condition.	Identifier	Required	FK	No	Must reference an existing listing.
LAT-333	BlockingTypeCode	Controlled blocking category.	Enumerated/reference value	Required	—	No	Values deferred; no blocking type invented.
LAT-334	BlockingStatusCode	Lifecycle state of the blocking condition.	Enumerated/reference value	Required	—	No	Content deferred; no resolution workflow invented.
LAT-335	ReasonNote	Reason the condition was recorded.	Long text	Required	—	No	Must not assert an adjudication outcome.
LAT-336	OpenedAt	Time the condition was opened.	Timestamp	Required	—	No	Not earlier than listing creation.
LAT-337	ResolvedAt	Time the condition was resolved.	Timestamp	Optional	—	No	Not earlier than OpenedAt.
LAT-338	ResolutionNote	How the condition was	Long text	Optional	—	No	Must not record compensation

		resolved.					or payment.
LAT-339	SourceReference	Origin of the blocking report.	Short text	Optional	—	No	Must not expose unnecessary personal data.
LAT-340	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-32 - PriceRecord

Source: CE-32. *Primary key:* PriceRecordId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-341	PriceRecordId	Stable price record identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-342	PriceTypeCode	Approved price type.	Enumerated/reference value	Required	—	No	Asking or Reference only.
LAT-343	ResaleListingId	Listing scope for an asking price.	Identifier	Optional	FK	No	Exactly one of ResaleListingId or ProductId must be populated.
LAT-344	ProductId	Product scope for a reference price.	Identifier	Optional	FK	No	Exactly one of ResaleListingId or ProductId must be populated.
LAT-345	Amount	Price amount.	Decimal	Required	—	No	Must be greater than or equal to zero; missing price is not zero.
LAT-346	CurrencyCode	Currency of the amount.	Short text	Required	—	No	Must be an approved currency code when currency policy exists.
LAT-347	SourceReference	Origin of the price value.	Short text	Optional	—	No	Must not imply a market feed that does not exist.
LAT-348	ObservedAt	Time the price was observed	Timestamp	Required	—	No	Not earlier than the listing creation for

		or declared.					asking prices.
LAT-349	ConditionContext	Condition or variant context of the price.	Short text	Optional	—	No	Must not substitute for a condition assessment.
LAT-350	PriceStatus	Lifecycle state of the price record.	Enumerated/reference value	Optional	—	No	Content deferred.
LAT-351	Limitations	Known limits of the price value.	Long text	Optional	—	No	Required when the source is not an approved market source.
LAT-352	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-33 - DecisionCriterion

Source: CE-33. *Primary key:* DecisionCriterionId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-353	DecisionCriterionId	Stable criterion identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-354	CriterionCode	Stable criterion code.	Short text	Required	—	Yes	Unique; not reused after retirement.
LAT-355	CriterionName	Human-readable criterion name.	Short text	Required	—	No	Not blank.
LAT-356	Description	Meaning of the criterion.	Long text	Optional	—	No	Must not embed a weight or formula.
LAT-357	PreferenceDirection	Whether higher or lower values are preferred.	Enumerated/reference value	Optional	—	No	Content deferred; no direction semantics invented.
LAT-358	ValueTypeCode	Logical value family of the criterion.	Enumerated/reference value	Required	—	No	One of the documented logical type families.

LAT-359	CriterionStatus	Lifecycle state of the criterion.	Enumerated/reference value	Optional	—	No	Content deferred.
LAT-360	ProvenanceNote	Approval or origin reference for the criterion.	Short text	Optional	—	No	Content pending D-06.
LAT-361	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.
LAT-362	UpdatedAt	Last permitted criterion update.	Timestamp	Optional	—	No	Not earlier than CreatedAt.

RL-34 - CriterionWeight

Source: CE-34. *Primary key:* CriterionWeightId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-363	CriterionWeightId	Stable weight identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-364	RuleVersionId	Rule version owning the weight.	Identifier	Required	FK	No	Must reference an existing rule version.
LAT-365	DecisionCriterionId	Criterion being weighted.	Identifier	Required	FK	No	Must reference an existing criterion.
LAT-366	WeightValue	Weight value within the declared scale.	Decimal	Required	—	No	Must conform to the declared scale; no sum or normalization rule is assumed.
LAT-367	WeightScaleNote	Declared scale or unit context.	Short text	Optional	—	No	Required until a single approved weight scale exists.
LAT-368	Rationale	Reason for the weight.	Long text	Optional	—	No	Content pending D-06.

LAT-369	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the rule version creation.

RL-35 - ScoringRule

Source: CE-35. *Primary key:* ScoringRuleId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-370	ScoringRuleId	Stable scoring rule identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-371	RuleCode	Stable rule code.	Short text	Required	—	Yes	Unique; not reused after retirement.
LAT-372	RuleName	Human-readable rule name.	Short text	Required	—	No	Not blank.
LAT-373	Purpose	Intended decision purpose.	Long text	Optional	—	No	Must not embed a formula.
LAT-374	Description	Rule description.	Long text	Optional	—	No	Must not contain executable code.
LAT-375	RuleStatus	Lifecycle state of the rule.	Enumerated/reference value	Optional	—	No	Content deferred.
LAT-376	OwnerNote	Steward or owning role reference.	Short text	Optional	—	No	Content pending D-06.
LAT-377	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.
LAT-378	UpdatedAt	Last permitted rule update.	Timestamp	Optional	—	No	Not earlier than CreatedAt.

RL-36 - RuleVersion

Source: CE-36. *Primary key:* RuleVersionId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-	RuleVersionId	Stable rule version	Identifier	Required	PK	Yes	Unique and

379		identifier.					immutable.
LAT-380	ScoringRuleId	Parent scoring rule.	Identifier	Required	FK	No	Must reference an existing rule.
LAT-381	VersionLabel	Version label unique within the rule.	Short text	Required	—	No	Not blank; unique within ScoringRuleId.
LAT-382	EffectiveFrom	Start of version applicability.	Date	Optional	—	No	Not later than EffectiveTo when both present.
LAT-383	EffectiveTo	End of version applicability.	Date	Optional	—	No	Not earlier than EffectiveFrom.
LAT-384	VersionStatus	Lifecycle state of the version.	Enumerated/reference value	Optional	—	No	Content deferred; published versions are not silently mutated.
LAT-385	ChangeSummary	Summary of changes from the prior version.	Long text	Optional	—	No	Required for a version after the first.
LAT-386	ApprovalReference	Approval or provenance reference.	Short text	Optional	—	No	Content pending D-06; no approval is claimed.
LAT-387	PublishedAt	Time the version became effective for use.	Timestamp	Optional	—	No	Not earlier than EffectiveFrom when both present.
LAT-388	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the scoring rule creation.

RL-37 - DecisionRequest

Source: CE-37. *Primary key:* DecisionRequestId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-389	DecisionRequestId	Stable request	Identifier	Required	PK	Yes	Unique and immutable.

		identifier.					
LAT-390	RequestingUserId	User who created the request.	Identifier	Required	FK	No	Must reference an existing user.
LAT-391	RequestType	Type of decision request.	Enumerated/reference value	Required	—	No	Content deferred; no decision scope invented.
LAT-392	RequirementSnapshot	Input requirements and preferences captured for the request.	Structured value	Required	—	No	Must distinguish missing input from a negative preference (IC-30).
LAT-393	InputLimitations	Known gaps or assumptions in the input context.	Long text	Optional	—	No	Required when the snapshot is incomplete.
LAT-394	RuleVersionId	Rule version requested for evaluation.	Identifier	Required	FK	No	Must reference an existing rule version.
LAT-395	SavedComparisonId	Saved comparison that originated or references the request.	Identifier	Optional	FK	No	Must belong to RequestingUserId.
LAT-396	RequestedAt	Time the request was created.	Timestamp	Required	—	No	Not earlier than the requesting user creation.
LAT-397	RequestStatus	Lifecycle state of the request.	Enumerated/reference value	Optional	—	No	Content deferred; no result is implied.
LAT-398	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-38 - DecisionResult

Source: CE-38. *Primary key:* DecisionResultId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-399	DecisionResultId	Stable result identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-400	DecisionRequestId	Request that produced	Identifier	Required	FK	No	Must reference an existing

		the result.					request.
LAT-401	RuleVersionId	Rule version used to produce the result.	Identifier	Required	FK	No	Must reference an existing rule version; must match the request version at generation.
LAT-402	GeneratedAt	Time the result was generated.	Timestamp	Required	—	No	Not earlier than the request time.
LAT-403	ResultStatus	Lifecycle state of the result.	Enumerated/reference value	Optional	—	No	Content deferred.
LAT-404	InputSnapshotReference	Pointer or embedded snapshot of the inputs used.	Structured value	Optional	—	No	Must not duplicate authoritative product facts as new truth.
LAT-405	ValidationStatus	Validation state of the result.	Enumerated/reference value	Optional	—	No	Content deferred; no validation is claimed.
LAT-406	Limitations	Known limits of the result.	Long text	Optional	—	No	Required when inputs are incomplete.
LAT-407	SupersedesResultId	Prior result replaced by this one.	Identifier	Optional	FK	No	Must concern the same request and rule version.
LAT-408	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than GeneratedAt.

RL-39 - DecisionExplanation

Source: CE-39. *Primary key:* DecisionExplanationId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-409	DecisionExplanationId	Stable explanation identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-410	DecisionResultId	Result being explained.	Identifier	Required	FK	No	Must reference an existing result.

LAT-411	ProducerTypeCode	Class of producer of the explanation.	Enumerated/reference value	Required	—	No	BackendRule, Human, AI-assisted, or SystemProcess only.
LAT-412	ExplanationContent	Explanation text or structured explanation.	Long text	Required	—	No	Must be labelled and must not override the deterministic result.
LAT-413	GeneratedAt	Time the explanation was produced.	Timestamp	Required	—	No	Not earlier than the result generation.
LAT-414	ValidationStatus	Validation state of the explanation.	Enumerated/reference value	Optional	—	No	Content deferred; unvalidated text remains labelled.
LAT-415	SourceMetadata	Producer or model metadata where disclosure is approved.	Structured value	Optional	—	No	Must not expose provider secrets or unapproved model details.
LAT-416	Limitations	Known limits or uncertainty of the explanation.	Long text	Optional	—	No	Required for AI-assisted explanations with incomplete input.
LAT-417	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-40 - UsageRecord

Source: CE-40. *Primary key:* UsageRecordId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-418	UsageRecordId	Stable usage record identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-419	ProductAssetId	Asset whose usage is recorded.	Identifier	Required	FK	No	Must reference an existing asset.
LAT-	UsageMetricCode	Controlled usage	Enumerated/reference	Required	—	No	Content deferred; no

420		metric.	value				metric invented.
LAT-421	UsageValue	Measured or declared usage value.	Decimal	Optional	—	No	Must not be defaulted to zero when unknown.
LAT-422	UsageUnitCode	Unit of the usage value.	Short text	Optional	—	No	Required when UsageValue is present.
LAT-423	PeriodStart	Start of the observed usage period.	Timestamp	Optional	—	No	Not later than PeriodEnd when both present.
LAT-424	PeriodEnd	End of the observed usage period.	Timestamp	Optional	—	No	Not earlier than PeriodStart.
LAT-425	SourceNote	Origin of the usage observation.	Short text	Optional	—	No	Must not imply verification.
LAT-426	RecordedAt	Time the observation was recorded.	Timestamp	Required	—	No	Not earlier than the asset enrollment.
LAT-427	ConfidenceCode	Confidence classification of the observation.	Enumerated/reference value	Optional	—	No	Low, Medium, High, or Insufficient Evidence only.
LAT-428	EvidenceSubmissionId	Evidence submission supporting the usage observation.	Identifier	Optional	FK	No	Must concern the same asset when present.
LAT-429	Limitations	Known limits of the observation.	Long text	Optional	—	No	Required when confidence is Insufficient Evidence.
LAT-430	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-41 - SavedComparison

Source: CE-41. *Primary key:* SavedComparisonId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-431	SavedComparisonId	Stable saved comparison identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-432	UserId	Owner of the saved comparison.	Identifier	Required	FK	No	Must reference an existing user.
LAT-433	Title	User-provided comparison title.	Short text	Required	—	No	Not blank.
LAT-434	ComparisonContext	Saved context such as selected attributes or notes.	Structured value	Optional	—	No	Must not freeze authoritative product facts.
LAT-435	SnapshotPolicyNote	Statement of snapshot behaviour.	Short text	Optional	—	No	Content pending D-16.
LAT-436	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the user creation.
LAT-437	UpdatedAt	Last permitted comparison update.	Timestamp	Optional	—	No	Not earlier than CreatedAt.

RL-42 - AuditRecord

Source: CE-42. *Primary key:* AuditRecordId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-438	AuditRecordId	Stable audit record identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-439	ActorUserId	User who performed the action.	Identifier	Optional	FK	No	Required unless the action was performed by a system process.
LAT-440	ActorProcessNote	System process or component	Short text	Optional	—	No	Required when ActorUserId

		identifier.					is empty.
LAT-441	ActionTypeCode	Controlled audit action category.	Enumerated/reference value	Required	—	No	Content pending D-10; no event catalogue invented.
LAT-442	TargetTypeCode	Type of the target record.	Enumerated/reference value	Required	—	No	Content pending D-10.
LAT-443	TargetIdentifier	Identifier of the target record.	Short text	Optional	—	No	Stored as a reference value, not a polymorphic foreign key (A-12).
LAT-444	EventAt	Time of the audited action.	Timestamp	Required	—	No	Not earlier than the platform epoch.
LAT-445	OutcomeCode	Outcome of the action.	Enumerated/reference value	Required	—	No	Success, Failure, or Denied only.
LAT-446	ReasonNote	Reason or context of the action.	Long text	Optional	—	No	Must not contain secrets or unnecessary personal data.
LAT-447	CorrelationReference	Correlation identifier for related events.	Short text	Optional	—	No	Must not embed sensitive request payloads.
LAT-448	PrivacySafeMetadata	Additional context permitted by audit policy.	Structured value	Optional	—	No	Must exclude secrets and full business-record copies.
LAT-449	CreatedAt	Record creation time.	Timestamp	Required	—	No	Equal to or later than EventAt.

RL-43 - ProductAssetIdentifier

Source: CE-14 (identifier attribute group). *Primary key:* ProductAssetIdentifierId.

AT ID	Attribute	Meaning	Logical type	Required	Key rol	Unique	Domain / validation

					e		
LAT-450	ProductAssetIdentifierId	Stable identifier-value record identifier.	Identifier	Required	PK	Yes	Unique and immutable.
LAT-451	ProductAssetId	Asset to which the external identifier belongs.	Identifier	Required	FK	No	Must reference an existing asset.
LAT-452	IdentifierTypeCode	Controlled identifier type.	Enumerated/reference value	Required	—	No	Content pending D-09; no identifier type invented.
LAT-453	IdentifierValue	Identifier value as recorded.	Short text	Required	—	No	Not blank; display masking policy deferred.
LAT-454	IdentifierValueNormalized	Normalized form used for matching and uniqueness.	Short text	Required	—	No	Must be derived deterministically from IdentifierValue.
LAT-455	IssuingAuthority	Authority that issued the identifier.	Short text	Optional	—	No	Content pending D-09.
LAT-456	RecordedAt	Time the identifier was recorded.	Timestamp	Required	—	No	Not earlier than the asset enrollment.
LAT-457	RetiredAt	Time the identifier stopped being current.	Timestamp	Optional	—	No	Not earlier than RecordedAt.
LAT-458	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the platform epoch.

RL-44 - AttributeAllowedValue

Source: CE-10 (value-domain attribute group). *Primary key:* AttributeDefinitionId, AllowedValueCode.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-459	AttributeDefinitionId	Definition that owns	Identifier	Required	PK, FK	No	Must reference an

		the permitted value.					existing definition.
LAT-460	AllowedValueCode	Stable permitted value code.	Short text	Required	PK	No	Unique within the attribute definition.
LAT-461	AllowedValueLabel	Human-readable value label.	Short text	Required	—	No	Not blank.
LAT-462	DisplaySequence	Presentation order of the value.	Integer	Optional	—	No	Must be unique within the definition when present.
LAT-463	IsActive	Whether the value may be used for new facts.	Boolean	Required	—	No	Default true; historical facts remain valid.
LAT-464	EffectiveFrom	Start of value applicability.	Date	Optional	—	No	Not later than EffectiveTo when both present.
LAT-465	EffectiveTo	End of value applicability.	Date	Optional	—	No	Not earlier than EffectiveFrom.

RL-45 - FactEvidenceLink

Source: R-19 conceptual M:N. *Primary key:* ProductFactId, EvidenceSubmissionId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-466	ProductFactId	Supported product fact.	Identifier	Required	PK, FK	No	Must reference an existing fact.
LAT-467	EvidenceSubmissionId	Evidence submission providing support.	Identifier	Required	PK, FK	No	Must reference an existing submission.
LAT-468	LinkedAt	Time the link was recorded.	Timestamp	Required	—	No	Not earlier than both linked records.

RL-46 - ValidationEvidenceLink

Source: R-31 conceptual M:N. *Primary key:* OwnershipValidationId, EvidenceSubmissionId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation

LAT-469	OwnershipValidationId	Ownership validation relying on evidence.	Identifier	Required	PK, FK	No	Must reference an existing validation.
LAT-470	EvidenceSubmissionId	Evidence submission used.	Identifier	Required	PK, FK	No	Must reference an existing submission.
LAT-471	LinkedAt	Time the link was recorded.	Timestamp	Required	—	No	Not earlier than both linked records.

RL-47 - SubmissionRequirementLink

Source: R-33 conceptual M:N. *Primary key:* EvidenceSubmissionId, EvidenceRequirementId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-472	EvidenceSubmissionId	Submission addressing the requirement.	Identifier	Required	PK, FK	No	Must reference an existing submission.
LAT-473	EvidenceRequirementId	Requirement addressed.	Identifier	Required	PK, FK	No	Must reference an existing requirement.
LAT-474	LinkedAt	Time the link was recorded.	Timestamp	Required	—	No	Not earlier than both linked records.

RL-48 - EligibilityEvidenceBasis

Source: R-47 conceptual M:N. *Primary key:* ListingEligibilityAssessmentId, EvidenceAssessmentId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-475	ListingEligibilityAssessmentId	Eligibility determination informed.	Identifier	Required	PK, FK	No	Must reference an existing determination.
LAT-476	EvidenceAssessmentId	Evidence assessment used as basis.	Identifier	Required	PK, FK	No	Must reference an existing assessment.
LAT-477	LinkedAt	Time the basis was recorded.	Timestamp	Required	—	No	Not earlier than both linked records.

RL-49 - EligibilityOwnershipBasis

Source: R-48 conceptual M:N. *Primary key:* ListingEligibilityAssessmentId, OwnershipValidationId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-478	ListingEligibilityAssessmentId	Eligibility determination informed.	Identifier	Required	PK, FK	No	Must reference an existing determination.
LAT-479	OwnershipValidationId	Ownership validation used as basis.	Identifier	Required	PK, FK	No	Must reference an existing validation.
LAT-480	LinkedAt	Time the basis was recorded.	Timestamp	Required	—	No	Not earlier than both linked records.

RL-50 - EligibilityConditionBasis

Source: R-49 conceptual M:N. *Primary key:* ListingEligibilityAssessmentId, ConditionAssessmentId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-481	ListingEligibilityAssessmentId	Eligibility determination informed.	Identifier	Required	PK, FK	No	Must reference an existing determination.
LAT-482	ConditionAssessmentId	Condition assessment used as basis.	Identifier	Required	PK, FK	No	Must reference an existing assessment.
LAT-483	LinkedAt	Time the basis was recorded.	Timestamp	Required	—	No	Not earlier than both linked records.

RL-51 - SavedComparisonItem

Source: R-74 conceptual M:N. *Primary key:* SavedComparisonId, ProductId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-484	SavedComparisonId	Saved comparison containing the product.	Identifier	Required	PK, FK	No	Must reference an existing comparison.
LAT-485	ProductId	Included product.	Identifier	Required	PK, FK	No	Must reference an existing product.
LAT-486	DisplaySequence	Display order of the product in	Integer	Required	—	No	Must be unique within

		the comparison.					SavedComparisonId.
LAT-487	AddedAt	Time the product was added.	Timestamp	Required	—	No	Not earlier than the comparison creation.

RL-52 - DecisionRequestCandidateProduct

Source: R-76 conceptual M:N. *Primary key:* DecisionRequestId, ProductId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-488	DecisionRequestId	Decision request considering the product.	Identifier	Required	PK, FK	No	Must reference an existing request.
LAT-489	ProductId	Candidate product.	Identifier	Required	PK, FK	No	Must reference an existing product.
LAT-490	CandidateSequence	Order in which the candidate was supplied.	Integer	Required	—	No	Must be unique within DecisionRequestId.
LAT-491	AddedAt	Time the candidate was added.	Timestamp	Required	—	No	Not earlier than the request time.

RL-53 - DecisionRequestCandidateAsset

Source: R-77 conceptual M:N. *Primary key:* DecisionRequestId, ProductAssetId.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-492	DecisionRequestId	Decision request considering the asset.	Identifier	Required	PK, FK	No	Must reference an existing request.
LAT-493	ProductAssetId	Candidate or owned asset.	Identifier	Required	PK, FK	No	Must reference an existing asset.
LAT-494	CandidateSequence	Order in which the asset was supplied.	Integer	Required	—	No	Must be unique within DecisionRequestId.
LAT-495	AddedAt	Time the asset was added.	Timestamp	Required	—	No	Not earlier than the request time.

RL-54 - DecisionResultItem

Source: CE-38 (ranking attribute group). *Primary key:* DecisionResultId, CandidateSequence.

AT ID	Attribute	Meaning	Logical type	Required	Key role	Unique	Domain / validation
LAT-496	DecisionResultId	Result that owns the item.	Identifier	Required	PK, FK	No	Must reference an existing result.
LAT-497	CandidateSequence	Stable sequence of the candidate within the result.	Integer	Required	PK	No	Unique within DecisionResultId.
LAT-498	CandidateProductId	Product evaluated in this result line.	Identifier	Required	FK	No	Must reference an existing product.
LAT-499	CandidateProductAssetId	Specific asset evaluated, where applicable.	Identifier	Optional	FK	No	Must be an asset of CandidateProductId.
LAT-500	RankPosition	Produced rank position.	Integer	Optional	—	No	Unique within DecisionResultId when present; no tie-break rule is invented.
LAT-501	ItemOutcomeCode	Outcome classification for the candidate.	Enumerated/reference value	Optional	—	No	Content pending D-06.
LAT-502	CreatedAt	Record creation time.	Timestamp	Required	—	No	Not earlier than the result generation.

5. Domain Reference

5.1 Documented controlled vocabularies

ID	Vocabulary	Values	Status	Extensibility	Source
CV-01	Listing state	Draft, Pending Review, Active, Rejected, Reserved, Sold, Suspended, Withdrawn, Expired	Documented	Closed at Phase 02.2 baseline; any change needs OD-14 review	OD-14; SM-01
CV-02	Ownership period state	Active, Ended	Documented conceptual	Closed for this phase	OD-04; SM-03
CV-03	Evidence	Low, Medium, High, Insufficient	Documented	Closed for this	OD-06

	confidence	Evidence		phase	
CV-04	Condition grade	5 - Excellent, 4 - Very Good, 3 - Good, 2 - Fair, 1 - Poor, Not Assessed, Unknown	Documented	Closed for this phase	OD-13
CV-05	Compatibility outcome	Compatible, Incompatible, Conditional, Unknown	Documented	Closed for this phase	OD-08
CV-06	Evidence review status	Submitted, Under Review, Accepted, Rejected, Needs More Information	Working conceptual	Extensible only with OD-06 confirmation	OD-06
CV-07	Eligibility outcome	Unknown, Under Review, Eligible, Ineligible, Incomplete	Mixed documented and working	Extensible only with OD-06 and OD-14 confirmation	OD-06; OD-14
CV-08	Price type	Asking, Reference	Working conceptual	Market observation deferred	OD-15
CV-09	Producer type	BackendRule, Human, AI-assisted, SystemProcess	Working conceptual	Closed for this phase	OD-11; OD-16
CV-10	Audit outcome	Success, Failure, Denied	Working conceptual	Extensible with OD-18 confirmation	OD-18
CV-11	Verification status	Unverified, Supported, Verified, Conflicted	Working conceptual	Extensible only with source policy	OD-09; OD-11
CV-12	Conflict status	Open, Under Review, Resolved	Working conceptual	Extensible only with review policy	OD-09
CV-13	Evidence item type	Content deferred	Deferred	Content not supplied	OD-05; D-02
CV-14	Blocking condition type	Content deferred	Deferred	Content not supplied	OD-07; OD-14
CV-15	Compatibility type	Content deferred	Deferred	Content not supplied	OD-08

5.2 Deferred controlled domains

The following domains are referenced by attributes but have no approved value list in the inspected sources. No value is invented. An attribute in one of these domains is structurally constrained but its permitted values remain a documented dependency.

ID	Domain	Values	Dependency
DV-16	Account status	Content deferred	D-17
DV-17	Role assignment status	Content deferred	OD-02; D-11
DV-18	Permission grant status	Content deferred	OD-02; D-11
DV-19	Preference type	Content deferred	OD-03
DV-20	Product catalogue status	Content deferred	OD-09
DV-21	Variant status	Content deferred	OD-09
DV-22	Logical value type family	Identifier, Short text, Long text, Integer, Decimal, Boolean, Date, Timestamp, Structured value, Binary/file reference, Enumerated/reference value	Technology-neutral vocabulary of this document
DV-23	Product source type	Content deferred	OD-09; source policy
DV-24	Source freshness status	Content deferred	OD-09; freshness policy
DV-25	Conflict type	Content deferred	OD-09
DV-26	Acquisition channel	Content deferred	OD-07
DV-27	Purchase record status	Content deferred	OD-07
DV-28	Ownership transfer status	Content deferred	OD-04; OD-07
DV-29	Ownership validation outcome	Content deferred	OD-04; D-04
DV-30	Evidence requirement purpose	Content deferred	OD-05; D-02
DV-31	Evidence submission purpose	Content deferred	OD-05
DV-32	Evidence privacy classification	Content deferred	OD-12; D-12
DV-33	Evidence assessment outcome	Content deferred	OD-06; D-11
DV-34	Compatibility rule status	Content deferred	OD-08
DV-35	Compatibility condition type	Content deferred	OD-08
DV-36	Blocking condition status	Content deferred	OD-07; OD-14

DV-37	Price record status	Content deferred	OD-15
DV-38	Criterion preference direction	Content deferred	OD-16; D-06
DV-39	Decision criterion status	Content deferred	OD-16; D-06
DV-40	Scoring rule status	Content deferred	OD-16; D-06
DV-41	Rule version status	Content deferred	OD-16; D-06
DV-42	Decision request type	Content deferred	OD-16
DV-43	Decision request status	Content deferred	OD-16
DV-44	Decision result status	Content deferred	OD-16
DV-45	Validation status	Content deferred	OD-11; OD-16
DV-46	Usage metric	Content deferred	OD-17
DV-47	Audit action type	Content deferred	OD-18; D-10
DV-48	Audit target type	Content deferred	OD-18; D-10
DV-49	Asset identifier type	Content deferred	D-09
DV-50	Decision result item outcome	Content deferred	OD-16; D-06

6. Attribute Count Reconciliation

Relation	Name	Attributes	Required	Optional	FKs
RL-01	User	10	3	7	0
RL-02	Role	7	5	2	0
RL-03	Permission	8	4	4	0
RL-04	UserRoleAssignment	8	5	3	3
RL-05	RolePermissionAssignment	8	5	3	3
RL-06	UserPreference	8	6	2	1
RL-07	Product	11	4	7	1
RL-08	ProductVariant	7	4	3	1
RL-09	Category	10	5	5	1
RL-10	AttributeDefinition	17	10	7	1
RL-11	ProductFact	23	8	15	4
RL-12	ProductSource	12	6	6	0
RL-13	InformationConflict	11	7	4	3
RL-14	ProductAsset	8	4	4	2

RL-15	Purchase	10	5	5	3
RL-16	OwnershipPeriod	13	6	7	5
RL-17	OwnershipTransfer	11	5	6	5
RL-18	OwnershipValidation	13	9	4	4
RL-19	EvidenceRequirement	11	8	3	1
RL-20	EvidenceSubmission	10	6	4	4
RL-21	EvidenceItem	10	4	6	1
RL-22	EvidenceAssessment	12	8	4	3
RL-23	ConditionClaim	10	6	4	3
RL-24	ConditionAssessment	13	6	7	4
RL-25	CompatibilityRelationship	15	7	8	6
RL-26	CompatibilityRule	11	6	5	1
RL-27	CompatibilityCondition	11	5	6	2
RL-28	ResaleListing	10	6	4	2
RL-29	ListingStatusEvent	11	6	5	3
RL-30	ListingEligibilityAssessment	11	8	3	3
RL-31	BlockingCondition	10	7	3	1
RL-32	PriceRecord	12	6	6	2
RL-33	DecisionCriterion	10	5	5	0
RL-34	CriterionWeight	7	5	2	2
RL-35	ScoringRule	9	4	5	0
RL-36	RuleVersion	10	4	6	1
RL-37	DecisionRequest	10	7	3	3
RL-38	DecisionResult	10	5	5	3
RL-39	DecisionExplanation	9	6	3	1
RL-40	UsageRecord	13	5	8	2
RL-41	SavedComparison	7	4	3	1
RL-42	AuditRecord	12	6	6	1
RL-43	ProductAssetIdentifier	9	7	2	1
RL-44	AttributeAllowedValue	7	4	3	1
RL-45	FactEvidenceLink	3	3	0	2

RL-46	ValidationEvidenceLink	3	3	0	2
RL-47	SubmissionRequirementLink	3	3	0	2
RL-48	EligibilityEvidenceBasis	3	3	0	2
RL-49	EligibilityOwnershipBasis	3	3	0	2
RL-50	EligibilityConditionBasis	3	3	0	2
RL-51	SavedComparisonItem	4	4	0	2
RL-52	DecisionRequestCandidateProduct	4	4	0	2
RL-53	DecisionRequestCandidateAsset	4	4	0	2
RL-54	DecisionResultItem	7	4	3	3

Total attributes	Required	Optional	Foreign keys
502	286	216	110

End of Logical Data Dictionary.

Appendix H: Logical Normalization and Integrity Report

Smart E-Commerce & Product Intelligence Platform

System Analysis & Design - Task 02, Phase 02.3 Logical Database Design

1. Document Control

Field	Value
Document Title	TechWise Logical Normalization and Integrity Report
Version	1.0
Preparation Date	4 November 2026
Status	Logical baseline; technology-neutral
Scope	54 relations, 502 attributes, 50 logical constraints

2. Purpose and Method

This report evaluates the logical model against First, Second, and Third Normal Form, records the functional dependencies that can actually be identified, classifies every integrity constraint by enforcement mechanism, and documents temporal rules and remaining risks.

Normalization was assessed from the logical specification: keys, declared dependencies, and the business meaning of each attribute. No workload measurements exist, so no performance-driven denormalization is claimed or proposed.

3. Normal Form Analysis

3.1 First Normal Form

Check	Result	Evidence
All attributes are atomic	PASS	Every attribute has one logical type and one meaning; no attribute stores a list or delimited set.
No repeating groups	PASS	Multivalued concepts are separate relations: RL-43, RL-44, RL-45 to RL-53, RL-54.
No unconstrained EAV	PASS	RL-11 ProductFact uses a governed AttributeDefinitionId and a typed value group; RL-44 constrains enumerated values.
Typed value group is not a repeating group	PASS	The value columns are alternative representations of one value, governed by the definition value type; exactly one is populated per row.
Associative relations have atomic attributes	PASS	Only parent keys and a linkage timestamp, plus a legitimate sequence where order is meaningful.

3.2 Second Normal Form

Second Normal Form applies to relations with composite keys: the nine associative relations, RL-44 AttributeAllowedValue, and RL-54 DecisionResultItem. In each case every non-key attribute depends on the whole key.

Relation	Composite key	Partial dependency found	Assessment
RL-44	(AttributeDefinitionId, AllowedValueCode)	None	Label and status describe the permitted value identified by the full pair.
RL-45	(ProductFactId, EvidenceSubmissionId)	None	LinkedAt describes the link, not either parent alone.
RL-46	(OwnershipValidationId, EvidenceSubmissionId)	None	LinkedAt describes the link.
RL-47	(EvidenceSubmissionId, EvidenceRequirementId)	None	LinkedAt describes the link.
RL-48	(ListingEligibilityAssessmentId, EvidenceAssessmentId)	None	LinkedAt describes the link.
RL-49	(ListingEligibilityAssessmentId, OwnershipValidationId)	None	LinkedAt describes the link.
RL-50	(ListingEligibilityAssessmentId, ConditionAssessmentId)	None	LinkedAt describes the link.
RL-51	(SavedComparisonId, ProductId)	None	DisplaySequence and AddedAt describe the product within the comparison.
RL-52	(DecisionRequestId, ProductId)	None	CandidateSequence and AddedAt describe the candidate within the request.
RL-53	(DecisionRequestId, ProductAssetId)	None	CandidateSequence and AddedAt describe the candidate asset within the request.
RL-54	(DecisionResultId, CandidateSequence)	None	Rank and outcome describe the specific result line.

All other relations have a single-attribute primary key, so partial dependency cannot occur.

3.3 Third Normal Form

No non-key attribute was found to determine another non-key attribute within the same relation. The following dependencies were examined because they are plausible sources of transitive dependency.

Relation	Potential dependency	Finding	Treatment
RL-07 Product	PrimaryCategoryId determines category name	Category attributes are not	No transitive dependency.

	or hierarchy	copied into Product.	
RL-08 ProductVariant	ProductId determines product name or category	Only the key is stored; product attributes remain in RL-07.	No transitive dependency.
RL-11 ProductFact	AttributeDefinitionId determines value type, unit, or allowed values	Definition metadata is not copied into the fact; validation resolves it by reference.	No transitive dependency; conformance is a constraint.
RL-16 OwnershipPeriod	PeriodEndAt determines PeriodStatusCode	Both are stored for query and lifecycle support.	Documented controlled redundancy in Section 4.
RL-28 ResaleListing	Latest event determines CurrentStateCode	Both current state and event history are stored.	Documented controlled redundancy in Section 4.
RL-38 DecisionResult	DecisionRequestId determines RuleVersionId	Both request and result store the rule version.	Documented reproducibility redundancy in Section 4.
RL-32 PriceRecord	PriceTypeCode determines the permitted scope	Scope is enforced by a check constraint, not by copied attributes.	No transitive dependency.
RL-13 InformationConflict	Fact pair determines conflict type	Type is independently recorded and may be refined during review.	No functional dependency asserted.

3.4 Identifiable functional dependencies

The following dependencies are asserted. Where a natural determinant is conditional or policy-dependent, that is stated rather than enforced.

Relation	Determinant	Determined attributes	Status
RL-01	UserId	all non-key attributes	Confirmed
RL-01	ContactEmail	UserId	Conditional (D-17)
RL-02	RoleCode	RoleId and all non-key attributes	Confirmed
RL-03	PermissionCode	PermissionId and all non-key attributes	Confirmed
RL-04	(UserId, RoleId) where EndedAt is absent	open assignment attributes	Conditional (partial uniqueness)
RL-05	(RoleId, PermissionId) where EndedAt is absent	open grant attributes	Conditional (partial uniqueness)
RL-06	(UserId, PreferenceType, PreferenceScope)	PreferenceValue	Conditional (scope default undefined)
RL-07	(BrandReference, ModelNumber)	ProductId	Conditional (D-01)
RL-08	(ProductId, VariantCode)	ProductVariantId and all non-key attributes	Confirmed

RL-09	CategoryCode	CategoryId and all non-key attributes	Confirmed
RL-10	(PrimaryCategoryId, AttributeCode)	AttributeDefinitionId and all non-key attributes	Confirmed
RL-11	ProductFactId	all non-key attributes	Confirmed
RL-12	(SourceTypeCode, SourceReference, RetrievedAt)	ProductSourceId	Conditional
RL-13	InformationConflictId	all non-key attributes	Confirmed
RL-14	ProductAssetId	all non-key attributes	Confirmed
RL-15	PurchaseId	all non-key attributes	Confirmed
RL-16	(ProductAssetId, PeriodStartAt)	OwnershipPeriodId and all non-key attributes	Confirmed
RL-17	PriorOwnershipPeriodId	OwnershipTransferId	Confirmed by 0..1 cardinality
RL-26	(RuleCode, VersionLabel)	CompatibilityRuleId and all non-key attributes	Confirmed
RL-33	CriterionCode	DecisionCriterionId and all non-key attributes	Confirmed
RL-34	(RuleVersionId, DecisionCriterionId)	WeightValue and scale context	Confirmed
RL-35	RuleCode	ScoringRuleId and all non-key attributes	Confirmed
RL-36	(ScoringRuleId, VersionLabel)	RuleVersionId and all non-key attributes	Confirmed
RL-44	(AttributeDefinitionId, AllowedValueCode)	label, status, and applicability	Confirmed
RL-54	(DecisionResultId, CandidateSequence)	candidate, rank, and outcome	Confirmed

For all remaining relations, the primary key is the only determinant asserted. No additional natural key is invented, and no multi-valued dependency requiring Fourth Normal Form analysis was identified outside the associative relations already decomposed.

4. Controlled Redundancy and Denormalization Register

Four deliberate redundancies exist. Each is documented, bounded, and given an enforcement rule. None is justified by performance measurement, because no workload evidence exists.

ID	Redundancy	Justification	Enforcement	Risk
CR-01	ResaleListing.CurrentStateCode duplicates the latest ListingStatusEvent.NewStateCode.	Active listing discovery and state gating require a single current-state value without	Every transition writes the event and updates the listing in one transaction. A reconciliation	Divergence if a writer bypasses the transition procedure.

		aggregating history on every read. The conceptual model records a current state on the listing.	query can detect divergence.	
CR-02	OwnershipPeriod.PeriodStatusCode duplicates the meaning of PeriodEndAt being absent or present.	The conceptual model defines Active and Ended states, and eligibility checks need an explicit state predicate.	Both fields are written together; a check rejects Active with an end time and Ended without one.	Divergence if the pair is updated independently.
CR-03	DecisionResult.RuleVersionId repeats the rule version recorded on DecisionRequest.	Reproducibility requires the result to name the exact rule version used, even if request metadata is later superseded or archived.	The values must match at generation. A request is immutable once a result exists; re-evaluation creates a new request and result.	A late rule-version change on a request would break reproducibility; the immutability rule prevents it.
CR-04	ProductAsset.FirstAcquisitionDate is related to, but not derived from, Purchase.AcquisitionAt.	The original first-purchase date must survive resale and may predate platform records (BR-08, DBR-05). It cannot be recomputed from the earliest stored purchase when history is imported.	The date is set from approved origin evidence once and is not updated by transfers. Corrections require a controlled audited change.	An incorrect initial value persists until corrected; the audit trail records the correction.

Product.Description is not a redundancy of ProductFact. It is catalogue copy, not an authoritative fact, and must not be promoted into product truth without a governed attribute definition.

5. Logical Constraint Catalogue

Fifty logical constraints are defined. LC-01 to LC-40 correspond directly to the Phase 02.2 constraints IC-01 to IC-40. LC-41 to LC-50 are structural logical constraints required by the associative relations and the ranking decomposition.

ID	Constraint statement	Class	Enforcement	Source	Implementation note
LC-01	A ProductAsset retains one stable platform identity across purchase, ownership, listing, resale, and transfer; resale never creates a new ProductAsset.	Source-backed	Transactional/application-level	IC-01	Enforced by prohibiting identity replacement and by never inserting a replacement asset on resale.
LC-02	Every ProductAsset references exactly one Product, and any ProductVariant reference must	Derived conceptual	Standard relational + transactional check	IC-02	FK plus a cross-row variant-product check.

	belong to that Product.				
LC-03	Every ProductVariant references exactly one Product and cannot exist without it.	Derived conceptual	Standard relational	IC-03	FK with mandatory participation and Restrict deletion.
LC-04	Every AttributeDefinition declares a value type, an allowed domain, a scope, and a unit where applicable; category-specific definitions are not unrestricted free text.	Source-backed	Standard relational + policy-dependent	IC-04	NOT NULL columns enforce declarations; domain content depends on D-01.
LC-05	Every ProductFact has exactly one Product subject and exactly one AttributeDefinition; a variant-scoped fact uses a variant of that Product and a value conforming to the definition.	Derived conceptual	Standard relational + transactional check	IC-05	FKs and NOT NULL plus value-type conformance logic.
LC-06	A ProductFact retains provenance class and recorded time; a source is optional only when the fact is explicitly marked unsourced; AI-generated content is not authoritative.	Source-backed	Standard relational + transactional check	IC-06	ProducerTypeCode and RecordedAt are mandatory; IsUnsourced is explicit.
LC-07	An InformationConflict references two distinct ProductFacts, and resolution never deletes or overwrites either fact.	Source-backed	Standard relational (check) + transactional	IC-07	CHECK ConflictFactAId <> ConflictFactBId; no destructive resolution.
LC-08	A ProductAsset may exist with zero ownership periods until an acquisition or origin is recorded; an initial ownership start references a purchase or accepted origin evidence.	Derived conceptual	Transactional/application-level	IC-08	Participation is not forced at asset creation.
LC-09	A ProductAsset has at most one Active OwnershipPeriod at a time, and active periods do not overlap.	Source-backed / conditional	Transactional/application-level	IC-09	Requires serialized overlap checking; not expressible as a simple unique constraint.
LC-10	An OwnershipPeriod end time does not precede its start time; an ended period is retained and corrected only through a controlled, auditable correction.	Derived conceptual	Standard relational (check) + transactional	IC-10	CHECK PeriodEndAt >= PeriodStartAt.
LC-11	An OwnershipTransfer references the same ProductAsset through its prior and new periods, closes the prior period, opens the new period, and never creates overlapping active ownership.	Derived conceptual	Transactional/application-level	IC-11	Multi-row consistency plus the LC-09 overlap rule.
LC-	Current ownership is determined from the applicable Active	Source-	Transactional/application-	IC-12	Derived query rule; no owner column

12	OwnershipPeriod; a role, listing, purchase, or claim is not by itself proof of ownership.	backed	level		exists.
LC-13	A Purchase references a recorded buyer and a Product, may reference a ProductAsset, and does not by itself prove current ownership.	Source-backed / derived	Standard relational + transactional	IC-13	Mandatory FKs; semantic rule prevents ownership inference.
LC-14	A ResaleListing references exactly one ProductAsset and one recorded seller; seller authorization is validated and not inferred from the listing.	Derived conceptual	Standard relational + policy-dependent	IC-14	FKs are standard; authorization policy depends on D-04.
LC-15	A ProductAsset has at most one Active or Reserved ResaleListing at the same time.	Source-backed design rule / conditional	Transactional/application-level	IC-15	Partial uniqueness over a state set; confirm marketplace policy A-06.
LC-16	An Active listing has an applicable Eligible ListingEligibilityAssessment and no unresolved blocking condition.	Source-backed / conditional	Transactional/application-level	IC-16	Cross-relation state check; policy inputs depend on D-02, D-03, D-04.
LC-17	A Sold listing is not proof that a sale, payment, or ownership transfer completed; listing state and ownership state remain separate.	Source-backed	Transactional/application-level	IC-17	No transfer or payment record is created from a listing state.
LC-18	An EvidenceSubmission has a submitter and an explicit context; a complete submission contains at least one EvidenceItem, and each item belongs to exactly one submission.	Derived conceptual	Standard relational + transactional	IC-18	Submitter and purpose are mandatory; completeness is a lifecycle check.
LC-19	Every EvidenceAssessment references one EvidenceSubmission and records an outcome, rationale, time, and reviewer or process; prior assessments are retained when superseded.	Source-backed / derived	Standard relational + transactional	IC-19	Mandatory FKs and fields; supersession never deletes history.
LC-20	Evidence acceptance is not listing eligibility; ownership validation and eligibility determination remain separate records.	Source-backed	Structural (separate relations) + transactional	IC-20	Separate relations plus basis links, not copied outcome columns.
LC-21	Evidence confidence uses Low, Medium, High, or Insufficient Evidence; condition uses 5 to 1, Not Assessed, or Unknown; the two classifications are never conflated.	Source-backed	Standard relational (domain check)	IC-21	Two independent enumerated attributes.

LC-22	A ConditionClaim records claimed condition; a ConditionAssessment records assessed condition; neither overwrites the other.	Source-backed	Structural + transactional	IC-22	Separate relations and separate grade attributes.
LC-23	A CompatibilityRelationship has a subject and target, uses Compatible, Incompatible, Conditional, or Unknown, and preserves direction; absence means Unknown, not Incompatible.	Source-backed	Standard relational + transactional	IC-23	Mandatory endpoints and outcome check; absence semantics are query rules.
LC-24	A Conditional compatibility outcome has attributable conditions, and rule and source attribution is retained where used.	Derived conceptual	Transactional/application-level	IC-24	Requires existence checking against the governing rule conditions.
LC-25	A role or permission does not establish ownership, product-truth authority, or transaction completion.	Source-backed	Policy-dependent	IC-25	Authorization checks are application policy; not a data constraint.
LC-26	Prior-owner personal information is not exposed to subsequent owners through ordinary ownership or listing views.	Source-backed	Transactional/application-level + policy-dependent	IC-26	Access-control and view rules; implementation deferred.
LC-27	AuditRecord is append-oriented for sensitive or lifecycle-critical events, contains no secrets or unnecessary personal data, and is not proof of transaction completion.	Source-backed / derived	Transactional/application-level + policy-dependent	IC-27	Append-only behaviour and privacy rules; catalogue D-10.
LC-28	A DecisionResult references the exact RuleVersion used, criterion weights are version-scoped, and no weight, formula, threshold, normalization, or tie-break rule is invented.	Source-backed / conditional	Standard relational + policy-dependent	IC-28	FKs are standard; scoring policy depends on D-06.
LC-29	An AI-assisted DecisionExplanation is labelled, cannot override deterministic output or authoritative facts, and cannot be the sole basis for a product-truth claim.	Source-backed	Standard relational (check) + transactional	IC-29	ProducerTypeCode check; semantic precedence rule.
LC-30	Unknown, Not Assessed, missing, and Insufficient Evidence are not converted into a negative fact, a zero value, or Incompatible.	Source-backed / derived	Transactional/application-level	IC-30	Nullable attributes and explicit Unknown states; query and validation rules.
LC-31	A PriceRecord has exactly one approved scope, a non-negative amount, a currency, and a source context; a missing price is not zero and market observation is not	Derived conceptual / conditional	Standard relational (check)	IC-31	CHECK exactly-one-scope and Amount >= 0; currency is

	implied.				mandatory.
LC-32	An EvidenceRequirement identifies its scope, evidence type, mandatory or optional intent, and policy or version context; no category matrix is invented.	Source-backed / conditional	Standard relational + policy-dependent	IC-32	Mandatory columns; matrix content depends on D-02.
LC-33	An Eligible determination requires the applicable ownership, evidence, condition, and blocking checks to pass; Incomplete and Ineligible are distinct and never treated as eligible.	Source-backed / conditional	Transactional/application-level + policy-dependent	IC-33	Basis links plus state and blocking checks; policy depends on D-04.
LC-34	A BlockingCondition is limited to approved blocking types and creates no payment, escrow, refund, remedy, or full dispute subsystem.	Source-backed	Standard relational + policy-dependent	IC-34	FK and type domain; taxonomy depends on D-10 and OD-07.
LC-35	RetentionCategory is a controlled classification, not a relation, and no retention period is invented.	Source-backed / conditional	Policy-dependent	IC-35	No retention column or period is created.
LC-36	A UserRoleAssignment is unique for an open (UserId, RoleId) pair; a RolePermissionAssignment is unique for an open (RoleId, PermissionId) pair; historical assignments remain distinct.	Derived conceptual	Transactional/application-level	IC-36	Partial uniqueness where EndedAt is absent.
LC-37	A UserPreference is user-scoped and is not a verified fact, a decision result, or a permanent requirement.	Derived conceptual	Standard relational + transactional	IC-37	Mandatory user FK; semantic separation.
LC-38	A ProductSource retains attributable source context; freshness is optional only when explicitly unknown; a source record does not make its claim true.	Derived conceptual	Standard relational + transactional	IC-38	Mandatory SourceReference and RetrievedAt.
LC-39	A UsageRecord is optional and evidence-limited; unknown usage is not zero and does not override assessed condition.	Derived conceptual	Standard relational + transactional	IC-39	Nullable value with explicit limitations; no default zero.
LC-40	A SavedComparison belongs to one user and references selected products; it is not a DecisionResult and does not freeze authoritative product facts.	Derived conceptual	Standard relational + transactional	IC-40	Mandatory user FK; items reference products without copying facts.
LC-41	FactEvidenceLink is unique on (ProductFactId, EvidenceSubmissionId).	Derived structural	Standard relational (composite PK)	R-19	Composite primary key.
LC-42	ValidationEvidenceLink is unique on (OwnershipValidationId,	Derived structural	Standard relational (composite PK)	R-31	Composite primary key.

	EvidenceSubmissionId).				
LC-43	SubmissionRequirementLink is unique on (EvidenceSubmissionId, EvidenceRequirementId).	Derived structural	Standard relational (composite PK)	R-33	Composite primary key.
LC-44	EligibilityEvidenceBasis is unique on (ListingEligibilityAssessmentId, EvidenceAssessmentId).	Derived structural	Standard relational (composite PK)	R-47	Composite primary key.
LC-45	EligibilityOwnershipBasis is unique on (ListingEligibilityAssessmentId, OwnershipValidationId).	Derived structural	Standard relational (composite PK)	R-48	Composite primary key.
LC-46	EligibilityConditionBasis is unique on (ListingEligibilityAssessmentId, ConditionAssessmentId).	Derived structural	Standard relational (composite PK)	R-49	Composite primary key.
LC-47	SavedComparisonItem is unique on (SavedComparisonId, ProductId) and display order is unique within the comparison.	Derived structural	Standard relational + transactional	R-74	Composite PK plus sequence uniqueness.
LC-48	DecisionRequestCandidateProduct is unique on (DecisionRequestId, ProductId) and supply order is unique within the request.	Derived structural	Standard relational + transactional	R-76	Composite PK plus sequence uniqueness.
LC-49	DecisionRequestCandidateAsset is unique on (DecisionRequestId, ProductAssetId) and supply order is unique within the request.	Derived structural	Standard relational + transactional	R-77	Composite PK plus sequence uniqueness.
LC-50	DecisionResultItem is unique on (DecisionResultId, CandidateSequence); a rank position, where present, is unique within the result and no tie-break rule is invented.	Derived structural	Standard relational + transactional	CE-38 ranking	Composite PK plus conditional rank uniqueness.

6. Enforcement Classification Summary

Enforcement classification	Count
Policy-dependent	2
Standard relational	1
Standard relational (check)	1
Standard relational (check) + transactional	3
Standard relational (composite PK)	6
Standard relational (domain check)	1
Standard relational + policy-dependent	5
Standard relational + transactional	12
Standard relational + transactional check	3

Structural (separate relations) + transactional	1
Structural + transactional	1
Transactional/application-level	11
Transactional/application-level + policy-dependent	3

Constraint class	Count
Derived conceptual	14
Derived conceptual / conditional	1
Derived structural	10
Source-backed	14
Source-backed / conditional	6
Source-backed / derived	4
Source-backed design rule / conditional	1

6.1 What is enforceable by standard relational constraints

Primary keys, foreign keys, mandatory participation, single-attribute uniqueness, composite-key uniqueness, and simple domain or check rules are enforceable by standard relational mechanisms. These cover identity, basic referential integrity, the ten enumerated domains, the price scope check, the distinct-conflict-facts check, and the period end-versus-start check.

6.2 What requires transactional or application-level enforcement

Non-overlapping active ownership, listing exclusivity per asset, active-listing eligibility gating, transfer consistency across prior and new periods, open-assignment uniqueness, cross-relation state consistency, current-state reconciliation, and soft business rules such as "a role does not prove ownership" require multi-row or multi-relation logic. They cannot be enforced by a simple unique constraint.

6.3 What requires a policy decision

Retention and deletion, reviewer authority, separation of duties, acceptable ownership proof, category evidence content, condition rubric, scoring weights and formulas, market-price sources, identifier visibility, audit event catalogue, and transaction completion semantics remain policy-dependent. The logical model constrains their structure but does not decide them.

7. Temporal and Lifecycle Rules

Rule	Relations	Enforcement
A current ownership period has no end time and the state Active.	RL-16	Check plus transactional state rule (LC-09, LC-10).
No two active ownership periods overlap for one asset.	RL-16	Transactional overlap check; cannot be a simple unique constraint.
A transfer closes one prior period and opens one new period for the same asset.	RL-16, RL-17	Transactional consistency (LC-11).
A listing state change is recorded as an event and reflected in the current	RL-28, RL-29	Transactional write pair and

state.		reconciliation (CR-01).
A Sold listing does not create or complete an ownership transfer.	RL-28, RL-17	No automatic transition; policy constraint LC-17.
Assessments are superseded, never deleted or overwritten.	RL-22, RL-24, RL-30, RL-18	Append-oriented writes with a self-reference (LC-19).
Facts retain observed and recorded time and are never silently overwritten by a conflicting claim.	RL-11, RL-13	Append plus conflict record (LC-06, LC-07).
A decision result remains bound to the rule version used at generation.	RL-36, RL-38	Mandatory FK plus immutability rule (CR-03).
Audit records are append-oriented and are not used as proof of transaction completion.	RL-42	Append and privacy rules (LC-27).

8. Integrity Risks and Limitations

Risk	Affected area	Mitigation	Residual limitation
Overlap check race under concurrent writes	Ownership periods	Serialize writes per asset in the application transaction.	Requires implementation discipline; no DBMS is selected.
Current listing state divergence	Listing state	Single transition procedure plus reconciliation query.	Manual or out-of-band updates can diverge.
Missing natural keys for facts and compatibility assertions	Product truth, compatibility	Attributes remain nullable and conflicts are first-class.	Duplicate assertions are permitted by design; deduplication policy is deferred.
Partial uniqueness depends on null semantics	Assignments, preferences, rank	Document the null-handling rule before physical enforcement.	Not enforceable until policy is fixed.
Referential actions are recommendations	All FKs	Physical design must translate them consistently.	No DBMS behaviour is claimed.
No deletion policy	All relations	No cascade into history; retention deferred.	Data volume and legal retention remain open.
Score components are not modeled	Decision result	Only candidate identity and rank position are persisted.	Criterion-level scoring persistence depends on D-06.

No performance-driven denormalization is claimed, because no query workload or measurement exists. The four documented redundancies are driven by lifecycle semantics and reproducibility, not by performance.

9. Conclusion

The logical model satisfies First, Second, and Third Normal Form for the dependencies that can be established. Multivalued attributes are decomposed, associative relations have no partial dependencies, and no non-key transitive dependency was found. Four controlled redundancies are

explicitly registered with enforcement rules. Integrity constraints are classified honestly: standard relational, transactional, and policy-dependent.

End of Logical Normalization and Integrity Report.

Appendix I: Full Problem Statement and Project Objectives

The complete existing Problem Statement & Objectives deliverable is included below, with its source status and substantive wording preserved. Local links are rebased for this consolidated file.

Original Problem Statement & Objectives Deliverable

System Analysis & Design — Deliverable 01

Document Control

Field	Value
Project Name	TechWise
Official Description	Smart E-Commerce & Product Intelligence Platform
Team	Falcon Team
Program	Digital Egypt Pioneers Initiative (DEPI) — Round 5
Document Title	Problem Statement & Project Objectives
Document Type	System Analysis & Design Deliverable 01
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Repository	https://github.com/DEPI-5-Falcon-Project/TechWise_E-Commerce

This document is Falcon Team's final submission version of the Problem Statement & Project Objectives deliverable for TechWise. It defines the problem the platform addresses, the solution direction, the project objectives, the target users, the documented scope, the expected outcomes, and the criteria by which achievement can be evaluated.

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1. Introduction

1.1 Project Overview

TechWise is a Smart E-Commerce & Product Intelligence Platform designed to support users across the complete lifecycle of a technology product rather than only at the point of purchase. The platform is designed to combine e-commerce capabilities with structured product information, compatibility intelligence, AI-assisted shopping, deterministic purchase decision support, lifecycle-aware ownership records, and a second-life marketplace governed by platform-origin and ownership eligibility rules.

The platform's distinguishing concern is continuity: the connection between an informed purchasing decision and the subsequent ownership, upgrade, resale, and transfer history of the same physical product. TechWise is a graduation project developed by Falcon Team within the Digital Egypt

Pioneers Initiative (DEPI), Round 5. This document defines the intended scope and direction of the project; it does not claim that the described capabilities are already implemented or deployed.

1.2 Document Purpose

This document is the first deliverable of the System Analysis & Design phase. It establishes a shared, defensible understanding of:

75. The problem that TechWise addresses.
76. The users for whom the problem is relevant.
77. The solution direction and the confirmed core capabilities.
78. The project objectives and the outcomes expected from them.
79. The documented scope and its boundaries.
80. The criteria by which achievement of the objectives can be evaluated.

The document provides the foundation for the subsequent System Analysis & Design deliverables listed below. It does not perform those deliverables.

Subsequent deliverable	Contribution of this document
System Scope	Confirmed in-scope capabilities, exclusions, and scope boundaries.
Actors and Roles	Documented user groups, stakeholders, and role considerations.
Functional and Non-Functional Requirements	Problem-to-objective basis for requirement traceability.
Use Cases	Lifecycle stages and user challenges that use cases must address.
Database Design	Terminology for Product, Product Asset, Purchase, and Ownership, plus lifecycle invariants.
Software Architecture	AI/backend responsibility boundary and source-of-truth principles.
Process Modeling	Lifecycle stages and core capability interactions at a conceptual level.
UI/UX Design	User-facing outcomes and decision-support expectations.

This document consolidates the current project documentation listed in the References section. It does not replace the Requirements Specification; where the two differ, the Requirements Specification prevails.

1.3 Project Background

Electronic commerce has developed from a channel for displaying and selling products into an information-intensive environment in which consumers must evaluate numerous alternatives before making a purchasing decision. Online platforms provide product descriptions, technical specifications, prices, images, customer reviews, seller information, and recommendations. These resources can improve product discovery, but their usefulness depends on their relevance, completeness, consistency, and ability to support the consumer's actual decision.

Technology products make this evaluation more demanding. A laptop, smartphone, peripheral, or other electronic device may need to satisfy constraints involving performance, budget, connectivity, supported standards, existing equipment, intended workload, and future requirements. A product that

performs well in isolation may still be unsuitable because it does not work with equipment the user already owns, or because another alternative provides better overall value. The purchasing decision also does not end when a transaction is completed: products may subsequently be maintained, upgraded, resold, or transferred to another owner.

The TechWise Literature Review records that many of the individual challenges relevant to the project are addressed separately in existing research. Recommender systems support discovery, explainable recommendation methods study how systems communicate reasons, product information quality research examines the usefulness and completeness of presented information, semantic and ontology-based methods model product relationships, decision-support methods structure the comparison of alternatives, second-hand commerce research examines trust and uncertainty, and product lifecycle and digital product passport research investigates information availability and traceability across lifecycle stages. The review also observes that these areas should not be assumed to form one unified solution automatically, because integration requires a consistent representation of products, explicit rules for validating product facts, appropriate handling of uncertainty, and a clear separation between generated explanations and authoritative system decisions.

TechWise is proposed in response to that integration challenge. The project context is therefore not the absence of product listings or recommendations, but the absence of a coherent connection between discovery, understanding, comparison, decision support, purchase, ownership, upgrade, resale, and transfer.

1.4 Basis of the Document

This document is based on the current TechWise Requirements Specification, the Project Proposal, and the Literature Review, together with confirmed project decisions. It describes intended behavior and scope rather than implementation status.

The six core capabilities, the product lifecycle, the AI/backend responsibility boundary, and the product-ownership principles are confirmed project scope. Design and implementation details identified for later design are listed in Section 9.4; they are not presented as commitments and do not affect the completeness of this deliverable.

1.5 Key Terminology

Consistent terminology is essential to the later analysis and design work. The following definitions are used throughout this document.

Term	Definition
Product	A product model or SKU representing a type of item.
Product Asset	A specific physical device or item with its own identity and lifecycle history.
Purchase	A transaction through which a product or asset is acquired.
Ownership	A particular user's period of ownership of a Product Asset.
Product Lifecycle	The sequence of product discovery, acquisition, usage, upgrade, resale, and transfer activities.
Product Truth	The structured presentation of product facts, their sources, and their confidence or verification status.

Compatibility	A determination of whether products or components can work together under defined criteria.
Seller-Reported Information	Information supplied by a seller that has not necessarily been independently verified.
Evidence	Supporting material used to assess a claim, such as screenshots, purchase records, or product test videos.
Platform Confidence	The level of confidence supported by the platform's actual assessment process, not by the mere submission of a claim.
Second-Life Marketplace	The marketplace for eligible previously owned products that satisfy platform-origin and ownership requirements.
Hybrid Search	The confirmed search direction in which system analysis and AI-assisted interpretation are reconciled before backend validation.
Advanced Smart Filtering	A cross-cutting filtering capability used across discovery, comparison, and decision workflows; not a separate core feature.

A Product, a Product Asset, a Purchase, and an Ownership period are distinct concepts. This distinction is preserved in all subsequent analysis and design work.

2. Problem Analysis

2.1 Problem Statement

Users of technology products must make purchase and ownership decisions, yet the information and services needed to support those decisions are fragmented across disconnected product listings, specification pages, reviews, compatibility sources, and resale markets. Product information may be incomplete, inconsistent, difficult to compare, or unverifiable; comparisons may not use consistent attributes; compatibility with equipment the user already owns may require manual interpretation; and conventional transaction records rarely connect a purchased item to its later ownership history. As a result, users may select products that do not satisfy their actual requirements, may be unable to determine whether a candidate works with their existing setup, may be unable to distinguish documented facts from seller claims, and may be unable to evaluate whether buying, upgrading, or reselling represents the better course of action. For previously owned products, the absence of a reliable relationship between a physical asset, its platform purchase origin, its successive ownership periods, and its listing eligibility further reduces confidence in second-hand transactions. TechWise is designed to address this problem by integrating product discovery, structured product intelligence, compatibility assessment, deterministic decision support, lifecycle-aware ownership records, and an eligibility-controlled second-life marketplace within a single platform, in which AI assists interpretation and explanation while authoritative product data and backend business rules govern facts and decisions.

The problem can be decomposed into four dimensions that the platform must address together:

81. **Discovery and interpretation.** Users may not know which technical criteria express their needs, and ordinary keyword search may not translate a stated need into applicable product constraints.
82. **Understanding and comparison.** Users need consistent product facts, visible information sources, and comparable attributes in order to evaluate alternatives rather than isolated listings.

83. **Decision support.** Users must reconcile mandatory constraints, preferences, budget, compatibility, and long-term ownership considerations, and they need explanations of the resulting trade-offs.
84. **Lifecycle continuity.** Ownership, upgrade, resale, and transfer are treated as separate events rather than as stages in the history of one physical product.

2.2 Existing User Challenges

The challenges below are derived from the user difficulties documented in the Requirements Specification and the Project Proposal, and are consistent with the problem context recorded in the Literature Review. They describe the project's motivating user difficulties; they are not presented as findings from a market survey or as measured market-wide statistics.

Lifecycle stage	User challenge	Related project response
Discover	Expressing a need in natural language or translating it into relevant technical criteria; obtaining relevant rather than merely similar results.	AI Shopping Intelligence; Advanced Smart Filtering; structured search.
Understand	Interpreting technical specifications and their practical implications; judging whether information is complete and trustworthy.	Product Truth Layer; structured product information.
Compare	Comparing candidates on consistent attributes when values are missing, described differently, or not directly comparable.	Product comparison; Product Truth Layer.
Decide	Reconciling mandatory constraints, preferences, budget, compatibility, and trade-offs without a transparent basis for the choice.	Purchase Decision Engine; explainable results.
Buy	Proceeding from evaluation to a purchase workflow without losing the criteria and reasoning behind the decision.	E-commerce scope; decision-support integration.
Own	Tracking a specific physical product, its purchase origin, and its ownership period separately from the product model.	Product Asset and Ownership records.
Upgrade	Determining whether a new purchase or an upgrade of existing equipment better addresses current requirements.	Buy vs Upgrade vs Resell Engine; compatibility assessment.
Resell	Establishing whether an owned product is eligible for resale and what evidence is required.	Verified Second-Life Marketplace eligibility rules.
Transfer	Changing ownership without erasing the asset's history, resetting its age, or exposing the previous owner's personal information.	Ownership-transfer rules and lifecycle invariants.

A recurring difficulty across these stages is that information and decisions are separated by system boundaries. A user may discover a product in one service, interpret its specifications in another, seek compatibility information in a third, and evaluate resale value in a fourth. Each service may use different attributes, different assumptions, and different levels of evidence, and none of them necessarily preserves the identity of the physical item across the sequence.

2.3 Limitations of Traditional E-Commerce Experiences

Conventional e-commerce experiences are primarily organized around listing, browsing, cart, and transaction functionality. That organization is appropriate for completing a sale, but it does not by itself provide the integrated product intelligence and lifecycle support that TechWise intends to offer. We do not claim that every existing e-commerce platform exhibits all of the limitations below, and we acknowledge that individual capabilities such as recommendations, reviews, comparison views, or trade-in programmes may exist in separate products.

85. **Transaction-centred rather than decision-centred design.** A listing presents an item for purchase. It may not structure the user's requirements, distinguish mandatory constraints from preferences, or explain why one candidate is more suitable than another.
86. **Listing-level rather than decision-level information.** Specifications may be incomplete, inconsistently formatted, or unsupported by a stated source. Missing values may be displayed as blank fields without clarifying their effect on the decision.
87. **Inconsistent comparison bases.** Comparison views may align attributes by label while leaving semantic differences, units, conditions, or missing values unresolved. A comparison can therefore appear complete while remaining unsuitable for a decision.
88. **Compatibility left to the user.** Compatibility relationships may be described in prose, scattered across support pages, or omitted. Users may have to interpret connector types, supported standards, power requirements, or generation constraints themselves.
89. **Ownership ending at checkout.** Transaction records typically establish that a purchase occurred. They do not necessarily establish which physical asset was purchased, which user owns it now, or how its ownership history has changed.
90. **Resale and trust signals disconnected from origin.** Second-hand listings may describe condition and provenance without a consistent relationship to the original purchase, the current owner, or prior ownership periods. Seller statements, submitted evidence, and platform assessment may also be presented without a clear indication of what has actually been established.

The integration gap is therefore not merely a missing feature. It is a missing consistent model that connects product facts, user requirements, compatibility rules, decision criteria, ownership state, and resale eligibility without allowing any one component, including AI, to become an uncontrolled source of truth.

2.4 Problem Impact

The following table distinguishes problems documented in the project sources from consequences that we reason are likely if those problems remain unaddressed. Reasoned consequences are project motivations, not measured findings.

Impact area	Potential consequence	Nature
Product fit	A user selects a product that does not satisfy a mandatory requirement, such as budget, capacity, supported standard, or intended workload.	Documented problem

Compatibility	A purchase proves incompatible with existing equipment, requiring replacement, return, or additional expenditure.	Documented problem
Information trust	A user cannot distinguish documented facts, seller claims, submitted evidence, and unknown information, and may rely on an unsupported claim.	Documented problem
Lifecycle continuity	Ownership history, original purchase date, and asset identity are lost or reset when a product is resold.	Documented problem
Second-hand confidence	Buyers cannot assess origin, condition, evidence, and ownership continuity, reducing willingness to consider eligible second-hand products.	Documented problem
Privacy	Previous owners' personal information may be exposed through lifecycle or listing records.	Documented constraint

3. Proposed Solution

3.1 TechWise Solution Overview

TechWise proposes an integrated platform that connects product information, decision support, purchasing, ownership records, and eligible second-life transactions. The solution is organized around four connected ideas:

91. **Structured product intelligence.** Product facts are represented as structured data with available source, evidence, and confidence context so that they can be compared, filtered, and evaluated consistently.
92. **Deterministic decision support.** Mandatory constraints, preferences, compatibility rules, eligibility rules, and calculations are evaluated by backend logic using validated inputs, producing results that can be explained and reproduced.
93. **Assistive AI.** AI interprets natural-language input, extracts candidate requirements, explains results, and summarizes available review information. AI output is treated as untrusted input until validated, and it does not replace authoritative product records or business decisions.
94. **Lifecycle-aware records.** A Product Asset retains its identity across ownership changes, and ownership periods are recorded separately from the asset and from the original purchase. This allows resale and transfer to occur without resetting the product's age or erasing its history.

The solution is a scope definition, not a claim that every capability is already implemented. The descriptions below define intended behavior and provide the basis for subsequent requirements, design, and verification work.

3.2 Product Lifecycle Approach

TechWise is designed around a product lifecycle rather than a sequence of isolated shopping transactions:

Discover → Understand → Compare → Decide → Buy → Own → Upgrade → Resell → Transfer

The lifecycle expresses the intended continuity of the user's relationship with a product. It is a scope statement, not a claim that every stage has already been implemented.

Stage	Meaning at a high level	Principal platform support
Discover	Identify candidate products that may satisfy a stated need.	AI Shopping Intelligence; Advanced Smart Filtering; structured search.
Understand	Interpret specifications, sources, evidence, and unknowns.	Product Truth Layer; structured product information.
Compare	Evaluate candidates on consistent attributes and visible differences.	Product comparison; Product Truth Layer.
Decide	Reconcile mandatory constraints, preferences, and trade-offs.	Purchase Decision Engine.
Buy	Complete an acquisition within the implemented e-commerce scope.	Purchase and transaction records.
Own	Associate the acquired physical item with an owner and an ownership period.	Product Asset and Ownership records.
Upgrade	Evaluate whether a new purchase or an improvement to existing equipment is preferable.	Buy vs Upgrade vs Resell Engine; compatibility assessment.
Resell	Determine eligibility and list an owned asset for second-life sale.	Verified Second-Life Marketplace.
Transfer	Change ownership while preserving asset identity and lifecycle history.	Ownership-transfer rules; privacy controls.

The lifecycle model relies on four distinctions that must remain stable throughout design and implementation: a Product is not a Product Asset; a Purchase is not an Ownership period; an Ownership period does not alter the identity of the Product Asset; and a resale does not reset the product's age or original first-purchase date.

3.3 Core Capabilities

The confirmed core feature set consists of six capabilities. They are presented separately below, but they are intended to operate together within the lifecycle described in Section 3.2.

3.3.1 AI Shopping Intelligence

Purpose. Support natural-language shopping requests, interpretation of user needs, extraction of requirements, and understandable explanations of relevant product options.

Contribution. AI assistance is intended to reduce the effort of translating a stated need into structured search or decision criteria and to make product differences easier to understand. It does not establish product facts, eligibility, compatibility, or ownership state. Its output is validated before it affects filtering, ranking, or business decisions.

3.3.2 Verified Second-Life Marketplace

Purpose. Support the discovery and listing of eligible previously owned products, subject to platform-origin, current ownership, evidence, listing-state, and other applicable eligibility checks.

Contribution. The marketplace is designed to provide a controlled resale workflow in which eligibility is derived from authoritative records and applicable rules, and in which seller-reported information, submitted evidence, and platform assessment remain distinguishable.

Verification boundary. The term "verified" refers to the platform's eligibility and evidence assessment process. It must not be interpreted as proof of physical inspection or independent verification unless such a process is actually implemented. Submitting a listing or evidence does not, by itself, establish that the platform has physically inspected the product.

3.3.3 Product Compatibility Graph

Purpose. Represent compatibility relationships between products and components, and support compatibility assessments based on available structured facts and defined rules.

Contribution. The capability is designed to support compatibility-aware discovery and evaluation, and to present compatible, incompatible, and unknown or insufficient-data outcomes as distinct results. Unknown compatibility is never represented as confirmed compatibility.

3.3.4 Purchase Decision Engine

Purpose. Evaluate eligible product candidates against user requirements, mandatory constraints, preferences, and relevant trade-offs.

Contribution. The engine is designed to produce ranked, reproducible, and explainable results, to evaluate mandatory constraints separately from optional preferences, and not to present a candidate that violates a mandatory constraint as a full match.

3.3.5 Product Truth Layer

Purpose. Organize product information, distinguish factual records from seller-reported claims, and present relevant sources or supporting evidence and confidence information where available.

Contribution. The layer is designed to improve the reliability and traceability of product information, to identify missing or conflicting information where it can be detected, and to prevent AI-generated or inferred values from being presented as authoritative product facts.

3.3.6 Buy vs Upgrade vs Resell Engine

Purpose. Support lifecycle-related decisions by comparing available options against user requirements, product condition, relevant costs, and lifecycle information.

Contribution. The engine is designed to compare applicable options using defined calculations and validated inputs, to disclose important assumptions and missing information, and not to present unsupported estimates as guaranteed outcomes.

3.4 Intelligent Search and Filtering

Advanced Smart Filtering is a cross-cutting system capability rather than an independent core feature. It supports product discovery, comparison, and decision workflows by applying structured criteria, mandatory constraints, and user-selected preferences. Filtering is to be performed by deterministic backend logic, and the interface must accurately represent the filters that are active.

Product discovery is designed to combine structured input with AI assistance under the confirmed Hybrid Search direction:

95. The system performs an initial analysis of the user's input.
96. AI assistance is applied when ambiguity, uncertainty, or a possible interpretation error is detected.
97. The system compares and reconciles its own interpretation with the AI-assisted interpretation.
98. Backend validation and business rules remain authoritative.

99. Database records remain the source of factual product information.
100. The system may ask the user for clarification when ambiguity cannot be reliably resolved.

Under this direction, AI is not the sole search parser and does not make the final search decision. The system retains its own interpretation, compares that interpretation with the AI-assisted result, and resolves or escalates discrepancies through validation and, where necessary, user clarification. The detailed reconciliation mechanism, confidence thresholds, and clarification behavior belong to later design work.

4. Project Objectives

4.1 General Objective

To establish an integrated smart e-commerce and product intelligence platform that enables users to discover, understand, compare, purchase, own, upgrade, resell, and transfer technology products with consistent product information, explicit decision criteria, preserved ownership continuity, and responsible AI assistance.

4.2 Specific Objectives

101. **OBJ-01 — Improve product discovery.** Combine structured search and filtering with AI-assisted interpretation so that users can express requirements and reach relevant product candidates without needing to know every technical specification in advance.
102. **OBJ-02 — Improve understanding of product information.** Present structured product facts together with available source, evidence, and confidence context so that users can distinguish documented facts, seller claims, and unknown information.
103. **OBJ-03 — Enable consistent product comparison.** Present candidate products using consistent attributes and visible differences, including the explicit identification of missing values, so that comparison supports a decision rather than merely displaying data.
104. **OBJ-04 — Support compatibility assessment.** Evaluate whether products or components can work together using explicit criteria and available structured facts, and distinguish compatible, incompatible, and unknown outcomes.
105. **OBJ-05 — Assist purchase decisions.** Evaluate candidates against mandatory constraints, preferences, and relevant trade-offs, and provide explainable, reproducible decision support that does not present a non-compliant candidate as a full match.
106. **OBJ-06 — Preserve product identity and lifecycle continuity.** Maintain the identity of a Product Asset and its ownership history across resale and transfer, preserve the original first-purchase date, and protect previous owners' personal information.
107. **OBJ-07 — Establish an eligibility-controlled second-life marketplace.** Enable previously owned products to be listed and discovered only when the applicable platform-origin, current-ownership, evidence, listing-state, and dispute-related eligibility conditions are satisfied.
108. **OBJ-08 — Support buy, upgrade, and resell evaluation.** Help owners compare purchasing a replacement, upgrading existing equipment, and reselling an eligible product using available cost, condition, requirement, and lifecycle information, with assumptions and limitations disclosed.
109. **OBJ-09 — Maintain a responsible AI boundary.** Use AI to interpret intent, extract candidate requirements, explain results, and summarize available information, while ensuring that authoritative product records and deterministic business rules govern facts, validation, eligibility, compatibility, calculations, and ownership state.

4.3 Objective-to-Capability Mapping

Objective ID	Objective	Related Core Capability	Expected Outcome
OBJ-01	Improve product discovery through structured search, filtering, and AI-assisted interpretation.	AI Shopping Intelligence; Advanced Smart Filtering (cross-cutting)	Users reach relevant candidates using explicit criteria, with informative empty or clarification states.
OBJ-02	Improve understanding and traceability of product information.	Product Truth Layer	Facts, sources, evidence, seller claims, and unknowns are distinguishable.
OBJ-03	Enable consistent, evidence-aware product comparison.	Purchase Decision Engine; Product Truth Layer	Differences and missing values are visible on a consistent comparison basis.
OBJ-04	Support compatibility assessment using explicit criteria and structured data.	Product Compatibility Graph	Compatible, incompatible, and unknown outcomes are distinguished and explained.
OBJ-05	Assist purchase decisions against constraints, preferences, and trade-offs.	Purchase Decision Engine	Ranked, explainable results; mandatory constraints are not silently relaxed.
OBJ-06	Preserve product identity and lifecycle continuity across ownership changes.	Verified Second-Life Marketplace; Buy vs Upgrade vs Resell Engine	Asset identity, first-purchase date, and ownership history survive resale and transfer.
OBJ-07	Establish an eligibility-controlled second-life marketplace.	Verified Second-Life Marketplace	Only eligible assets are listed; claims, evidence, and platform assessment remain distinct.
OBJ-08	Support buy, upgrade, and resell evaluation.	Buy vs Upgrade vs Resell Engine	Options are compared transparently, with assumptions and missing inputs disclosed.
OBJ-09	Maintain a responsible AI boundary.	AI Shopping Intelligence (cross-cutting architectural principle)	AI remains assistive; validated data and deterministic results remain authoritative.

Lifecycle continuity (OBJ-06) is a cross-cutting concern rather than the exclusive responsibility of one capability. It is mapped to the two capabilities whose workflows most directly depend on ownership state, while remaining a constraint on discovery, decision support, and administration.

5. Target Users and Stakeholders

5.1 Target Users

The user groups below are documented in the Requirements Specification and the Project Proposal. The final role and permission model, including whether these groups correspond to distinct system roles, will be defined in the Actors and Roles deliverable.

User group	Primary need	Documented basis
Prospective shopper	Find relevant products and understand available options.	Requirements Specification, Section 4; Project Proposal, Section 6.1.
User with specific requirements	Identify products that satisfy defined budgets, technical requirements, preferences, or compatibility constraints.	Project Proposal, Section 6.1.
Registered shopper	Compare candidates, evaluate compatibility, and make informed purchase decisions.	Requirements Specification, Sections 4 and 5.
Product owner	Track a specific physical product, its ownership period, and relevant lifecycle history.	Requirements Specification, Sections 4 and 7.9.
Second-life seller	Determine resale eligibility and provide the required listing information and evidence.	Requirements Specification, Sections 4 and 7.8.
Second-life buyer	Evaluate a previously owned product, its available evidence, and its ownership-transfer status before acquiring it.	Requirements Specification, Sections 4 and 6.
Platform administrator or reviewer	Manage permitted records and review cases within authorized permissions.	Requirements Specification, Section 4; role subject to validation in the Actors and Roles deliverable.

A single individual may act in more than one group at different times. For example, a registered shopper may later become a product owner and then a second-life seller.

5.2 Stakeholders

Stakeholder	Interest in the project	Relationship to the system
Falcon Team	Convert requirements into an implementable, testable, and documented system.	Project team and system provider.
DEPI programme and project supervisor	Assess scope, documentation quality, technical reasoning, and project progress.	Academic and program oversight.
Intended platform users	Obtain reliable product information, decision support, and lifecycle continuity.	Primary beneficiaries and users.
Academic evaluators and project reviewers	Evaluate the deliverable against program expectations and technical quality.	External review.

Product data providers and relevant third-party service providers	Supply data or services where an integration is confirmed.	Potential external dependency; not a confirmed commitment.

Administrative permissions, external organizations, and commercial partners are not presented as confirmed. Specific external integrations and service providers remain subject to confirmation.

5.3 User Groups and System Actors

This document identifies user groups and stakeholders; it does not define the final system actor model. A user group is a category of person with a documented need. A system actor is an entity that interacts with the system through defined responsibilities and permissions. The mapping between the two, including authorization boundaries and administrative capabilities, will be established in the Actors and Roles deliverable.

6. Project Scope

6.1 In-Scope

The documented project scope includes:

110. **Product discovery and structured search**, including AI-assisted interpretation of shopping requests and Advanced Smart Filtering as a cross-cutting capability.
111. **Structured product information and product detail presentation**, including source, evidence, and confidence context where available.
112. **Product comparison** using consistent attributes and explicit identification of missing values.
113. **Product compatibility evaluation** using structured product or component relationships and defined rules.
114. **Purchase decision support** that evaluates mandatory constraints, preferences, and trade-offs and explains the result.
115. **Product information transparency**, including the distinction between seller-reported claims, submitted evidence, platform assessment, and unknown information.
116. **Platform-origin second-life marketplace workflows**, including eligibility evaluation, evidence submission, listing status, and listing management within authorized permissions.
117. **Ownership and lifecycle continuity**, including Product Asset identity, ownership periods, ownership transfer, and protection of previous owners' personal information.
118. **Buy, upgrade, and resale decision support** using available cost, condition, requirement, and lifecycle information.
119. **Responsive desktop and mobile interfaces**, backend validation and business rules, relevant data modeling, and the testing and documentation activities required by the project and the program.

6.2 Out-of-Scope

The following items are not part of the confirmed project scope. They remain out of scope unless the project owner explicitly approves them and records the decision.

Excluded or unconfirmed item	Basis
A specific payment provider, payment-processing integration, or guaranteed real-money payment processing.	Requirements Specification, Section 3.3; Project Proposal, Section 16.2.
A specific AI model, AI provider, orchestration framework, or hosting provider.	Requirements Specification, Section 3.3; Project Proposal, Section 16.2.
A specific backend framework, frontend framework,	Requirements Specification, Section 3.3; Project

database engine, or deployment architecture.	Proposal, Section 14.4.
Physical inspection of every second-life product by the platform, or automatic verification of every seller claim.	Requirements Specification, Section 3.3; Project Proposal, Section 16.2.
Guaranteed accuracy of AI-generated explanations or recommendations.	Requirements Specification, Section 3.3.
Specific escrow, refund, or dispute-resolution mechanisms that are not defined as approved business rules.	Requirements Specification, Section 3.3.
Guaranteed access to commercial product databases, live inventory, or market-price integrations.	Project Proposal, Section 16.2.
Automated repair, refurbishment, or a fully automated depreciation and resale-price prediction model.	Project Proposal, Section 16.2.
Guaranteed third-party marketplace integrations or any external service not selected and validated by the team.	Project Proposal, Section 16.2.
Production-scale operational guarantees, benchmarks, or security certifications that have not been demonstrated.	Requirements Specification, Section 3.3; Project Proposal, Section 16.2.

6.3 Proposed Enhancements and Future Considerations

The following items have been discussed or identified as possible extensions. They are not confirmed scope and do not form part of this deliverable's commitments:

120. A fourth "Keep" option in the buy, upgrade, and resell decision workflow, allowing users to continue using an existing product when no change is justified.
121. Automated depreciation or resale-price prediction beyond defined calculations on available inputs.
122. Physical inspection or refurbishment services for second-life products.
123. Additional external integrations, such as commercial product databases, live inventory, market-price feeds, or third-party marketplaces.

These items may be evaluated after feasibility review and explicit project-owner approval.

6.4 Scope Boundaries and Invariants

The following boundaries are architectural and analytical constraints on the project. They must remain consistent across requirements, design, implementation, and testing.

124. **AI is not the source of truth.** AI may interpret intent, extract candidate requirements, explain results, summarize available information, and support review intelligence. It does not establish product facts, prices, inventory, compatibility, eligibility, authorization, ownership state, or calculations. AI output is untrusted until validated.
125. **Backend logic is authoritative.** Validation, authorization, filtering, ranking, compatibility evaluation, eligibility assessment, calculations, and ownership changes are enforced by deterministic backend logic or other explicitly defined authoritative systems, not by front-end controls or AI output alone.
126. **Database records and supported evidence are the factual basis.** Product information presented as factual derives from database records or documented evidence, with source and confidence context where available.

127. **"Verified" has a bounded meaning.** In the second-life context, "verified" refers to the platform's eligibility and evidence assessment. It does not assert physical inspection or independent verification unless such a process is implemented.
128. **Seller claims, evidence, and platform assessment are distinct.** A listing must not present seller-reported information as an independently established fact.
129. **Product, Product Asset, Purchase, and Ownership are distinct.** A Product is a model or SKU; a Product Asset is a specific physical item; a Purchase is an acquisition transaction; Ownership is a user's ownership period. Design and implementation preserve these distinctions.
130. **Resale does not reset the product's lifecycle.** The Product Asset retains its identity and original first-purchase date. Previous ownership records remain part of the lifecycle history, while previous owners' personal information remains protected.
131. **Unknown or missing data is not a positive result.** An unknown attribute must not be treated as proof that a product satisfies a filter, a constraint, or a compatibility condition.

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7. Expected Outcomes

The outcomes below describe what the project will produce if the confirmed scope is implemented and validated. They are expectations, not achieved results.

7.1 User-Facing Outcomes

132. Users will be able to express shopping needs in natural language and receive relevant candidates based on structured criteria.
133. Users will be able to inspect product information with visible source, evidence, and confidence context.
134. Users will be able to compare candidates on consistent attributes and see missing values rather than inferred equivalence.
135. Users will be able to assess compatibility and receive a clear distinction between compatible, incompatible, and unknown outcomes.
136. Users will be able to obtain an explainable purchase recommendation that respects mandatory constraints and to compare buying, upgrading, and reselling with disclosed assumptions and limitations.

7.2 Platform and Business Outcomes

137. The project will establish a coherent integration of e-commerce workflows with product intelligence and lifecycle records.
138. Backend-enforced eligibility, authorization, and ownership rules will not depend on client-side controls or AI output.
139. The second-life marketplace will trace listings to eligible Product Assets and platform-origin records, keeping seller claims, evidence, and platform assessment distinct.
140. The project will maintain a documented, testable boundary between AI assistance and deterministic business logic that supports later architecture, data modeling, interface design, and verification work.

7.3 Product Lifecycle Outcomes

141. A Product Asset will remain identifiable as the same physical item across ownership changes.
142. Ownership periods will be recorded separately from asset identity and from the original purchase.
143. The original first-purchase date and product age will be preserved after resale.
144. Previous ownership records will remain available for authorized lifecycle purposes, and ownership transfer will update the current owner without erasing prior history.

145. Resale eligibility will be derived from authoritative records and applicable business rules.

7.4 Information Quality and Decision-Support Outcomes

146. Product facts will be structured consistently enough to support filtering, comparison, and evaluation.
147. Missing, conflicting, or unsupported information will be identified rather than silently resolved.
148. Decision results will be reproducible for the same validated inputs and rule version.
149. Explanations will remain consistent with the underlying facts and evaluation results.
150. Users will understand the principal reasons for a recommendation, compatibility result, or eligibility outcome, together with its limitations.
151. AI-generated content will not become an uncontrolled source of product facts or business decisions.

8. Success Criteria

8.1 Evaluation Approach

Achievement of the objectives is evaluated through a combination of requirements inspection, traceability review, scenario-based testing, and demonstration using representative inputs. The Requirements Specification defines representative acceptance scenarios (AT-01 to AT-10) that provide a starting point for this evaluation.

Each criterion below defines an evaluation target, a measurement method, and the evidence that would demonstrate achievement. Where a quantitative threshold is appropriate, the project test plan will define it before evaluation; this document does not fabricate numerical targets.

8.2 Success Criteria

ID	Objective	Evaluation target	Measurement method	Evidence required
SC-01	OBJ-01	Discovery results satisfy the stated structured criteria in every evaluated scenario, and empty or ambiguous requests produce an informative response rather than an invented result.	Scenario-based testing with representative queries and filter combinations.	Test cases, inputs, expected results, and recorded outcomes.
SC-02	OBJ-02	Every evaluated product fact is presented with available source, evidence, or confidence context, and seller claims and unknowns are never presented as verified facts.	Inspection of product information records and presentation; targeted test cases.	Reviewed records, screenshots or demonstrations, and test results.
SC-03	OBJ-03	Comparisons use consistent attributes and explicitly identify unavailable values	Comparison test cases including missing and semantically	Test cases and comparison outputs.

		in every evaluated comparison case.	different values.	
SC-04	OBJ-04	Compatibility outcomes match the documented reference rules, and unknown or insufficient-data cases are never reported as compatible.	Rule-based test cases with expected compatible, incompatible, and unknown results.	Reference rules, test cases, and results.
SC-05	OBJ-05	Mandatory constraints are evaluated separately from preferences, no violating candidate is labelled a full match, and every result is explainable from its stated inputs.	Decision-scenario testing with conflicting constraints and incomplete data.	Decision inputs, rule version, outputs, and explanations.
SC-06	OBJ-06	Asset identity, original first-purchase date, and prior ownership records are preserved across every evaluated transfer, and private data is not exposed.	Lifecycle and authorization test cases across simulated ownership changes.	Before-and-after records and access-control test results.
SC-07	OBJ-07	Eligibility decisions match the documented rules for origin, ownership, evidence, listing state, and blocking disputes in every evaluated case.	Eligibility test cases including ineligible and incomplete submissions.	Test cases, eligibility outputs, and listing status records.
SC-08	OBJ-08	Buy, upgrade, and resell comparisons use defined inputs, disclose assumptions, and limit conclusions when required data is missing.	Scenario testing with complete and incomplete input sets.	Inputs, calculation records, assumptions, and outputs.
SC-09	OBJ-09	AI assistance never overrides validated product records, rules, eligibility, or ownership state in any evaluated conflict case.	Adversarial and consistency testing where AI output conflicts with validated results.	Test cases, AI outputs, validated backend results, and resolution records.

8.3 Measurement Notes

152. A criterion is satisfied only when the required evidence has been produced; design intent alone is not sufficient.
153. Target values, tolerances, dataset sizes, and test environments will be defined in the project test plan before evaluation.
154. Results are reported for the conditions under which they were obtained and are not generalized beyond those conditions.
155. AI-related evaluation distinguishes language interpretation quality from factual correctness, because the latter depends on validated data and deterministic rules.

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9. Assumptions, Constraints, and Future Design Considerations

9.1 Project Assumptions

The assumptions below are consistent with the current project documentation and are required for the analysis in this document. They are recorded so that they are not mistaken for new requirements.

ID	Assumption	Status
AS-01	TechWise addresses technology products, including electronic devices and related components.	Documented; product categories to be confirmed in design.
AS-02	Users have requirements, preferences, and constraints that can be expressed and evaluated.	Documented.
AS-03	Structured product data can be obtained or curated for the supported categories.	Requires validation during implementation planning.
AS-04	Second-life eligibility can be anchored to platform purchase-origin and current-ownership records.	Confirmed project direction.
AS-05	Required seller evidence can be submitted and stored; the current direction for electronics includes screenshots and a product test video.	Documented; per-category requirements to be defined in design.
AS-06	Users access the platform through interfaces that support desktop and mobile screen sizes.	Documented.
AS-07	The six confirmed core capabilities remain the project's core scope, with Advanced Smart Filtering as a cross-cutting capability.	Confirmed.
AS-08	AI output can be validated before it affects filtering, ranking, eligibility, or ownership state.	Documented architectural principle.